```
# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py                启动并自动打开浏览器
    python run.py --no-browser   只启动服务，不打开浏览器
    python run.py --dev          开发模式（开启代码热重载与调试器）
    python run.py --port 5001    指定端口
默认以 127.0.0.1:5000 启动，就绪后自动打开浏览器。
"""
import os
import sys
import threading
import time
import webbrowser
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")   # 控制台中文正常显示
from app import create_app
from app.models import db
PORT = 5000
for i, a in enumerate(sys.argv):
    if a == "--port" and i + 1 < len(sys.argv):
        PORT = int(sys.argv[i + 1])
URL = "http://127.0.0.1:%d" % PORT
NO_BROWSER = "--no-browser" in sys.argv
DEV = "--dev" in sys.argv
app = create_app()
def open_browser_later(delay=1.2):
    """等服务工作起来再打开浏览器，避免打开时页面还没就绪。
    放在后台线程里，不用 sleep 阻塞服务启动；线程设为守护线程，
    用户 Ctrl+C 停止服务时不会残留。
    """
    def run():
        time.sleep(delay)
        try:
            webbrowser.open(URL)
        except Exception:
            pass            # 打开失败不影响服务本身
    t = threading.Thread(target=run, daemon=True)
    t.start()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()     # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)     # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("   历史遗留废弃矿山自然恢复适宜性评价系统  Ｖ１．０")
    print("=" * 58)
```

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    print("   访问地址：%s" % URL)
    print("   演示账号：admin      密码：admin123")
    print("   停止运行：按 Ctrl+C，或直接关闭本窗口")
    print("=" * 58)
    if not NO_BROWSER and not DEV:
        print("   正在打开浏览器……")
    print()
    # 开发模式才启用热重载（重载会二次启动进程，故此时不自动开浏览器）
    if not NO_BROWSER and not DEV:
        open_browser_later()
    app.run(host="127.0.0.1", port=PORT, debug=DEV,
    use_reloader=DEV)
# ==== app/__init__.py ====
# -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from datetime import date
from flask import Flask
from flask_login import LoginManager
from app.config import Config
from app.models import db, User
from app.utils.constants import ROLE_LABELS
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    # 注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report, search
    app.register_blueprint(auth.bp)
    app.register_blueprint(project.bp)
    app.register_blueprint(unit.bp)
    app.register_blueprint(standard.bp)
    app.register_blueprint(constraint.bp)
    app.register_blueprint(ahp.bp)
    app.register_blueprint(indicator.bp)
    app.register_blueprint(evaluate.bp)
    app.register_blueprint(report.bp)
    app.register_blueprint(search.bp)
    # 全局登录保护：除登录页与静态资源外，均需登录
    from flask import request, redirect, url_for,
    send_from_directory
    from flask_login import current_user, login_user
```

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PUBLIC_ENDPOINTS = {"auth.login", "static", "favicon"}
#  浏览器默认会请求根路径下的 /favicon.ico。
#  页面里虽已用 <link rel="icon"> 指向 static 目录，但出错页等场景仍会
#  回退到这个路径，补一条路由免得日志里一直刷 404。
@app.route("/favicon.ico")
def favicon():
    return send_from_directory(app.static_folder,
"favicon.ico",
                               mimetype="image/vnd.micro
soft.icon")
@app.before_request
def require_login():
    #  截图模式（仅用于生成说明书配图）：自动以管理员身份登录
    if app.config.get("SCREENSHOT_MODE"):
        if (request.endpoint not in ("auth.login",
"static")
                and not current_user.is_authenticated):
            admin = User.query.filter_by(role="admin").f
irst()
            if admin:
                login_user(admin)
        return None
    if request.endpoint in PUBLIC_ENDPOINTS or
request.endpoint is None:
        return None
    if not current_user.is_authenticated:
        return redirect(url_for("auth.login"))
    return None
#  侧边栏导航：按论文评价流程的顺序编排，分为「评价流程」与「系统管理」两组。
#  不使用模块编号，避免出现与实际流程不一致的代号。
#  每项带图标名（内联 SVG，见 templates/_macros.html 的 ICONS）
#  与可选的待办角标（该步骤还没做完的图斑数）。
FLOW_LINKS = [
    ("项目与矿山档案", "/project/", "folder"),
    ("政策与标准知识库", "/standard/", "book"),
    ("限制性指标判定", "/constraint/", "shield"),
    ("微地质单元", "/unit/", "layers"),
    ("指标体系与权重", "/ahp/", "sliders"),
    ("指标数据采集", "/indicator/", "database"),
    ("适宜性定量评价", "/evaluate/", "target"),
    ("报告与结果输出", "/report/", "file"),
]
ADMIN_LINKS = [
    ("用户与操作日志", "/users", "users"),
]
@app.context_processor
def inject_nav():
    path = request.path or "/"
    def hit(prefix):
        """当前路径是否属于某个入口（/unit/ 也匹配 /unit/3）。"""
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            return path.startswith(prefix.rstrip("/"))
      badges = {}
      if current_user.is_authenticated:
            from app.services import dashboard as dash
            badges = dash.nav_badges()
      def build(links):
            return [{"label": label, "href": href, "icon":
icon,
                     "active": hit(href), "badge":
badges.get(href, 0)}
                    for label, href, icon in links]
      groups = [
            {"title": "评价流程", "links": build(FLOW_LINKS)},
            {"title": "系统管理", "links": build(ADMIN_LINKS)},
      ]
      #  面包屑用的当前页名：按路径命中的入口取，未命中时给个通用名
      if path == "/":
            page_title = "工作台"
      elif path.startswith(("/users", "/logs")):
            page_title = "用户与操作日志"
      elif path.startswith("/search"):
            page_title = "搜索结果"
      else:
            page_title = "工作台"
            for label, href, _icon in FLOW_LINKS:
                  if hit(href):
                        page_title = label
                        break
      #  层级引导用：项目是归档单位，图斑才是评价对象。
      #  下游各页面按图斑组织数据，因此需要提示「哪些项目下还没有图斑」。
      from app.models import Project, MinePatch
      projects = Project.query.order_by(Project.id).all()
      return {
            "nav_groups": groups,
            "home_active": path == "/",
            "page_title": page_title,
            "today": date.today().strftime("%Y-%m-%d"),
            "search_q": request.args.get("q", ""),
            "role_labels": ROLE_LABELS,
            "project_count": len(projects),
            "patch_count": MinePatch.query.count(),
            "projects_without_patch": [p for p in projects
                                       if p.patches.count()
== 0],
      }
#  操作审计：统一在响应阶段记录成功的写操作（操作日志）
from app.services import audit
@app.after_request
def audit_writes(response):
      if request.method in ("POST", "PUT", "PATCH",
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    "DELETE"):
            audit.record(request.endpoint,
    response.status_code)
        return response
    # 错误页：给出可读提示，而不是默认的英文错误页
    from flask import render_template
    @app.errorhandler(403)
    def forbidden(e):
        return render_template("error.html", code=403,
                               message="没有权限执行该操作"), 403
    @app.errorhandler(404)
    def not_found(e):
        return render_template("error.html", code=404,
                               message="请求的页面不存在"), 404
    return app
# ==== app/config.py ====
# -*- coding: utf-8 -*-
"""应用配置。"""
import os
from datetime import timedelta
BASE_DIR = os.path.abspath(os.path.dirname(__file__))
class Config:
    SECRET_KEY = os.environ.get("SECRET_KEY",
    "mine-restoration-dev-key")
    # 单机部署，SQLite 数据库文件放在应用目录下
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE_DIR, "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项
    WTF_CSRF_ENABLED = True
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录
    REMEMBER_COOKIE_DURATION = timedelta(days=30)
    REMEMBER_COOKIE_HTTPONLY = True
    REMEMBER_COOKIE_SAMESITE = "Lax"
    # 影像与数据文件上传（遥感影像挂接、批量导入）
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限
    32 MB
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}
# ==== app/models.py ====
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from datetime import datetime
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from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（用户与权限管理）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
    nullable=False)
    # 角色：管理员 admin / 评价员 evaluator / 查看者 viewer
    role = db.Column(db.String(20), nullable=False,
    default="viewer")
    active = db.Column(db.Boolean, nullable=False,
    default=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    def to_dict(self):
        return {"id": self.id, "username": self.username,
                "role": self.role, "active": self.active}
class OperationLog(db.Model):
    """操作日志。"""
    __tablename__ = "operation_logs"
    id = db.Column(db.Integer, primary_key=True)
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"
    ), nullable=True)
    action = db.Column(db.String(128), nullable=False)
    # 操作类型
    target_type = db.Column(db.String(64), nullable=True)
    # 对象类型
    target_id = db.Column(db.Integer, nullable=True)
    # 对象主键
    detail = db.Column(db.Text, nullable=True)
    # 操作详情
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow, index=True)
class Project(db.Model):
    """评价项目（项目与矿山档案）。"""
    __tablename__ = "projects"
    id = db.Column(db.Integer, primary_key=True)
    name = db.Column(db.String(128), nullable=False)
    code = db.Column(db.String(64), nullable=True)
    # 项目编号
    description = db.Column(db.Text, nullable=True)
    status = db.Column(db.String(20), default="进行中")
    # 进行中 / 已完成 / 已归档
    created_by = db.Column(db.Integer,
    db.ForeignKey("users.id"), nullable=True)
    created_at = db.Column(db.DateTime,
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    default=datetime.utcnow)                                          301
    patches = db.relationship("MinePatch",                            302
    backref="project", lazy="dynamic",                                303
                                    cascade="all, delete-orphan")     304
class MinePatch(db.Model):                                            305
    """矿山图斑。"""                                                   306
    __tablename__ = "mine_patches"                                    307
    id = db.Column(db.Integer, primary_key=True)                      308
    project_id = db.Column(db.Integer,                                309
    db.ForeignKey("projects.id"), nullable=False)                     310
    name = db.Column(db.String(128), nullable=False)                  311
    admin_region = db.Column(db.String(128), nullable=True)           312
    # 所在行政区                                                      313
    center_x = db.Column(db.Float, nullable=True)                     314
    # 中心经度 / 横坐标                                               315
    center_y = db.Column(db.Float, nullable=True)                     316
    # 中心纬度 / 纵坐标                                               317
    area_ha = db.Column(db.Float, nullable=True)                      318
    # 图斑面积（公顷）                                                319
    mineral_type = db.Column(db.String(64), nullable=True)            320
    # 开采矿种                                                        321
    mining_method = db.Column(db.String(64), nullable=True)           322
    # 开采方式                                                        323
    stop_years = db.Column(db.Float, nullable=True)                   324
    # 停产年限                                                        325
    responsible_entity_missing = db.Column(db.Boolean,                326
    default=True)   # 责任主体是否缺失                               327
    # 遥感底图 / 无人机正射影像挂接                                   328
    image_path = db.Column(db.String(256), nullable=True)             329
    # 相对 static 的路径                                              330
    image_note = db.Column(db.String(256), nullable=True)             331
    # 影像说明（如来源、时相）                                        332
    image_uploaded_at = db.Column(db.DateTime,                        333
    nullable=True)                                                    334
    created_at = db.Column(db.DateTime,                               335
    default=datetime.utcnow)                                          336
    units = db.relationship("MicroUnit", backref="patch",             337
    lazy="dynamic",                                                   338
                                    cascade="all, delete-orphan")     339
class ReferenceEcosystem(db.Model):                                   340
    """参照生态系统：标准化基准值，与图斑绑定。"""                   341
    __tablename__ = "reference_ecosystems"                            342
    id = db.Column(db.Integer, primary_key=True)                      343
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat          344
    ches.id"), nullable=False)                                        345
    coverage_benchmark = db.Column(db.Float, default=75.0)            346
    # 植被覆盖度基准 %                                                347
    shannon_benchmark = db.Column(db.Float, default=0.7)              348
    # Shannon-Wiener 基准                                             349
    dominant_species = db.Column(db.String(256),                      350
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    nullable=True)   #  建群种                                                    351
    soil_benchmark = db.Column(db.String(256),                                   352
    nullable=True)      #  土壤基准                                              353
class Standard(db.Model):                                                        354
    """地方标准 / 政策文件（政策与标准知识库）。"""                                  355
    __tablename__ = "standards"                                                  356
    id = db.Column(db.Integer, primary_key=True)                                 357
    province = db.Column(db.String(32), nullable=False)                          358
    #  省份                                                                      359
    name = db.Column(db.String(256), nullable=False)                             360
    #  名称                                                                      361
    doc_number = db.Column(db.String(128), nullable=True)                        362
    #  标准编号 / 文号                                                           363
    publish_date = db.Column(db.String(32), nullable=True)                       364
    implement_date = db.Column(db.String(32), nullable=True)                     365
    #  类别：地方标准  /  政策文件                                               366
    category = db.Column(db.String(32), nullable=False,                          367
    default="地方标准")                                                          368
    summary = db.Column(db.Text, nullable=True)                                  369
    #  特点概述                                                                  370
    rules = db.relationship("StandardRule",                                      371
    backref="standard",                                                          372
                                    cascade="all, delete-orphan")                373
class StandardRule(db.Model):                                                    374
    """结构化规则：将标准条文拆解为可执行规则。"""                                   375
    __tablename__ = "standard_rules"                                             376
    id = db.Column(db.Integer, primary_key=True)                                 377
    standard_id = db.Column(db.Integer,                                          378
    db.ForeignKey("standards.id"), nullable=False)                               379
    item = db.Column(db.String(128), nullable=False)                             380
    #  判定项                                                                    381
    operator = db.Column(db.String(8), nullable=False)                           382
    #  比较符：>  >=  <  <=  ==                                                  383
    threshold = db.Column(db.String(64), nullable=True)                          384
    #  阈值                                                                      385
    condition_type = db.Column(db.String(32),                                    386
    default="必要条件")    #  必要条件 / 充分条件                               387
    region = db.Column(db.String(64), nullable=True)                             388
    #  适用分区（宁夏分区差异化）                                               389
    note = db.Column(db.Text, nullable=True)                                     390
class ConstraintCheck(db.Model):                                                 391
    """限制性判定记录。"""                                                       392
    __tablename__ = "constraint_checks"                                          393
    id = db.Column(db.Integer, primary_key=True)                                 394
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat                     395
    ches.id"), nullable=False)                                                   396
    policy_pass = db.Column(db.Boolean, nullable=True)                           397
    #  政策约束                                                                  398
    location_pass = db.Column(db.Boolean, nullable=True)                         399
    #  区位约束                                                                  400
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    pollution_pass = db.Column(db.Boolean, nullable=True)
    # 土壤污染
    geo_pass = db.Column(db.Boolean, nullable=True)
    # 地质灾害
    farmland_pass = db.Column(db.Boolean, nullable=True)
    # 耕地属性
    passed = db.Column(db.Boolean, nullable=True)
    # 总体是否通过
    reason = db.Column(db.Text, nullable=True)
    # 不通过原因
    checked_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class MicroUnit(db.Model):
    """微地质单元：岩质边坡 / 碎石边坡 / 平台。"""
    __tablename__ = "micro_units"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    unit_type = db.Column(db.String(32), nullable=False)
    # 单元类型
    area = db.Column(db.Float, nullable=True)
    # 面积
    ratio = db.Column(db.Float, nullable=True)
    # 面积占比（自动计算）
    description = db.Column(db.Text, nullable=True)
    # 现场描述
class Indicator(db.Model):
    """指标定义：指标树，含角色与启用单元类型。"""
    __tablename__ = "indicators"
    id = db.Column(db.Integer, primary_key=True)
    code = db.Column(db.String(32), unique=True,
    nullable=False)  # 指标编码
    name = db.Column(db.String(64), nullable=False)
    # 指标名称
    level = db.Column(db.Integer, nullable=False)
    # 1/2/3 级
    parent_code = db.Column(db.String(32), nullable=True)
    # 父指标编码
    # 角色：土壤污染指标同时作为限制性指标与评价指标
    is_constraint = db.Column(db.Boolean, default=False)
    is_evaluative = db.Column(db.Boolean, default=True)
    unit = db.Column(db.String(32), nullable=True)
    # 单位
    ideal_value = db.Column(db.String(64), nullable=True)
    # 理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    # benefit/cost/mid/discrete
    # 启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
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    # 取值依据说明
    def units_list(self):
        return [s for s in (self.enabled_units or
    "").split(",") if s]
class WeightScheme(db.Model):
    """权重方案：按微地质单元类型分别保存。"""
    __tablename__ = "weight_schemes"
    id = db.Column(db.Integer, primary_key=True)
    unit_type = db.Column(db.String(32), nullable=False)
    # 所属单元类型
    name = db.Column(db.String(64), nullable=True)
    # 方案名称
    level = db.Column(db.Integer, nullable=False,
    default=1)          # 1=一级指标 2=二级指标
    is_default = db.Column(db.Boolean, default=False)
    # 是否默认（论文表 3）
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class WeightItem(db.Model):
    """权重明细：方案内的各项权重。"""
    __tablename__ = "weight_items"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=False)
    indicator_code = db.Column(db.String(32),
    nullable=False)        # 指标编码
    weight = db.Column(db.Float, nullable=False)
    # 权重值
class ExpertMatrix(db.Model):
    """专家判断矩阵：多专家打分的原始矩阵。"""
    __tablename__ = "expert_matrices"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=True)
    # 直接标注适用对象，便于无需先建权重方案即可录入专家矩阵
    unit_type = db.Column(db.String(32), nullable=True)
    # 微地质单元类型
    l1_name = db.Column(db.String(64), nullable=True)
    # 所属一级指标（二级矩阵用）
    expert_name = db.Column(db.String(64), nullable=True)
    # 专家姓名
    level = db.Column(db.Integer, nullable=False,
    default=1)          # 矩阵对应指标层级
    matrix_json = db.Column(db.Text, nullable=False)
    # 判断矩阵（JSON 二维数组）
    cr = db.Column(db.Float, nullable=True)
    # 该专家矩阵的一致性比例
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class IndicatorData(db.Model):
```

```
    """指标实测数据：按微地质单元组织。"""                                    501
    __tablename__ = "indicator_data"                                        502
    id = db.Column(db.Integer, primary_key=True)                            503
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni                504
    ts.id"), nullable=False, unique=True)                                   505
    soil_ratio = db.Column(db.Float, nullable=True)                         506
    # 土石比                                                                507
    thickness = db.Column(db.Float, nullable=True)                          508
    # 土层厚度 cm                                                           509
    ph = db.Column(db.Float, nullable=True)                                 510
    organic_matter = db.Column(db.Float, nullable=True)                     511
    # 有机质 g/kg                                                           512
    # 重金属含量，JSON 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null      513
    heavy_metals = db.Column(db.Text, nullable=True)                        514
    coverage = db.Column(db.Float, nullable=True)                           515
    # 植被覆盖度 %                                                          516
    shannon = db.Column(db.Float, nullable=True)                            517
    # Shannon-Wiener 多样性指数                                             518
    slope = db.Column(db.Float, nullable=True)                              519
    # 坡度 °                                                                520
    aspect = db.Column(db.String(16), nullable=True)                        521
    # 坡向：阳坡/阴阳坡/阴坡                                                522
    def heavy_metals_dict(self):                                            523
        return json.loads(self.heavy_metals) if                             524
    self.heavy_metals else {}                                               525
class EvaluationResult(db.Model):                                           526
    """评价结果。"""                                                        527
    __tablename__ = "evaluation_results"                                    528
    id = db.Column(db.Integer, primary_key=True)                            529
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat                530
    ches.id"), nullable=False)                                              531
    final_score = db.Column(db.Float, nullable=True)                        532
    # 最终评价因子 F                                                        533
    grade = db.Column(db.String(32), nullable=True)                         534
    # 适宜性等级                                                            535
    measures = db.Column(db.Text, nullable=True)                            536
    # 推荐措施                                                              537
    unit_scores = db.Column(db.Text, nullable=True)                         538
    # 各单元得分 JSON                                                       539
    created_at = db.Column(db.DateTime,                                     540
    default=datetime.utcnow)                                                541
class ScoreDetail(db.Model):                                                542
    """得分明细：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""    543
    __tablename__ = "score_details"                                         544
    id = db.Column(db.Integer, primary_key=True)                            545
    result_id = db.Column(db.Integer,                                       546
    db.ForeignKey("evaluation_results.id"), nullable=False)                 547
    unit_id = db.Column(db.Integer, nullable=True)                          548
    l1_name = db.Column(db.String(64), nullable=True)                       549
    # 一级指标                                                              550
```

```
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
from wtforms.validators import DataRequired, Length
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES
class AHPSelectForm(FlaskForm):
    """选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  # 动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """登录表单。支持「记住我」以在关闭浏览器后保持登录状态。"""
    username = StringField("用户名", validators=[DataRequired()
    , Length(max=64)])
    password = PasswordField("密码", validators=[DataRequired(
    )])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
# ==== app/utils/constants.py ====
# -*- coding: utf-8 -*-
"""评价指标常量表。
全部数值取自论文，作为系统计算的唯一基准：
```

```
- 论文 ４．２．２ 节：指标标准化理想极值与取值依据
- 论文 ４．２．３ 节 / 表 ４：适宜性等级划分
- 论文表 ３：按微地质单元分别确定的一级、二级指标权重
- 论文 ４．２．１ 节：平均随机一致性指标 ＲＩ
"""
# ----------------------------------------------------------------
    ----------
# 一、微地质单元类型
# ----------------------------------------------------------------
    ----------
# 论文 ４．１ 节："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡
# 三种地质微单元类型"
UNIT_TYPES = ("岩质边坡", "碎石边坡", "平台")
# 每个单元类型启用的指标类别（论文表 ３ 隐含：岩质边坡无土壤指标，平台无地形指标）
# 供数据录入界面与评价引擎判断"该单元是否需要某类指标"
UNIT_ENABLED_INDICATORS = {
    "岩质边坡": ("植被指标", "地形指标"),                    # 不设土壤指标
    "碎石边坡": ("土壤指标", "植被指标", "地形指标"),
    "平台": ("土壤指标", "植被指标"),                        # 不设地形指标
}
# ----------------------------------------------------------------
    ----------
# 二、指标标准化理想极值（论文 ４．２．２ 节）
# ----------------------------------------------------------------
    ----------
IDEAL_SOIL_RATIO = 0.8          # 土石比理想值
IDEAL_THICKNESS = 20.0          # 土层厚度理想值 （ｃｍ）
IDEAL_PH_RANGE = (5.0, 9.0)     # ｐＨ 理想区间（适中型指标）
IDEAL_ORGANIC = 30.0            # 有机质理想值
    （ｇ/ｋｇ），全国第二次土壤普查养分等级划分标准二等最低值
IDEAL_COVERAGE = 75.0           # 植被覆盖度参照生态系统 （％）
IDEAL_SHANNON = 0.7             # Ｓｈａｎｎｏｎ-Ｗｉｅｎｅｒ 多样性指数参照生态系统
# 重金属极值 （ｍｇ/ｋｇ），《绿化种植土壤》ＣＪ/Ｔ ３４０—２０１６ 土壤重金属含量 ＩＶ 级最高值
# 论文原文：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、５３０、４００、４５、２２０、６００、８００
HEAVY_METALS = ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu",
    "Zn")
HEAVY_METAL_LIMIT = {
    "Cd": 2.0, "Hg": 2.0, "Pb": 530.0, "Cr": 400.0,
    "As": 45.0, "Ni": 220.0, "Cu": 600.0, "Zn": 800.0,
}
# 污染指标下 8 种重金属"重要性相当，权重平均分配"（论文 ４．２．１ 节）
HEAVY_METAL_EQUAL_WEIGHT = 1.0 / 8.0
# 坡度分界，论文 ４．２．２ 节引《新一轮退耕还林还草总体方案》发改西部〔２０１４〕１７７２ 号
SLOPE_BREAK = 25.0
# 坡向离散赋值，论文 ４．２．２ 节："阳坡、阴阳坡、阴坡分别赋值 ０．２、０．６、１"
ASPECT_SCORES = {"阳坡": 0.2, "阴阳坡": 0.6, "阴坡": 1.0}
ASPECT_TYPES = ("阳坡", "阴阳坡", "阴坡")
# ｐＨ 适中型标准化的容差边界（区间外线性衰减，避免得分突变）
PH_TOL_LOW = 4.0
PH_TOL_HIGH = 10.0
```

```
# ---------------------------------------------------------------- 651
    ----------------                                               652
# 三、适宜性等级划分（论文表 4）                                   653
# ---------------------------------------------------------------- 654
    ----------------                                               655
GRADE_LEVELS = (                                                   656
    {"grade": "不适宜", "min": None, "max": 0.3,                   657
     "meaning": "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复",  658
     "measure": "生态重建、人工辅助"},                             659
    {"grade": "基本适宜", "min": 0.3, "max": 0.7,                  660
     "meaning": "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求",  661
     "measure": "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"},  662
    {"grade": "适宜", "min": 0.7, "max": None,                     663
     "meaning": "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快",  664
     "measure": "自然恢复"},                                       665
)                                                                  666
def grade_of(score):                                               667
    """按论文表 4 判定适宜性等级。"""                              668
    for g in GRADE_LEVELS:                                         669
        lo, hi = g["min"], g["max"]                                670
        if lo is None and score < hi:                              671
            return g                                               672
        if hi is None and score >= lo:                             673
            return g                                               674
        if lo is not None and hi is not None and lo <=             675
    score < hi:                                                    676
            return g                                               677
    # score 恰好等于 0.7 时落入"适宜"                              678
    return GRADE_LEVELS[-1]                                        679
# ---------------------------------------------------------------- 680
    ----------------                                               681
# 四、平均随机一致性指标 RI（论文 4.2.1 节给出 n=2,3,4,5，其余为常用扩展值）  682
# ---------------------------------------------------------------- 683
    ----------------                                               684
RI_TABLE = {1: 0.0, 2: 0.0, 3: 0.58, 4: 0.90, 5: 1.12, 6:          685
    1.24,                                                          686
            7: 1.32, 8: 1.41, 9: 1.45, 10: 1.49}                   687
# 一致性比例阈值（论文：当 CR<0.1 时，判断矩阵满足一致性要求）     688
CR_THRESHOLD = 0.1                                                 689
# ---------------------------------------------------------------- 690
    ----------------                                               691
# 五、默认权重方案（论文表 3）                                     692
# ---------------------------------------------------------------- 693
    ----------------                                               694
# 论文 4.2.2 节："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，  695
# 分别计算评价指标权重"（论文摘要、图 2 流程与表 5 均作"碎石边坡"，  696
# 系统统一采用"碎石边坡"）。权重由 10 位专家打分后经 AHP 方法确定。  697
# 下表为论文表 3 所载的归一化权重，作为系统默认方案。             698
DEFAULT_WEIGHTS = {                                                699
    "岩质边坡": {                                                  700
```

```
            "ｌ1": {"植被指标": 0.45, "地形指标": 0.55},
            "ｌ2": {
                "植被指标": {"植被覆盖度": 0.67, "生物多样性": 0.33},
                "地形指标": {"坡度": 0.64, "坡向": 0.36},
            },
        },
        "碎石边坡": {
            "ｌ1": {"土壤指标": 0.44, "植被指标": 0.31, "地形指标": 0.25},
            "ｌ2": {
                "土壤指标": {"土石比": 0.25, "土层厚度": 0.25, "pH": 0.17,
                             "有机质": 0.17, "污染指标": 0.17},
                "植被指标": {"植被覆盖度": 0.64, "生物多样性": 0.36},
                "地形指标": {"坡度": 0.67, "坡向": 0.33},
            },
        },
        "平台": {
            "ｌ1": {"土壤指标": 0.54, "植被指标": 0.46},
            "ｌ2": {
                "土壤指标": {"土石比": 0.25, "土层厚度": 0.25, "pH": 0.17,
                             "有机质": 0.17, "污染指标": 0.17},
                "植被指标": {"植被覆盖度": 0.61, "生物多样性": 0.39},
            },
        },
}
# ----------------------------------------------------------------------
    ------------------
# 六、限制性指标（论文表 2 + 4.1 节流程）
# ----------------------------------------------------------------------
    ------------------
# 论文摘要与 4.1 节：四项限制性指标；流程段另明确"是否为耕地"为约束条件
CONSTRAINT_ITEMS = ("政策", "区位约束", "土壤污染", "地质灾害", "耕地属性")
# ----------------------------------------------------------------------
    ------------------
# 七、用户角色
# ----------------------------------------------------------------------
    ------------------
ROLE_LABELS = {"admin": "管理员", "evaluator": "评价员",
    "viewer": "查看者"}
# ==== app/utils/schema.py ====
# -*- coding: utf-8 -*-
"""轻量结构迁移。
SQLite 下 `db.create_all()` 只建缺失的表，不会给已有表补列。
本模块在启动时检查目标列是否存在，缺失则 ALTER TABLE 补上，
使旧数据库无需重建即可用上新字段。
"""
from sqlalchemy import text
# （表名， 列名， 列定义）
REQUIRED_COLUMNS = [
    # 遥感影像挂接
    ("mine_patches", "image_path", "VARCHAR(256)"),
```

```
    ("mine_patches", "image_note", "VARCHAR(256)"),
    ("mine_patches", "image_uploaded_at", "DATETIME"),
    # 指标维护
    ("indicators", "enabled_units", "VARCHAR(128)"),
    ("indicators", "description", "TEXT"),
    # 多专家矩阵
    ("expert_matrices", "unit_type", "VARCHAR(32)"),
    ("expert_matrices", "l1_name", "VARCHAR(64)"),
    ("expert_matrices", "cr", "FLOAT"),
]
def ensure_schema(db):
    """为已有表补齐缺失列。返回补列清单。"""
    added = []
    try:
        with db.engine.connect() as conn:
            for table, column, ddl in REQUIRED_COLUMNS:
                rows = conn.execute(text("PRAGMA
    table_info(%s)" % table)).fetchall()
                if not rows:
                    continue                                #
    表还不存在，create_all 会建
                existing = {r[1] for r in rows}
                if column in existing:
                    continue
                conn.execute(text(
                    "ALTER TABLE %s ADD COLUMN %s %s" %
    (table, column, ddl)))
                conn.commit()
                added.append("%s.%s" % (table, column))
    except Exception as e:                                  #
    迁移失败不应阻断启动
        print("[schema] 结构检查跳过：%s" % e)
    return added
# ==== app/utils/uploads.py ====
# -*- coding: utf-8 -*-
"""上传文件处理（遥感影像、批量导入）。"""
import os
import re
import uuid
from datetime import datetime
from werkzeug.utils import secure_filename
def _safe_ext(filename):
    return os.path.splitext(filename)[1].lower().lstrip(".")
def ext_allowed(filename, allowed):
    return _safe_ext(filename) in allowed
def save_upload(file_storage, subdir, allowed_ext,
    upload_root):
    """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。
    文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。
    """
```

```
    if file_storage is None or not file_storage.filename:
        return None, "未选择文件"
    ext = _safe_ext(file_storage.filename)
    if ext not in allowed_ext:
        return None, "不支持的文件类型：.%s" % ext
    target_dir = os.path.join(upload_root, subdir)
    os.makedirs(target_dir, exist_ok=True)
    # 仅保留安全字符，并加时间戳与随机串防重名
    base = secure_filename(os.path.splitext(file_storage.fil
    ename)[0]) or "file"
    base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or
    "file"
    name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M
    %S"),
                         uuid.uuid4().hex[:6], ext)
    abs_path = os.path.join(target_dir, name)
    file_storage.save(abs_path)
    rel = os.path.join("uploads", subdir,
    name).replace("\\", "/")
    return rel, None
def remove_upload(rel_path, upload_root):
    """删除已保存的上传文件（相对 static 的路径）。"""
    if not rel_path:
        return
    abs_path = os.path.join(upload_root,
    rel_path.replace("uploads/", "", 1))
    try:
        if os.path.isfile(abs_path):
            os.remove(abs_path)
    except OSError:
        pass
# ==== app/utils/charts.py ====
# -*- coding: utf-8 -*-
"""轻量图表。
用纯 SVG 输出，不依赖任何绘图库或前端脚本：
- 服务端渲染，页面与打印（PDF）中都能稳定显示
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）
- 单色方案，符合界面「一个视图一个强调色」的约定
"""
from html import escape
ACCENT = "#1f3a5f"
ACCENT_SOFT = "#7f93ad"
INK = "#23262b"
MUTED = "#6b7280"
LINE = "#dfe3e8"
# 灰度梯度：黑白打印时相邻扇区 / 条柱仍可区分
GRAYS = ["#1f3a5f", "#456188", "#6d86a8", "#96a9c4",
    "#bcc9da", "#dbe3ec"]
def _fmt(v, digits=4):
    try:
```

```
            return ("%%.%df" % digits) % float(v)                    851
        except (TypeError, ValueError):                              852
            return str(v)                                            853
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,           854
    value_w=76):                                                     855
    """水平条形图。items: [(标签, 数值, 显示文本)]"""                  856
    if not items:                                                    857
        return '<p class="muted">暂无数据</p>'                       858
    n = len(items)                                                   859
    height = n * (bar_h + gap) + 10                                  860
    max_v = max((abs(float(v)) for _, v, _ in items),                861
    default=1) or 1                                                  862
    body_w = width - label_w - value_w                               863
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"                 864
    height="%d" '                                                    865
              'role="img" aria-label="条形图" '                      866
              'style="max-width:%dpx;font-family:inherit;">'         867
    % (width, height, height, width)]                                868
    for i, (label, value, disp) in enumerate(items):                 869
        y = i * (bar_h + gap) + 5                                    870
        frac = (abs(float(value)) / max_v) if max_v else 0           871
        w = max(2.0, body_w * frac)                                  872
        fill = GRAYS[i % len(GRAYS)]                                 873
        parts.append(                                                874
            '<text x="%d" y="%d" text-anchor="end"                   875
    font-size="12" fill="%s">%s</text>'                              876
            % (label_w - 10, y + bar_h * 0.72, INK,                  877
    escape(str(label))))                                             878
        parts.append(                                                879
            '<rect x="%d" y="%d" width="%.1f" height="%d"            880
    rx="3" fill="%s"/>'                                              881
            % (label_w, y, w, bar_h, fill))                          882
        parts.append(                                                883
            '<text x="%d" y="%d" font-size="12" fill="%s" '          884
            'style="font-variant-numeric:tabular-nums;">%s</         885
    text>'                                                           886
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,             887
    escape(str(disp))))                                              888
    parts.append('</svg>')                                           889
    return "".join(parts)                                            890
def donut(items, size=210, thickness=34):                            891
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]           892
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。           893
    """                                                              894
    vals = [(l, float(v), d) for l, v, d in items if                 895
    float(v) > 0]                                                    896
    if not vals:                                                     897
        return '<p class="muted">暂无数据</p>'                       898
    total = sum(v for _, v, _ in vals) or 1                          899
    r = (size - thickness) / 2                                       900
```

```
    cx = cy = size / 2                                                  901
    import math                                                         902
    parts = ['<svg viewBox="0 0 %d %d" width="%d"                       903
    height="%d" role="img" '                                            904
              'aria-label="权重分布环形图">' % (size, size, size,       905
    size)]                                                              906
    angle = -math.pi / 2                                                907
    for i, (label, v, disp) in enumerate(vals):                         908
        sweep = 2 * math.pi * (v / total)                               909
        a0, a1 = angle, angle + sweep                                   910
        x0, y0 = cx + r * math.cos(a0), cy + r *                        911
    math.sin(a0)                                                        912
        x1, y1 = cx + r * math.cos(a1), cy + r *                        913
    math.sin(a1)                                                        914
        large = 1 if sweep > math.pi else 0                             915
        parts.append(                                                   916
            '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f               917
    %.2f" '                                                             918
            'fill="none" stroke="%s" stroke-width="%d"/>'               919
           % (x0, y0, r, r, large, x1, y1, GRAYS[i %                    920
    len(GRAYS)], thickness))                                            921
        angle = a1                                                      922
    parts.append('<text x="%d" y="%d" text-anchor="middle"              923
    font-size="12" '                                                    924
                 'fill="%s">合计</text>' % (cx, cy - 2,                 925
    MUTED))                                                             926
    parts.append('<text x="%d" y="%d" text-anchor="middle"              927
    font-size="15" '                                                    928
                 'fill="%s" style="font-variant-numeric:tabu            929
    lar-nums;">%s</text>'                                               930
                 % (cx, cy + 18, INK, _fmt(total, 4)))                  931
    parts.append('</svg>')                                              932
    legend = ['<table style="width:auto;margin-top:10px;"><t            933
    head><tr>'                                                          934
              '<th scope="col">项目</th><th                             935
    scope="col">数值</th>'                                              936
              '<th scope="col">占比</th></tr></thead><tbody>']          937
    for i, (label, v, disp) in enumerate(vals):                         938
        legend.append(                                                  939
            '<tr><td style="text-align:left;">'                         940
            '<span style="display:inline-block;width:10px;he            941
    ight:10px;'                                                         942
            'border-radius:2px;background:%s;margin-right:6p            943
    x;"></span>%s</td>'                                                 944
            '<td class="mono">%s</td><td                                945
    class="mono">%.1f%%</td></tr>'                                      946
           % (GRAYS[i % len(GRAYS)], escape(str(label)),                947
    escape(str(disp)),                                                  948
               100.0 * v / total))                                      949
    legend.append('</tbody></table>')                                   950
```

```
        return ('<div style="display:flex;flex-wrap:wrap;gap:20p
        x;align-items:center;">'
                + "".join(parts) + "".join(legend) + '</div>')
# ==== app/utils/browser.py ====
# -*- coding: utf-8 -*-
"""HTML 转 PDF（报告导出）。
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，
避免中文路径在 file:// URL 下出问题。
"""
import os
import shutil
import subprocess
import tempfile
import uuid
CANDIDATES = [
    r"C:\Program Files\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe",
    r"C:\Program Files\Microsoft\Edge\Application\msedge.exe",
]
# 允许通过环境变量覆盖，便于在其他机器上指定浏览器
ENV_KEY = "DSH_BROWSER_PATH"
def browser_path():
    """定位可用的浏览器可执行文件，找不到返回 None。"""
    override = os.environ.get(ENV_KEY)
    if override and os.path.isfile(override):
        return override
    for p in CANDIDATES:
        if os.path.isfile(p):
            return p
    for name in ("chrome", "msedge", "chromium"):
        found = shutil.which(name)
        if found:
            return found
    return None
def html_to_pdf(html, timeout=90):
    """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""
    exe = browser_path()
    if exe is None:
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")
    tag = uuid.uuid4().hex[:8]
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)
```

```
    try:
        with open(html_path, "w", encoding="utf-8") as f:
            f.write(html)
        cmd = [
            exe, "--headless=new", "--disable-gpu",
    "--no-sandbox",
            "--no-first-run", "--no-default-browser-check",
            "--no-pdf-header-footer",
            "--virtual-time-budget=4000",
            "--print-to-pdf=%s" % pdf_path,
            "file:///" + html_path.replace("\\", "/"),
        ]
        p = subprocess.run(cmd, capture_output=True,
    text=True,
                           timeout=timeout,
    encoding="utf-8", errors="replace")
        if not os.path.isfile(pdf_path) or
    os.path.getsize(pdf_path) < 800:
            return None, "浏览器未能生成 PDF（%s）" % (p.stderr or
    "").strip()[:200]
        with open(pdf_path, "rb") as f:
            return f.read(), None
    except subprocess.TimeoutExpired:
        return None, "生成 PDF 超时，请稍后重试"
    except Exception as e:
        return None, "生成 PDF 失败：%s" % e
    finally:
        shutil.rmtree(tmpdir, ignore_errors=True)
# ==== app/services/ahp_service.py ====
# -*- coding: utf-8 -*-
"""AHP 权重引擎。
严格按论文 4.2.1 节实现：
- 式(2)(3)(4)：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式(5)(6)(7)：一致性检验（λmax、CI、CR）
- 论文要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
    """由判断矩阵上三角构造完整互反矩阵。
    参数
    ----
    upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
        例：3 阶时传入 [[a12, a13], [a23]]。
        对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
    """
    upper = np.asarray(upper, dtype=float)
    k = upper.shape[0]
    n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
    A = np.ones((n, n), dtype=float)
```

```
    idx = 0
    for i in range(n):
        for j in range(i + 1, n):
            A[i, j] = upper[idx]
            A[j, i] = 1.0 / upper[idx]
            idx += 1
    return A
def weights_and_consistency(A):
    """由判断矩阵 A 计算权重与一致性指标。
    返回
    ----
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式(2) 行元素乘积
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
    return {
        "weights": weights,
        "lambda_max": lambda_max,
        "ci": ci,
        "cr": cr,
        "n": n,
        "consistent": cr < CR_THRESHOLD,
    }
def find_most_inconsistent(A):
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
    ----
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)
    及对应取值。
    """
    A = np.asarray(A, dtype=float)
    r = weights_and_consistency(A)
    w = r["weights"]
```

```
    n = A.shape[0]
    best = None
    for i in range(n):
        for j in range(i + 1, n):
            expected = w[i] / w[j]
            if expected == 0:
                continue
            deviation = abs(A[i, j] - expected) / expected
            if best is None or deviation >
    best["deviation"]:
                best = {"i": i, "j": j, "value": A[i, j],
                        "expected": expected, "deviation":
    deviation}
    return best
def aggregate_matrices(matrices):
    """多专家判断矩阵聚合：几何平均（保持互反性）。
    参数
    ----
    matrices : list[np.ndarray]，多个同阶判断矩阵。
    """
    stack = np.stack([np.asarray(m, dtype=float) for m in
    matrices])
    return np.prod(stack, axis=0) ** (1.0 / len(stack))
def judge_from_weights(weights):
    """由权重向量构造完全一致判断矩阵并检验。
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。
    """
    w = np.asarray(weights, dtype=float)
    n = w.shape[0]
    A = np.outer(w, 1.0 / w)
    r = weights_and_consistency(A)
    r["matrix"] = A
    return r
# ==== app/services/standardization.py ====
# -*- coding: utf-8 -*-
"""指标标准化（的一部分）。
按论文 4.2.2 节式(8) 实现：
    x' = (x - x_min) / (x_max - x_min)
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。
所有理想极值取自 app.utils.constants，可配置。
"""
from app.utils.constants import (
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,
    PH_TOL_LOW, PH_TOL_HIGH,
)
def std_benefit(value, ideal):
```

```
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, value / ideal))
def std_cost(value, extreme):
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, 1.0 - value / extreme))
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,
    tol_hi=PH_TOL_HIGH):
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    论文 4.2.2 节：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（论文 4.2.2 节）。"""
    if aspect is None:
        return None
    return ASPECT_SCORES.get(aspect)
def pollution_score(heavy_metals):
    """污染指标：8 种重金属等权（各 0.125）合成。
    论文注：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染（=1）处理。
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。
    """
    total = 0.0
    for metal, extreme in HEAVY_METAL_LIMIT.items():
        value = heavy_metals.get(metal)
        total += std_cost(value, extreme) if value is not
    None else 1.0
    return total / len(HEAVY_METAL_LIMIT)
def standardize_unit(unit_type, data, weights=None):
    """对单个微地质单元的全部指标做标准化。
    参数
    ----
    unit_type : str，岩质边坡 / 碎石边坡 / 平台
```

```
    data : dict，实测数据，字段见接口契约：
        soil_ratio, thickness, ph, organic_matter,
    heavy_metals(dict),
        coverage, shannon, slope, aspect
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIGHTS 结构），
                未提供时用默认值。
    返回
    -----
    dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }
    标准化值可能为 None，表示该指标不适用或方法待定。
    """
    from app.utils.constants import DEFAULT_WEIGHTS
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    l1_enabled = weights["l1"].keys()
    l2w = weights["l2"]
    result = {}
    # ----- 土壤指标 -----
    if "土壤指标" in l1_enabled:
        result["土壤指标"] = {
            "土石比": (data.get("soil_ratio"),
                       std_benefit(data.get("soil_ratio"), IDEAL_SOIL_RATIO),
                       l2w["土壤指标"]["土石比"]),
            "土层厚度": (data.get("thickness"),
                         std_benefit(data.get("thickness"), IDEAL_THICKNESS),
                         l2w["土壤指标"]["土层厚度"]),
            "pH": (data.get("ph"),
                   std_mid_ph(data.get("ph"), *IDEAL_PH_RANGE),
                   l2w["土壤指标"]["pH"]),
            "有机质": (data.get("organic_matter"),
                       std_benefit(data.get("organic_matter"), IDEAL_ORGANIC),
                       l2w["土壤指标"]["有机质"]),
            "污染指标": (data.get("heavy_metals") or {},
                         pollution_score(data.get("heavy_metals") or {}),
                         l2w["土壤指标"]["污染指标"]),
        }
    # ----- 植被指标 -----
    if "植被指标" in l1_enabled:
        result["植被指标"] = {
            "植被覆盖度": (data.get("coverage"),
                           std_benefit(data.get("coverage"), IDEAL_COVERAGE),
                           l2w["植被指标"]["植被覆盖度"]),
            "生物多样性": (data.get("shannon"),
```

```
                                    std_benefit(data.get("shannon"),
    IDEAL_SHANNON),
                                    l2w["植被指标"]["生物多样性"]),
            }
        # ---- 地形指标 ----
        if "地形指标" in l1_enabled:
            result["地形指标"] = {
                "坡度": (data.get("slope"),
                         std_slope(data.get("slope")),
                         l2w["地形指标"]["坡度"]),
                "坡向": (data.get("aspect"),
                         aspect_score(data.get("aspect")),
                         l2w["地形指标"]["坡向"]),
            }
        return result
# ==== app/services/weight_service.py ====
# -*- coding: utf-8 -*-
"""权重方案的存取与生效权重解析。
系统有两套权重来源：
1. 论文表 3 给出的默认权重（app.utils.constants.DEFAULT_WEIGHTS），
   是**权威取值来源**，案例复算与报告结论以此为准；
2. 通过层次分析法在本系统中实际算出的权重，保存在
   weight_schemes / weight_items 两张表中，可标记为「生效」。
本模块负责把这两者合成为计算引擎直接可用的嵌套权重结构：
以论文表 3 为底，用生效方案中已保存的条目覆盖。
没有任何生效方案时，即完全使用论文表 3 的权重。
这样既保证默认行为与论文一致，又使 AHP 计算结果真正参与评价，
而不是算完即弃。
"""
import copy
from app.models import db, WeightScheme, WeightItem
from app.utils.constants import DEFAULT_WEIGHTS
from app.services.indicator_service import TREE
# 方案名称：每个微地质单元类型固定一个可生效的方案
ACTIVE_NAME = "自定义权重方案（AHP 计算结果）"
def _code_maps():
    """由指标树导出 编码 → 名称 的映射，避免名称在两处各写一遍。"""
    l1, l2 = {}, {}
    for row in TREE:
        code, name, level, parent = row[0], row[1], row[2],
    row[3]
        if level == 1:
            l1[code] = name
        elif level == 2:
            l2[code] = (l1.get(parent, parent), name)
    return l1, l2
def effective_weights(unit_type):
    """返回该单元类型当前生效的权重。
    返回 (weights, source_label, scheme)
        weights        嵌套权重结构，可直接交给标准化与评价引擎
```

```
        source_label  权重来源说明，用于界面与报告标注                                1301
        scheme        生效的权重方案对象，无自定义方案时为 None                       1302
    """                                                                              1303
    base = DEFAULT_WEIGHTS.get(unit_type)                                            1304
    if base is None:                                                                 1305
        return None, "未定义该微地质单元类型的权重", None                            1306
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ                         1307
    e,                                                                               1308
                                                                                     1309
    name=ACTIVE_NAME).first()                                                        1310
    if scheme is None:                                                               1311
        return copy.deepcopy(base), "论文表 3 默认权重", None                        1312
    merged = copy.deepcopy(base)                                                     1313
    l1_map, l2_map = _code_maps()                                                    1314
    applied = 0                                                                      1315
    for item in WeightItem.query.filter_by(scheme_id=scheme.                         1316
    id).all():                                                                       1317
        code, w = item.indicator_code, item.weight                                   1318
        if w is None or w <= 0:                                                      1319
            continue                                                                 1320
        if code in l1_map and l1_map[code] in merged["l1"]:                          1321
            merged["l1"][l1_map[code]] = w                                           1322
            applied += 1                                                             1323
        elif code in l2_map:                                                         1324
            l1_name, l2_name = l2_map[code]                                          1325
            if l1_name in merged["l2"] and l2_name in                                1326
    merged["l2"][l1_name]:                                                           1327
                merged["l2"][l1_name][l2_name] = w                                   1328
                applied += 1                                                         1329
    if applied == 0:                                                                 1330
        return copy.deepcopy(base), "论文表 3 默认权重", None                        1331
    return merged, "自定义方案（%d 项覆盖论文表 3）" % applied, scheme               1332
def save_weights(unit_type, code_weight_pairs, note=None):                           1333
    """把一组「指标编码 → 权重」写入生效方案（已存在则更新对应条目）。              1334
    仅覆盖本次传入的条目，其余条目保留，便于分层次（一级 / 二级）                   1335
    分次录入而无需一次填完。                                                        1336
    """                                                                              1337
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ                         1338
    e,                                                                               1339
                                                                                     1340
    name=ACTIVE_NAME).first()                                                        1341
    if scheme is None:                                                               1342
        scheme = WeightScheme(unit_type=unit_type,                                   1343
    name=ACTIVE_NAME,                                                                1344
                                  level=1, is_default=True)                          1345
        db.session.add(scheme)                                                       1346
        db.session.flush()                                                           1347
    if note:                                                                         1348
        scheme.level = scheme.level or 1                                             1349
    if not scheme.is_default:                                                        1350
```

```
            scheme.is_default = True                                    1351
        written = 0                                                     1352
        for code, w in code_weight_pairs:                               1353
            if w is None or w <= 0:                                     1354
                continue                                                1355
            item = WeightItem.query.filter_by(scheme_id=scheme.i        1356
        d,                                                              1357
                                                                        1358
        indicator_code=code).first()                                    1359
            if item is None:                                            1360
                item = WeightItem(scheme_id=scheme.id,                  1361
        indicator_code=code)                                            1362
                db.session.add(item)                                    1363
            item.weight = float(w)                                      1364
            written += 1                                                1365
        db.session.commit()                                             1366
        return scheme, written                                          1367
def reset_weights(unit_type):                                           1368
        """删除该单元类型的自定义方案，恢复为论文表 3 的默认权重。"""                1369
        schemes = WeightScheme.query.filter_by(unit_type=unit_ty        1370
        pe,                                                             1371
                                                                        1372
        name=ACTIVE_NAME).all()                                         1373
        n = 0                                                           1374
        for s in schemes:                                               1375
            WeightItem.query.filter_by(scheme_id=s.id).delete(          1376
                synchronize_session=False)                              1377
            db.session.delete(s)                                        1378
            n += 1                                                      1379
        db.session.commit()                                             1380
        return n                                                        1381
def scheme_items(unit_type):                                            1382
        """返回该单元类型已保存的自定义权重条目 {指标编码： 权重}。"""                1383
        scheme = WeightScheme.query.filter_by(unit_type=unit_typ        1384
        e,                                                              1385
                                                                        1386
        name=ACTIVE_NAME).first()                                       1387
        if scheme is None:                                              1388
            return {}, None                                             1389
        items = {i.indicator_code: i.weight                             1390
                 for i in WeightItem.query.filter_by(scheme_id=s        1391
        cheme.id).all()}                                                1392
        return items, scheme                                            1393
def summary():                                                          1394
        """各单元类型的权重来源概览，供权重首页展示。"""                          1395
        rows = []                                                       1396
        for unit_type in DEFAULT_WEIGHTS:                               1397
            weights, label, scheme = effective_weights(unit_type        1398
        )                                                               1399
            rows.append({                                               1400
```

```
                "unit_type": unit_type,
                "label": label,
                "custom": scheme is not None,
            })
        return rows
# ==== app/services/evaluation_service.py ====
# -*- coding: utf-8 -*-
"""适宜性评价引擎。
按论文式(９) 实现：
- 单元级：F' = Σ（二级指标权重 × 该二级指标下三级指标标准化值）
- 矿山级：F = Σ（Si × F'i），Si 为各微地质单元面积占比
- 等级判定按论文表 4。
"""
from app.utils.constants import DEFAULT_WEIGHTS, grade_of
from app.services import standardization as std
def score_unit(unit_type, data, weights=None):
    """计算单个微地质单元的评价因子 F'。
    返回
    ----
    dict：{ l1_scores: {一级指标: 得分}, unit_score: F', details:
    [...] }
    若某一级指标下存在"不适用"（标准化值为 None）的二级指标，
    则按其余项权重归一化后计算，保证单元得分始终有定义。
    """
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    std_result = std.standardize_unit(unit_type, data,
    weights)
    l1_weights = weights["l1"]
    l1_scores = {}
    details = []
    for l1_name, items in std_result.items():
        wsum = 0.0
        acc = 0.0
        for l2_name, (raw, std_value, w2) in items.items():
            if std_value is None:
                # 不适用项：不计入
                details.append({
                    "l1": l1_name, "l2": l2_name, "raw":
    raw,
                    "std_value": None, "weight": w2,
    "contribution": None,
                })
                continue
            contribution = std_value * w2
            acc += contribution
            wsum += w2
            details.append({
                "l1": l1_name, "l2": l2_name, "raw": raw,
                "std_value": std_value, "weight": w2,
```

```
                    "contribution": contribution,
                })
            l1_scores[l1_name] = acc / wsum if wsum > 0 else 0.0
        # 一级指标权重按实际参与计算的指标归一化。
        # 论文表 3 的权重本身已归一（和为 1），此处为恒等变换；
        # 若某层指标缺失或自定义权重不完整，可避免单元得分被整体压低。
        lw_sum = sum(l1_weights[k] for k in l1_scores if k in
        l1_weights)
        if lw_sum > 0:
            unit_score = sum(l1_weights[k] * l1_scores[k]
                             for k in l1_scores if k in
        l1_weights) / lw_sum
        else:
            unit_score = 0.0
        return {"l1_scores": l1_scores, "unit_score":
        unit_score, "details": details}
def score_mine(units, weights_map=None):
        """计算整个矿山的评价因子 F 与适宜性等级。
        参数
        ----
        units : list[dict]，每个元素：
            {unit_type, ratio, data}
            ratio 为该单元面积占比（0~1），data 为实测数据 dict。
        weights_map : dict，{单元类型: 权重结构}。未提供的单元类型
            使用论文表 3 的默认权重。权重可来自 AHP 计算并经
            weight_service 保存生效，从而真正参与评价。
        返回
        ----
        dict:{ final_score, grade, grade_info, unit_results }
        """
        weights_map = weights_map or {}
        total = 0.0
        unit_results = []
        for u in units:
            r = score_unit(u["unit_type"], u["data"],
                           weights_map.get(u["unit_type"]))
            weighted = u["ratio"] * r["unit_score"]
            total += weighted
            unit_results.append({
                "unit_type": u["unit_type"],
                "ratio": u["ratio"],
                "unit_score": r["unit_score"],
                "weighted": weighted,
                "l1_scores": r["l1_scores"],
                "details": r["details"],
            })
        grade_info = grade_of(total)
        return {
            "final_score": total,
            "grade": grade_info["grade"],
```

```
            "grade_info": grade_info,
            "unit_results": unit_results,
        }
# ==== app/services/report_service.py ====
# -*- coding: utf-8 -*-
"""报告生成：生成 Word 评价报告。
依赖 python-docx。报告结构：
标题 → 图斑信息 → 评价结论 → 各微地质单元得分 → 完整得分明细 → 生成时间。
"""
import io
from datetime import datetime
from docx import Document
from docx.shared import Pt, Cm
from docx.oxml.ns import qn
from docx.enum.text import WD_ALIGN_PARAGRAPH
def _set_run(run, text=None, size=12, bold=False):
    if text is not None:
        run.text = text
    run.font.name = "Times New Roman"
    run.font.size = Pt(size)
    run.font.bold = bold
    # 中文字体
    run._element.rPr.rFonts.set(qn("w:eastAsia"), "宋体")
    return run
def _para(doc, text, size=12, bold=False, align=None,
    space_after=6):
    p = doc.add_paragraph()
    if align is not None:
        p.alignment = align
    p.paragraph_format.space_after = Pt(space_after)
    _set_run(p.add_run(), text, size, bold)
    return p
def _heading(doc, text, level=1):
    h = doc.add_heading(text, level=level)
    for run in h.runs:
        run.font.name = "Times New Roman"
        run._element.rPr.rFonts.set(qn("w:eastAsia"), "黑体")
    return h
def _table(doc, headers, rows, widths=None):
    t = doc.add_table(rows=1, cols=len(headers))
    t.style = "Table Grid"
    hdr = t.rows[0].cells
    for i, htext in enumerate(headers):
        _set_run(hdr[i].paragraphs[0].add_run(), htext,
    10.5, bold=True)
    for row in rows:
        cells = t.add_row().cells
        for i, val in enumerate(row):
            _set_run(cells[i].paragraphs[0].add_run(),
    str(val), 10.5)
```

```
    return t                                                                    1551
def build_report(patch, mine_result):                                          1552
    """生成评价报告 Word 文档，返回 BytesIO。                                    1553
    参数                                                                        1554
    ----                                                                        1555
    patch : MinePatch，图斑对象                                                 1556
    mine_result : evaluation_service.score_mine 的返回 dict                     1557
    """                                                                         1558
    doc = Document()                                                            1559
    # 页边距                                                                    1560
    for section in doc.sections:                                                1561
        section.top_margin = Cm(2.5)                                            1562
        section.bottom_margin = Cm(2.5)                                         1563
        section.left_margin = Cm(2.5)                                           1564
        section.right_margin = Cm(2.5)                                          1565
    _heading(doc, "历史遗留废弃矿山自然恢复适宜性评价报告", 0)                   1566
    _para(doc, "软件：历史遗留废弃矿山自然恢复适宜性评价系统 V1.0",              1567
          size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER)                           1568
    _para(doc, "生成时间：%s" % datetime.now().strftime("%Y-%m-%d               1569
%H:%M:%S"),                                                                     1570
          size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER,                           1571
space_after=12)                                                                 1572
    # 图斑信息                                                                  1573
    _heading(doc, "一、图斑基本信息", 1)                                        1574
    info = [                                                                    1575
        ("图斑名称", patch.name),                                               1576
        ("所在行政区", patch.admin_region or "—"),                              1577
        ("图斑面积（公顷）", str(patch.area_ha) if patch.area_ha is             1578
not None else "—"),                                                             1579
        ("开采矿种", patch.mineral_type or "—"),                                1580
        ("开采方式", patch.mining_method or "—"),                               1581
    ]                                                                           1582
    _table(doc, ["项目", "内容"], info)                                         1583
    # 评价结论                                                                  1584
    _heading(doc, "二、评价结论", 1)                                            1585
    gi = mine_result["grade_info"]                                              1586
    concl = [                                                                   1587
        ("最终评价因子 F", "%.4f" % mine_result["final_score"]),                1588
        ("适宜性等级", mine_result["grade"]),                                   1589
        ("判定依据", gi["meaning"]),                                            1590
        ("推荐措施", gi["measure"]),                                            1591
    ]                                                                           1592
    _table(doc, ["项目", "结果"], concl)                                        1593
    # 各单元得分                                                                1594
    _heading(doc, "三、各微地质单元得分", 1)                                    1595
    rows = []                                                                   1596
    for u in mine_result["unit_results"]:                                       1597
        rows.append((u["unit_type"], "%.2f%%" % (u["ratio"]                     1598
* 100),                                                                         1599
                     "%.4f" % u["unit_score"], "%.4f" %                         1600
```

```
        u["weighted"]))
        _table(doc, ["微地质单元", "面积占比", "单元评价因子 F'", "加权贡献"],
        rows)
        # 完整得分明细
        _heading(doc, "四、得分明细", 1)
        for u in mine_result["unit_results"]:
            _para(doc, "%s（占比 %.2f%%）" % (u["unit_type"],
        u["ratio"] * 100),
                  size=11, bold=True, space_after=4)
            detail_rows = []
            for d in u["details"]:
                raw = "—" if d["raw"] is None else d["raw"]
                std = "—" if d["std_value"] is None else "%.4f"
        % d["std_value"]
                w = "%.4f" % d["weight"]
                c = "—" if d["contribution"] is None else
        "%.4f" % d["contribution"]
                detail_rows.append((d["l1"], d["l2"], raw, std,
        w, c))
            _table(doc, ["一级指标", "二级指标", "实测值", "标准化值", "权重",
        "贡献"],
                   detail_rows)
            # 一级指标得分
            l1_row = []
            for k, v in u["l1_scores"].items():
                l1_row.append("%s=%.4f" % (k, v))
            _para(doc, "一级指标得分：" + ",".join(l1_row), size=10.5,
        space_after=10)
        buf = io.BytesIO()
        doc.save(buf)
        buf.seek(0)
        return buf
# ==== app/services/indicator_service.py ====
# -*- coding: utf-8 -*-
"""指标定义维护。
把论文表 2/表 3 的指标体系写入 indicators 表，使指标名称、单位、
理想值与取值依据可在界面上查看维护，而不再只存在于代码常量中。
代码常量（app.utils.constants）仍是计算时的**权威取值来源**；
本表用于展示、核对与记录取值依据。二者由 seed_indicators() 保持同步。
"""
from app.models import db, Indicator
from app.utils.constants import (
    UNIT_TYPES, UNIT_ENABLED_INDICATORS, IDEAL_SOIL_RATIO,
    IDEAL_THICKNESS,
    IDEAL_PH_RANGE, IDEAL_ORGANIC, IDEAL_COVERAGE,
    IDEAL_SHANNON,
    HEAVY_METAL_LIMIT, SLOPE_BREAK, ASPECT_SCORES,
)
ALL_UNITS = ",".join(UNIT_TYPES)
# (code, name, level, parent, unit, ideal_value,
```

```
    ideal_type, enabled_units, desc, is_constraint)
TREE = [
    # ---- 一级指标 ----
    ("soil", "土壤指标", 1, None, None, None, None, "碎石边坡,平台",
    "论文表 2 评价指标之一级指标", False),
    ("veg", "植被指标", 1, None, None, None, None, ALL_UNITS,
    "论文表 2 评价指标之一级指标", False),
    ("terrain", "地形指标", 1, None, None, None, None,
    "岩质边坡,碎石边坡", "论文表 2 评价指标之一级指标", False),
    # ---- 二级指标：土壤 ----
    ("soil_ratio", "土石比", 2, "soil", "—",
    str(IDEAL_SOIL_RATIO), "benefit", "碎石边坡,平台",
     "论文 4.2.2 节：理想值 0.8，根据矿山修复实际情况确定", False),
    ("thickness", "土层厚度", 2, "soil", "cm",
    str(IDEAL_THICKNESS), "benefit", "碎石边坡,平台",
     "论文 4.2.2 节：理想值 20 cm", False),
    ("ph", "pH", 2, "soil", "—", "%g~%g" % IDEAL_PH_RANGE,
    "mid", "碎石边坡,平台",
     "论文 4.2.2 节：pH 用适中指标的标准化方法，理想区间 %g~%g" %
    IDEAL_PH_RANGE, False),
    ("organic_matter", "有机质", 2, "soil", "g/kg",
    str(IDEAL_ORGANIC), "benefit", "碎石边坡,平台",
     "《全国第二次土壤普查养分等级划分标准》二等（较丰富）最低值 30 g/kg", False),
    ("pollution", "污染指标", 2, "soil", "mg/kg", "见三级指标",
    "cost", "碎石边坡,平台",
     "同时作为限制性指标与评价指标；8 种重金属等权合成（各 0.125）", True),
    # ---- 二级指标：植被 ----
    ("coverage", "植被覆盖度", 2, "veg", "%",
    str(IDEAL_COVERAGE), "benefit", ALL_UNITS,
     "论文 4.2.2 节：参照生态系统取值 75%", False),
    ("shannon", "生物多样性", 2, "veg", "—", str(IDEAL_SHANNON),
    "benefit", ALL_UNITS,
     "Shannon-Wiener 多样性指数，参照生态系统取值 0.7", False),
    # ---- 二级指标：地形 ----
    ("slope", "坡度", 2, "terrain", "°", "%g° 分界" %
    SLOPE_BREAK, "cost", "岩质边坡,碎石边坡",
     "论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号：25°
    为陡坡重要分界", False),
    ("aspect", "坡向", 2, "terrain", "—",
    "阳坡0.2/阴阳坡0.6/阴坡1.0", "discrete", "岩质边坡,碎石边坡",
     "论文 4.2.2 节：阴坡及半阴坡水分条件更优，故赋值高于阳坡", False),
]
# 污染指标下的三级指标（8 种重金属）
for _m, _limit in HEAVY_METAL_LIMIT.items():
    TREE.append((
        "hm_" + _m, _m, 3, "pollution", "mg/kg",
    str(_limit), "cost",
        "碎石边坡,平台",
        "《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值 %g mg/kg" %
    _limit,
```

```
            False,
        ))
def seed_indicators():
    """写入/更新指标定义（幂等）。返回写入条数。"""
    code_map = {
        "benefit": "越大越优", "cost": "越小越优",
        "mid": "适中型", "discrete": "离散赋值",
    }
    n = 0
    for (code, name, level, parent, unit, ideal, itype,
    units, desc, is_con) in TREE:
        ind = Indicator.query.filter_by(code=code).first()
        if ind is None:
            ind = Indicator(code=code)
            db.session.add(ind)
        ind.name = name
        ind.level = level
        ind.parent_code = parent
        ind.unit = unit
        ind.ideal_value = ideal
        ind.ideal_type = itype
        ind.enabled_units = units
        ind.description = desc
        ind.is_constraint = is_con
        ind.is_evaluative = (level in (2, 3))
        n += 1
    db.session.commit()
    return n
def ideal_type_label(code):
    return {"benefit": "越大越优", "cost": "越小越优",
            "mid": "适中型", "discrete": "离散赋值"}.get(code,
    code or "—")
# ==== app/services/import_service.py ====
# -*- coding: utf-8 -*-
"""指标数据批量导入。
支持 CSV 与 Excel（.xlsx/.xls）。一行对应一个微地质单元，
按「单元类型」匹配图斑内已有的微地质单元后写入指标数据。
表头约定（缺失列按不适用处理）：
    单元类型， 土石比， 土层厚度， pH， 有机质，
    Hg, As, Ni, Cu, Cd, Pb, Cr, Zn,
    植被覆盖度， Shannon， 坡度， 坡向
"""
import csv
import io
import json
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_TYPES
# 表头（顺序即导出模板顺序）
HEADERS = (["单元类型", "土石比", "土层厚度", "pH", "有机质"]
           + list(HEAVY_METALS)
```

```
                + ["植被覆盖度", "Shannon", "坡度", "坡向"])
FIELD_ALIASES = {
    "单元类型": "unit_type",
    "土石比": "soil_ratio",
    "土层厚度": "thickness",
    "pH": "ph", "PH": "ph", "ph": "ph",
    "有机质": "organic_matter",
    "植被覆盖度": "coverage",
    "Shannon": "shannon", "shannon": "shannon", "生物多样性":
    "shannon",
    "坡度": "slope",
    "坡向": "aspect",
}
for _m in HEAVY_METALS:
    FIELD_ALIASES[_m] = "hm_" + _m
def build_template_csv():
    """生成导入模板（含表头与示例行）。"""
    buf = io.StringIO()
    w = csv.writer(buf)
    w.writerow(HEADERS)
    # 示例：碎石边坡一行，说明每列含义
    row = {h: "" for h in HEADERS}
    row.update({"单元类型": "碎石边坡", "土石比": "0.34", "土层厚度": "10",
                "pH": "9.24", "有机质": "4.89", "Hg": "0.05",
    "As": "5.17",
                "Ni": "12.60", "Cu": "11.70", "Cd": "0.04",
    "Pb": "6.50",
                "植被覆盖度": "87.92", "Shannon": "0.60",
                "坡度": "28.75", "坡向": "阳坡"})
    w.writerow([row[h] for h in HEADERS])
    return buf.getvalue()
def _read_rows(filename, stream):
    """读取上传文件为「表头 -> 值」的字典列表。"""
    ext = filename.rsplit(".", 1)[-1].lower()
    if ext == "csv":
        text = stream.read().decode("utf-8-sig",
    errors="replace")
        reader = csv.DictReader(io.StringIO(text))
        return [dict(r) for r in reader]
    if ext in ("xlsx", "xls"):
        from openpyxl import load_workbook
        wb = load_workbook(io.BytesIO(stream.read()),
    data_only=True)
        ws = wb.active
        rows = list(ws.iter_rows(values_only=True))
        if not rows:
            return []
        header = [str(c).strip() if c is not None else ""
    for c in rows[0]]
        out = []
```

```
        for r in rows[1:]:
            if all(c is None or str(c).strip() == "" for c in r):
                continue
            out.append({header[i]: (r[i] if i < len(r) else None)
                        for i in range(len(header)) if header[i]})
        return out
    raise ValueError("不支持的文件类型：.%s" % ext)
def _num(value):
    if value is None:
        return None, None
    s = str(value).strip()
    if s == "":
        return None, None
    try:
        return float(s), None
    except ValueError:
        return None, "「%s」不是有效数值" % s
def parse_upload(filename, stream, unit_type_map):
    """解析上传文件。
    参数
    ----
    unit_type_map : dict, {单元类型: MicroUnit 对象}
    返回
    ----
    (parsed, errors)
      parsed : list[dict], 每项 {unit_type, fields{...}, heavy_metals{}}
      errors : list[str], 逐行问题说明
    """
    rows = _read_rows(filename, stream)
    parsed, errors = [], []
    if not rows:
        return [], ["文件为空或没有数据行"]
    # 表头校验
    unknown = [h for h in (rows[0].keys() if rows else []) if h not in FIELD_ALIASES]
    if unknown:
        errors.append("存在无法识别的列，将被忽略：%s" %
                      "、".join(map(str, unknown)))
    for idx, raw in enumerate(rows, start=2):       # 第 1 行是表头
        rec = {}
        for k, v in raw.items():
            key = FIELD_ALIASES.get(str(k).strip())
            if key:
                rec[key] = v
        unit_type = str(rec.get("unit_type") or "").strip()
        if not unit_type:
```

```
            errors.append("第 %d 行：缺少「单元类型」" % idx)
            continue
        if unit_type not in unit_type_map:
            errors.append("第 %d 行：图斑内不存在单元类型「%s」（现有：%s）"
                          % (idx, unit_type,
"、".join(unit_type_map) or "无"))
            continue
        if unit_type not in UNIT_TYPES:
            errors.append("第 %d 行：单元类型「%s」不是系统支持的三种之一"
                          % (idx, unit_type))
            continue
        fields, heavy = {}, {}
        ok = True
        for key, val in rec.items():
            if key in ("unit_type",):
                continue
            if key.startswith("hm_"):
                n, err = _num(val)
                if err:
                    errors.append("第 %d 行：%s %s" % (idx,
key[3:], err))
                    ok = False
                elif n is not None:
                    heavy[key[3:]] = n
                continue
            if key == "aspect":
                s = str(val).strip() if val is not None
else ""
                if s == "":
                    fields["aspect"] = None
                elif s in ASPECT_TYPES:
                    fields["aspect"] = s
                else:
                    errors.append("第 %d 行：坡向「%s」无效（应为 %s）"
                                  % (idx, s,
"、".join(ASPECT_TYPES)))
                    ok = False
                continue
            n, err = _num(val)
            if err:
                errors.append("第 %d 行：%s %s" % (idx, key,
err))
                ok = False
            else:
                fields[key] = n
        if ok:
            parsed.append({"unit_type": unit_type,
"fields": fields,
                           "heavy_metals": heavy})
return parsed, errors
```

```
# ==== app/services/workflow.py ====
# -*- coding: utf-8 -*-
"""评价流程前置条件（定量评价与报告输出的流程门禁）。
论文 4.1 节确定的评价流程存在严格的先后关系：
    建立图斑 → 限制性条件判定（通过） → 划分微地质单元并计算面积占比
    → 逐单元录入指标数据 → 计算评价因子并判定等级 → 输出报告
论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
废弃矿山，按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡三种
地质微单元类型，并计算各种地形面积占比。」限制性条件判定（一票否决）
在前，微地质单元划分在后，顺序不可颠倒。
本模块把该顺序固化为可检查的状态。后续模块在入口处调用它，
前置步骤未完成时直接拦截并给出「缺什么、去哪补」的提示，
避免在条件不具备时产出看似合理的评价结论。
"""
from app.models import (db, MicroUnit, IndicatorData,
    ConstraintCheck,
                        EvaluationResult)
# 指标数据非空字段（用于判断某单元是否已录入数据）
DATA_FIELDS = ("soil_ratio", "thickness", "ph",
    "organic_matter",
               "coverage", "shannon", "slope", "aspect")
def _has_data(ind):
    if ind is None:
        return False
    if any(getattr(ind, f, None) is not None for f in
    DATA_FIELDS):
        return True
    return bool(ind.heavy_metals_dict())
def patch_progress(patch_id):
    """检查某图斑的评价流程完成情况。
    返回 dict：
        units       {done, count, ratio_sum, detail}
        constraint  {done, passed, detail}
        data        {done, ready, missing, detail}
        blocking    未满足的前置步骤列表（按流程顺序）
        ready       是否可进入定量评价
    """
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    ratio_sum = sum(u.ratio or 0.0 for u in units)
    # ---- 步骤一：微地质单元 ----
    units_step = {
        "done": bool(units),
        "count": len(units),
        "ratio_sum": ratio_sum,
        "detail": ("已划分 %d 个微地质单元，面积占比合计 %.2f%%"
                   % (len(units), ratio_sum * 100))
                  if units else "尚未划分微地质单元",
    }
    # ---- 步骤二：限制性条件判定 ----
```

```
    cc = ConstraintCheck.query.filter_by(patch_id=patch_id)
    \
        .order_by(ConstraintCheck.id.desc()).first()
    if cc is None:
        constraint_step = {"done": False, "passed": False,
                           "detail": "尚未执行限制性条件判定"}
    elif cc.passed:
        constraint_step = {"done": True, "passed": True,
                           "detail": "限制性条件判定已通过"}
    else:
        constraint_step = {"done": True, "passed": False,
                           "detail": "限制性条件判定未通过（%s）" %
    (cc.reason or "")}
    # ---- 步骤三：指标数据 ----
    missing = []
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi
    rst()
        if not _has_data(ind):
            missing.append(u.unit_type)
    if not units:
        data_step = {"done": False, "ready": False,
    "missing": [],
                     "detail": "尚无可录入数据的微地质单元"}
    elif missing:
        data_step = {"done": False, "ready": False,
    "missing": missing,
                     "detail": "以下微地质单元尚未录入指标数据：%s"
                               % "、".join(missing)}
    else:
        data_step = {"done": True, "ready": True,
    "missing": [],
                     "detail": "全部 %d 个微地质单元的指标数据已录入" %
    len(units)}
    ratio_ok = ratio_sum > 0.999 if units else False
    if units and not ratio_ok:
        units_step["detail"] += "（占比合计不等于 100%，请检查面积录入）"
    # ---- 汇总未满足的前置条件（按论文 4.1 节的流程顺序：限制性判定优先） ----
    # reason 只写「缺什么」这一个事实，不写为什么重要的说明 ——
    # 缺少什么由 label 指明，怎么补由 fix_label 给出，页面不必再解释一遍。
    blocking = []
    if not constraint_step["done"]:
        blocking.append({
            "step": "constraint", "label": "限制性条件判定未执行",
            "reason": "五项限制性条件尚未逐项判定。",
            "fix_url": "/constraint/%d" % patch_id,
    "fix_label": "去执行判定",
        })
    elif not constraint_step["passed"]:
        blocking.append({
```

```
                "step": "constraint", "label": "限制性条件判定未通过",
                "reason": cc.reason or "",
                "fix_url": "/constraint/%d" % patch_id,
        "fix_label": "去复核判定结果",
            })
        if not units_step["done"]:
            blocking.append({
                "step": "units", "label": "未划分微地质单元",
                "reason": "该图斑下没有任何微地质单元。",
                "fix_url": "/unit/%d" % patch_id, "fix_label":
        "去划分",
            })
        elif not ratio_ok:
            blocking.append({
                "step": "units", "label": "面积占比合计不为 100%",
                "reason": "当前合计 %.2f%%。" % (ratio_sum * 100),
                "fix_url": "/unit/%d" % patch_id, "fix_label":
        "去核对面积",
            })
        if units_step["done"] and not data_step["ready"]:
            blocking.append({
                "step": "data", "label": "指标数据未录入完整",
                "reason": data_step["detail"],
                "fix_url": "/indicator/", "fix_label": "去录入",
            })
        return {
            "units": units_step,
            "constraint": constraint_step,
            "data": data_step,
            "blocking": blocking,
            "ready": not blocking,
        }
    def latest_result(patch_id):
        """取该图斑最近一次的评价记录。"""
        return EvaluationResult.query.filter_by(patch_id=patch_i
        d) \
            .order_by(EvaluationResult.id.desc()).first()
    def report_progress(patch_id):
        """报告输出的前置条件：评价流程齐备，且已有评价记录。
        报告本质上是评价结论的呈现，因此必须先真实执行过定量评价，
        导出时读取已保存的评价结果，而不是在导出环节重新计算。
        否则评价完成后再修改指标数据，报告会与系统记录的评价结论不一致，
        同一份评价出现两个数字，结论无法追溯。
        返回 dict：{ready, blocking, result, flow}
            result 为最近一次评价记录（EvaluationResult），无记录时为空。
        """
        flow = patch_progress(patch_id)
        blocking = list(flow["blocking"])
        record = latest_result(patch_id)
        if not blocking and record is None:
```

```
            blocking.append({
                "step": "evaluate", "label": "尚未执行适宜性评价",
                "reason": "该图斑还没有评价记录。",
                "fix_url": "/evaluate/%d" % patch_id,
                "fix_label": "去执行评价",
            })
        return {
            "ready": not blocking,
            "blocking": blocking,
            "result": record,
            "flow": flow,
        }
# ==== app/services/dashboard.py ====
# -*- coding: utf-8 -*-
"""工作台数据：统计数字、评价流程步骤状态、侧边栏待办角标。
首页（工作台）需要把「现在有哪些数据」「流程走到哪一步」「哪一步还差活」
一次算清楚，供模板直接渲染。集中在此，避免在视图与模板里各算一遍。
"""
from flask import g
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    Standard)
from app.services import workflow as wf
#  评价流程的八个步骤（顺序即论文 4.1 节的作业顺序）
#    key          步骤标识
#    no           步骤编号（01~08）
#    title        名称
#    icon         图标名（见 templates/_macros.html 的 ICONS）
#    desc         一句话说明
#    href         入口地址
STEPS = [
    dict(key="project", no="01", title="项目与矿山档案",
    icon="folder",
         href="/project/",
         desc="建立评价项目、录入矿山图斑、挂接遥感影像、设置参照生态系统"),
    dict(key="standard", no="02", title="政策与标准知识库",
    icon="book",
         href="/standard/",
         desc="收录地方标准与政策文件，结构化判定规则，支持省际口径比对"),
    dict(key="constraint", no="03", title="限制性指标判定",
    icon="shield",
         href="/constraint/",
         desc="政策、区位约束、土壤污染、地质灾害、耕地属性五项一票否决"),
    dict(key="unit", no="04", title="微地质单元", icon="layers",
         href="/unit/",
         desc="划分岩质边坡、碎石边坡、平台三类单元，自动计算面积占比"),
    dict(key="ahp", no="05", title="指标体系与权重",
    icon="sliders",
         href="/ahp/",
```

```
            desc="指标定义维护，判断矩阵录入、权重计算与一致性检验、多专家汇总"),
        dict(key="indicator", no="06", title="指标数据采集",
        icon="database",
            href="/indicator/",
            desc="按微地质单元录入土壤、植被、地形指标，支持 CSV / Excel 批量导入"),
        dict(key="evaluate", no="07", title="适宜性定量评价",
        icon="target",
            href="/evaluate/",
            desc="指标标准化、分单元赋分、面积加权合成，判定适宜性等级"),
        dict(key="report", no="08", title="报告与结果输出",
        icon="file",
            href="/report/",
            desc="结果图表与逐层得分明细，导出 PDF 与 Word 评价报告"),
]
def _has_data(ind):
    return wf._has_data(ind)
# 系统的九个功能模块（登录页介绍用，含不在评价流程内的用户与权限管理）
MODULES = [
    ("01", "用户与权限管理", "登录注销、三种角色权限、写操作自动留痕"),
    ("02", "项目与矿山档案", "建立项目、录入图斑、挂接遥感影像、设置参照生态系统"),
    ("03", "政策与标准知识库", "标准判定条件结构化为规则，支持省际口径比对"),
    ("04", "限制性指标判定", "政策、区位、污染、地质灾害、耕地五项一票否决"),
    ("05", "微地质单元管理", "划分岩质边坡、碎石边坡、平台，自动算面积占比"),
    ("06", "指标体系与权重", "指标定义维护、AHP 判断矩阵与一致性检验、多专家汇总"),
    ("07", "指标数据采集", "按单元类型录入土壤、植被、地形指标，支持批量导入"),
    ("08", "适宜性定量评价", "四类标准化、分单元赋分、面积加权、等级判定"),
    ("09", "报告与结果输出", "结果图表与逐层得分明细，导出 PDF 与 Word 报告"),
]
def snapshot():
    """一次算出工作台需要的全部数据（同一请求内只算一次）。
    返回 dict：
        stats      {projects, patches, unit_types, standards}
        steps      八个步骤，附 available / pending / note
        badges     {key: 待办数量}，仅含有待办的项
        progress   {patches, units, data_units, ready,
    evaluated}
    """
    cached = getattr(g, "_dash_snapshot", None)
    if cached is None:
        cached = _compute()
        g._dash_snapshot = cached
    return cached
def _compute():
    projects = Project.query.count()
    patches = MinePatch.query.all()
    patch_ids = [p.id for p in patches]
    units = MicroUnit.query.filter(
        MicroUnit.patch_id.in_(patch_ids)).all() if
    patch_ids else []
    unit_by_patch = {}
```

```
    for u in units:
        unit_by_patch.setdefault(u.patch_id, []).append(u)
    unit_ids = [u.id for u in units]
    data_unit_ids = set()
    if unit_ids:
        for ind in IndicatorData.query.filter(
                IndicatorData.unit_id.in_(unit_ids)).all():
            if _has_data(ind):
                data_unit_ids.add(ind.unit_id)
    passed_constraint = set()
    if patch_ids:
        for c in ConstraintCheck.query.filter(
                ConstraintCheck.patch_id.in_(patch_ids)).all
    ():
            if c.passed:
                passed_constraint.add(c.patch_id)
    evaluated = set()
    if patch_ids:
        for r in EvaluationResult.query.filter(
                EvaluationResult.patch_id.in_(patch_ids)).al
    l():
            evaluated.add(r.patch_id)
    # ---- 各步骤的待办数量 ----
    pending = {
        "constraint": sum(1 for p in patches if p.id not in
    passed_constraint),
        "unit": sum(1 for p in patches if not
    unit_by_patch.get(p.id)),
        "indicator": 0,
        "evaluate": 0,
    }
    ready_patches = 0
    for p in patches:
        us = unit_by_patch.get(p.id, [])
        if us and any(u.id not in data_unit_ids for u in
    us):
            pending["indicator"] += 1
        pr = wf.patch_progress(p.id)
        if pr["ready"]:
            ready_patches += 1
            if p.id not in evaluated:
                pending["evaluate"] += 1
    # ---- 步骤可用性：前置数据不具备时置灰 ----
    avail = {
        "project": True,
        "standard": True,
        "constraint": bool(patches),
        "unit": bool(patches),
        "ahp": True,
        "indicator": bool(units),
```

```
            "evaluate": ready_patches > 0,                                  2201
            "report": bool(evaluated),                                      2202
        }                                                                   2203
        notes = {                                                           2204
            "constraint": "需先建立矿山图斑",                                2205
            "unit": "需先建立矿山图斑",                                      2206
            "indicator": "需先划分微地质单元",                               2207
            "evaluate": "需前置步骤齐备的图斑",                              2208
            "report": "需先执行适宜性评价",                                  2209
        }                                                                   2210
        steps = []                                                          2211
        for s in STEPS:                                                     2212
            steps.append(dict(                                              2213
                s,                                                          2214
                available=avail[s["key"]],                                  2215
                pending=pending.get(s["key"], 0),                           2216
                note=None if avail[s["key"]] else                           2217
        notes.get(s["key"], ""),                                            2218
            ))                                                              2219
        unit_types = len({u.unit_type for u in units})                      2220
        stats = {                                                           2221
            "projects": projects,                                           2222
            "patches": len(patches),                                        2223
            "unit_types": unit_types,                                       2224
            "standards": Standard.query.count(),                            2225
        }                                                                   2226
        return {                                                            2227
            "stats": stats,                                                 2228
            "steps": steps,                                                 2229
            "badges": {k: v for k, v in pending.items() if v},              2230
            "progress": {                                                   2231
                "patches": len(patches),                                    2232
                "units": len(units),                                        2233
                "data_units": len(data_unit_ids),                           2234
                "ready": ready_patches,                                     2235
                "evaluated": len(evaluated),                                2236
            },                                                              2237
        }                                                                   2238
def nav_badges():                                                           2239
    """侧边栏待办角标：{入口地址前缀： 数量}，只含有待办的项。"""                  2240
    snap = snapshot()                                                       2241
    return {                                                                2242
        "/constraint/": snap["badges"].get("constraint", 0),                2243
        "/unit/": snap["badges"].get("unit", 0),                            2244
        "/indicator/": snap["badges"].get("indicator", 0),                  2245
        "/evaluate/": snap["badges"].get("evaluate", 0),                    2246
    }                                                                       2247
# ==== app/services/audit.py ====                                           2248
# -*- coding: utf-8 -*-                                                     2249
"""操作审计。                                                               2250
```

```
在应用工厂中以 after_request 钩子统一埋点：凡是成功的写操作（POST/PUT/DELETE）且已登录，即写入 operation_logs 表，无需在每个视图里手工调用。
动作名称由「端点 → 中文动作」映射表给出，未收录的端点回退为方法 + 路径。
"""
from flask import request
from flask_login import current_user
from app.models import db, OperationLog
# 端点 → 可读动作名
ACTION_NAMES = {
    "auth.login": "登录系统",
    "auth.logout": "退出登录",
    "auth.add_user": "新增用户",
    "auth.edit_user": "编辑用户",
    "auth.delete_user": "删除用户",
    "auth.reset_password": "重置用户密码",
    "project.add": "新建项目",
    "project.edit": "编辑项目",
    "project.delete": "删除项目",
    "project.patch_add": "新建矿山图斑",
    "project.patch_edit": "编辑矿山图斑",
    "project.patch_delete": "删除矿山图斑",
    "project.upload_image": "上传遥感影像",
    "unit.add": "添加微地质单元",
    "unit.edit": "编辑微地质单元",
    "unit.delete": "删除微地质单元",
    "standard.add": "添加标准/政策文件",
    "standard.edit": "编辑标准/政策文件",
    "standard.delete": "删除标准/政策文件",
    "standard.rules": "维护标准规则",
    "standard.rule_delete": "删除标准规则",
    "constraint.check": "执行限制性条件判定",
    "ahp.save_expert": "录入专家判断矩阵",
    "ahp.delete_expert": "删除专家判断矩阵",
    "ahp.aggregate": "汇总多专家矩阵",
    "indicator.edit": "保存指标数据",
    "indicator.import_data": "批量导入指标数据",
    "indicator.update_meta": "维护指标定义",
    "evaluate.run": "执行适宜性评价",
}
# 无需记录的操作（查看类或纯导航）
SKIP_ENDPOINTS = {"static", "auth.index"}
def action_name(endpoint):
    if endpoint in ACTION_NAMES:
        return ACTION_NAMES[endpoint]
    return "写操作 %s %s" % (request.method, request.path)
def extract_target_id():
    """从视图参数中取出被操作对象的主键，取不到则返回 None。"""
    args = request.view_args or {}
    for key in ("pid", "patch_id", "unit_id", "sid", "rid",
    "result_id", "user_id"):
```

```
            if key in args:                                                    2301
                try:                                                           2302
                    return int(args[key])                                      2303
                except (TypeError, ValueError):                                2304
                    return None                                                2305
        return None                                                            2306
def record(endpoint, status_code):                                             2307
        """记录一次写操作。失败（4xx/5xx）也记录，便于追溯。"""                    2308
        if endpoint in SKIP_ENDPOINTS:                                         2309
            return                                                             2310
        try:                                                                   2311
            detail = "路径 %s" % request.path                                  2312
            if status_code >= 400:                                             2313
                detail += "；结果 HTTP %d" % status_code                       2314
            db.session.add(OperationLog(                                       2315
                user_id=current_user.id if                                     2316
        current_user.is_authenticated else None,                               2317
                action=action_name(endpoint),                                  2318
                target_type=(endpoint or "").split(".")[0] or                  2319
        None,                                                                  2320
                target_id=extract_target_id(),                                 2321
                detail=detail,                                                 2322
            ))                                                                 2323
            db.session.commit()                                                2324
        except Exception:                                                      2325
            # 审计失败不应影响业务响应                                         2326
            db.session.rollback()                                              2327
# ==== app/views/auth.py ====                                                  2328
# -*- coding: utf-8 -*-                                                        2329
"""用户与权限管理 —— 登录、注销、用户维护、操作日志。"""                        2330
from flask import (Blueprint, render_template, redirect,                       2331
        url_for, request,                                                      2332
                           flash, abort)                                       2333
from flask_login import login_user, logout_user,                               2334
        login_required, current_user                                           2335
from werkzeug.security import check_password_hash,                             2336
        generate_password_hash                                                 2337
from app.forms import LoginForm, EmptyForm                                     2338
from app.models import db, User, OperationLog                                  2339
from app.utils.constants import ROLE_LABELS                                    2340
bp = Blueprint("auth", __name__)                                               2341
def require_admin():                                                           2342
        """非管理员直接 403，避免误操作基础数据。"""                           2343
        if current_user.role != "admin":                                       2344
            abort(403)                                                         2345
@bp.route("/")                                                                 2346
def index():                                                                   2347
        if current_user.is_authenticated:                                      2348
            from app.services import dashboard as dash                         2349
            return render_template("auth/home.html",                           2350
```

```
    dash=dash.snapshot())                                                    2351
    return redirect(url_for("auth.login"))                                   2352
@bp.route("/login", methods=["GET", "POST"])                                 2353
def login():                                                                 2354
    if current_user.is_authenticated:                                        2355
        return redirect(url_for("auth.index"))                               2356
    form = LoginForm()                                                       2357
    if form.validate_on_submit():                                            2358
        user = User.query.filter_by(username=form.username.d                 2359
    ata).first()                                                             2360
        if user and check_password_hash(user.password_hash,                  2361
    form.password.data):                                                     2362
            if not user.active:                                              2363
                flash("该账号已停用，请联系管理员", "error")                  2364
                return render_template("auth/login.html",                    2365
    form=form,                                                               2366
                                                    **_intro_data())         2367
            login_user(user, remember=form.remember.data)                    2368
            # 用 toast 类别：渲染为页面正中的浮层，淡入停留后淡出。          2369
            # 登录成功只是告知性反馈，不需要用户停下来阅读。                 2370
            flash("登录成功", "toast")                                       2371
            return redirect(url_for("auth.index"))                           2372
        flash("用户名或密码错误", "error")                                   2373
    return render_template("auth/login.html", form=form,                     2374
    **_intro_data())                                                         2375
def _intro_data():                                                           2376
    """登录页下方的系统介绍所用的静态内容。"""                               2377
    from app.services import dashboard as dash                               2378
    return {"modules": dash.MODULES}                                         2379
@bp.route("/logout")                                                         2380
@login_required                                                              2381
def logout():                                                                2382
    logout_user()                                                            2383
    flash("已退出登录", "ok")                                                2384
    return redirect(url_for("auth.login"))                                   2385
# ---------------------------- 用户维护 ----------------------------        2386
    ----                                                                     2387
@bp.route("/users")                                                          2388
@login_required                                                              2389
def users():                                                                 2390
    all_users = User.query.order_by(User.id).all()                           2391
    return render_template("auth/users.html",                                2392
    users=all_users,                                                         2393
                                    role_labels=ROLE_LABELS,                 2394
    form=EmptyForm())                                                        2395
@bp.route("/users/add", methods=["GET", "POST"])                             2396
@login_required                                                              2397
def add_user():                                                              2398
    require_admin()                                                          2399
    if request.method == "POST":                                             2400
```

```
        username = request.form.get("username", "").strip()  2401
        password = request.form.get("password", "")          2402
        role = request.form.get("role", "viewer")            2403
        if not username or not password:                     2404
            flash("用户名和密码不能为空", "error")                2405
            return redirect(url_for("auth.add_user"))        2406
        if User.query.filter_by(username=username).first():  2407
            flash("用户名已存在", "error")                      2408
            return redirect(url_for("auth.add_user"))        2409
        db.session.add(User(username=username, role=role,    2410
                            password_hash=generate_password_2411
    hash(password)))                                         2412
        db.session.commit()                                  2413
        flash("用户已添加", "ok")                               2414
        return redirect(url_for("auth.users"))               2415
    return render_template("auth/add_user.html")             2416
@bp.route("/users/<int:user_id>/edit", methods=["GET",       2417
    "POST"])                                                 2418
@login_required                                              2419
def edit_user(user_id):                                      2420
    require_admin()                                          2421
    user = db.get_or_404(User, user_id)                      2422
    if request.method == "POST":                             2423
        role = request.form.get("role", user.role)           2424
        active = request.form.get("active") == "1"           2425
        # 保护：不能把最后一个管理员降级或停用，否则无人能管理系统          2426
        if user.role == "admin" and (role != "admin" or not  2427
    active):                                                 2428
            others = User.query.filter(User.role == "admin",2429
                                       User.id != user.id,   2430
                                                             2431
    User.active.is_(True)).count()                           2432
            if others == 0:                                  2433
                flash("系统至少需保留一名启用状态的管理员", "error")      2434
                return redirect(url_for("auth.edit_user",    2435
    user_id=user.id))                                        2436
        user.role = role                                     2437
        user.active = active                                 2438
        new_password = request.form.get("password",          2439
    "").strip()                                              2440
        if new_password:                                     2441
            user.password_hash = generate_password_hash(new_2442
    password)                                                2443
        db.session.commit()                                  2444
        flash("用户信息已保存", "ok")                           2445
        return redirect(url_for("auth.users"))               2446
    return render_template("auth/edit_user.html", user=user,2447
                           role_labels=ROLE_LABELS,          2448
    form=EmptyForm())                                        2449
@bp.route("/users/<int:user_id>/delete", methods=["POST"])   2450
```

```
@login_required                                                    2451
def delete_user(user_id):                                          2452
    require_admin()                                                2453
    user = db.get_or_404(User, user_id)                            2454
    if user.id == current_user.id:                                 2455
        flash("不能删除当前登录的账号", "error")                    2456
        return redirect(url_for("auth.users"))                     2457
    if user.role == "admin":                                       2458
        others = User.query.filter(User.role == "admin",           2459
                                   User.id != user.id,             2460
                                   User.active.is_(True)).co       2461
    unt()                                                          2462
        if others == 0:                                            2463
            flash("系统至少需保留一名启用状态的管理员", "error")    2464
            return redirect(url_for("auth.users"))                 2465
    db.session.delete(user)                                        2466
    db.session.commit()                                            2467
    flash("用户「%s」已删除" % user.username, "ok")                 2468
    return redirect(url_for("auth.users"))                         2469
# ------------------------------ 操作日志 ------------------------2470
    ----                                                           2471
@bp.route("/logs")                                                 2472
@login_required                                                    2473
def logs():                                                        2474
    page = request.args.get("page", 1, type=int)                   2475
    per_page = 50                                                  2476
    q = OperationLog.query.order_by(OperationLog.id.desc())        2477
    total = q.count()                                              2478
    records = q.offset((page - 1) * per_page).limit(per_page       2479
    ).all()                                                        2480
    users_map = {u.id: u.username for u in User.query.all()}       2481
    pages = max(1, (total + per_page - 1) // per_page)             2482
    return render_template("auth/logs.html",                       2483
    records=records, page=page,                                    2484
                                pages=pages, total=total,          2485
    users_map=users_map)                                           2486
# ==== app/views/project.py ====                                   2487
# -*- coding: utf-8 -*-                                            2488
"""项目与矿山档案 —— 项目、图斑、参照生态系统、遥感影像挂接。"""    2489
from datetime import datetime                                      2490
from flask import (Blueprint, render_template, request,            2491
    redirect, url_for,                                             2492
                      flash, current_app)                          2493
from flask_login import login_required, current_user               2494
from app.forms import EmptyForm                                    2495
from app.models import db, Project, MinePatch,                     2496
    ReferenceEcosystem                                             2497
bp = Blueprint("project", __name__, url_prefix="/project")         2498
@bp.route("/")                                                     2499
@login_required                                                    2500
```

```
def index():
    projects = Project.query.order_by(Project.id.desc()).all()
    return render_template("project/index.html",
    projects=projects,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit():
        p = Project(
            name=request.form.get("name", "").strip(),
            code=request.form.get("code", "").strip() or None,
            description=request.form.get("description", "").strip() or None,
            created_by=current_user.id,
        )
        if not p.name:
            flash("项目名称不能为空", "error")
        else:
            db.session.add(p)
            db.session.commit()
            flash("项目已创建", "ok")
            return redirect(url_for("project.index"))
    return render_template("project/edit.html", form=form, project=None)
@bp.route("/<int:pid>/edit", methods=["GET", "POST"])
@login_required
def edit(pid):
    p = db.get_or_404(Project, pid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit():
        p.name = request.form.get("name", "").strip()
        p.code = request.form.get("code", "").strip() or None
        p.description = request.form.get("description", "").strip() or None
        p.status = request.form.get("status", "进行中")
        db.session.commit()
        flash("项目已保存", "ok")
        return redirect(url_for("project.index"))
    return render_template("project/edit.html", form=form, project=p)
@bp.route("/<int:pid>/delete", methods=["POST"])
@login_required
def delete(pid):
```

```
    p = db.get_or_404(Project, pid)                                    2551
    db.session.delete(p)                                               2552
    db.session.commit()                                                2553
    flash("项目已删除", "ok")                                           2554
    return redirect(url_for("project.index"))                          2555
# ---------------------------- 矿山图斑 ----------------------------2556
    ----                                                               2557
@bp.route("/<int:pid>/patches")                                        2558
@login_required                                                        2559
def patches(pid):                                                      2560
    p = db.get_or_404(Project, pid)                                    2561
    return render_template("project/patches.html",                     2562
    project=p,                                                         2563
                                  form=EmptyForm())                    2564
@bp.route("/patch/<int:patch_id>/delete", methods=["POST"])            2565
@login_required                                                        2566
def patch_delete(patch_id):                                            2567
    """删除矿山图斑，及其微地质单元与指标数据。"""                          2568
    from app.models import MicroUnit, IndicatorData,                   2569
    EvaluationResult, \                                                2570
        ScoreDetail, ConstraintCheck, ReferenceEcosystem               2571
    patch = db.get_or_404(MinePatch, patch_id)                         2572
    pid = patch.project_id                                             2573
    unit_ids = [u.id for u in MicroUnit.query.filter_by(patc           2574
    h_id=patch.id).all()]                                              2575
    if unit_ids:                                                       2576
        IndicatorData.query.filter(IndicatorData.unit_id.in_           2577
    (unit_ids)).delete(                                                2578
            synchronize_session=False)                                 2579
    for r in EvaluationResult.query.filter_by(patch_id=patch           2580
    .id).all():                                                        2581
        ScoreDetail.query.filter_by(result_id=r.id).delete(            2582
            synchronize_session=False)                                 2583
    EvaluationResult.query.filter_by(patch_id=patch.id).dele           2584
    te(                                                                2585
        synchronize_session=False)                                     2586
    ConstraintCheck.query.filter_by(patch_id=patch.id).delet           2587
    e(                                                                 2588
        synchronize_session=False)                                     2589
    MicroUnit.query.filter_by(patch_id=patch.id).delete(sync           2590
    hronize_session=False)                                             2591
    ReferenceEcosystem.query.filter_by(patch_id=patch.id).de           2592
    lete(                                                              2593
        synchronize_session=False)                                     2594
    db.session.delete(patch)                                           2595
    db.session.commit()                                                2596
    flash("图斑「%s」及其数据已删除" % patch.name, "ok")                  2597
    return redirect(url_for("project.patches", pid=pid))               2598
@bp.route("/<int:pid>/patch/add", methods=["GET", "POST"])             2599
@login_required                                                        2600
```

```
def patch_add(pid):
    p = db.get_or_404(Project, pid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        patch = MinePatch(
            project_id=p.id,
            name=request.form.get("name", "").strip(),
            admin_region=fv("admin_region"),
            area_ha=fv("area_ha", float),
            mineral_type=fv("mineral_type"),
            mining_method=fv("mining_method"),
            stop_years=fv("stop_years", float),
            responsible_entity_missing=bool(request.form.get
    ("responsible_entity_missing")),
        )
        if not patch.name:
            flash("图斑名称不能为空", "error")
        else:
            db.session.add(patch)
            db.session.flush()
            # 参照生态系统默认取论文值
            db.session.add(ReferenceEcosystem(
                patch_id=patch.id, coverage_benchmark=75.0,
    shannon_benchmark=0.7))
            db.session.commit()
            flash("图斑已创建", "ok")
            return redirect(url_for("project.patches",
    pid=p.id))
    return render_template("project/patch_edit.html",
    form=form, project=p, patch=None)
@bp.route("/patch/<int:patch_id>/edit", methods=["GET",
    "POST"])
@login_required
def patch_edit(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        patch.name = request.form.get("name", "").strip()
        patch.admin_region = fv("admin_region")
        patch.area_ha = fv("area_ha", float)
        patch.mineral_type = fv("mineral_type")
        patch.mining_method = fv("mining_method")
```

```
        patch.stop_years = fv("stop_years", float)
        patch.responsible_entity_missing =
    bool(request.form.get("responsible_entity_missing"))
        # 参照生态系统
        ref = ReferenceEcosystem.query.filter_by(patch_id=pa
    tch.id).first()
        if ref is None:
            ref = ReferenceEcosystem(patch_id=patch.id)
            db.session.add(ref)
        ref.coverage_benchmark = fv("coverage_benchmark",
    float) or 75.0
        ref.shannon_benchmark = fv("shannon_benchmark",
    float) or 0.7
        ref.dominant_species = fv("dominant_species")
        patch.image_note = fv("image_note")
        db.session.commit()
        flash("图斑已保存", "ok")
        return redirect(url_for("project.patches",
    pid=patch.project_id))
    return render_template("project/patch_edit.html",
    form=form,
                           project=patch.project,
    patch=patch,
                           reference=ReferenceEcosystem.quer
    y.filter_by(
                               patch_id=patch.id).first())
# ---------------------------- 遥感影像挂接
    ----------------------------
@bp.route("/patch/<int:patch_id>/image", methods=["POST"])
@login_required
def upload_image(patch_id):
    """上传遥感底图 / 无人机正射影像并挂接到图斑。"""
    from app.utils import uploads as up
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if not form.validate_on_submit():
        flash("表单校验失败，请重试", "error")
        return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
    rel, err = up.save_upload(request.files.get("image"),
    "patches",
                              current_app.config["ALLOWED_IM
    AGE_EXT"],
                              current_app.config["UPLOAD_FOL
    DER"])
    if err:
        flash(err, "error")
        return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
    # 替换旧影像时清理原文件，避免残留
```

```
    if patch.image_path:                                              2701
        up.remove_upload(patch.image_path,                            2702
    current_app.config["UPLOAD_FOLDER"])                             2703
    patch.image_path = rel                                            2704
    patch.image_uploaded_at = datetime.utcnow()                       2705
    note = request.form.get("image_note", "").strip()                 2706
    if note:                                                          2707
        patch.image_note = note                                       2708
    db.session.commit()                                               2709
    flash("遥感影像已挂接", "ok")                                      2710
    return redirect(url_for("project.patch_edit",                     2711
    patch_id=patch.id))                                               2712
@bp.route("/patch/<int:patch_id>/image/delete",                       2713
    methods=["POST"])                                                 2714
@login_required                                                       2715
def delete_image(patch_id):                                           2716
    from app.utils import uploads as up                               2717
    patch = db.get_or_404(MinePatch, patch_id)                        2718
    form = EmptyForm()                                                2719
    if form.validate_on_submit():                                     2720
        up.remove_upload(patch.image_path,                            2721
    current_app.config["UPLOAD_FOLDER"])                             2722
        patch.image_path = None                                       2723
        patch.image_uploaded_at = None                                2724
        db.session.commit()                                           2725
        flash("遥感影像已移除", "ok")                                  2726
    return redirect(url_for("project.patch_edit",                     2727
    patch_id=patch.id))                                               2728
# ==== app/views/unit.py ====                                         2729
# -*- coding: utf-8 -*-                                               2730
"""微地质单元管理 —— 岩质边坡 / 碎石边坡 / 平台的划分与面积占比。"""          2731
from flask import Blueprint, render_template, request,                2732
    redirect, url_for, flash                                          2733
from flask_login import login_required                                2734
from app.forms import EmptyForm                                       2735
from app.models import db, MinePatch, MicroUnit,                      2736
    IndicatorData                                                     2737
from app.utils.constants import UNIT_TYPES                            2738
bp = Blueprint("unit", __name__, url_prefix="/unit")                  2739
def recalc_ratio(patch_id):                                           2740
    """按面积重算该图斑各单元面积占比。"""                                2741
    units = MicroUnit.query.filter_by(patch_id=patch_id).all          2742
    ()                                                                2743
    total = sum(u.area or 0.0 for u in units)                         2744
    if total > 0:                                                     2745
        for u in units:                                               2746
            u.ratio = round((u.area or 0.0) / total, 6)               2747
    return total                                                      2748
@bp.route("/")                                                        2749
@login_required                                                       2750
```

```
def list_patches():
    """微地质单元入口：列出各图斑及其单元划分情况。"""
    patches = MinePatch.query.order_by(MinePatch.id.desc()).all()
    rows = []
    for p in patches:
        units = MicroUnit.query.filter_by(patch_id=p.id).all()
        rows.append({
            "patch": p,
            "count": len(units),
            "ratio_sum": sum(u.ratio or 0.0 for u in units),
        })
    return render_template("unit/list.html", rows=rows)
@bp.route("/<int:patch_id>")
@login_required
def index(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    units = MicroUnit.query.filter_by(patch_id=patch_id).all()
    total = sum(u.area or 0.0 for u in units)
    ratio_sum = sum(u.ratio or 0.0 for u in units)
    return render_template("unit/index.html", patch=patch,
    units=units,
                           total_area=total,
    ratio_sum=ratio_sum,
                           unit_types=UNIT_TYPES,
    form=EmptyForm())
@bp.route("/<int:patch_id>/add", methods=["POST"])
@login_required
def add(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if form.validate_on_submit():
        unit_type = request.form.get("unit_type")
        if unit_type not in UNIT_TYPES:
            flash("微地质单元类型不正确", "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        # 论文 4.1 节将图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
        # 允许重复会让面积占比与按单元类型组织的指标数据对不上号。
        if MicroUnit.query.filter_by(patch_id=patch.id,
                                     unit_type=unit_type).fi
    rst():
            flash("该图斑已存在「%s」，同一类型只能划分一个" % unit_type,
    "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        area = request.form.get("area", "").strip()
        u = MicroUnit(patch_id=patch.id,
```

```
    unit_type=unit_type,                                                2801
                        area=float(area) if area else None,             2802
                        description=request.form.get("descript          2803
    ion", "").strip() or None)                                          2804
        db.session.add(u)                                               2805
        db.session.flush()                                              2806
        recalc_ratio(patch.id)                                          2807
        db.session.commit()                                             2808
        flash("微地质单元已添加", "ok")                                  2809
    return redirect(url_for("unit.index",                               2810
    patch_id=patch_id))                                                 2811
@bp.route("/edit/<int:unit_id>", methods=["GET", "POST"])               2812
@login_required                                                         2813
def edit(unit_id):                                                      2814
    """单个微地质单元的编辑页。"""                                       2815
    u = db.get_or_404(MicroUnit, unit_id)                               2816
    form = EmptyForm()                                                  2817
    if request.method == "POST" and form.validate_on_submit(            2818
    ):                                                                  2819
        new_type = request.form.get("unit_type",                        2820
    u.unit_type)                                                        2821
        if new_type not in UNIT_TYPES:                                  2822
            flash("微地质单元类型不正确", "error")                       2823
            return redirect(url_for("unit.edit",                        2824
    unit_id=u.id))                                                      2825
        #  改类型时同样不允许与本图斑已有的其他单元重复                  2826
        dup = MicroUnit.query.filter_by(patch_id=u.patch_id,            2827
                                                                        2828
    unit_type=new_type).first()                                         2829
        if dup is not None and dup.id != u.id:                          2830
            flash("该图斑已存在「%s」，同一类型只能划分一个" % new_type,  2831
    "error")                                                            2832
            return redirect(url_for("unit.edit",                        2833
    unit_id=u.id))                                                      2834
        u.unit_type = new_type                                          2835
        area = request.form.get("area", "").strip()                     2836
        u.area = float(area) if area else None                          2837
        u.description = request.form.get("description",                 2838
    "").strip() or None                                                 2839
        recalc_ratio(u.patch_id)                                        2840
        db.session.commit()                                             2841
        flash("微地质单元已保存", "ok")                                  2842
        return redirect(url_for("unit.index",                           2843
    patch_id=u.patch_id))                                               2844
    return render_template("unit/edit.html", unit=u,                    2845
    patch=u.patch,                                                      2846
                                  unit_types=UNIT_TYPES, form=form)     2847
@bp.route("/delete/<int:unit_id>", methods=["POST"])                    2848
@login_required                                                         2849
def delete(unit_id):                                                    2850
```

```
    u = db.get_or_404(MicroUnit, unit_id)
    pid = u.patch_id
    name = u.unit_type
    # 一并删除该单元的指标数据，避免残留
    IndicatorData.query.filter_by(unit_id=u.id).delete(synch
    ronize_session=False)
    db.session.delete(u)
    db.session.flush()
    recalc_ratio(pid)
    db.session.commit()
    flash("微地质单元「%s」已删除" % name, "ok")
    return redirect(url_for("unit.index", patch_id=pid))
# ==== app/views/standard.py ====
# -*- coding: utf-8 -*-
"""政策与标准知识库 —— 标准 / 政策文件的结构化规则与省际口径比对。"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, Standard, StandardRule
bp = Blueprint("standard", __name__, url_prefix="/standard")
@bp.route("/")
@login_required
def index():
    standards = Standard.query.order_by(Standard.category,
    Standard.province).all()
    local = [s for s in standards if s.category == "地方标准"]
    policy = [s for s in standards if s.category != "地方标准"]
    return render_template("standard/index.html",
    local=local, policy=policy,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        s = Standard(
            province=request.form.get("province",
    "").strip(),
            name=request.form.get("name", "").strip(),
            doc_number=request.form.get("doc_number",
    "").strip() or None,
            publish_date=request.form.get("publish_date",
    "").strip() or None,
            implement_date=request.form.get("implement_date"
    , "").strip() or None,
            category=request.form.get("category", "地方标准"),
            summary=request.form.get("summary", "").strip()
    or None,
```

```
        )                                                                    2901
        if not s.province or not s.name:                                     2902
            flash("省份与名称不能为空", "error")                              2903
        else:                                                                2904
            db.session.add(s)                                                2905
            db.session.commit()                                              2906
            flash("标准已添加", "ok")                                         2907
            return redirect(url_for("standard.index"))                       2908
    return render_template("standard/edit.html", form=form,                  2909
    standard=None)                                                           2910
@bp.route("/<int:sid>/edit", methods=["GET", "POST"])                        2911
@login_required                                                              2912
def edit(sid):                                                               2913
    s = db.get_or_404(Standard, sid)                                         2914
    form = EmptyForm()                                                       2915
    if request.method == "POST" and form.validate_on_submit(                 2916
    ):                                                                       2917
        s.province = request.form.get("province",                            2918
    "").strip()                                                              2919
        s.name = request.form.get("name", "").strip()                        2920
        s.doc_number = request.form.get("doc_number",                        2921
    "").strip() or None                                                      2922
        s.publish_date = request.form.get("publish_date",                    2923
    "").strip() or None                                                      2924
        s.implement_date = request.form.get("implement_date"                 2925
    , "").strip() or None                                                    2926
        s.category = request.form.get("category", "地方标准")                 2927
        s.summary = request.form.get("summary", "").strip()                  2928
    or None                                                                  2929
        db.session.commit()                                                  2930
        flash("已保存", "ok")                                                2931
        return redirect(url_for("standard.rules", sid=s.id))                 2932
    return render_template("standard/edit.html", form=form,                  2933
    standard=s)                                                              2934
@bp.route("/<int:sid>/delete", methods=["POST"])                             2935
@login_required                                                              2936
def delete(sid):                                                             2937
    s = db.get_or_404(Standard, sid)                                         2938
    StandardRule.query.filter_by(standard_id=sid).delete()                   2939
    db.session.delete(s)                                                     2940
    db.session.commit()                                                      2941
    flash("已删除", "ok")                                                    2942
    return redirect(url_for("standard.index"))                               2943
@bp.route("/<int:sid>/rules", methods=["GET", "POST"])                       2944
@login_required                                                              2945
def rules(sid):                                                              2946
    s = db.get_or_404(Standard, sid)                                         2947
    form = EmptyForm()                                                       2948
    if request.method == "POST" and form.validate_on_submit(                 2949
    ):                                                                       2950
```

```
            r = StandardRule(                                                   2951
                standard_id=s.id,                                               2952
                item=request.form.get("item", "").strip(),                      2953
                operator=request.form.get("operator", ">="),                    2954
                threshold=request.form.get("threshold",                         2955
        "").strip() or None,                                                    2956
                condition_type=request.form.get("condition_type"                2957
        , "必要条件"),                                                          2958
                region=request.form.get("region", "").strip()                   2959
        or None,                                                                2960
                note=request.form.get("note", "").strip() or                    2961
        None,                                                                   2962
            )                                                                   2963
            if not r.item:                                                      2964
                flash("判定项不能为空", "error")                                2965
            else:                                                               2966
                db.session.add(r)                                               2967
                db.session.commit()                                             2968
                flash("规则已添加", "ok")                                       2969
            return redirect(url_for("standard.rules", sid=s.id))                2970
        rs = StandardRule.query.filter_by(standard_id=s.id).all(                2971
        )                                                                       2972
        return render_template("standard/rules.html",                           2973
        form=form, standard=s, rules=rs)                                        2974
@bp.route("/rule/<int:rid>/delete", methods=["POST"])                           2975
@login_required                                                                 2976
def rule_delete(rid):                                                           2977
        r = db.get_or_404(StandardRule, rid)                                    2978
        sid = r.standard_id                                                     2979
        db.session.delete(r)                                                    2980
        db.session.commit()                                                     2981
        flash("规则已删除", "ok")                                               2982
        return redirect(url_for("standard.rules", sid=sid))                     2983
@bp.route("/compare")                                                           2984
@login_required                                                                 2985
def compare():                                                                  2986
        """省际口径比对：同一指标在不同省份的阈值横向对比。"""                  2987
        rules = StandardRule.query.join(Standard).all()                         2988
        groups = {}                                                             2989
        for r in rules:                                                         2990
            groups.setdefault(r.item, []).append(r)                             2991
        return render_template("standard/compare.html",                         2992
        groups=groups)                                                          2993
#  ==== app/views/constraint.py ====                                            2994
#  -*- coding: utf-8 -*-                                                        2995
"""限制性指标判定 —— 一票否决判定。                                             2996
按论文表 2 与 4.1 节流程，共 5 项：政策、区位约束、土壤污染、地质灾害、耕地属性。 2997
任一项不通过即判定为"人工辅助或生态重建"，不再进入定量评价。                   2998
"""                                                                             2999
from flask import Blueprint, render_template, request,                          3000
```

```
    redirect, url_for, flash                                          3001
from flask_login import login_required                                3002
from app.forms import EmptyForm                                       3003
from app.models import db, MinePatch, ConstraintCheck                 3004
from app.utils.constants import CONSTRAINT_ITEMS                      3005
bp = Blueprint("constraint", __name__,                                3006
    url_prefix="/constraint")                                         3007
# 字段名 -> 中文名                                                    3008
FIELDS = [                                                            3009
    ("policy_pass", "政策约束"),                                      3010
    ("location_pass", "区位约束"),                                    3011
    ("pollution_pass", "土壤污染"),                                   3012
    ("geo_pass", "地质灾害"),                                         3013
    ("farmland_pass", "耕地属性"),                                    3014
]                                                                     3015
@bp.route("/")                                                        3016
@login_required                                                       3017
def index():                                                          3018
    patches = MinePatch.query.order_by(MinePatch.id.desc()).3019
    all()                                                             3020
    latest = {}                                                       3021
    for p in patches:                                                 3022
        latest[p.id] = ConstraintCheck.query.filter_by(patch3023
    _id=p.id) \                                                       3024
            .order_by(ConstraintCheck.id.desc()).first()              3025
    return render_template("constraint/index.html",                   3026
    patches=patches, latest=latest)                                   3027
@bp.route("/<int:patch_id>", methods=["GET", "POST"])                 3028
@login_required                                                       3029
def check(patch_id):                                                  3030
    patch = db.get_or_404(MinePatch, patch_id)                        3031
    form = EmptyForm()                                                3032
    record = ConstraintCheck.query.filter_by(patch_id=patch_3033
    id) \                                                             3034
        .order_by(ConstraintCheck.id.desc()).first()                  3035
    if request.method == "POST" and form.validate_on_submit(3036
    ):                                                                3037
        if record is None:                                            3038
            record = ConstraintCheck(patch_id=patch.id)               3039
            db.session.add(record)                                    3040
        for field, _ in FIELDS:                                       3041
            setattr(record, field, request.form.get(field)    3042
    == "1")                                                           3043
        record.passed = all(getattr(record, f) for f, _ in    3044
    FIELDS)                                                           3045
        failed = [name for f, name in FIELDS if not               3046
    getattr(record, f)]                                               3047
        record.reason = ("全部通过" if record.passed                  3048
                         else "未通过：" + "、".join(failed))         3049
        db.session.commit()                                           3050
```

```
            flash("判定结果已保存", "ok")
            return redirect(url_for("constraint.check",
        patch_id=patch.id))
        # 预处理各条件的当前取值，供模板直接使用（Jinja 无 getattr）
        field_states = [(f, name, None if record is None else
        getattr(record, f))
                        for f, name in FIELDS]
        return render_template("constraint/check.html",
        patch=patch, form=form,
                               fields=FIELDS,
        field_states=field_states,
                               record=record)
# ==== app/views/ahp.py ====
# -*- coding: utf-8 -*-
"""指标体系与 AHP 权重 —— 路由与页面逻辑。"""
import json
from flask import Blueprint, render_template, request,
     redirect, url_for, flash
from app.forms import AHPSelectForm, EmptyForm
from app.models import db, Indicator, ExpertMatrix,
     WeightScheme
from app.utils.constants import DEFAULT_WEIGHTS, UNIT_TYPES
from app.services import ahp_service as ahp
from app.services import indicator_service as ind_svc
from app.services import weight_service as ws
bp = Blueprint("ahp", __name__, url_prefix="/ahp")
# 一次计算的权重结果对应的指标编码：名称 → 编码
_L1_CODE = {"土壤指标": "soil", "植被指标": "veg", "地形指标":
     "terrain"}
_L2_CODE = {
     ("土壤指标", "土石比"): "soil_ratio",
     ("土壤指标", "土层厚度"): "thickness",
     ("土壤指标", "pH"): "ph",
     ("土壤指标", "有机质"): "organic_matter",
     ("土壤指标", "污染指标"): "pollution",
     ("植被指标", "植被覆盖度"): "coverage",
     ("植被指标", "生物多样性"): "shannon",
     ("地形指标", "坡度"): "slope",
     ("地形指标", "坡向"): "aspect",
}
def to_code_pairs(unit_type, level, l1, names, weights):
     """把一次 AHP 计算的权重结果转成「指标编码 → 权重」，便于落库。"""
     pairs = []
     for name, w in zip(names, weights):
          if level == 1:
               code = _L1_CODE.get(name)
          else:
               code = _L2_CODE.get((l1, name))
          if code:
               pairs.append((code, w))
```

```
    return pairs
def normalize_l1(unit_type, level, l1):
    """把 l1 参数规整为该单元类型下确实存在的一级指标名。
    level=2 的页面（判断矩阵、多专家矩阵）都按 l1 取二级指标。
    直接带 ?level=2 访问而没给 l1 时，原先会 KeyError: None 直接 500；
    这里退回该单元类型的第一个一级指标，页面照常可用。
    """
    if level != 2:
        return None
    valid = list(DEFAULT_WEIGHTS.get(unit_type,
    {}).get("l2", {}))
    if not valid:
        return None
    return l1 if l1 in valid else valid[0]
def indicator_list(unit_type, level, l1=None):
    """返回 （指标名称列表， 默认权重列表）。
    level=1 时返回一级指标；level=2 时返回指定一级指标下的二级指标。
    l1 缺失或不属于该单元类型时，退回第一个一级指标，不抛异常。
    """
    weights = DEFAULT_WEIGHTS[unit_type]
    if level == 1:
        names = list(weights["l1"].keys())
        vec = [weights["l1"][n] for n in names]
        return names, vec
    l2_all = weights["l2"]
    if l1 not in l2_all:
        l1 = next(iter(l2_all))
    l2 = l2_all[l1]
    names = list(l2.keys())
    vec = [l2[n] for n in names]
    return names, vec
def l1_choices(unit_type):
    """某单元类型的一级指标名称列表（用于二级指标录入时选择父指标）。"""
    return list(DEFAULT_WEIGHTS[unit_type]["l1"].keys())
@bp.route("/", methods=["GET", "POST"])
def index():
    form = AHPSelectForm()
    # 先根据请求中的单元类型填充 l1 选项，再进行表单校验
    unit = request.form.get("unit_type") or
    form.unit_type.choices[0][0]
    form.l1.choices = [(n, n) for n in l1_choices(unit)]
    if form.validate_on_submit():
        unit_type = form.unit_type.data
        level = int(form.level.data)
        if level == 1:
            return redirect(url_for("ahp.matrix",
    unit_type=unit_type, level=1))
        if not form.l1.data:
            flash("选择「二级指标」时，请同时选择所属的一级指标", "error")
            return render_template("ahp/index.html",
```

```
    form=form,                                                    3151
                                          weight_rows=ws.summary()) 3152
        return redirect(url_for("ahp.matrix",                     3153
    unit_type=unit_type,                                          3154
                                      level=2, l1=form.l1.data))  3155
    return render_template("ahp/index.html", form=form,           3156
                           weight_rows=ws.summary())              3157
@bp.route("/matrix", methods=["GET", "POST"])                     3158
def matrix():                                                     3159
    unit_type = request.values.get("unit_type")                   3160
    level = int(request.values.get("level", 1))                   3161
    l1 = normalize_l1(unit_type, level,                           3162
    request.values.get("l1"))                                     3163
    if unit_type not in DEFAULT_WEIGHTS:                          3164
        flash("无效的微地质单元类型", "error")                    3165
        return redirect(url_for("ahp.index"))                     3166
    names, vec = indicator_list(unit_type, level, l1)             3167
    n = len(names)                                                3168
    form = EmptyForm()                                            3169
    # 预填默认值：论文表 3 权重对应的完全一致矩阵 a_ij = w_i / w_j 3170
    default = [[round(vec[i] / vec[j], 4) for j in                3171
    range(n)] for i in range(n)]                                  3172
    def compute(upper):                                           3173
        A = ahp.matrix_from_upper(upper)                          3174
        r = ahp.weights_and_consistency(A)                        3175
        bad = ahp.find_most_inconsistent(A)                       3176
        return {                                                  3177
            "weights": [round(v, 4) for v in r["weights"]],       3178
            "lambda_max": round(r["lambda_max"], 4),              3179
            "ci": r["ci"],                                        3180
            "cr": r["cr"],                                        3181
            "consistent": r["consistent"],                        3182
            "matrix": [[round(A[i, j], 4) for j in                3183
    range(n)] for i in range(n)],                                 3184
            "most_inconsistent": bad,                             3185
        }                                                         3186
    result = None                                                 3187
    if request.method == "POST" and form.validate_on_submit(      3188
    ):                                                            3189
        action = request.form.get("action", "calc")               3190
        try:                                                      3191
            upper = [float(request.form.get("m_%d_%d" % (i,       3192
    j), "1"))                                                     3193
                     for i in range(n) for j in range(i +         3194
    1, n)]                                                        3195
            result = compute(upper)                               3196
        except (ValueError, ZeroDivisionError):                   3197
            flash("矩阵输入有误：请填入正数（对角线自动为 1，下三角自动取倒数）",  3198
    "error")                                                      3199
            return redirect(url_for("ahp.matrix",                 3200
```

```
    unit_type=unit_type,                                                3201
                                          level=level, l1=l1))          3202
        if action == "save":                                            3203
            if not result["consistent"]:                                3204
                flash("一致性比例 CR=%.4f ≥ 0.1，判断矩阵不满足一致性要求，"   3205
                      "请调整后再保存" % result["cr"], "error")           3206
            else:                                                       3207
                pairs = to_code_pairs(unit_type, level, l1,             3208
    names,                                                              3209
                                      result["weights"])                3210
                scheme, cnt = ws.save_weights(unit_type,                3211
    pairs)                                                              3212
                flash("已保存并为「%s」生效 %d 项权重（CR=%.4f）。"          3213
                      "此后该单元类型的评价将以此权重计算。"                3214
                      % (unit_type, cnt, result["cr"]),                 3215
    "ok")                                                               3216
                return redirect(url_for("ahp.matrix",                   3217
    unit_type=unit_type,                                                3218
                                            level=level, l1=l1))        3219
    elif request.values.get("calc") == "1":                             3220
        # 按默认（论文表 3）矩阵直接试算，便于一键查看权重与一致性                3221
        result = compute([default[i][j] for i in range(n)               3222
    for j in range(i + 1, n)])                                          3223
    _, source_label, custom_scheme = ws.effective_weights(un            3224
    it_type)                                                            3225
    return render_template(                                             3226
        "ahp/matrix.html",                                              3227
        form=form, unit_type=unit_type, level=level, l1=l1,             3228
        names=names, vec=vec, default=default,                          3229
    result=result,                                                      3230
        source_label=source_label, custom_scheme=custom_sche            3231
    me,                                                                 3232
    )                                                                   3233
@bp.route("/weights/reset", methods=["POST"])                           3234
def weights_reset():                                                    3235
    """删除自定义权重方案，恢复为该单元类型的论文表 3 默认权重。"""                3236
    unit_type = request.form.get("unit_type")                           3237
    form = EmptyForm()                                                  3238
    if form.validate_on_submit():                                       3239
        if unit_type not in DEFAULT_WEIGHTS:                            3240
            flash("无效的微地质单元类型", "error")                          3241
        else:                                                           3242
            n = ws.reset_weights(unit_type)                             3243
            if n:                                                       3244
                flash("已恢复「%s」为论文表 3 默认权重" % unit_type,           3245
    "ok")                                                               3246
            else:                                                       3247
                flash("「%s」本就是论文表 3 默认权重，无需恢复" % unit_type,   3248
    "ok")                                                               3249
    return redirect(url_for("ahp.index"))                               3250
```

```
# ---------------------------- 指标维护 ----------------------------
# ----
@bp.route("/indicators")
def indicators():
    """指标定义维护：查看指标树、单位、理想值与取值依据。"""
    if Indicator.query.count() == 0:
        ind_svc.seed_indicators()
    all_ind = Indicator.query.order_by(Indicator.level,
    Indicator.id).all()
    by_parent = {}
    for i in all_ind:
        by_parent.setdefault(i.parent_code, []).append(i)
    return render_template("ahp/indicators.html",
    by_parent=by_parent,
                           all_ind=all_ind,
    unit_types=UNIT_TYPES,
                           type_label=ind_svc.ideal_type_lab
    el,
                           form=EmptyForm())
@bp.route("/indicators/<code>/edit", methods=["GET",
    "POST"])
def indicator_edit(code):
    ind = Indicator.query.filter_by(code=code).first_or_404(
    )
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        name = request.form.get("name", "").strip()
        if not name:
            flash("指标名称不能为空", "error")
            return redirect(url_for("ahp.indicator_edit",
    code=code))
        ind.name = name
        ind.unit = request.form.get("unit", "").strip() or
    None
        ind.ideal_value = request.form.get("ideal_value",
    "").strip() or None
        ind.ideal_type = request.form.get("ideal_type") or
    ind.ideal_type
        units = request.form.getlist("enabled_units")
        ind.enabled_units = ",".join(units) if units else
    None
        ind.is_constraint = request.form.get("is_constraint"
    ) == "1"
        ind.description = request.form.get("description",
    "").strip() or None
        db.session.commit()
        flash("指标「%s」已保存" % ind.name, "ok")
        return redirect(url_for("ahp.indicators"))
    return render_template("ahp/indicator_edit.html",
```

```
    ind=ind,                                                          3301
                                  unit_types=UNIT_TYPES,              3302
                                  type_label=ind_svc.ideal_type_lab   3303
    el, form=form)                                                    3304
@bp.route("/indicators/reset", methods=["POST"])                      3305
def indicators_reset():                                               3306
    """按论文取值重置指标定义。"""                                    3307
    form = EmptyForm()                                                3308
    if form.validate_on_submit():                                     3309
        n = ind_svc.seed_indicators()                                 3310
        flash("已按论文取值重置 %d 项指标定义" % n, "ok")              3311
    return redirect(url_for("ahp.indicators"))                        3312
# ---------------------------- 多专家矩阵 ------------------------------3313
-----                                                                 3314
def expert_scope():                                                   3315
    """从请求中取（并规整）单元类型、层级与所属一级指标。"""          3316
    unit_type = request.values.get("unit_type") or                    3317
    UNIT_TYPES[0]                                                     3318
    if unit_type not in DEFAULT_WEIGHTS:                              3319
        unit_type = UNIT_TYPES[0]                                     3320
    level = int(request.values.get("level", 1))                       3321
    l1 = normalize_l1(unit_type, level,                               3322
    request.values.get("l1"))                                         3323
    return unit_type, level, l1                                       3324
def expert_query(unit_type, level, l1):                               3325
    q = ExpertMatrix.query.filter_by(unit_type=unit_type,             3326
    level=level)                                                      3327
    if level == 2:                                                    3328
        q = q.filter_by(l1_name=l1)                                   3329
    else:                                                             3330
        q = q.filter(ExpertMatrix.l1_name.is_(None))                  3331
    return q.order_by(ExpertMatrix.id)                                3332
def default_scheme(unit_type, level, l1=None):                        3333
    """取（或建）该范围的权重方案，专家矩阵统一挂到它下面。"""       3334
    name = "%s · %s%s" % (unit_type,                                  3335
                          "一级指标" if level == 1 else "二级指标",   3336
                          "(%s)" % l1 if l1 else "")                  3337
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ          3338
    e,                                                                3339
                                          level=level,                3340
    name=name).first()                                                3341
    if scheme is None:                                                3342
        scheme = WeightScheme(unit_type=unit_type,                    3343
    level=level, name=name,                                           3344
                              is_default=False)                       3345
        db.session.add(scheme)                                        3346
        db.session.flush()                                            3347
    return scheme                                                     3348
@bp.route("/experts", methods=["GET", "POST"])                        3349
def experts():                                                        3350
```

```
    """多专家判断矩阵录入与汇总。"""                                    3351
    unit_type, level, l1 = expert_scope()                          3352
    if unit_type not in DEFAULT_WEIGHTS:                           3353
        flash("无效的微地质单元类型", "error")                          3354
        return redirect(url_for("ahp.index"))                      3355
    names, vec = indicator_list(unit_type, level, l1)              3356
    n = len(names)                                                 3357
    form = EmptyForm()                                             3358
    if request.method == "POST" and form.validate_on_submit(       3359
    ):                                                             3360
        action = request.form.get("action")                        3361
        if action == "add":                                        3362
            expert_name = request.form.get("expert_name",          3363
    "").strip() or "未具名专家"                                      3364
            try:                                                   3365
                upper = [float(request.form.get("m_%d_%d" %        3366
    (i, j), "1"))                                                  3367
                         for i in range(n) for j in range(i        3368
    + 1, n)]                                                       3369
            except ValueError:                                     3370
                flash("矩阵输入有误，请填入正数", "error")                  3371
                return redirect(url_for("ahp.experts",             3372
    unit_type=unit_type,                                           3373
                                            level=level, l1=l1))   3374
            A = ahp.matrix_from_upper(upper)                       3375
            r = ahp.weights_and_consistency(A)                     3376
            scheme = default_scheme(unit_type, level, l1)          3377
            db.session.add(ExpertMatrix(                           3378
                scheme_id=scheme.id,                               3379
                unit_type=unit_type, l1_name=(l1 if level          3380
    == 2 else None),                                               3381
                expert_name=expert_name, level=level,              3382
                matrix_json=json.dumps([list(map(float,            3383
    row)) for row in A]),                                          3384
                cr=round(r["cr"], 6)))                             3385
            db.session.commit()                                    3386
            flash("已录入专家「%s」的判断矩阵（CR=%.4f）" % (expert_name,      3387
    r["cr"]), "ok")                                                3388
        elif action == "save_agg":                                 3389
            #  把汇总权重落库并生效，使多专家结果真正参与评价计算              3390
            weights = _aggregate(unit_type, level, l1)             3391
            if weights is None:                                    3392
                flash("暂无可汇总的专家矩阵", "error")                      3393
            elif not weights["consistent"]:                        3394
                flash("汇总矩阵一致性比例 CR=%.4f ≥ 0.1，不满足要求，"          3395
                      "请先检查或调整专家矩阵" % weights["cr"],                3396
    "error")                                                       3397
            else:                                                  3398
                names_, _ = indicator_list(unit_type,              3399
    level, l1)                                                     3400
```

```
                pairs = to_code_pairs(unit_type, level, l1,
    names_,
                                      weights["weights"])
                _, cnt = ws.save_weights(unit_type, pairs)
                flash("已保存并为「%s」生效 %d 项汇总权重（CR=%.4f）"
                      % (unit_type, cnt, weights["cr"]),
    "ok")
        return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                                level=level, l1=l1))
    rows = expert_query(unit_type, level, l1).all()
    default = [[round(vec[i] / vec[j], 4) for j in
    range(n)] for i in range(n)]
    return render_template("ahp/experts.html",
    unit_type=unit_type, level=level,
                           l1=l1, names=names, rows=rows,
    default=default,
                           n=n, form=form)
@bp.route("/experts/<int:mid>/delete", methods=["POST"])
def expert_delete(mid):
    row = db.get_or_404(ExpertMatrix, mid)
    unit_type, level, l1 = row.unit_type, row.level,
    row.l1_name
    form = EmptyForm()
    if form.validate_on_submit():
        db.session.delete(row)
        db.session.commit()
        flash("已删除专家矩阵", "ok")
    return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                            level=level, l1=l1))
def _aggregate(unit_type, level, l1):
    """按几何平均汇总多专家矩阵并做一致性检验。无矩阵时返回 None。"""
    rows = expert_query(unit_type, level, l1).all()
    mats = []
    for r in rows:
        try:
            mats.append([list(map(float, line))
                         for line in json.loads(r.matrix_jso
    n)])
        except Exception:
            continue
    if not mats:
        return None
    agg = ahp.aggregate_matrices(mats)
    res = ahp.weights_and_consistency(agg)
    return {
        "count": len(mats),
        "weights": res["weights"],
        "lambda_max": res["lambda_max"],
```

```
            "ci": res["ci"], "cr": res["cr"],
            "consistent": res["consistent"],
            "most_inconsistent": ahp.find_most_inconsistent(agg)
        ,
            "expert_crs": [r.cr for r in rows if r.cr is not
        None],
        }
@bp.route("/experts/aggregate")
def experts_aggregate():
        """按几何平均汇总多专家矩阵，并做一致性检验。"""
        unit_type, level, l1 = expert_scope()
        if unit_type not in DEFAULT_WEIGHTS:
            flash("无效的微地质单元类型", "error")
            return redirect(url_for("ahp.index"))
        names, vec = indicator_list(unit_type, level, l1)
        rows = expert_query(unit_type, level, l1).all()
        result = _aggregate(unit_type, level, l1)
        if result is not None:
            result["weights"] = [round(v, 4) for v in
        result["weights"]]
            result["lambda_max"] = round(result["lambda_max"],
        4)
        return render_template("ahp/experts_aggregate.html",
        unit_type=unit_type,
                               level=level, l1=l1, names=names,
        rows=rows,
                               result=result, form=EmptyForm())
# ==== app/views/indicator.py ====
# -*- coding: utf-8 -*-
"""指标数据采集与管理 —— 路由与页面逻辑。
按微地质单元组织录入；系统按单元类型自动决定应录入哪些指标
（岩质边坡无土壤指标、平台无地形指标）。
"""
import io
import json
from flask import (Blueprint, render_template, request,
        redirect, url_for,
                   flash, send_file)
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
        IndicatorData
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
        UNIT_ENABLED_INDICATORS
from app.services import import_service as imp
bp = Blueprint("indicator", __name__,
        url_prefix="/indicator")
# 指标字段定义：字段名 -> (中文名, 单位, 是否可选)
SOIL_FIELDS = [
        ("soil_ratio", "土石比", ""),
        ("thickness", "土层厚度", "cm"),
```

```
    ("ph", "pH", ""),
    ("organic_matter", "有机质", "g/kg"),
]
VEG_FIELDS = [
    ("coverage", "植被覆盖度", "%"),
    ("shannon", "Shannon-Wiener 多样性指数", ""),
]
TERRAIN_FIELDS = [
    ("slope", "坡度", "°"),
    ("aspect", "坡向", ""),
]
def unit_field_config(unit_type):
    """根据单元类型返回需录入的字段分组。"""
    enabled = UNIT_ENABLED_INDICATORS.get(unit_type, ())
    groups = []
    if "土壤指标" in enabled:
        groups.append(("土壤指标", SOIL_FIELDS, True))  # True = 含重金属
    if "植被指标" in enabled:
        groups.append(("植被指标", VEG_FIELDS, False))
    if "地形指标" in enabled:
        groups.append(("地形指标", TERRAIN_FIELDS, False))
    return groups
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).all()
    return render_template("indicator/index.html", patches=patches)
@bp.route("/<int:unit_id>", methods=["GET", "POST"])
def edit(unit_id):
    unit = db.get_or_404(MicroUnit, unit_id)
    ind = IndicatorData.query.filter_by(unit_id=unit.id).first()
    form = EmptyForm()
    groups = unit_field_config(unit.unit_type)
    if request.method == "POST" and form.validate_on_submit():
        def f(name, cast=float):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        heavy = {}
        for m in HEAVY_METALS:
            v = f("hm_" + m)
            if v is not None:
                heavy[m] = v
        if ind is None:
            ind = IndicatorData(unit_id=unit.id)
            db.session.add(ind)
        ind.soil_ratio = f("soil_ratio")
```

```
            ind.thickness = f("thickness")                                  3551
            ind.ph = f("ph")                                                3552
            ind.organic_matter = f("organic_matter")                        3553
            ind.heavy_metals = json.dumps(heavy,                            3554
        ensure_ascii=False) if heavy else None                              3555
            ind.coverage = f("coverage")                                    3556
            ind.shannon = f("shannon")                                      3557
            ind.slope = f("slope")                                          3558
            ind.aspect = request.form.get("aspect", "").strip()             3559
        or None                                                             3560
            db.session.commit()                                             3561
            flash("指标数据已保存", "ok")                                    3562
            return redirect(url_for("indicator.index"))                     3563
        # 组装当前值供模板回显                                              3564
        current = {}                                                        3565
        if ind:                                                             3566
            current = {                                                     3567
                "soil_ratio": ind.soil_ratio, "thickness":                  3568
        ind.thickness,                                                      3569
                "ph": ind.ph, "organic_matter":                             3570
        ind.organic_matter,                                                 3571
                "coverage": ind.coverage, "shannon":                        3572
        ind.shannon,                                                        3573
                "slope": ind.slope, "aspect": ind.aspect,                   3574
            }                                                               3575
            current["heavy"] = ind.heavy_metals_dict()                      3576
        return render_template("indicator/edit.html",                       3577
                               unit=unit, form=form,                        3578
        groups=groups,                                                      3579
                               heavy_metals=HEAVY_METALS,                   3580
                               aspect_types=ASPECT_TYPES,                   3581
        current=current)                                                    3582
# ------------------------------ 批量导入 ------------------------------3583
    ----                                                                    3584
@bp.route("/import", methods=["GET", "POST"])                               3585
def import_data():                                                          3586
    """批量导入指标数据（CSV / Excel）。"""                                  3587
    patches = MinePatch.query.order_by(MinePatch.id.desc()).3588
    all()                                                                   3589
    form = EmptyForm()                                                      3590
    report = None                                                           3591
    if request.method == "POST" and form.validate_on_submit(3592
    ):                                                                      3593
        patch_id = request.form.get("patch_id", type=int)                   3594
        patch = db.session.get(MinePatch, patch_id) if                      3595
    patch_id else None                                                      3596
        if patch is None:                                                   3597
            flash("请选择矿山图斑", "error")                                3598
            return redirect(url_for("indicator.import_data")3599
    )                                                                       3600
```

```
    file = request.files.get("datafile")
    if file is None or not file.filename:
        flash("请选择要导入的文件", "error")
        return redirect(url_for("indicator.import_data")
)
    ext = file.filename.rsplit(".", 1)[-1].lower()
    if ext not in ("csv", "xlsx", "xls"):
        flash("仅支持 CSV 或 Excel(.xlsx/.xls)文件", "error")
        return redirect(url_for("indicator.import_data")
)
    units = MicroUnit.query.filter_by(patch_id=patch.id)
.all()
    unit_map = {}
    for u in units:
        unit_map.setdefault(u.unit_type, u)
    if not unit_map:
        flash("该图斑尚未划分微地质单元，请先划分后再导入", "error")
        return redirect(url_for("indicator.import_data")
)
    try:
        parsed, errors = imp.parse_upload(file.filename,
file.stream, unit_map)
    except Exception as e:
        flash("文件解析失败：%s" % e, "error")
        return redirect(url_for("indicator.import_data")
)
    saved = 0
    for rec in parsed:
        unit = unit_map[rec["unit_type"]]
        ind = IndicatorData.query.filter_by(unit_id=unit
.id).first()
        if ind is None:
            ind = IndicatorData(unit_id=unit.id)
            db.session.add(ind)
        f = rec["fields"]
        for key in ("soil_ratio", "thickness", "ph",
"organic_matter",
                    "coverage", "shannon", "slope",
"aspect"):
            if key in f:
                setattr(ind, key, f[key])
        if rec["heavy_metals"]:
            ind.heavy_metals = json.dumps(rec["heavy_met
als"],

ensure_ascii=False)
        saved += 1
    db.session.commit()
    report = {"patch": patch.name, "saved": saved,
"errors": errors,
```

```
                    "total": len(parsed) + len(                    3651
                        [e for e in errors if e.startswith("第     3652
    ")])}                                                          3653
        if saved:                                                  3654
            flash("已导入 %d 个微地质单元的指标数据" % saved, "ok")    3655
        else:                                                      3656
            flash("没有可导入的数据行，请检查文件内容", "error")      3657
    return render_template("indicator/import.html",                3658
    patches=patches,                                               3659
                                form=form, report=report,          3660
    headers=imp.HEADERS)                                           3661
@bp.route("/import/template")                                      3662
def import_template():                                             3663
    """下载导入模板（CSV，含一行示例）。"""                          3664
    csv_text = imp.build_template_csv()                            3665
    data = csv_text.encode("utf-8-sig")       # 带 BOM，便于       3666
    Excel 正确识别中文                                             3667
    return send_file(io.BytesIO(data), as_attachment=True,         3668
                     download_name="指标数据导入模板.csv",          3669
                     mimetype="text/csv")                          3670
# ==== app/views/evaluate.py ====                                  3671
# -*- coding: utf-8 -*-                                            3672
"""适宜性定量评价引擎 —— 路由与页面逻辑。                          3673
从数据库加载某图斑的微地质单元与指标数据，调用评价引擎计算，      3674
展示各单元得分、最终评价因子 F、适宜性等级与推荐措施。            3675
入口处按论文 4.1 节的流程顺序检查前置条件（见 app/services/workflow.py），3676
前置步骤未完成时不予计算，避免在条件不具备时产出评价结论。        3677
"""                                                                3678
import json                                                        3679
from flask import Blueprint, render_template, request,             3680
    redirect, url_for, flash                                       3681
from app.forms import EmptyForm                                    3682
from app.models import (db, MinePatch, MicroUnit,                  3683
    IndicatorData,                                                 3684
                        EvaluationResult, ScoreDetail)             3685
from app.services import evaluation_service as ev                  3686
from app.services import workflow as wf                            3687
bp = Blueprint("evaluate", __name__, url_prefix="/evaluate")       3688
def load_patch_input(patch_id):                                    3689
    """从数据库组装评价引擎输入。"""                               3690
    units = MicroUnit.query.filter_by(patch_id=patch_id).all       3691
    ()                                                             3692
    result = []                                                    3693
    for u in units:                                                3694
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi       3695
    rst()                                                          3696
        data = {                                                   3697
            "soil_ratio": ind.soil_ratio if ind else None,         3698
            "thickness": ind.thickness if ind else None,           3699
            "ph": ind.ph if ind else None,                         3700
```

```
                "organic_matter": ind.organic_matter if ind
    else None,
                "heavy_metals": ind.heavy_metals_dict() if ind
    else {},
                "coverage": ind.coverage if ind else None,
                "shannon": ind.shannon if ind else None,
                "slope": ind.slope if ind else None,
                "aspect": ind.aspect if ind else None,
            }
            result.append({
                "unit_id": u.id,
                "unit_type": u.unit_type,
                "ratio": u.ratio if u.ratio is not None else
    0.0,
                "data": data,
            })
    return result
def _round(v, n=6):
    return None if v is None else round(v, n)
def load_weights(units):
    """解析各单元类型当前生效的权重与来源说明。
    权重可来自 AHP 计算并保存的生效方案；未配置时用论文表 3 的默认值。
    来源一并记录，便于报告标注与事后复核。
    """
    from app.services import weight_service as ws
    weights, sources = {}, {}
    for u in units:
        ut = u["unit_type"]
        if ut not in weights:
            w, label, _ = ws.effective_weights(ut)
            weights[ut] = w
            sources[ut] = label
    return weights, sources
def pack_units(unit_results, sources=None):
    """把评价结果整理为可持久化的 JSON。
    完整保留各单元得分、一级指标得分与逐条二级指标明细，
    使结果页与报告能够复现「已记录的那一次评价」，
    而不是每次打开时重新推算。
    """
    sources = sources or {}
    out = []
    for u in unit_results:
        out.append({
            "unit_type": u["unit_type"],
            "ratio": _round(u["ratio"]),
            "unit_score": _round(u["unit_score"]),
            "weighted": _round(u["weighted"]),
            "weight_source": sources.get(u["unit_type"]),
            "l1_scores": {k: _round(v) for k, v in
    u["l1_scores"].items()},
```

```
                "details": [{                                                    3751
                    "l1": d["l1"], "l2": d["l2"],                                3752
                    "raw": None if d["raw"] is None else                         3753
    str(d["raw"]),                                                               3754
                    "std_value": _round(d["std_value"]),                         3755
                    "weight": _round(d["weight"]),                               3756
                    "contribution": _round(d["contribution"]),                   3757
                } for d in u["details"]],                                        3758
            })                                                                   3759
    return out                                                                   3760
def save_evaluation(patch_id, units):                                            3761
    """执行评价并把结果与逐层明细落库，返回评价记录。                                    3762
    权重取各单元类型当前生效的方案（AHP 计算结果），未配置时用                          3763
    论文表 3 的默认权重；来源一并记录。结果 JSON 保留各单元得分、                       3764
    一级指标得分与逐条二级指标明细，结果页与报告据此复现                               3765
    记录下来的那次评价，不再重新推算。                                               3766
    """                                                                          3767
    weights, sources = load_weights(units)                                       3768
    res = ev.score_mine(units, weights)                                          3769
    saved = EvaluationResult(                                                    3770
        patch_id=patch_id,                                                       3771
        final_score=round(res["final_score"], 4),                                3772
        grade=res["grade"],                                                      3773
        measures=res["grade_info"]["measure"],                                   3774
        unit_scores=json.dumps(pack_units(res["unit_results"]3775
    ], sources),                                                                 3776
                                    ensure_ascii=False),                         3777
    )                                                                            3778
    db.session.add(saved)                                                        3779
    db.session.flush()        # 先取得 result.id，再写明细                        3780
    unit_ids = {u["unit_type"]: u["unit_id"] for u in units}3781
    for u in res["unit_results"]:                                                3782
        for d in u["details"]:                                                   3783
            db.session.add(ScoreDetail(                                          3784
                result_id=saved.id,                                              3785
                unit_id=unit_ids.get(u["unit_type"]),                            3786
                l1_name=d["l1"], l2_name=d["l2"],                                3787
                raw_value=None if d["raw"] is None else                          3788
    str(d["raw"]),                                                               3789
                std_value=d["std_value"],                                        3790
    weight=d["weight"],                                                          3791
                contribution=d["contribution"]))                                 3792
    db.session.commit()                                                          3793
    return saved                                                                 3794
def block_reasons(progress):                                                     3795
    """把未满足的前置条件转成可读的拦截原因。"""                                        3796
    return ["%s：%s" % (b["label"], b["reason"]) for b in                        3797
    progress["blocking"]]                                                        3798
@bp.route("/")                                                                   3799
def index():                                                                     3800
```

```
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    latest = {}
    progress = {}
    for p in patches:
        latest[p.id] = EvaluationResult.query.filter_by(patc
    h_id=p.id) \
            .order_by(EvaluationResult.id.desc()).first()
        progress[p.id] = wf.patch_progress(p.id)
    return render_template("evaluate/index.html",
    patches=patches,
                           latest=latest, progress=progress)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
def run(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    progress = wf.patch_progress(patch_id)
    if request.method == "POST" and form.validate_on_submit(
    ):
        if not progress["ready"]:
            for reason in block_reasons(progress):
                flash("无法执行评价 —— " + reason, "error")
            return redirect(url_for("evaluate.run",
    patch_id=patch_id))
        units = load_patch_input(patch_id)
        saved = save_evaluation(patch.id, units)
        return redirect(url_for("evaluate.result",
    result_id=saved.id))
    # 展示将采用的权重来源，便于在执行前确认
    units_now = load_patch_input(patch_id)
    _, sources = load_weights(units_now) if units_now else
    ({}, {})
    return render_template("evaluate/run.html",
    patch=patch, form=form,
                           progress=progress,
    weight_sources=sources)
@bp.route("/result/<int:result_id>")
def result(result_id):
    r = db.get_or_404(EvaluationResult, result_id)
    patch = db.session.get(MinePatch, r.patch_id)
    unit_scores = json.loads(r.unit_scores) if
    r.unit_scores else []
    return render_template("evaluate/result.html", r=r,
    patch=patch,
                           unit_scores=unit_scores)
# ==== app/views/report.py ====
# -*- coding: utf-8 -*-
"""报告与结果输出 —— 得分明细、图表、Word / PDF 导出。"""
import io
import json
```

```
from datetime import datetime
from flask import Blueprint, render_template, send_file,
    flash, redirect, url_for
from app.models import db, MinePatch, ReferenceEcosystem
from app.services import report_service as rs
from app.services import workflow as wf
from app.utils import charts
from app.utils import browser
from app.utils.constants import grade_of
bp = Blueprint("report", __name__, url_prefix="/report")
def _prepare(patch_id):
    """加载图斑并取出已记录的评价结果，供明细页与导出复用。
    报告是评价结论的呈现，因此这里读取最近一次保存的评价记录，
    而不是重新计算：评价完成后再改动指标数据，也不会让报告与
    系统记录的评价结论对不上。没有任何评价记录时，返回流程检查
    结果，由调用方拦截并提示先去执行评价。
    """
    patch = db.get_or_404(MinePatch, patch_id)
    prog = wf.report_progress(patch_id)
    if not prog["ready"]:
        return patch, None, None, None, prog
    rec = prog["result"]
    result = {
        "final_score": rec.final_score,
        "grade": rec.grade,
        "grade_info": grade_of(rec.final_score),
        "unit_results": json.loads(rec.unit_scores) if
    rec.unit_scores else [],
    }
    reference = ReferenceEcosystem.query.filter_by(patch_id=
    patch_id).first()
    return patch, reference, result, rec, prog
def _deny(progress, patch_id, endpoint="report.detail"):
    """前置条件未满足时的统一处理：逐条提示并退回报告列表。"""
    for b in progress["blocking"]:
        flash("暂不能生成报告 —— %s：%s" % (b["label"],
    b["reason"]), "error")
    if progress["blocking"]:
        flash("请先完成上述步骤后再生成报告。", "error")
    return redirect(url_for("report.index"))
def _charts(result):
    """生成两张图表：单元得分对比 + 一级指标平均得分分布。"""
    units = [(u["unit_type"], u["unit_score"],
              "%.4f" % u["unit_score"]) for u in
    result["unit_results"]]
    chart_units = charts.hbars(units)
    # 各一级指标在各单元上的平均得分，反映其对评价结果的实际影响
    acc, cnt = {}, {}
    for u in result["unit_results"]:
        for k, v in (u.get("l1_scores") or {}).items():
```

```
            acc[k] = acc.get(k, 0.0) + v                                    3901
            cnt[k] = cnt.get(k, 0) + 1                                      3902
    items = [(k, acc[k] / cnt[k], "%.4f" % (acc[k] /                        3903
    cnt[k])) for k in acc]                                                  3904
    items.sort(key=lambda x: -x[1])                                         3905
    chart_weight = charts.donut(items)                                      3906
    return chart_units, chart_weight                                        3907
@bp.route("/")                                                              3908
def index():                                                                3909
    patches = MinePatch.query.order_by(MinePatch.id.desc()).                3910
    all()                                                                   3911
    progress = {p.id: wf.report_progress(p.id) for p in                     3912
    patches}                                                                3913
    return render_template("report/index.html",                             3914
    patches=patches,                                                        3915
                                progress=progress)                          3916
@bp.route("/<int:patch_id>")                                                3917
def detail(patch_id):                                                       3918
    patch, reference, result, record, prog =                                3919
    _prepare(patch_id)                                                      3920
    if result is None:                                                      3921
        return _deny(prog, patch_id)                                        3922
    chart_units, chart_weight = _charts(result)                             3923
    return render_template("report/detail.html",                            3924
    patch=patch, result=result,                                             3925
                                record=record,                              3926
    reference=reference,                                                    3927
                                chart_units=chart_units,                    3928
    chart_weight=chart_weight)                                              3929
@bp.route("/<int:patch_id>/word")                                           3930
def word(patch_id):                                                         3931
    patch, reference, result, record, prog =                                3932
    _prepare(patch_id)                                                      3933
    if result is None:                                                      3934
        return _deny(prog, patch_id)                                        3935
    buf = rs.build_report(patch, result)                                    3936
    filename = "自然恢复适宜性评价报告_%s.docx" % patch.name                 3937
    return send_file(buf, as_attachment=True,                               3938
    download_name=filename,                                                 3939
                        mimetype="application/vnd.openxmlformat             3940
    s-officedocument."                                                      3941
                                "wordprocessingml.document")                3942
@bp.route("/<int:patch_id>/pdf")                                            3943
def pdf(patch_id):                                                          3944
    """导出 PDF 报告：服务端渲染打印模板，再交浏览器无头打印。"""                3945
    patch, reference, result, record, prog =                                3946
    _prepare(patch_id)                                                      3947
    if result is None:                                                      3948
        return _deny(prog, patch_id)                                        3949
    chart_units, chart_weight = _charts(result)                             3950
```

```
    html = render_template(                                          3951
        "report/print.html", patch=patch,                            3952
    reference=reference, result=result,                              3953
        chart_units=chart_units, chart_weight=chart_weight,          3954
        generated_at=datetime.now().strftime("%Y-%m-%d               3955
    %H:%M"))                                                         3956
    data, err = browser.html_to_pdf(html)                            3957
    if err:                                                          3958
        flash("PDF 导出失败：%s" % err, "error")                      3959
        return redirect(url_for("report.detail",                     3960
    patch_id=patch_id))                                              3961
    return send_file(io.BytesIO(data), as_attachment=True,           3962
                     download_name="自然恢复适宜性评价报告_%s.pdf" %   3963
    patch.name,                                                      3964
                     mimetype="application/pdf")                     3965
# ==== app/views/search.py ====                                      3966
# -*- coding: utf-8 -*-                                              3967
"""全局搜索：在评价项目、矿山图斑、标准与政策文件中查找关键词。                  3968
顶栏的搜索框指向本模块。只做关键词包含匹配，不做分词，                          3969
但覆盖名称、编号、行政区、矿种、文号、摘要等常用检索字段。                      3970
"""                                                                  3971
from sqlalchemy import or_                                           3972
from flask import Blueprint, render_template, request                3973
from app.models import db, Project, MinePatch, Standard              3974
bp = Blueprint("search", __name__)                                   3975
LIMIT = 50              # 每类结果上限，避免一次渲染过多                 3976
@bp.route("/search")                                                 3977
def index():                                                         3978
    q = (request.args.get("q") or "").strip()                        3979
    projects, patches, standards = [], [], []                        3980
    if q:                                                            3981
        like = "%" + q + "%"                                         3982
        projects = Project.query.filter(or_(                         3983
            Project.name.like(like),                                 3984
            Project.code.like(like),                                 3985
            Project.description.like(like),                          3986
        )).order_by(Project.id.desc()).limit(LIMIT).all()            3987
        patches = MinePatch.query.filter(or_(                        3988
            MinePatch.name.like(like),                               3989
            MinePatch.admin_region.like(like),                       3990
            MinePatch.mineral_type.like(like),                       3991
        )).order_by(MinePatch.id.desc()).limit(LIMIT).all()          3992
        standards = Standard.query.filter(or_(                       3993
            Standard.name.like(like),                                3994
            Standard.doc_number.like(like),                          3995
            Standard.province.like(like),                            3996
            Standard.summary.like(like),                             3997
        )).order_by(Standard.id).limit(LIMIT).all()                  3998
    total = len(projects) + len(patches) + len(standards)            3999
    return render_template("search.html", q=q,                       4000
```

```
    projects=projects,                                                  4001
                                  patches=patches,                      4002
    standards=standards, total=total)                                   4003
# ==== app/templates/base.html ====                                     4004
{% import "_macros.html" as ui %}<!DOCTYPE html>                        4005
<html lang="zh-CN">                                                     4006
<head>                                                                  4007
  <meta charset="UTF-8">                                                4008
  <meta name="viewport" content="width=device-width,                    4009
    initial-scale=1.0">                                                 4010
  <link rel="icon" href="{{ url_for('static',                           4011
    filename='favicon.ico') }}">                                        4012
  <title>{% block title %}历史遗留废弃矿山自然恢复适宜性评价系统{% endblock      4013
    %}</title>                                                          4014
  <style>                                                               4015
    /* ---------- 设计变量：绿色主色 + 中性灰阶 ---------- */                4016
    :root {                                                             4017
      --accent: #1f6b45;                  /* 主色：深绿，呼应自然恢复主题 */      4018
      --accent-hover: #185738;                                          4019
      --accent-soft: #e9f4ee;                                           4020
      --side-1: #17503a;                  /* 侧边栏渐变起止色 */              4021
      --side-2: #113c2b;                                                4022
      --danger: #b3261e;                                                4023
      --danger-soft: #fdecea;                                           4024
      --ok: #1e6b3a;                                                    4025
      --ok-soft: #eaf6ef;                                               4026
      --todo: #d98324;                    /* 待办角标 */                  4027
      --ink: #23262b;                                                   4028
      --ink-soft: #6b7280;                                              4029
      --line: #e3e7ea;                                                  4030
      --surface: #ffffff;                                               4031
      --canvas: #f4f7f5;                                                4032
      --radius: 10px;                                                   4033
      --shadow: 0 1px 2px rgba(0, 0, 0, .05);                           4034
      --header-h: 60px;                                                 4035
      --nav-w: 226px;                                                   4036
    }                                                                   4037
    * { box-sizing: border-box; }                                       4038
    html, body { margin: 0; padding: 0; }                               4039
    body {                                                              4040
      font-family: "Microsoft YaHei", "PingFang SC",                    4041
    "SimSun", sans-serif;                                               4042
      font-size: 14px;                                                  4043
      line-height: 1.6;                                                 4044
      color: var(--ink);                                                4045
      background: var(--canvas);                                        4046
    }                                                                   4047
    /* ---------- 跳到主要内容（键盘可达性） ---------- */                    4048
    .skip-link {                                                        4049
      position: absolute;                                               4050
```

```
    left: 12px;
    top: -60px;
    z-index: 30;
    padding: 8px 14px;
    background: var(--accent);
    color: #fff;
    border-radius: var(--radius);
    text-decoration: none;
    transition: top .15s ease-out;
}
.skip-link:focus { top: 12px; }
/* ---------- 整体骨架：左侧深绿导航 + 右侧内容 ---------- */
.app-shell {
    display: flex;
    align-items: stretch;
    min-height: 100dvh;
}
/* ---------- 侧边栏 ---------- */
.app-side {
    flex: 0 0 var(--nav-w);
    width: var(--nav-w);
    display: flex;
    flex-direction: column;
    background: linear-gradient(180deg, var(--side-1) 0%,
var(--side-2) 100%);
    color: #fff;
}
.app-brand {
    display: flex;
    align-items: center;
    gap: 11px;
    padding: 17px 16px 15px;
    border-bottom: 1px solid rgba(255, 255, 255, .10);
}
.app-brand__mark {
    flex: none;
    display: grid;
    place-items: center;
    width: 36px;
    height: 36px;
    border-radius: 10px;
    background: rgba(255, 255, 255, .15);
}
.app-brand__name {
    font-size: 14.5px;
    font-weight: 600;
    line-height: 1.35;
    letter-spacing: .2px;
}
.app-brand__sub {
```

```
    font-size: 11.5px;
    line-height: 1.35;
    color: rgba(255, 255, 255, .58);
}
.app-nav { flex: 1; padding: 6px 12px 14px; overflow-y:
auto; }
.app-nav__group {
   margin: 12px 0 5px;
    padding: 0 10px;
    font-size: 11.5px;
    letter-spacing: .05em;
    color: rgba(255, 255, 255, .48);
}
.app-nav ul { margin: 0; padding: 0; list-style: none; }
.app-nav a {
    display: flex;
    align-items: center;
    gap: 10px;
    padding: 9px 11px;
   margin-bottom: 2px;
    border-radius: 8px;
    color: rgba(255, 255, 255, .80);
    text-decoration: none;
    transition: background-color .16s ease, color .16s
ease;
}
.app-nav a:hover { background: rgba(255, 255, 255,
.10); color: #fff; }
.app-nav a:focus-visible {
    outline: 2px solid rgba(255, 255, 255, .60);
    outline-offset: 1px;
}
.app-nav a[aria-current="page"] {
    background: rgba(255, 255, 255, .17);
    color: #fff;
    font-weight: 600;
}
.app-nav .ico { flex: none; opacity: .92; }
.app-nav__badge {
    margin-left: auto;
    min-width: 20px;
    padding: 0 6px;
    font-size: 11px;
    font-weight: 600;
    line-height: 18px;
    text-align: center;
    border-radius: 9px;
    background: var(--todo);
    color: #fff;
    font-variant-numeric: tabular-nums;
```

```
}
.app-user {
  display: flex;
  align-items: center;
  gap: 10px;
  padding: 12px 16px;
  border-top: 1px solid rgba(255, 255, 255, .10);
}
.app-user__avatar {
  flex: none;
  display: grid;
  place-items: center;
  width: 32px;
  height: 32px;
  border-radius: 50%;
  background: rgba(255, 255, 255, .16);
  font-size: 13px;
  font-weight: 600;
}
.app-user__name { font-size: 13px; line-height: 1.3; }
.app-user__state {
  display: flex;
  align-items: center;
  gap: 5px;
  font-size: 11.5px;
  line-height: 1.3;
  color: rgba(255, 255, 255, .60);
}
.app-user__dot {
  width: 6px;
  height: 6px;
  border-radius: 50%;
  background: #6ed49a;
}
.app-user a {
  margin-left: auto;
  padding: 5px;
  border-radius: 6px;
  color: rgba(255, 255, 255, .68);
  text-decoration: none;
}
.app-user a:hover { color: #fff; background: rgba(255,
255, 255, .12); }
/* ---------- 右侧内容区 ---------- */
.app-content {
  flex: 1;
  min-width: 0;
  display: flex;
  flex-direction: column;
}
```

```
/* ---------- 顶栏：面包屑 + 搜索 ---------- */
.app-header {
  display: flex;
  align-items: center;
  gap: 14px;
  height: var(--header-h);
  padding: 0 22px;
  background: var(--surface);
  border-bottom: 1px solid var(--line);
}
.crumb {
  display: flex;
  align-items: center;
  gap: 7px;
  font-size: 13px;
  color: var(--ink-soft);
  white-space: nowrap;
}
.crumb a { color: var(--ink-soft); text-decoration: none; }
.crumb a:hover { color: var(--accent); }
.crumb strong { color: var(--ink); font-weight: 600; }
.crumb .ico { color: #c3c9d0; }
.app-header__spacer { flex: 1; }
.search-box {
  display: flex;
  align-items: center;
  gap: 8px;
  width: 268px;
  padding: 7px 12px;
  background: var(--canvas);
  border: 1px solid var(--line);
  border-radius: 9px;
  color: var(--ink-soft);
  transition: border-color .16s ease, background-color .16s ease;
}
.search-box:focus-within { border-color: var(--accent); background: #fff; }
.search-box input {
  flex: 1;
  min-width: 0;
  border: none;
  outline: none;
  background: transparent;
  font-family: inherit;
  font-size: 13px;
  color: var(--ink);
}
.search-box button {
```

```
    display: grid;                                                    4251
    place-items: center;                                              4252
    padding: 0;                                                       4253
    border: none;                                                     4254
    background: transparent;                                          4255
    color: inherit;                                                   4256
    cursor: pointer;                                                  4257
}                                                                     4258
.icon-btn {                                                           4259
    display: grid;                                                    4260
    place-items: center;                                              4261
    width: 34px;                                                      4262
    height: 34px;                                                     4263
    border-radius: 9px;                                               4264
    border: 1px solid transparent;                                    4265
    color: var(--ink-soft);                                           4266
    text-decoration: none;                                            4267
    transition: background-color .16s ease, color .16s                4268
ease,                                                                 4269
                border-color .16s ease;                               4270
}                                                                     4271
.icon-btn:hover {                                                     4272
    color: var(--accent);                                             4273
    background: var(--accent-soft);                                   4274
    border-color: rgba(31, 107, 69, .18);                             4275
}                                                                     4276
.app-main {                                                           4277
    flex: 1;                                                          4278
   min-width: 0;                                                      4279
    padding: 22px 26px 26px;                                          4280
}                                                                     4281
.container { max-width: 1120px; }                                     4282
/* ---------- 页脚 ---------- */                                      4283
.app-foot {                                                           4284
    display: flex;                                                    4285
    align-items: center;                                              4286
    justify-content: space-between;                                   4287
    gap: 14px;                                                        4288
    padding: 0 26px 18px;                                             4289
    font-size: 12px;                                                  4290
    color: var(--ink-soft);                                           4291
}                                                                     4292
/* ---------- 未登录时的裸页面（登录页） ---------- */                4293
.login-shell {                                                        4294
    display: flex;                                                    4295
    flex-direction: column;                                           4296
   min-height: 100dvh;                                                4297
}                                                                     4298
/* ---------- 卡片 ---------- */                                      4299
.card {                                                               4300
```

```
    background: var(--surface);
    border: 1px solid var(--line);
    border-radius: var(--radius);
    box-shadow: var(--shadow);
    padding: 22px 24px;
    margin-bottom: 18px;
}
.card > h2 {
    margin: 0 0 14px;
    font-size: 16px;
    font-weight: 600;
    color: var(--accent);
    text-wrap: balance;
}
.card > h3 {
    margin: 20px 0 10px;
    font-size: 14px;
    font-weight: 600;
    color: var(--accent);
}
/* ---------- 表格 ---------- */
table {
    width: 100%;
    border-collapse: collapse;
    margin: 10px 0 16px;
    font-variant-numeric: tabular-nums;
}
caption {
    text-align: left;
    color: var(--ink-soft);
    font-size: 12px;
    padding-bottom: 6px;
}
th, td {
    border: 1px solid var(--line);
    padding: 8px 12px;
    text-align: center;
    vertical-align: middle;
}
th {
    background: var(--accent-soft);
    font-weight: 600;
    white-space: nowrap;
}
tbody tr:hover { background: #fafbfc; }
/* ---------- 表单控件 ---------- */
label { display: inline-block; }
.field { margin-bottom: 12px; }
.field__label {
    display: block;
```

```
    margin-bottom: 4px;
    color: var(--ink-soft);
    font-size: 13px;
}
input[type="text"], input[type="number"],
input[type="password"],
select, textarea {
    padding: 6px 9px;
    font-family: inherit;
    font-size: 14px;
    color: var(--ink);
    background: #fff;
    border: 1px solid #c6ccd4;
    border-radius: 4px;
    max-width: 100%;
}
input[type="number"] {
    width: 104px;
    text-align: center;
    font-variant-numeric: tabular-nums;
}
textarea { width: 100%; min-height: 64px; resize:
vertical; }
input:focus-visible, select:focus-visible,
textarea:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 1px;
    border-color: var(--accent);
}
.checkbox {
    display: inline-flex;
    align-items: center;
    gap: 7px;
    cursor: pointer;
    user-select: none;
}
.checkbox input { width: 16px; height: 16px; margin: 0;
cursor: pointer; }
/* ---------- 按钮 ----------
   统一为圆角矩形：默认半透明底色，鼠标悬停时填充为实色。 */
.btn {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    gap: 6px;
    padding: 8px 18px;
    font-family: inherit;
    font-size: 14px;
    line-height: 1.4;
    color: #fff;
```

```
  background: rgba(31, 107, 69, .80);
  border: 1px solid rgba(31, 107, 69, .30);
  border-radius: 8px;
  cursor: pointer;
  text-decoration: none;
  transition: background-color .16s ease, border-color .16s ease,
              color .16s ease, box-shadow .16s ease, transform .16s ease;
}
.btn:hover {
  background: var(--accent);
  border-color: var(--accent);
  box-shadow: 0 3px 12px rgba(31, 107, 69, .22);
}
.btn:active { transform: translateY(1px); box-shadow: none; }
.btn:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.btn--ghost {
  color: var(--accent);
  background: rgba(31, 107, 69, .06);
  border-color: rgba(31, 107, 69, .22);
}
.btn--ghost:hover {
  color: #fff;
  background: var(--accent);
  border-color: var(--accent);
}
.btn--danger {
  color: var(--danger);
  background: rgba(179, 38, 30, .07);
  border-color: rgba(179, 38, 30, .26);
}
.btn--danger:hover {
  color: #fff;
  background: var(--danger);
  border-color: var(--danger);
  box-shadow: 0 3px 12px rgba(179, 38, 30, .20);
}
.btn--sm { padding: 5px 13px; font-size: 13px; border-radius: 7px; }
.btn:disabled { opacity: .45; cursor: not-allowed; }
.btn:disabled:hover {
  background: rgba(31, 107, 69, .80);
  border-color: rgba(31, 107, 69, .30);
  box-shadow: none;
}
.btn--ghost:disabled:hover { background: rgba(31, 107, 69, .06); color: var(--accent); }
```

```
/*  ----------  表格操作列的行内小按钮  ----------
    取代原来「文字 | 文字 | 文字」的写法：
    圆角矩形 + 半透明底色，悬停时变实色。 */
.row-actions {
  display: inline-flex;
  flex-wrap: nowrap;            /* 按钮不折行；宁可让旁边长名称换行 */
  align-items: center;
  gap: 6px;
  white-space: nowrap;
}
.row-actions form { display: inline-flex; margin: 0; }
.link-btn {
  display: inline-flex;
  align-items: center;
  padding: 4px 11px;
  font-family: inherit;
  font-size: 13px;
  line-height: 1.5;
  color: var(--accent);
  background: rgba(31, 107, 69, .07);
  border: 1px solid rgba(31, 107, 69, .20);
  border-radius: 7px;
  cursor: pointer;
  text-decoration: none;
  white-space: nowrap;
  transition: background-color .16s ease, border-color .16s ease,
              color .16s ease, box-shadow .16s ease, transform .16s ease;
}
.link-btn:hover {
  color: #fff;
  background: var(--accent);
  border-color: var(--accent);
  box-shadow: 0 2px 8px rgba(31, 107, 69, .20);
}
.link-btn:active { transform: translateY(1px); box-shadow: none; }
.link-btn:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.link-btn--danger {
  color: var(--danger);
  background: rgba(179, 38, 30, .07);
  border-color: rgba(179, 38, 30, .24);
}
.link-btn--danger:hover {
  color: #fff;
  background: var(--danger);
  border-color: var(--danger);
  box-shadow: 0 2px 8px rgba(179, 38, 30, .22);
```

```
}
.actions { display: flex; flex-wrap: wrap; align-items: center; gap: 8px; }
.actions form { display: inline; }
/* ---------- 提示条 ---------- */
.flash {
  padding: 10px 14px;
  margin-bottom: 14px;
  border-radius: var(--radius);
  border-left: 4px solid;
}
.flash.error {
  color: var(--danger);
  background: var(--danger-soft);
  border-left-color: var(--danger);
}
.flash.ok {
  color: var(--ok);
  background: var(--ok-soft);
  border-left-color: var(--ok);
}
/* ---------- 居中浮层提示（登录成功、各类操作完成） ---------- */
    外层 toast-stack 固定在页面正中心并纵向排列，
    内层每条各自淡入 → 停留 → 淡出，多条提示会依次排开而不重叠。
    半透明底 + 背景模糊做出毛玻璃质感，不做成实心块；
    pointer-events: none 让它不拦截鼠标，出现期间也能正常操作。 */
.toast-stack {
  position: fixed;
  top: 50%;
  left: 50%;
  z-index: 100;
  display: flex;
  flex-direction: column;
  align-items: center;
  gap: 10px;
  transform: translate(-50%, -50%);
  pointer-events: none;
}
.toast-center {
  max-width: min(560px, 78vw);
  padding: 13px 28px;
  border-radius: 999px;
  background: rgba(31, 107, 69, .62);          /* 半透明，非实体 */
  color: #fff;
  font-size: 16px;
  font-weight: 600;
  line-height: 1.65;
  letter-spacing: .03em;
  text-align: center;
```

```
  text-wrap: pretty;
  border: 1px solid rgba(255, 255, 255, .28);
  backdrop-filter: blur(9px) saturate(1.3);
  -webkit-backdrop-filter: blur(9px) saturate(1.3);
  box-shadow: 0 10px 34px rgba(19, 36, 58, .18);
  animation: toast-fade 2.6s ease-in-out forwards;
}
@keyframes toast-fade {
  0%         { opacity: 0; transform: scale(.90); }
  14%, 74%   { opacity: 1; transform: scale(1); }
  100%       { opacity: 0; transform: scale(1); }
}
/* 系统设置里要求「减少动态效果」时，去掉缩放，只做淡入淡出 */
@media (prefers-reduced-motion: reduce) {
  .toast-center { animation: toast-fade-plain 2.6s
linear forwards; }
  @keyframes toast-fade-plain {
    0%         { opacity: 0; }
    14%, 74%   { opacity: 1; }
    100%       { opacity: 0; }
  }
}
/* 不支持背景模糊时，加深底色保证文字仍然清楚 */
@supports not (backdrop-filter: blur(1px)) {
  .toast-center { background: rgba(31, 107, 69, .88); }
}
/* ---------- 文字工具类 ---------- */
.muted { color: var(--ink-soft); font-size: 12.5px; }
.mono { font-family: Consolas, "Courier New",
monospace; font-variant-numeric: tabular-nums; }
.num { font-variant-numeric: tabular-nums; }
.warn { color: var(--danger); font-weight: 600; }
.ok { color: var(--ok); font-weight: 600; }
.nowrap { white-space: nowrap; }
/* 表格操作列里常把删除按钮放在 <form> 中。form 默认是块级元素，
   会把「删除」挤到下一行，导致操作列参差不齐，这里让它留在行内。 */
.nowrap form { display: inline; margin: 0; }
.stack { display: flex; flex-direction: column; gap:
14px; }
/* 状态标签：不依赖颜色单独表意 */
.tag {
  display: inline-block;
  padding: 1px 9px;
  font-size: 12.5px;
  border-radius: 10px;
  border: 1px solid;
  white-space: nowrap;
}
.tag--ok { color: var(--ok); background:
var(--ok-soft); border-color: #b8ddc6; }
```

```
.tag--warn { color: var(--danger); background:                    4601
var(--danger-soft); border-color: #eec4c0; }                      4602
.tag--muted { color: var(--ink-soft); background:                 4603
#f1f3f6; border-color: #dde1e6; }                                 4604
/* ---------- 登录页 ---------- */                                 4605
/* 仿 Apple 账户页：接近纯白的底，仅在顶部有一层极淡的绿色晕染，   4606
   让彩色徽标成为整页唯一的视觉重心。排版见 auth/login.html。 */   4607
.body--login {                                                    4608
  background:                                                     4609
    radial-gradient(1100px 640px at 50% 8%,                       4610
                    #edf6f1 0%, rgba(237, 246, 241, 0)            4611
72%),                                                             4612
    linear-gradient(180deg, #ffffff 0%, #fbfdfc 100%);            4613
}                                                                 4614
.body--login .app-main { padding: 0; }                            4615
.body--login .container { max-width: none; width: 100%;           4616
}                                                                 4617
/* 仅供读屏软件的隐藏文本 */                                       4618
.visually-hidden {                                                4619
  position: absolute !important;                                  4620
  width: 1px;                                                     4621
  height: 1px;                                                    4622
  margin: -1px;                                                   4623
  padding: 0;                                                     4624
  overflow: hidden;                                               4625
  clip: rect(0 0 0 0);                                            4626
  white-space: nowrap;                                            4627
  border: 0;                                                      4628
}                                                                 4629
/* ---------- 工作台：页首横幅 ---------- */                       4630
.hero {                                                           4631
  position: relative;                                             4632
  overflow: hidden;                                               4633
  min-height: 176px;                                              4634
  padding: 26px 28px;                                             4635
  margin-bottom: 16px;                                            4636
  border-radius: 12px;                                            4637
  /* 左侧压深保证文字可读，右侧透出远山插画；                     4638
     背景按底部对齐，避免远山被居中裁掉 */                        4639
  background:                                                     4640
    linear-gradient(100deg, rgba(9, 40, 27, .92) 0%,              4641
                            rgba(11, 48, 33, .68) 50%,            4642
                            rgba(11, 48, 33, .06) 100%),          4643
    url("{{ url_for('static', filename='hero.svg') }}")           4644
    center bottom / cover no-repeat;                              4645
  color: #fff;                                                    4646
}                                                                 4647
.hero__body { max-width: 720px; }                                 4648
.hero__title {                                                    4649
  margin: 0 0 10px;                                               4650
```

```
    font-size: 21px;
    font-weight: 700;
    line-height: 1.4;
    letter-spacing: .3px;
}
.hero__desc {
  margin: 0;
  max-width: 640px;
  font-size: 12.5px;
  line-height: 1.85;
  color: rgba(255, 255, 255, .86);
}
.hero__actions {
  display: flex;
  flex-wrap: wrap;
  gap: 10px;
  margin-top: 18px;
}
.hero-btn {
  display: inline-flex;
  align-items: center;
  gap: 6px;
  padding: 7px 15px;
  font-size: 13px;
  border-radius: 8px;
  text-decoration: none;
  transition: background-color .16s ease, border-color .16s ease,
              color .16s ease;
}
.hero-btn--solid {
  color: #14432f;
  background: rgba(255, 255, 255, .92);
  border: 1px solid rgba(255, 255, 255, .92);
}
.hero-btn--solid:hover { background: #fff; }
.hero-btn--ghost {
  color: #fff;
  background: rgba(255, 255, 255, .14);
  border: 1px solid rgba(255, 255, 255, .45);
}
.hero-btn--ghost:hover { background: rgba(255, 255, 255, .26); }
.hero-btn:focus-visible { outline: 2px solid #fff; outline-offset: 2px; }
/* ---------- 工作台：数据概览 ---------- */
.stat-cards {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(190px, 1fr));
```

```
    gap: 12px;
    margin-bottom: 22px;
}
.stat-card {
    display: block;
    padding: 15px 17px;
    background: var(--surface);
    border: 1px solid var(--line);
    border-radius: var(--radius);
    box-shadow: var(--shadow);
    color: var(--ink);
    text-decoration: none;
    transition: border-color .16s ease, box-shadow .16s
ease;
}
.stat-card:hover {
    border-color: rgba(31, 107, 69, .45);
    box-shadow: 0 3px 12px rgba(19, 60, 43, .08);
}
.stat-card__label {
    display: block;
    font-size: 12.5px;
    color: var(--ink-soft);
}
.stat-card__value {
    display: block;
    margin: 3px 0 2px;
    font-size: 26px;
    font-weight: 700;
    line-height: 1.25;
    color: var(--accent);
    font-variant-numeric: tabular-nums;
}
.stat-card__unit {
    margin-left: 3px;
    font-size: 13px;
    font-weight: 500;
    color: var(--ink-soft);
}
.stat-card__note {
    display: block;
    font-size: 12px;
    color: var(--ink-soft);
}
/* ---------- 工作台：分区标题 ---------- */
.section { margin-bottom: 22px; }
.section__head {
    display: flex;
    align-items: baseline;
    flex-wrap: wrap;
```

```
    gap: 6px 14px;                                                    4751
    margin-bottom: 12px;                                              4752
  }                                                                   4753
  .section__title { margin: 0; font-size: 16px; }                     4754
  .section__desc { margin: 0; font-size: 12.5px; color:               4755
  var(--ink-soft); }                                                  4756
  /* ---------- 工作台：流程步骤卡 ---------- */                        4757
  .step-grid {                                                        4758
    display: grid;                                                    4759
    grid-template-columns: repeat(auto-fill,                          4760
  minmax(238px, 1fr));                                                4761
    gap: 12px;                                                        4762
  }                                                                   4763
  .step-card {                                                        4764
    position: relative;                                               4765
    display: block;                                                   4766
    padding: 16px 16px 15px;                                          4767
    background: var(--surface);                                       4768
    border: 1px solid var(--line);                                    4769
    border-radius: var(--radius);                                     4770
    box-shadow: var(--shadow);                                        4771
    color: var(--ink);                                                4772
    text-decoration: none;                                            4773
    transition: border-color .16s ease, box-shadow .16s               4774
  ease,                                                               4775
                transform .16s ease;                                  4776
  }                                                                   4777
  .step-card:hover {                                                  4778
    border-color: rgba(31, 107, 69, .55);                             4779
    box-shadow: 0 5px 16px rgba(19, 60, 43, .10);                     4780
    transform: translateY(-1px);                                      4781
  }                                                                   4782
  .step-card:focus-visible { outline: 2px solid                       4783
  var(--accent); outline-offset: 2px; }                               4784
  .step-card__no {                                                    4785
    position: absolute;                                               4786
    top: 13px;                                                        4787
    right: 15px;                                                      4788
    font-size: 17px;                                                  4789
    font-weight: 700;                                                 4790
    color: #d7e3db;                                                   4791
    font-variant-numeric: tabular-nums;                               4792
  }                                                                   4793
  .step-card__icon {                                                  4794
    display: grid;                                                    4795
    place-items: center;                                              4796
    width: 32px;                                                      4797
    height: 32px;                                                     4798
    margin-bottom: 10px;                                              4799
    border-radius: 9px;                                               4800
```

```
    background: var(--accent-soft);
    color: var(--accent);
}
.step-card__title {
    display: block;
    margin-bottom: 5px;
    font-size: 14px;
    font-weight: 600;
    color: var(--accent);
}
.step-card__desc {
    display: block;
    font-size: 12.5px;
    line-height: 1.65;
    color: var(--ink-soft);
    text-wrap: pretty;
}
.step-card__badge {
    display: inline-block;
    margin-top: 10px;
    padding: 1px 9px;
    font-size: 11.5px;
    border-radius: 9px;
    background: #fdf1e2;
    color: #9a5a12;
    border: 1px solid #f0d3ab;
}
/* 前置数据不具备时置灰，并说明缺什么 */
.step-card--locked {
    background: #fafbfa;
    border-color: #eceff0;
    box-shadow: none;
    cursor: default;
}
.step-card--locked .step-card__no { color: #e6e9e8; }
.step-card--locked .step-card__icon {
    background: #f1f3f2;
    color: #a9b2ae;
}
.step-card--locked .step-card__title { color: #9aa3a0; }
.step-card--locked .step-card__desc { color: #b3bab7; }
.step-card__lock {
    display: inline-block;
    margin-top: 10px;
    padding: 1px 9px;
    font-size: 11.5px;
    border-radius: 9px;
    background: #f1f3f2;
    color: #8b948f;
}
```

```
/* ---------- 工作台：整行入口 ---------- */
.row-card {
  display: flex;
  align-items: center;
  gap: 14px;
  padding: 15px 18px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  color: var(--ink);
  text-decoration: none;
  transition: border-color .16s ease, box-shadow .16s ease;
}
.row-card:hover {
  border-color: rgba(31, 107, 69, .45);
  box-shadow: 0 3px 12px rgba(19, 60, 43, .08);
}
.row-card__icon {
  display: grid;
  place-items: center;
  width: 36px;
  height: 36px;
  flex: none;
  border-radius: 10px;
  background: var(--accent-soft);
  color: var(--accent);
}
.row-card__main { flex: 1; min-width: 0; }
.row-card__title {
  display: block;
  font-size: 14px;
  font-weight: 600;
  color: var(--accent);
}
.row-card__desc {
  display: block;
  font-size: 12.5px;
  color: var(--ink-soft);
}
.row-card__arrow { color: #c3c9d0; flex: none; }
/* ---------- 结果摘要 ---------- */
.stat-row {
  display: flex;
  flex-wrap: wrap;
  gap: 28px;
  margin: 4px 0 6px;
}
.stat__label { font-size: 12.5px; color:
```

```
var(--ink-soft); }
.stat__value { font-size: 20px; font-weight: 700;
font-variant-numeric: tabular-nums; }
/* ---------- 流程步骤条 ---------- */
.steps { display: flex; flex-wrap: wrap; gap: 8px;
margin: 4px 0 18px; }
.step {
  display: flex; align-items: center; gap: 7px;
  padding: 8px 12px; font-size: 13px;
  background: #fff; border: 1px solid var(--line);
  border-radius: var(--radius);
}
.step__no {
  display: inline-flex; align-items: center;
justify-content: center;
  width: 20px; height: 20px; border-radius: 50%;
  background: #eef1f5; color: var(--ink-soft);
  font-size: 12px; font-variant-numeric: tabular-nums;
}
.step__name { font-weight: 600; }
.step__state { font-size: 12.5px; color:
var(--ink-soft); }
.step--done { border-color: #b8ddc6; background:
var(--ok-soft); }
.step--done .step__no { background: var(--ok); color:
#fff; }
.step--done .step__state { color: var(--ok); }
.step--fail { border-color: #eec4c0; background:
var(--danger-soft); }
.step--fail .step__no { background: var(--danger);
color: #fff; }
.step--fail .step__state { color: var(--danger);
font-weight: 600; }
.step--next { border-color: var(--accent); }
.step--next .step__no { background: var(--accent);
color: #fff; }
/* ---------- 提示块 ---------- */
.notice {
  padding: 14px 16px; margin: 0 0 16px;
  border-radius: var(--radius); border-left: 4px solid;
}
.notice--warn {
  background: var(--danger-soft);
  border-left-color: var(--danger);
  color: #7a1c17;
}
.notice__title { margin: 0 0 8px; font-weight: 600; }
.notice__list { margin: 0; padding-left: 20px;
line-height: 1.9; }
.notice__list a { color: var(--accent); }
```

```
.notice__hint { margin: 8px 0 0; font-size: 12.5px;
color: var(--ink-soft); }
/* ---------- 图斑面板（按图斑分块展示，如指标数据采集） ---------- */
.patch-panel {
  border: 1px solid var(--line);
  border-radius: var(--radius);
  background: var(--surface);
  padding: 14px 16px;
  margin: 0 0 14px;
}
.patch-panel:last-child { margin-bottom: 0; }
.patch-panel__head {
  display: flex;
  align-items: baseline;
  flex-wrap: wrap;
  gap: 6px 14px;
  margin: 0 0 12px;
  padding-bottom: 9px;
  border-bottom: 1px solid var(--line);
}
.patch-panel__name {
  margin: 0;
  font-size: 15px;
  font-weight: 600;
}
.patch-panel__meta {
  font-size: 12.5px;
  color: var(--ink-soft);
}
.patch-panel__meta a { color: var(--accent); }
.patch-panel__tail { margin-left: auto; }
.patch-panel .notice { margin: 0; }
.patch-panel table { margin: 0; }
/* ---------- 窄屏适配 ---------- */
@media (max-width: 960px) {
  .app-shell { flex-direction: column; }
  .app-side { width: auto; flex: none; }
  .app-brand { padding: 12px 14px; }
  .app-nav { padding: 4px 10px 10px; }
  .app-nav ul { display: flex; flex-wrap: wrap; gap:
4px; }
  .app-nav a { margin-bottom: 0; }
  .app-nav__group { margin: 8px 0 4px; }
  .app-user { padding: 10px 14px; }
  .app-header { padding: 0 14px; gap: 10px; }
  .search-box { width: 180px; }
  .app-main { padding: 16px 14px 20px; }
  .app-foot { padding: 0 14px 14px; }
}
@media (max-width: 640px) {
```

```
        .crumb { display: none; }                                          5001
        .search-box { width: auto; flex: 1; }                              5002
      }                                                                    5003
    </style>                                                               5004
</head>                                                                    5005
<body class="{% if not current_user.is_authenticated                       5006
    %}body--login{% endif %}">                                             5007
    <a class="skip-link" href="#main">跳到主要内容</a>                      5008
    {# 取一次即可（同一请求内重复调用返回相同内容）。                      5009
       类别为 toast 的渲染成居中浮层，其余仍是页面顶部的提示条 ——          5010
       错误提示需要停下来细读，不适合做成会消失的浮层。 #}                 5011
    {% set flash_msgs = get_flashed_messages(with_categories=t             5012
    rue) %}                                                                5013
    {# 类别为 toast / ok 的渲染成页面正中的浮层（操作完成的即时反馈）；   5014
       其余（尤其 error）仍是页面顶部的提示条 ——                          5015
       错误提示需要停下来细读，不适合做成会消失的浮层。                   5016
       没有浮层时这个容器是空的，不占任何位置。 #}                        5017
    <div class="toast-stack">                                              5018
      {% for category, message in flash_msgs if category in                5019
      ("toast", "ok") %}                                                   5020
      <div class="toast-center" role="status"                              5021
      aria-live="polite">{{ message }}</div>                               5022
      {% endfor %}                                                         5023
    </div>                                                                 5024
    {% if current_user.is_authenticated %}                                 5025
    <div class="app-shell">                                                5026
      <aside class="app-side">                                             5027
        <div class="app-brand">                                            5028
          <span class="app-brand__mark">{{                                 5029
      ui.icon("mountain", 21) }}</span>                                    5030
          <span>                                                           5031
            <span class="app-brand__name">矿山自然恢复评价</span><br>      5032
            <span class="app-brand__sub">适宜性评价系统 Ｖ1．0</span>       5033
          </span>                                                          5034
        </div>                                                             5035
        <nav class="app-nav" aria-label="功能导航">                        5036
          <ul>                                                             5037
            <li>                                                           5038
              <a href="/"{% if home_active %}                              5039
      aria-current="page"{% endif %}>                                      5040
                {{ ui.icon("home") }}<span>工作台</span>                   5041
              </a>                                                         5042
            </li>                                                          5043
          </ul>                                                            5044
          {% for g in nav_groups %}                                        5045
          <p class="app-nav__group">{{ g.title }}</p>                      5046
          <ul>                                                             5047
            {% for item in g.links %}                                      5048
            <li>                                                           5049
              <a href="{{ item.href }}"{% if item.active %}                5050
```

```
aria-current="page"{% endif %}>                                    5051
            {{ ui.icon(item.icon) }}<span>{{ item.label            5052
}}</span>                                                          5053
            {% if item.badge %}<span                               5054
class="app-nav__badge">{{ item.badge }}</span>{% endif             5055
%}                                                                 5056
          </a>                                                     5057
        </li>                                                      5058
        {% endfor %}                                               5059
      </ul>                                                        5060
      {% endfor %}                                                 5061
    </nav>                                                         5062
    <div class="app-user">                                         5063
      <span class="app-user__avatar">{{                            5064
current_user.username[0]|upper }}</span>                           5065
      <span>                                                       5066
        <span class="app-user__name">                              5067
          {{ role_labels.get(current_user.role,                    5068
current_user.role) }} · {{ current_user.username }}                5069
        </span><br>                                                5070
        <span class="app-user__state">                             5071
          <span class="app-user__dot"                              5072
aria-hidden="true"></span>在线                                     5073
        </span>                                                    5074
      </span>                                                      5075
      <a href="/logout" title="退出登录" aria-label="退出登录">{{  5076
ui.icon("logout", 17) }}</a>                                       5077
    </div>                                                         5078
</aside>                                                           5079
<div class="app-content">                                          5080
    <header class="app-header">                                    5081
      <nav class="crumb" aria-label="当前位置">                    5082
        <a href="/">首页</a>                                       5083
        {{ ui.icon("chevron", 13) }}                               5084
        <strong>{{ page_title }}</strong>                          5085
      </nav>                                                       5086
      <span class="app-header__spacer"></span>                     5087
      <form class="search-box" method="get"                        5088
action="/search" role="search">                                    5089
        {{ ui.icon("search", 16) }}                                5090
        <label for="q" class="visually-hidden">搜索项目、图斑、标准<5091
/label>                                                            5092
        <input id="q" type="search" name="q"                       5093
autocomplete="off"                                                 5094
               placeholder="搜索项目、图斑、标准" value="{{        5095
search_q }}">                                                      5096
        <button type="submit" aria-label="搜索">{{                 5097
ui.icon("chevron", 15) }}</button>                                 5098
      </form>                                                      5099
      <a class="icon-btn" href="/logs" title="操作日志"            5100
```

```
      aria-label="操作日志">                                                    5101
            {{ ui.icon("bell", 18) }}                                          5102
          </a>                                                                 5103
        </header>                                                              5104
    {% else %}                                                                 5105
    <div class="login-shell">                                                  5106
    {% endif %}                                                                5107
      <main id="main" class="app-main">                                        5108
        <div class="container">                                                5109
          {% for category, message in flash_msgs if category                   5110
      not in ("toast", "ok") %}                                                5111
          <p class="flash {{ category }}" role="status">{{                     5112
      message }}</p>                                                           5113
          {% endfor %}                                                         5114
          {% block content %}{% endblock %}                                    5115
        </div>                                                                 5116
      </main>                                                                  5117
    {% if current_user.is_authenticated %}                                     5118
        <footer class="app-foot">                                              5119
          <span>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0</span>                5120
          <span>数据截止 {{ today }}</span>                                    5121
        </footer>                                                              5122
      </div>                                                                   5123
    </div>                                                                     5124
    {% else %}                                                                 5125
    </div>                                                                     5126
    {% endif %}                                                                5127
    {% block scripts %}{% endblock %}                                          5128
</body>                                                                        5129
</html>                                                                        5130
#  ==== app/templates/_macros.html ====                                       5131
{#  共用宏：图标、流程状态面板  #}                                             5132
{#  ============================================================             5133
      ==================                                                       5134
    图标：内联 SVG 线性图标（24×24 视图框，随文字颜色走）                     5135
    不引图标字体或外部图片，避免额外请求与资源依赖；                          5136
    stroke 用 currentColor，在深色侧边栏与浅色内容区都能自适应。              5137
    ============================================================             5138
      ==================  #}                                                   5139
{% set ICONS = {                                                               5140
  "dot": '<circle cx="12" cy="12" r="3"/>',                                    5141
  "home": '<path d="M3 9.6 12 3l9 6.6V20a2 2 0 0 1-2 2H5a2                     5142
    2 0 0 1-2-2z"/><path d="M9.5 22v-7h5v7"/>',                                5143
  "folder": '<path d="M3 7.5A2 2 0 0 1 5 5.5h3.6l2 2.2H19a2                   5144
    2 0 0 1 2 2v7.8a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2z"/>',                        5145
  "book": '<path d="M4 4.8A2.8 2.8 0 0 1 6.8                                   5146
    2H20v17.5H6.8A2.8 2.8 0 0 0 4 22z"/><path d="M4                            5147
    19.5A2.8 2.8 0 0 1 6.8 16.7H20"/>',                                        5148
  "shield": '<path d="M12 22s8-4 8-10V5.2L12 2 4 5.2V12c0 6                    5149
    8 10 8 10z"/><path d="m9 12 2.2 2.2L15.5 10"/>',                           5150
```

```
  "layers": '<path d="m12 2.8 9 4.7-9 4.7-9-4.7z"/><path
    d="m3 12.5 9 4.7 9-4.7"/><path d="m3 17 9 4.7 9-4.7"/>',
  "sliders": '<path d="M4 21v-6M4 11V3M12 21v-9M12 8V3M20
    21v-4M20 13V3"/><path d="M1.5 15h5M9.5 12h5M17.5
    17h5"/>',
  "database": '<ellipse cx="12" cy="5.5" rx="8"
    ry="3.2"/><path d="M4 5.5v13c0 1.8 3.6 3.2 8 3.2s8-1.4
    8-3.2v-13"/><path d="M4 12c0 1.8 3.6 3.2 8 3.2s8-1.4
    8-3.2"/>',
  "target": '<circle cx="12" cy="12" r="9"/><circle cx="12"
    cy="12" r="4.6"/><circle cx="12" cy="12" r="1"/>',
  "file": '<path d="M14 2.8H7a2 2 0 0 0-2 2v14.4a2 2 0 0 0
    2 2h10a2 2 0 0 0 2-2V7.8z"/><path d="M14
    2.8V7.8h5"/><path d="M8.8 13h6.4M8.8 17h6.4"/>',
  "users": '<circle cx="9.5" cy="8" r="3.6"/><path d="M3.5
    20.5v-1.4A3.6 3.6 0 0 1 7.1 15.5h4.8a3.6 3.6 0 0 1 3.6
    3.6v1.4"/><path d="M17 4.9a3.6 3.6 0 0 1 0 6.9"/><path
    d="M19 15.8a3.6 3.6 0 0 1 2.4 3.4v1.3"/>',
  "search": '<circle cx="11" cy="11" r="7"/><path d="m20.5
    20.5-4-4"/>',
  "bell": '<path d="M18 9.5a6 6 0 1 0-12 0c0 5-2.2 6.2-2.2
    6.2h16.4S18 14.5 18 9.5"/><path d="M13.8 19.5a2 2 0 0
    1-3.6 0"/>',
  "plus": '<path d="M12 5v14M5 12h14"/>',
  "upload": '<path d="M12 16V4"/><path d="m7.5 8.5 4.5-4.5
    4.5 4.5"/><path d="M4 16.5v2A2.5 2.5 0 0 0 6.5
    21h11a2.5 2.5 0 0 0 2.5-2.5v-2"/>',
  "logout": '<path d="M15 4h3.5A2.5 2.5 0 0 1 21 6.5v11A2.5
    2.5 0 0 1 18.5 20H15"/><path d="M10 16.5 5.5 12 10
    7.5"/><path d="M5.5 12H15"/>',
  "chevron": '<path d="m9.5 5.5 6.5 6.5-6.5 6.5"/>',
  "warn": '<path d="M12 3.5 21.5 20H2.5z"/><path d="M12
    10v4.2"/><path d="M12 17.4h.01"/>',
  "mountain": '<path d="m3 20 6.5-11L14 17l2.5-4L21
    20z"/><circle cx="17" cy="6" r="2.2"/>',
} %}
{% macro icon(name, size=18) %}
<svg class="ico" width="{{ size }}" height="{{ size }}"
     viewBox="0 0 24 24"
      fill="none" stroke="currentColor" stroke-width="1.7"
      stroke-linecap="round" stroke-linejoin="round"
      aria-hidden="true" focusable="false">{{
     ICONS.get(name, ICONS.dot)|safe }}</svg>
{% endmacro %}
{# 评价流程步骤条：按论文 4.1 节顺序显示各前置步骤的完成情况
   顺序为：限制性条件判定 → 划分微地质单元 → 录入指标数据 → 执行评价。
   论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
   废弃矿山，按照地形地貌划分……三种地质微单元类型」，故判定在前。 #}
{% macro workflow_steps(progress) %}
<div class="steps">
```

```
  <div class="step {% if progress.constraint.passed                5201
    %}step--done{% elif progress.constraint.done                   5202
    %}step--fail{% else %}step--todo{% endif %}">                  5203
    <span class="step__no">1</span>                                5204
    <span class="step__name">限制性条件判定</span>                  5205
    <span class="step__state">                                     5206
      {% if progress.constraint.passed %}已通过                    5207
      {% elif progress.constraint.done %}未通过                    5208
      {% else %}待完成{% endif %}                                  5209
    </span>                                                        5210
  </div>                                                           5211
  <div class="step {% if progress.units.done %}step--done{%        5212
    else %}step--todo{% endif %}">                                 5213
    <span class="step__no">2</span>                                5214
    <span class="step__name">划分微地质单元</span>                  5215
    <span class="step__state">                                     5216
      {% if progress.units.done %}已完成{% else %}待完成{% endif   5217
    %}                                                             5218
    </span>                                                        5219
  </div>                                                           5220
  <div class="step {% if progress.data.ready %}step--done{%        5221
    else %}step--todo{% endif %}">                                 5222
    <span class="step__no">3</span>                                5223
    <span class="step__name">录入指标数据</span>                    5224
    <span class="step__state">                                     5225
      {% if progress.data.ready %}已完成{% else %}待完成{% endif   5226
    %}                                                             5227
    </span>                                                        5228
  </div>                                                           5229
  <div class="step step--next">                                    5230
    <span class="step__no">4</span>                                5231
    <span class="step__name">执行适宜性评价</span>                  5232
    <span class="step__state">当前步骤</span>                      5233
  </div>                                                           5234
</div>                                                             5235
{% endmacro %}                                                     5236
{# 前置条件未满足时的提示块：只列缺什么，并给出处理入口 #}          5237
{% macro workflow_blocking(progress) %}                            5238
{% if progress.blocking %}                                         5239
<div class="notice notice--warn">                                  5240
  <p class="notice__title">尚不能执行本步骤，需要先完成：</p>      5241
  <ol class="notice__list">                                        5242
    {% for b in progress.blocking %}                               5243
    <li>                                                           5244
      <strong>{{ b.label }}</strong>{% if b.reason %}：{{          5245
    b.reason }}{% endif %}                                         5246
      <a class="link-btn" href="{{ b.fix_url }}">{{                5247
    b.fix_label }}</a>                                             5248
    </li>                                                          5249
    {% endfor %}                                                   5250
```

```
    </ol>
</div>
{% endif %}
{% endmacro %}
{# 一行式状态摘要，用于列表页 #}
{% macro workflow_badge(progress) %}
{% if progress.ready %}
<span class="tag tag--ok">可执行评价</span>
{% elif progress.blocking %}
<span class="tag tag--warn">{{ progress.blocking[0].label }}</span>
<span class="muted">{{ progress.blocking[0].reason }}</span>
{% endif %}
{% endmacro %}
{# 层级引导：本系统是「项目 → 图斑」两级结构，评价针对的是图斑而非项目。
   下游各页按图斑组织数据，故只要存在「还没建图斑的项目」，就直接列出并给出入口。 #}
{% macro patch_guide(projects_without_patch) %}
{% if projects_without_patch %}
<div class="notice notice--warn">
  <p class="notice__title">
    以下 {{ projects_without_patch|length }}
    个项目下还没有图斑，本页不会出现它们：
  </p>
  <ul class="notice__list">
    {% for p in projects_without_patch %}
    <li>
      {{ p.name }}
      <a class="link-btn" href="/project/{{ p.id }}/patch/add">新建图斑</a>
    </li>
    {% endfor %}
  </ul>
</div>
{% endif %}
{% endmacro %}
# ==== app/templates/_login_badge.svg ====
<svg class="lp-badge__svg" xmlns="http://www.w3.org/2000/svg"
     viewBox="0 0 240 240" role="img"
     aria-label="历史遗留废弃矿山自然恢复适宜性评价系统徽标">
  <title>系统徽标</title>
  <g class="lp-ring" style="--k: 2.7">
    <circle cx="162.00" cy="120.00" r="2.60"
    fill="#c4f5ab" />
    <circle cx="157.84" cy="138.22" r="2.97"
    fill="#b8f5ab" />
    <circle cx="146.19" cy="152.84" r="3.06"
    fill="#acf5ab" />
    <circle cx="129.35" cy="160.95" r="2.80"
    fill="#abf5b7" />
    <circle cx="110.65" cy="160.95" r="2.40"
```

```
    fill="#abf5c2" />                                                    5301
    <circle cx="93.81" cy="152.84" r="2.14" fill="#abf5ce" /> 5302
    <circle cx="82.16" cy="138.22" r="2.23" fill="#abf5da" /> 5303
    <circle cx="78.00" cy="120.00" r="2.60" fill="#abf5e6" /> 5304
    <circle cx="82.16" cy="101.78" r="2.97" fill="#abf5da" /> 5305
    <circle cx="93.81" cy="87.16" r="3.06" fill="#abf5ce" /> 5306
    <circle cx="110.65" cy="79.05" r="2.80" fill="#abf5c2" /> 5307
    <circle cx="129.35" cy="79.05" r="2.40" fill="#abf5b7" /> 5308
    <circle cx="146.19" cy="87.16" r="2.14" fill="#acf5ab" /> 5309
    <circle cx="157.84" cy="101.78" r="2.23"                             5310
    fill="#b8f5ab" />                                                    5311
  </g>                                                                   5312
  <g class="lp-ring" style="--k: 2.15">                                  5313
    <circle cx="177.29" cy="129.07" r="3.92"                             5314
    fill="#a4e28d" />                                                    5315
    <circle cx="171.68" cy="146.33" r="4.01"                             5316
    fill="#9be28d" />                                                    5317
    <circle cx="161.01" cy="161.01" r="3.87"                             5318
    fill="#91e28d" />                                                    5319
    <circle cx="146.33" cy="171.68" r="3.55"                             5320
    fill="#8de292" />                                                    5321
    <circle cx="129.07" cy="177.29" r="3.17"                             5322
    fill="#8de29c" />                                                    5323
    <circle cx="110.93" cy="177.29" r="2.88"                             5324
    fill="#8de2a6" />                                                    5325
    <circle cx="93.67" cy="171.68" r="2.79" fill="#8de2af" /> 5326
    <circle cx="78.99" cy="161.01" r="2.93" fill="#8de2b9" /> 5327
    <circle cx="68.32" cy="146.33" r="3.25" fill="#8de2c2" /> 5328
    <circle cx="62.71" cy="129.07" r="3.63" fill="#8de2cc" /> 5329
    <circle cx="62.71" cy="110.93" r="3.92" fill="#8de2cc" /> 5330
    <circle cx="68.32" cy="93.67" r="4.01" fill="#8de2c2" /> 5331
    <circle cx="78.99" cy="78.99" r="3.87" fill="#8de2b9" /> 5332
    <circle cx="93.67" cy="68.32" r="3.55" fill="#8de2af" /> 5333
    <circle cx="110.93" cy="62.71" r="3.17" fill="#8de2a6" /> 5334
    <circle cx="129.07" cy="62.71" r="2.88" fill="#8de29c" /> 5335
    <circle cx="146.33" cy="68.32" r="2.79" fill="#8de292" /> 5336
    <circle cx="161.01" cy="78.99" r="2.93" fill="#91e28d" /> 5337
    <circle cx="171.68" cy="93.67" r="3.25" fill="#9be28d" /> 5338
    <circle cx="177.29" cy="110.93" r="3.63"                             5339
    fill="#a4e28d" />                                                    5340
  </g>                                                                   5341
  <g class="lp-ring" style="--k: 1.85">                                  5342
    <circle cx="194.00" cy="120.00" r="4.94"                             5343
    fill="#91cf72" />                                                    5344
    <circle cx="191.85" cy="137.71" r="4.92"                             5345
    fill="#88cf72" />                                                    5346
    <circle cx="185.52" cy="154.39" r="4.73"                             5347
    fill="#80cf72" />                                                    5348
    <circle cx="175.39" cy="169.07" r="4.42"                             5349
    fill="#78cf72" />                                                    5350
```

```
    <circle cx="162.04" cy="180.90" r="4.06"                          5351
    fill="#72cf73"/>                                                  5352
    <circle cx="146.24" cy="189.19" r="3.73"                          5353
    fill="#72cf7b"/>                                                  5354
    <circle cx="128.92" cy="193.46" r="3.51"                          5355
    fill="#72cf83"/>                                                  5356
    <circle cx="111.08" cy="193.46" r="3.45"                          5357
    fill="#72cf8b"/>                                                  5358
    <circle cx="93.76" cy="189.19" r="3.56" fill="#72cf93"/>          5359
    <circle cx="77.96" cy="180.90" r="3.81" fill="#72cf9c"/>          5360
    <circle cx="64.61" cy="169.07" r="4.16" fill="#72cfa4"/>          5361
    <circle cx="54.48" cy="154.39" r="4.51" fill="#72cfac"/>          5362
    <circle cx="48.15" cy="137.71" r="4.80" fill="#72cfb4"/>          5363
    <circle cx="46.00" cy="120.00" r="4.94" fill="#72cfbc"/>          5364
    <circle cx="48.15" cy="102.29" r="4.92" fill="#72cfb4"/>          5365
    <circle cx="54.48" cy="85.61" r="4.73" fill="#72cfac"/>           5366
    <circle cx="64.61" cy="70.93" r="4.42" fill="#72cfa4"/>           5367
    <circle cx="77.96" cy="59.10" r="4.06" fill="#72cf9c"/>           5368
    <circle cx="93.76" cy="50.81" r="3.73" fill="#72cf93"/>           5369
    <circle cx="111.08" cy="46.54" r="3.51" fill="#72cf8b"/>          5370
    <circle cx="128.92" cy="46.54" r="3.45" fill="#72cf83"/>          5371
    <circle cx="146.24" cy="50.81" r="3.56" fill="#72cf7b"/>          5372
    <circle cx="162.04" cy="59.10" r="3.81" fill="#72cf73"/>          5373
    <circle cx="175.39" cy="70.93" r="4.16" fill="#78cf72"/>          5374
    <circle cx="185.52" cy="85.61" r="4.51" fill="#80cf72"/>          5375
    <circle cx="191.85" cy="102.29" r="4.80"                          5376
    fill="#88cf72"/>                                                  5377
  </g>                                                                5378
  <g class="lp-ring" style="--k: 1.62">                               5379
    <circle cx="209.57" cy="128.82" r="5.68"                          5380
    fill="#76bb59"/>                                                  5381
    <circle cx="206.12" cy="146.13" r="5.40"                          5382
    fill="#6fbb59"/>                                                  5383
    <circle cx="199.37" cy="162.43" r="5.06"                          5384
    fill="#68bb59"/>                                                  5385
    <circle cx="189.57" cy="177.10" r="4.71"                          5386
    fill="#61bb59"/>                                                  5387
    <circle cx="177.10" cy="189.57" r="4.41"                          5388
    fill="#5abb59"/>                                                  5389
    <circle cx="162.43" cy="199.37" r="4.19"                          5390
    fill="#59bb5f"/>                                                  5391
    <circle cx="146.13" cy="206.12" r="4.10"                          5392
    fill="#59bb66"/>                                                  5393
    <circle cx="128.82" cy="209.57" r="4.15"                          5394
    fill="#59bb6d"/>                                                  5395
    <circle cx="111.18" cy="209.57" r="4.32"                          5396
    fill="#59bb73"/>                                                  5397
    <circle cx="93.87" cy="206.12" r="4.60" fill="#59bb7a"/>          5398
    <circle cx="77.57" cy="199.37" r="4.94" fill="#59bb81"/>          5399
    <circle cx="62.90" cy="189.57" r="5.29" fill="#59bb88"/>          5400
```

```
    <circle cx="50.43" cy="177.10" r="5.59" fill="#59bb8f"/>5401
    <circle cx="40.63" cy="162.43" r="5.81" fill="#59bb96"/>5402
    <circle cx="33.88" cy="146.13" r="5.90" fill="#59bb9d"/>5403
    <circle cx="30.43" cy="128.82" r="5.85" fill="#59bba4"/>5404
    <circle cx="30.43" cy="111.18" r="5.68" fill="#59bba4"/>5405
    <circle cx="33.88" cy="93.87" r="5.40" fill="#59bb9d"/> 5406
    <circle cx="40.63" cy="77.57" r="5.06" fill="#59bb96"/> 5407
    <circle cx="50.43" cy="62.90" r="4.71" fill="#59bb8f"/> 5408
    <circle cx="62.90" cy="50.43" r="4.41" fill="#59bb88"/> 5409
    <circle cx="77.57" cy="40.63" r="4.19" fill="#59bb81"/> 5410
    <circle cx="93.87" cy="33.88" r="4.10" fill="#59bb7a"/> 5411
    <circle cx="111.18" cy="30.43" r="4.15" fill="#59bb73"/>5412
    <circle cx="128.82" cy="30.43" r="4.32" fill="#59bb6d"/>5413
    <circle cx="146.13" cy="33.88" r="4.60" fill="#59bb66"/>5414
    <circle cx="162.43" cy="40.63" r="4.94" fill="#59bb5f"/>5415
    <circle cx="177.10" cy="50.43" r="5.29" fill="#5abb59"/>5416
    <circle cx="189.57" cy="62.90" r="5.59" fill="#61bb59"/>5417
    <circle cx="199.37" cy="77.57" r="5.81" fill="#68bb59"/>5418
    <circle cx="206.12" cy="93.87" r="5.90" fill="#6fbb59"/>5419
    <circle cx="209.57" cy="111.18" r="5.85"                 5420
    fill="#76bb59"/>                                         5421
  </g>                                                       5422
  <g class="lp-ring" style="--k: 1.46">                      5423
    <circle cx="226.00" cy="120.00" r="5.94"                 5424
    fill="#65a843"/>                                         5425
    <circle cx="224.55" cy="137.45" r="5.62"                 5426
    fill="#5fa843"/>                                         5427
    <circle cx="220.26" cy="154.42" r="5.29"                 5428
    fill="#59a843"/>                                         5429
    <circle cx="213.22" cy="170.45" r="5.00"                 5430
    fill="#53a843"/>                                         5431
    <circle cx="203.65" cy="185.11" r="4.77"                 5432
    fill="#4da843"/>                                         5433
    <circle cx="191.79" cy="197.99" r="4.63"                 5434
    fill="#47a843"/>                                         5435
    <circle cx="177.98" cy="208.74" r="4.59"                 5436
    fill="#43a846"/>                                         5437
    <circle cx="162.58" cy="217.07" r="4.67"                 5438
    fill="#43a84c"/>                                         5439
    <circle cx="146.02" cy="222.76" r="4.85"                 5440
    fill="#43a852"/>                                         5441
    <circle cx="128.75" cy="225.64" r="5.10"                 5442
    fill="#43a858"/>                                         5443
    <circle cx="111.25" cy="225.64" r="5.42"                 5444
    fill="#43a85e"/>                                         5445
    <circle cx="93.98" cy="222.76" r="5.75" fill="#43a864"/>5446
    <circle cx="77.42" cy="217.07" r="6.06" fill="#43a86a"/>5447
    <circle cx="62.02" cy="208.74" r="6.33" fill="#43a870"/>5448
    <circle cx="48.21" cy="197.99" r="6.52" fill="#43a876"/>5449
    <circle cx="36.35" cy="185.11" r="6.60" fill="#43a87c"/>5450
```

```
      <circle cx="26.78" cy="170.45" r="6.58" fill="#43a882"/>
      <circle cx="19.74" cy="154.42" r="6.45" fill="#43a888"/>
      <circle cx="15.45" cy="137.45" r="6.23" fill="#43a88e"/>
      <circle cx="14.00" cy="120.00" r="5.94" fill="#43a894"/>
      <circle cx="15.45" cy="102.55" r="5.62" fill="#43a88e"/>
      <circle cx="19.74" cy="85.58" r="5.29" fill="#43a888"/>
      <circle cx="26.78" cy="69.55" r="5.00" fill="#43a882"/>
      <circle cx="36.35" cy="54.89" r="4.77" fill="#43a87c"/>
      <circle cx="48.21" cy="42.01" r="4.63" fill="#43a876"/>
      <circle cx="62.02" cy="31.26" r="4.59" fill="#43a870"/>
      <circle cx="77.42" cy="22.93" r="4.67" fill="#43a86a"/>
      <circle cx="93.98" cy="17.24" r="4.85" fill="#43a864"/>
      <circle cx="111.25" cy="14.36" r="5.10" fill="#43a85e"/>
      <circle cx="128.75" cy="14.36" r="5.42" fill="#43a858"/>
      <circle cx="146.02" cy="17.24" r="5.75" fill="#43a852"/>
      <circle cx="162.58" cy="22.93" r="6.06" fill="#43a84c"/>
      <circle cx="177.98" cy="31.26" r="6.33" fill="#43a846"/>
      <circle cx="191.79" cy="42.01" r="6.52" fill="#47a843"/>
      <circle cx="203.65" cy="54.89" r="6.60" fill="#4da843"/>
      <circle cx="213.22" cy="69.55" r="6.58" fill="#53a843"/>
      <circle cx="220.26" cy="85.58" r="6.45" fill="#59a843"/>
      <circle cx="224.55" cy="102.55" r="6.23"
      fill="#5fa843"/>
    </g>
    <g class="lp-badge__mark">
      <circle cx="120" cy="120" r="27" fill="#14432f"/>
      <g transform="translate(120 120) scale(1.5)
      translate(-12 -12)" fill="none" stroke="#ffffff"
      stroke-width="1.9" stroke-linecap="round"
      stroke-linejoin="round">
        <path d="m3 20 6.5-11L14 17l2.5-4L21 20z"/>
        <circle cx="17" cy="6" r="2.2"/>
      </g>
    </g>
  </svg>
  # ==== app/templates/error.html ====
  {% extends "base.html" %}
  {% block title %}{{ code }} · {{ message }}{% endblock %}
  {% block content %}
  <div class="card" style="max-width:560px;">
    <h2>{{ code }} · {{ message }}</h2>
    <p class="muted">
      请返回上一页重试；若确认该操作应当可用，请检查当前账号的角色权限。
    </p>
    <p class="actions" style="margin-top:16px;">
      <a class="btn" href="/">返回首页</a>
    </p>
  </div>
  {% endblock %}
  # ==== app/templates/auth/login.html ====
```

```
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}登录 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 登录页（仿 Apple 账户页） ================== */
     纯白底、居中、大量留白；色彩只用在徽标与按钮上。
     首屏放徽标、标题、一句话说明与登录按钮；系统介绍在下方，
     随滚动逐条弹入、往回滚时弹回。 */
  .lp { color: #1d1d1f; }
  /* ---------- 首屏 ---------- */
  .lp-hero {
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    text-align: center;
    min-height: 100dvh;
    padding: 56px 20px 72px;
  }
  /* 徽标：滚动时各圈向四周散开并淡出。
     --p 为 0~1 的滚动进度，由页面脚本写入；
     每圈自带 --k（散开倍数，由内向外递减），使各圈逐渐拉开距离 */
  .lp-badge {
    width: 212px;
    height: 212px;
    margin-bottom: 36px;
    --p: 0;
    animation: lp-in .7s ease both;
  }
  .lp-badge__svg {
    display: block;
    width: 100%;
    height: 100%;
    overflow: visible;                    /* 散开时圆点不被画布裁掉 */
    opacity: calc(1 - var(--p) * .94);
    filter: blur(calc(var(--p) * 2.6px));
  }
  .lp-ring {
    transform-box: fill-box;
    transform-origin: center;
    transform: scale(calc(1 + (var(--k) - 1) * var(--p)));
  }
  .lp-badge__mark {
    transform-box: fill-box;
    transform-origin: center;
    transform: scale(calc(1 + .18 * var(--p)));
  }
  .lp-title {
    margin: 0 0 18px;
```

```
    font-size: clamp(28px, 4.2vw, 42px);
    font-weight: 700;
    line-height: 1.3;
    letter-spacing: -.4px;
    animation: lp-in .7s ease .06s both;
  }
  .lp-cta {
    padding: 13px 52px;
    font-family: inherit;
    font-size: 17px;
    font-weight: 500;
    color: #fff;
    background: var(--accent);
    border: none;
    border-radius: 980px;
    cursor: pointer;
    animation: lp-in .7s ease .18s both;
    transition: background-color .2s ease, box-shadow .2s
    ease,
                transform .2s ease;
  }
  .lp-cta:hover {
    background: var(--accent-hover);
    box-shadow: 0 10px 26px rgba(31, 107, 69, .26);
  }
  .lp-cta:active { transform: translateY(1px); box-shadow:
    none; }
  .lp-cta:focus-visible { outline: 3px solid rgba(31, 107,
    69, .40); outline-offset: 3px; }
  .lp-more {
    margin-top: 38px;
    font-size: 13px;
    color: #86868b;
    text-decoration: none;
    animation: lp-in .7s ease .24s both;
  }
  .lp-more:hover { color: var(--accent); }
  @keyframes lp-in {
    from { opacity: 0; transform: translateY(12px); }
    to   { opacity: 1; transform: none; }
  }
  /* ---------- 滚动进场 ---------- */
     进入视口时按方向弹入，离开视口时弹回，可反复。
     交错延迟只加在「进入」的规则上，退出时不延迟，否则会显得拖沓。 */
  .lp-reveal {
    opacity: 0;
    transition: opacity .62s ease,
                transform .62s cubic-bezier(.16, .84, .34,
    1),
                border-color .62s ease, background-color
```

```
    .62s ease;                                                    5601
    transition-delay: 0s;                                         5602
    will-change: opacity, transform;                              5603
  }                                                               5604
  .lp-reveal[data-dir="left"]  { transform:                       5605
    translate3d(-48px, 0, 0); }                                   5606
  .lp-reveal[data-dir="right"] { transform:                       5607
    translate3d(48px, 0, 0); }                                    5608
  .lp-reveal[data-dir="up"]    { transform: translate3d(0,        5609
    36px, 0); }                                                   5610
  .lp-reveal.is-in {                                              5611
    opacity: 1;                                                   5612
    transform: none;                                              5613
    transition-delay: calc(var(--i, 0) * 85ms);                   5614
  }                                                               5615
  /* 标题颜色由淡到深，强化「浮现」的层次感 */                    5616
  .lp-reveal h2,                                                  5617
  .lp-reveal h3,                                                  5618
  .lp-reveal .lp-mod__name { color: #c2d2c8; transition:          5619
    color .7s ease; }                                             5620
  .lp-reveal.is-in h3,                                            5621
  .lp-reveal.is-in .lp-mod__name { color: var(--accent); }        5622
  .lp-reveal.is-in h2 { color: #1d1d1f; }                         5623
  /* ---------- 系统介绍 ---------- */                           5624
  .lp-intro {                                                     5625
    max-width: 980px;                                             5626
    margin: 0 auto;                                               5627
    padding: 20px 24px 72px;                                      5628
  }                                                               5629
  .lp-intro__title {                                              5630
    margin: 0 0 26px;                                             5631
    font-size: clamp(22px, 2.6vw, 30px);                          5632
    text-align: center;                                           5633
    letter-spacing: -.3px;                                        5634
  }                                                               5635
  .lp-points {                                                    5636
    display: grid;                                                5637
    grid-template-columns: repeat(auto-fit, minmax(240px,         5638
    1fr));                                                        5639
    gap: 16px;                                                    5640
    margin-bottom: 46px;                                          5641
  }                                                               5642
  .lp-point {                                                     5643
    padding: 24px 22px;                                           5644
    background: #f5f5f7;                                          5645
    border-radius: 18px;                                          5646
  }                                                               5647
  .lp-point__icon {                                               5648
    display: grid;                                                5649
    place-items: center;                                          5650
```

```
    width: 38px;
    height: 38px;
    margin-bottom: 14px;
    border-radius: 11px;
    background: #fff;
    color: var(--accent);
  }
  .lp-point h3 { margin: 0 0 8px; font-size: 16px; }
  .lp-point p {
    margin: 0;
    font-size: 13.5px;
    line-height: 1.85;
    color: #6e6e73;
    text-wrap: pretty;
  }
  .lp-sub-title { margin: 0 0 20px; font-size: 17px; }
  .lp-mods {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(258px,
    1fr));
    gap: 12px;
    margin-bottom: 46px;
    align-items: start;
  }
  /* 模块卡：默认只显示编号与名称，鼠标悬停或键盘聚焦时才展开职责说明 */
  .lp-mod {
    position: relative;
    padding: 17px 18px 17px 48px;
    background: #fff;
    border: 1px solid #e6e6e8;
    border-radius: 14px;
    outline: none;
    transition: border-color .3s ease, box-shadow .3s ease;
  }
  .lp-mod:hover,
  .lp-mod:focus-within,
  .lp-mod:focus {
    border-color: rgba(31, 107, 69, .42);
    box-shadow: 0 8px 22px rgba(19, 60, 43, .08);
  }
  .lp-mod__no {
    position: absolute;
    left: 17px;
    top: 17px;
    font-size: 13px;
    font-weight: 700;
    color: #c8d6cd;
    font-variant-numeric: tabular-nums;
  }
  .lp-mod__name {
```

```
    display: block;
    font-size: 14px;
    font-weight: 600;
  }
  /* 用 grid-template-rows 0fr→1fr 展开，高度过渡自然 */
  .lp-mod__body {
    display: grid;
    grid-template-rows: 0fr;
    transition: grid-template-rows .34s ease, margin-top
    .34s ease;
  }
  .lp-mod:hover .lp-mod__body,
  .lp-mod:focus-within .lp-mod__body {
    grid-template-rows: 1fr;
    margin-top: 7px;
  }
  .lp-mod__desc {
    overflow: hidden;
    margin: 0;
    font-size: 12.5px;
    line-height: 1.7;
    color: #6e6e73;
    opacity: 0;
    transition: opacity .3s ease;
    text-wrap: pretty;
  }
  .lp-mod:hover .lp-mod__desc,
  .lp-mod:focus-within .lp-mod__desc { opacity: 1; }
  /* 触摸设备没有悬停，直接把说明摊开 */
  @media (hover: none) {
    .lp-mod__body { grid-template-rows: 1fr; margin-top:
    7px; }
    .lp-mod__desc { opacity: 1; }
  }
  .lp-foot {
    margin: 0;
    padding-top: 26px;
    border-top: 1px solid #e6e6e8;
    text-align: center;
    font-size: 12.5px;
    color: #86868b;
  }
  /* ---------- 登录面板：点击后渐变浮现 ---------- */
  .lp-auth {
    position: fixed;
    inset: 0;
    z-index: 60;
    display: grid;
    place-items: center;
    padding: 24px;
```

```
    opacity: 0;                                                          5751
    visibility: hidden;                                                  5752
    transition: opacity .42s ease, visibility 0s linear                  5753
    .42s;                                                                5754
  }                                                                      5755
  .lp-auth.is-open {                                                     5756
    opacity: 1;                                                          5757
    visibility: visible;                                                 5758
    transition: opacity .42s ease, visibility 0s;                        5759
  }                                                                      5760
  /* 遮罩用绿色径向渐变，形成「渐变渗出来」的观感 */                                  5761
  .lp-auth__scrim {                                                      5762
    position: absolute;                                                  5763
    inset: 0;                                                            5764
    background:                                                          5765
      radial-gradient(1000px 640px at 50% 0%,                            5766
                      rgba(31, 107, 69, .22), rgba(31, 107,              5767
    69, 0) 70%),                                                         5768
      linear-gradient(180deg, rgba(255, 255, 255, .84),                  5769
                              rgba(246, 250, 247, .95));                 5770
    backdrop-filter: blur(16px) saturate(1.25);                          5771
    -webkit-backdrop-filter: blur(16px) saturate(1.25);                  5772
  }                                                                      5773
  .lp-auth__panel {                                                      5774
    position: relative;                                                  5775
    width: 100%;                                                         5776
    max-width: 396px;                                                    5777
    padding: 36px 36px 26px;                                             5778
    background: #fff;                                                    5779
    border-radius: 20px;                                                 5780
    box-shadow: 0 28px 80px rgba(14, 46, 32, .18);                       5781
    opacity: 0;                                                          5782
    transform: translateY(18px) scale(.975);                             5783
    transition: transform .55s cubic-bezier(.16, .84, .34,               5784
    1) .05s,                                                             5785
                opacity .55s ease .05s;                                  5786
  }                                                                      5787
  .lp-auth.is-open .lp-auth__panel { opacity: 1; transform:              5788
    none; }                                                              5789
  .lp-auth__close {                                                      5790
    position: absolute;                                                  5791
    top: 14px;                                                           5792
    right: 14px;                                                         5793
    display: grid;                                                       5794
    place-items: center;                                                 5795
    width: 30px;                                                         5796
    height: 30px;                                                        5797
    color: #86868b;                                                      5798
    background: transparent;                                             5799
    border: none;                                                        5800
```

```
    border-radius: 50%;                                                    5801
    cursor: pointer;                                                       5802
    transition: background-color .18s ease, color .18s ease;               5803
}                                                                          5804
.lp-auth__close:hover { background: #f5f5f7; color:                        5805
    #1d1d1f; }                                                             5806
.lp-auth__title {                                                          5807
    margin: 0 0 6px;                                                       5808
    text-align: center;                                                    5809
    font-size: 24px;                                                       5810
    font-weight: 700;                                                      5811
    letter-spacing: -.2px;                                                 5812
}                                                                          5813
.lp-auth__sub {                                                            5814
    margin: 0 0 26px;                                                      5815
    text-align: center;                                                    5816
    font-size: 13px;                                                       5817
    color: #86868b;                                                        5818
}                                                                          5819
.lp-field { margin-bottom: 15px; }                                         5820
.lp-field label {                                                          5821
    display: block;                                                        5822
    margin-bottom: 7px;                                                    5823
    font-size: 13px;                                                       5824
    color: #6e6e73;                                                        5825
}                                                                          5826
.lp-field input {                                                          5827
    width: 100%;                                                           5828
    padding: 13px 16px;                                                    5829
    font-family: inherit;                                                  5830
    font-size: 16px;              /* 不小于 16px，避免手机端聚焦时自动放大 */    5831
    color: #1d1d1f;                                                        5832
    background: #f5f5f7;                                                   5833
    border: 1px solid transparent;                                         5834
    border-radius: 12px;                                                   5835
    outline: none;                                                         5836
    transition: background-color .18s ease, border-color                   5837
    .18s ease,                                                             5838
                box-shadow .18s ease;                                      5839
}                                                                          5840
.lp-field input::placeholder { color: #a1a1a6; }                           5841
.lp-field input:focus {                                                    5842
    background: #fff;                                                      5843
    border-color: rgba(31, 107, 69, .55);                                  5844
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                          5845
}                                                                          5846
.lp-remember {                                                             5847
    display: flex;                                                         5848
    align-items: center;                                                   5849
    gap: 8px;                                                              5850
```

```
    margin: 4px 0 20px;
    font-size: 13px;
    color: #6e6e73;
    cursor: pointer;
}
.lp-remember input {
    width: 16px;
    height: 16px;
    margin: 0;
    accent-color: var(--accent);
    cursor: pointer;
}
.lp-submit {
    width: 100%;
    padding: 13px 20px;
    font-family: inherit;
    font-size: 16px;
    font-weight: 500;
    color: #fff;
    background: var(--accent);
    border: none;
    border-radius: 980px;
    cursor: pointer;
    transition: background-color .2s ease, box-shadow .2s
    ease,
                transform .2s ease;
}
.lp-submit:hover {
    background: var(--accent-hover);
    box-shadow: 0 10px 24px rgba(31, 107, 69, .26);
}
.lp-submit:active { transform: translateY(1px);
    box-shadow: none; }
.lp-submit:focus-visible { outline: 3px solid rgba(31,
    107, 69, .40); outline-offset: 3px; }
.lp-auth__hint {
    margin: 18px 0 0;
    text-align: center;
    font-size: 12.5px;
    color: #a1a1a6;
}
/* 系统设置里要求减少动态效果时，去掉位移、缩放与滚动特效 */
@media (prefers-reduced-motion: reduce) {
    .lp-badge, .lp-title, .lp-cta, .lp-more { animation:
    none; }
    .lp-badge__svg { opacity: 1; filter: none; }
    .lp-ring, .lp-badge__mark { transform: none; }
    .lp-reveal { opacity: 1; transform: none !important; }
    .lp-reveal h2, .lp-reveal h3, .lp-reveal .lp-mod__name
    { color: inherit; }
```

```
      .lp-reveal.is-in h2 { color: #1d1d1f; }                                  5901
      .lp-reveal.is-in h3, .lp-reveal.is-in .lp-mod__name {                    5902
      color: var(--accent); }                                                  5903
      .lp-auth, .lp-auth__panel { transition: opacity .2s                      5904
      linear; }                                                                5905
      .lp-auth__panel { transform: none; }                                     5906
   }                                                                           5907
   @media (max-width: 560px) {                                                 5908
      .lp-hero { padding: 36px 18px 48px; min-height: auto; }                  5909
      .lp-badge { width: 160px; height: 160px; margin-bottom:                  5910
      26px; }                                                                  5911
      .lp-auth__panel { padding: 30px 24px 22px; }                             5912
   }                                                                           5913
</style>                                                                       5914
<div class="lp">                                                               5915
   {# ---------------- 首屏：徽标 + 标题 + 说明 + 登录入口                      5916
      ---------------- #}                                                      5917
   <section class="lp-hero">                                                   5918
      <div class="lp-badge" id="lp-badge" aria-hidden="true">                  5919
         {% include "_login_badge.svg" %}                                      5920
      </div>                                                                   5921
      <h1 class="lp-title">历史遗留废弃矿山<br>自然恢复适宜性评价系统</h1>          5922
      <button type="button" class="lp-cta" id="lp-open"                        5923
              aria-expanded="false" aria-controls="lp-auth">登录               5924
      </button>                                                                5925
      <a class="lp-more" href="#intro">                                        5926
         向下滑动，了解系统 <span aria-hidden="true">↓</span>                  5927
      </a>                                                                     5928
   </section>                                                                  5929
   {# ---------------- 系统介绍：随滚动逐条弹入 ---------------- #}             5930
   <section class="lp-intro" id="intro">                                       5931
      <h2 class="lp-intro__title lp-reveal"                                    5932
      data-dir="up">一套把评价流程固定下来的工具</h2>                          5933
      <div class="lp-points">                                                  5934
         {% set pdirs = ['left', 'up', 'right'] %}                             5935
         {% set points = [                                                     5936
            ('sliders', '方法可溯',                                            5937
             '权重由层次分析法算出并做一致性检验，指标标准化、分单元赋分与面积加权合成的每一步都留有记录，可逐层回 5938
      看。'),                                                                  5939
            ('shield', '流程可控',                                             5940
             '政策、区位约束、土壤污染、地质灾害、耕地属性五项限制性条件一票否决；未通过即终止评价，直接给出生态重 5941
      建或人工辅助的结论。'),                                                 5942
            ('file', '结果可复算',                                             5943
             '评价结果与逐层得分明细一并留存，报告取自已保存的评价记录，事后修改数据不会让报告与结论对不上。')   5944
         ] %}                                                                  5945
         {% for ic, title, text in points %}                                   5946
         <div class="lp-point lp-reveal" data-dir="{{                          5947
      pdirs[loop.index0 % 3] }}"                                               5948
              style="--i: {{ loop.index0 }}">                                  5949
            <span class="lp-point__icon">{{ icon(ic, 20)                       5950
```

```
    }}</span>                                                          5951
        <h3>{{ title }}</h3>                                           5952
        <p>{{ text }}</p>                                              5953
      </div>                                                           5954
      {% endfor %}                                                     5955
    </div>                                                             5956
    <h3 class="lp-sub-title lp-reveal"                                 5957
    data-dir="up">九个功能模块</h3>                                    5958
    <div class="lp-mods">                                              5959
      {% set mdirs = ['left', 'up', 'right'] %}                        5960
      {% for no, name, desc in modules %}                              5961
      <div class="lp-mod lp-reveal" data-dir="{{                       5962
    mdirs[loop.index0 % 3] }}"                                         5963
          style="--i: {{ loop.index0 % 3 }}" tabindex="0">             5964
        <span class="lp-mod__no">{{ no }}</span>                       5965
        <span class="lp-mod__name">{{ name }}</span>                   5966
        <div class="lp-mod__body"><p class="lp-mod__desc">{{           5967
    desc }}</p></div>                                                  5968
      </div>                                                           5969
      {% endfor %}                                                     5970
    </div>                                                             5971
    <p class="lp-foot lp-reveal" data-dir="up">                        5972
      历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1．0 ·                  5973
      运行于浏览器，数据保存在本机 ＳＱＬｉｔｅ 数据库                  5974
    </p>                                                               5975
  </section>                                                           5976
  {# ---------------- 登录面板：点击「登录」后渐变浮现 ---------------- #}  5977
  <div class="lp-auth" id="lp-auth" aria-hidden="true">                5978
    <div class="lp-auth__scrim" data-lp-close></div>                   5979
    <div class="lp-auth__panel" role="dialog"                          5980
    aria-modal="true"                                                  5981
        aria-labelledby="lp-auth-title">                               5982
      <button type="button" class="lp-auth__close"                     5983
    data-lp-close                                                      5984
          aria-label="关闭登录框">                                     5985
        <svg width="16" height="16" viewBox="0 0 24 24"                5986
    fill="none"                                                        5987
          stroke="currentColor" stroke-width="2"                       5988
    stroke-linecap="round"                                             5989
          aria-hidden="true"><path d="M6 6l12 12M18 6L6                5990
    18"/></svg>                                                        5991
      </button>                                                        5992
      <h2 class="lp-auth__title" id="lp-auth-title">登录</h2>          5993
      <p class="lp-auth__sub">使用系统账号登录</p>                     5994
      <form method="post" novalidate>                                  5995
        {{ form.hidden_tag() }}                                        5996
        <div class="lp-field">                                         5997
          <label for="{{ form.username.id }}">用户名</label>           5998
          {{ form.username(id=form.username.id,                        5999
    autocomplete="username",                                           6000
```

```
                                placeholder="请输入用户名") }}
        </div>
        <div class="lp-field">
          <label for="{{ form.password.id }}">密码</label>
          {{ form.password(id=form.password.id,
    autocomplete="current-password",
                                placeholder="请输入密码") }}
        </div>
        <label class="lp-remember" for="{{ form.remember.id
    }}">
          {{ form.remember(id=form.remember.id) }}
          <span>记住我（下次自动登录，一个月内有效）</span>
        </label>
        {{ form.submit(class_="lp-submit") }}
      </form>
      <p class="lp-auth__hint">演示账号：admin 密码：admin123</p>
    </div>
  </div>
</div>
{% endblock %}
{% block scripts %}
<script>
(function () {
  var reduce = window.matchMedia &&
    window.matchMedia('(prefers-reduced-motion:
    reduce)').matches;
  /* ---------- 登录面板 ---------- */
  var btn = document.getElementById('lp-open');
  var auth = document.getElementById('lp-auth');
  var lastFocus = null;
  function openPanel() {
    lastFocus = document.activeElement;
    auth.classList.add('is-open');
    auth.setAttribute('aria-hidden', 'false');
    btn.setAttribute('aria-expanded', 'true');
    document.body.style.overflow = 'hidden';
    // 记到地址栏，直接访问 /login#login 也能展开登录框
    if (location.hash !== '#login') {
      history.replaceState(null, '', location.pathname +
    '#login');
    }
    var first = auth.querySelector('input');
    if (first) window.setTimeout(function () {
    first.focus(); }, 300);
  }
  function closePanel() {
    auth.classList.remove('is-open');
    auth.setAttribute('aria-hidden', 'true');
    btn.setAttribute('aria-expanded', 'false');
    document.body.style.overflow = '';
```

```
    if (location.hash === '#login') {
      history.replaceState(null, '', location.pathname);
    }
    if (lastFocus && lastFocus.focus) lastFocus.focus();
  }
  if (btn && auth) {
    btn.addEventListener('click', openPanel);
    Array.prototype.forEach.call(
      auth.querySelectorAll('[data-lp-close]'),
      function (el) { el.addEventListener('click',
    closePanel); }
    );
    document.addEventListener('keydown', function (e) {
      if (e.key === 'Escape' && auth.classList.contains('is-
    open')) closePanel();
    });
    //  登录失败回到本页、或直接带 #login 访问时，直接展开登录框
    if (location.hash === '#login' ||
    document.querySelector('.flash.error')) {
      openPanel();
    }
  }
  /* ---------- 内容进场 + 徽标散开：同一套滚动计算 ---------- 
     原先用 IntersectionObserver，实测在滚动到位置后回调并未生效，
     内容始终停在 opacity:0。改为直接按元素与视口的相对位置判断，
     与徽标用同一次滚动回调，行为可预期、也便于排查。 */
  var items = document.querySelectorAll('.lp-reveal');
  var badge = document.getElementById('lp-badge');
  var ticking = false;
  function applyScroll() {
    ticking = false;
    var vh = window.innerHeight || 800;
    /* 徽标：滚动前约三分之一屏内完成散开淡出 */
    if (badge) {
      var range = vh * 0.34;
      var p = range > 0 ? window.scrollY / range : 0;
      badge.style.setProperty('--p', Math.min(1,
    Math.max(0, p)).toFixed(3));
    }
    /* 内容：先批量测量，再改类名，避免读写交替触发反复重排 */
    if (reduce || !items.length) return;
    var box = [];
    for (var i = 0; i < items.length; i++) {
      box.push(items[i].getBoundingClientRect());
    }
    for (var j = 0; j < items.length; j++) {
      var r = box[j];
      //  上沿进入视口下部、下沿尚未越出视口上部 → 视为在视野内
      var shown = r.top < vh * 0.88 && r.bottom > vh * 0.06;
      if (shown) items[j].classList.add('is-in');
```

```
        else items[j].classList.remove('is-in');
      }
    }
    function onScroll() {
      if (!ticking) {
        ticking = true;
        window.requestAnimationFrame(applyScroll);
      }
    }
    // 减少动态效果时不做滚动特效，内容直接显示
    if (reduce) {
      Array.prototype.forEach.call(items, function (el) {
        el.classList.add('is-in');
      });
      if (badge) badge.style.setProperty('--p', '0');
    } else {
      window.addEventListener('scroll', onScroll, { passive:
      true });
      window.addEventListener('resize', onScroll);
      applyScroll();
    }
})();
</script>
{% endblock %}
# ==== app/templates/auth/home.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}工作台 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
{# ---------- 页首横幅 ---------- #}
<section class="hero">
  <div class="hero__body">
    <h1 class="hero__title">历史遗留废弃矿山自然恢复适宜性评价系统</h1>
    <p class="hero__desc">
      以论文《历史遗留废弃矿山自然恢复适宜性评价指标体系建设及典型案例》
      建立的评价方法为算法内核。下方按论文 4.1 节的评价流程顺序排列，
      系统会在每一步检查前置条件，前置步骤未完成时不会放行后续计算。
    </p>
    <div class="hero__actions">
      <a class="hero-btn hero-btn--ghost"
    href="/indicator/import">
        {{ icon("upload", 16) }}批量导入
      </a>
      <a class="hero-btn hero-btn--solid"
    href="/project/add">
        {{ icon("plus", 16) }}新建评价项目
      </a>
    </div>
  </div>
</section>
```

```
{# ---------- 数据概览 ---------- #}
<section class="stat-cards">
  <a class="stat-card" href="/project/">
    <span class="stat-card__label">评价项目</span>
    <span class="stat-card__value">{{ dash.stats.projects }}</span><span class="stat-card__unit">个</span></span>
    <span class="stat-card__note">
      {%- if projects_without_patch -%}
        其中 {{ projects_without_patch|length }} 个项目尚未建立图斑
      {%- elif dash.progress.patches -%}
        已建立 {{ dash.progress.patches }} 个矿山图斑
      {%- else -%}
        尚未建立矿山图斑
      {%- endif -%}
    </span>
  </a>
  <a class="stat-card" href="/project/">
    <span class="stat-card__label">矿山图斑</span>
    <span class="stat-card__value">{{ dash.stats.patches }}</span><span class="stat-card__unit">个</span></span>
    <span class="stat-card__note">
      {% if dash.progress.ready %}其中 {{ dash.progress.ready }} 个具备评价条件{% else %}暂无具备评价条件的图斑{% endif %}
    </span>
  </a>
  <a class="stat-card" href="/unit/">
    <span class="stat-card__label">微地质单元</span>
    <span class="stat-card__value">{{ dash.stats.unit_types }}</span><span class="stat-card__unit">类</span></span>
    <span class="stat-card__note">
      {% if dash.progress.units %}共划分 {{ dash.progress.units }} 个单元{% else %}尚未划分单元{% endif %}
    </span>
  </a>
  <a class="stat-card" href="/standard/">
    <span class="stat-card__label">标准与政策</span>
    <span class="stat-card__value">{{ dash.stats.standards }}</span><span class="stat-card__unit">项</span></span>
    <span class="stat-card__note">地方标准与政策文件</span>
  </a>
</section>
{# ---------- 评价流程 ---------- #}
<section class="section">
  <div class="section__head">
    <h2 class="section__title">评价流程</h2>
    <p class="section__desc">
      按论文 4.1 节顺序排列，前置步骤未完成时不会执行后续计算
    </p>
  </div>
  <div class="step-grid">
```

```
      {% for s in dash.steps %}                                          6201
      {% if s.available %}                                               6202
      <a class="step-card" href="{{ s.href }}">                          6203
        <span class="step-card__no">{{ s.no }}</span>                    6204
        <span class="step-card__icon">{{ icon(s.icon, 18)                6205
      }}</span>                                                          6206
        <span class="step-card__title">{{ s.title }}</span>              6207
        <span class="step-card__desc">{{ s.desc }}</span>                6208
        {% if s.pending %}                                               6209
        <span class="step-card__badge">待处理 {{ s.pending               6210
      }}</span>                                                          6211
        {% endif %}                                                      6212
      </a>                                                               6213
      {% else %}                                                         6214
      <div class="step-card step-card--locked"                           6215
      aria-disabled="true">                                              6216
        <span class="step-card__no">{{ s.no }}</span>                    6217
        <span class="step-card__icon">{{ icon(s.icon, 18)                6218
      }}</span>                                                          6219
        <span class="step-card__title">{{ s.title }}</span>              6220
        <span class="step-card__desc">{{ s.desc }}</span>                6221
        <span class="step-card__lock">{{ s.note }}</span>                6222
      </div>                                                             6223
      {% endif %}                                                        6224
      {% endfor %}                                                       6225
    </div>                                                               6226
</section>                                                               6227
{# ---------- 系统管理 ---------- #}                                     6228
<section class="section">                                                6229
    <div class="section__head">                                          6230
      <h2 class="section__title">系统管理</h2>                           6231
    </div>                                                               6232
    <a class="row-card" href="/users">                                   6233
      <span class="row-card__icon">{{ icon("users", 19)                  6234
      }}</span>                                                          6235
      <span class="row-card__main">                                      6236
        <span class="row-card__title">用户与操作日志</span>              6237
        <span class="row-card__desc">用户维护与角色权限；全部写操作自动留痕，可查询追溯6238
      </span>                                                            6239
      </span>                                                            6240
      <span class="row-card__arrow">{{ icon("chevron", 17)               6241
      }}</span>                                                          6242
    </a>                                                                 6243
</section>                                                               6244
{% endblock %}                                                           6245
# ==== app/templates/auth/users.html ====                                6246
{% extends "base.html" %}                                                6247
{% block title %}用户与权限管理{% endblock %}                            6248
{% block content %}                                                      6249
<div class="card">                                                       6250
```

```
<h2>用户管理</h2>
<p class="actions" style="margin-bottom:12px;">
  {% if current_user.role == "admin" %}
  <a class="btn btn--sm" href="/users/add">新增用户</a>
  {% endif %}
  <a class="btn btn--sm btn--ghost"
  href="/logs">查看操作日志</a>
</p>
{% if current_user.role != "admin" %}
<p class="muted">
  当前角色为「{{ role_labels.get(current_user.role,
  current_user.role) }}」，
  仅管理员可新增、编辑或删除用户。
</p>
{% endif %}
<table>
  <caption class="muted">共 {{ users|length }}
  个用户</caption>
  <thead>
    <tr>
      <th scope="col">ID</th>
      <th scope="col">用户名</th>
      <th scope="col">角色</th>
      <th scope="col">状态</th>
      {% if current_user.role == "admin" %}<th
  scope="col">操作</th>{% endif %}
    </tr>
  </thead>
  <tbody>
    {% for u in users %}
    <tr>
      <td class="num">{{ u.id }}</td>
      <td>
        {{ u.username }}
        {% if u.id == current_user.id %}<span
  class="muted">（当前登录）</span>{% endif %}
      </td>
      <td>{{ role_labels.get(u.role, u.role) }}</td>
      <td>
        {% if u.active %}<span class="tag
  tag--ok">启用</span>
        {% else %}<span class="tag tag--muted">停用</span>{%
  endif %}
      </td>
      {% if current_user.role == "admin" %}
      <td class="nowrap">
        <span class="row-actions">
          <a class="link-btn" href="/users/{{ u.id
  }}/edit">编辑</a>
          <form method="post" action="/users/{{ u.id
```

```
      }}/delete"                                                    6301
                        onsubmit="return confirm('确定删除用户「{{          6302
      u.username }}」吗？\n该操作无法恢复。');">                          6303
                  {{ form.hidden_tag() }}                           6304
                  <button type="submit" class="link-btn             6305
      link-btn--danger"                                             6306
                          {% if u.id == current_user.id             6307
      %}disabled                                                    6308
                          title="不能删除当前登录的账号"{% endif        6309
      %}>删除</button>                                              6310
                </form>                                             6311
              </span>                                               6312
            </td>                                                   6313
            {% endif %}                                             6314
          </tr>                                                     6315
          {% endfor %}                                              6316
        </tbody>                                                    6317
      </table>                                                      6318
</div>                                                              6319
{% endblock %}                                                      6320
# ==== app/templates/auth/add_user.html ====                        6321
{% extends "base.html" %}                                           6322
{% block title %}新增用户{% endblock %}                             6323
{% block content %}                                                 6324
<div class="card" style="max-width:560px;">                         6325
  <h2>新增用户</h2>                                                 6326
  <form method="post">                                              6327
    {{ form.hidden_tag() }}                                         6328
    <div class="field">                                             6329
      <label class="field__label" for="username">用户名</label>     6330
      <input id="username" type="text" name="username"              6331
    required                                                        6332
            autocomplete="off" style="width:280px;">                6333
    </div>                                                          6334
    <div class="field">                                             6335
      <label class="field__label" for="password">密码</label>       6336
      <input id="password" type="password" name="password"          6337
    required                                                        6338
            autocomplete="new-password"                             6339
    style="width:280px;">                                           6340
    </div>                                                          6341
    <div class="field">                                             6342
      <label class="field__label" for="role">角色</label>           6343
      <select id="role" name="role">                                6344
        <option value="viewer">查看者（仅可查阅）</option>          6345
        <option value="evaluator">评价员（可建项目、录数据、执行评价）</option6346
    >                                                               6347
        <option value="admin">管理员（可维护用户与基础数据）</option>  6348
      </select>                                                     6349
    </div>                                                          6350
```

```
      <p class="actions" style="margin-top:18px;">                    6351
        <button type="submit" class="btn">保存</button>               6352
        <a class="btn btn--ghost" href="/users">返回</a>               6353
      </p>                                                            6354
    </form>                                                           6355
</div>                                                                6356
{% endblock %}                                                        6357
# ==== app/templates/auth/edit_user.html ====                         6358
{% extends "base.html" %}                                             6359
{% block title %}编辑用户 · {{ user.username }}{% endblock %}          6360
{% block content %}                                                   6361
<div class="card" style="max-width:560px;">                           6362
  <h2>编辑用户 · {{ user.username }}</h2>                             6363
  <form method="post">                                                6364
    {{ form.hidden_tag() }}                                           6365
    <div class="field">                                               6366
      <label class="field__label" for="role">角色</label>             6367
      <select id="role" name="role">                                  6368
        {% for code, label in role_labels.items() %}                  6369
        <option value="{{ code }}" {% if user.role == code            6370
    %}selected{% endif %}>{{ label }}</option>                        6371
        {% endfor %}                                                  6372
      </select>                                                       6373
    </div>                                                            6374
    <div class="field">                                               6375
      <label class="checkbox" for="active">                           6376
        <input id="active" type="checkbox" name="active"              6377
    value="1"                                                         6378
               {% if user.active %}checked{% endif %}>                6379
        <span>启用该账号</span>                                       6380
      </label>                                                        6381
      <p class="muted" style="margin:6px 0 0;">                       6382
        停用后该账号无法登录，但其历史操作记录会保留。                6383
        系统至少需保留一名启用状态的管理员。                          6384
      </p>                                                            6385
    </div>                                                            6386
    <div class="field">                                               6387
      <label class="field__label" for="password">重置密码（留空则不修改6388
    ）</label>                                                        6389
      <input id="password" type="password" name="password"            6390
             autocomplete="new-password"                              6391
    style="width:280px;"                                              6392
             placeholder="留空表示保持原密码">                        6393
    </div>                                                            6394
    <p class="actions" style="margin-top:18px;">                      6395
      <button type="submit" class="btn">保存</button>                 6396
      <a class="btn btn--ghost" href="/users">返回</a>                 6397
    </p>                                                              6398
  </form>                                                             6399
</div>                                                                6400
```

```
{% endblock %}                                                          6401
# ==== app/templates/auth/logs.html ====                                6402
{% extends "base.html" %}                                               6403
{% block title %}操作日志{% endblock %}                                  6404
{% block content %}                                                     6405
<div class="card">                                                      6406
  <h2>操作日志</h2>                                                     6407
  <p class="actions" style="margin-bottom:12px;">                       6408
    <a class="btn btn--sm btn--ghost"                                   6409
    href="/users">返回用户管理</a>                                      6410
  </p>                                                                  6411
  {% if not records %}                                                  6412
  <p class="muted">暂无操作记录。</p>                                   6413
  {% else %}                                                            6414
  <table>                                                               6415
    <caption class="muted">共 {{ total }} 条记录，当前第 {{ page }}    6416
    / {{ pages }} 页</caption>                                          6417
    <thead>                                                             6418
      <tr>                                                              6419
        <th scope="col">时间</th>                                       6420
        <th scope="col">操作人</th>                                     6421
        <th scope="col">操作</th>                                       6422
        <th scope="col">对象</th>                                       6423
        <th scope="col">说明</th>                                       6424
      </tr>                                                             6425
    </thead>                                                            6426
    <tbody>                                                             6427
      {% for r in records %}                                            6428
      <tr>                                                              6429
        <td class="mono nowrap">{{ r.created_at.strftime("%Y            6430
    -%m-%d %H:%M:%S") if r.created_at else "—" }}</td>                  6431
        <td>{{ users_map.get(r.user_id, "—") }}</td>                    6432
        <td style="text-align:left;">{{ r.action }}</td>                6433
        <td>                                                            6434
          {% if r.target_type %}{{ r.target_type }}{% if                6435
    r.target_id %}#{{ r.target_id }}{% endif %}                         6436
          {% else %}—{% endif %}                                        6437
        </td>                                                           6438
        <td style="text-align:left;">{{ r.detail or "—"                 6439
    }}</td>                                                             6440
      </tr>                                                             6441
      {% endfor %}                                                      6442
    </tbody>                                                            6443
  </table>                                                              6444
  {% if pages > 1 %}                                                    6445
  <nav class="actions" aria-label="日志分页">                           6446
    {% if page > 1 %}                                                   6447
    <a class="btn btn--sm btn--ghost" href="/logs?page={{               6448
    page - 1 }}">上一页</a>                                             6449
    {% endif %}                                                         6450
```

```
      <span class="muted">第 {{ page }} / {{ pages }} 页</span>
      {% if page < pages %}
      <a class="btn btn--sm btn--ghost" href="/logs?page={{
      page + 1 }}">下一页</a>
      {% endif %}
   </nav>
   {% endif %}
   {% endif %}
</div>
{% endblock %}
# ==== app/templates/project/index.html ====
{% extends "base.html" %}
{% block title %}项目与矿山档案{% endblock %}
{% block content %}
<div class="card">
   <h2>项目与矿山档案</h2>
   <p class="actions" style="margin-bottom:12px;">
      <a class="btn btn--sm" href="/project/add">新建项目</a>
   </p>
   {% if not projects %}
   <p class="muted">
      暂无项目。点击上方「新建项目」开始建立评价项目，
      或运行 <span class="mono">python seed_demo.py</span>
      载入论文案例。
   </p>
   {% else %}
   <table>
      <caption class="muted">共 {{ projects|length }}
      个项目</caption>
      <thead>
         <tr>
            <th scope="col">项目名称</th>
            <th scope="col">编号</th>
            <th scope="col">状态</th>
            <th scope="col">图斑数</th>
            <th scope="col">操作</th>
         </tr>
      </thead>
      <tbody>
         {% for p in projects %}
         <tr>
            <td style="text-align:left;">{{ p.name }}</td>
            <td class="mono">{{ p.code or "—" }}</td>
            <td><span class="tag tag--muted">{{ p.status
      }}</span></td>
            <td class="num">
               {% if p.patches.count() %}
                  {{ p.patches.count() }}
               {% else %}
                  <span class="tag tag--warn">0</span>
```

```
            {% endif %}                                                    6501
          </td>                                                            6502
          <td class="nowrap">                                              6503
            <span class="row-actions">                                     6504
              <a class="link-btn" href="/project/{{ p.id                   6505
    }}/patches">{% if p.patches.count() %}矿山图斑{% else %}+              6506
    新建图斑{% endif %}</a>                                                6507
              <a class="link-btn" href="/project/{{ p.id                   6508
    }}/edit">编辑</a>                                                      6509
              <form method="post" action="/project/{{ p.id                 6510
    }}/delete"                                                             6511
                    onsubmit="return confirm('确定删除项目「{{             6512
    p.name }}」吗？\n该项目下的图斑、微地质单元与评价数据将一并删除，且无法恢复。');">  6513
                {{ form.hidden_tag() }}                                    6514
                <button type="submit" class="link-btn                      6515
    link-btn--danger">删除</button>                                        6516
              </form>                                                      6517
            </span>                                                        6518
          </td>                                                            6519
        </tr>                                                              6520
        {% endfor %}                                                       6521
      </tbody>                                                             6522
    </table>                                                               6523
    {% endif %}                                                            6524
</div>                                                                     6525
{% endblock %}                                                             6526
# ==== app/templates/project/edit.html ====                                6527
{% extends "base.html" %}                                                  6528
{% block title %}{{ "编辑项目" if project else "新建项目" }}{%             6529
    endblock %}                                                            6530
{% block content %}                                                        6531
<div class="card" style="max-width:640px;">                                6532
  <h2>{{ "编辑项目" if project else "新建项目" }}</h2>                     6533
  <form method="post">                                                     6534
    {{ form.hidden_tag() }}                                                6535
    <div class="field">                                                    6536
      <label class="field__label" for="name">项目名称</label>              6537
      <input id="name" type="text" name="name"                             6538
    style="width:100%;" required                                           6539
             value="{{ project.name if project else '' }}">                6540
    </div>                                                                 6541
    <div class="field">                                                    6542
      <label class="field__label" for="code">项目编号</label>              6543
      <input id="code" type="text" name="code"                             6544
    style="width:240px;"                                                   6545
             value="{{ project.code or '' if project else                  6546
    '' }}">                                                                6547
    </div>                                                                 6548
    {% if project %}                                                       6549
    <div class="field">                                                    6550
```

```
        <label class="field__label" for="status">状态</label>
        <select id="status" name="status">
          {% for s in ["进行中", "已完成", "已归档"] %}
          <option value="{{ s }}" {% if project.status == s
      %}selected{% endif %}>{{ s }}</option>
          {% endfor %}
        </select>
      </div>
      {% endif %}
      <div class="field">
        <label class="field__label" for="description">项目说明</la
      bel>
        <textarea id="description" name="description">{{
      project.description or '' if project else ''
      }}</textarea>
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/project/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/project/patches.html ====
{% extends "base.html" %}
{% block title %}矿山图斑 · {{ project.name }}{% endblock %}
{% block content %}
<div class="card">
  <h2>矿山图斑 · {{ project.name }}</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/project/{{ project.id
    }}/patch/add">新建图斑</a>
    <a class="btn btn--sm btn--ghost"
    href="/project/">返回项目列表</a>
  </p>
  {% if not project.patches.count() %}
  <p class="muted">该项目下暂无图斑。点击上方「新建图斑」建立第一个矿山图斑。</p>
  {% else %}
  <table>
    <caption class="muted">共 {{ project.patches.count() }}
    个图斑</caption>
    <thead>
      <tr>
        <th scope="col">图斑名称</th>
        <th scope="col">行政区</th>
        <th scope="col">面积（公顷）</th>
        <th scope="col">开采矿种</th>
        <th scope="col">微地质单元</th>
        <th scope="col">操作</th>
      </tr>
```

```
      </thead>                                                                  6601
      <tbody>                                                                   6602
        {% for p in project.patches %}                                          6603
        <tr>                                                                    6604
          <td style="text-align:left;">{{ p.name }}</td>                        6605
          <td>{{ p.admin_region or "—" }}</td>                                  6606
          <td class="num">{{ p.area_ha if p.area_ha is not                      6607
      none else "—" }}</td>                                                     6608
          <td>{{ p.mineral_type or "—" }}</td>                                  6609
          <td class="num">{{ p.units.count() }}</td>                            6610
          <td class="nowrap">                                                   6611
            <span class="row-actions">                                          6612
              <a class="link-btn" href="/unit/{{ p.id                           6613
      }}">微地质单元</a>                                                        6614
              <a class="link-btn" href="/constraint/{{ p.id                     6615
      }}">限制性判定</a>                                                        6616
              <a class="link-btn" href="/project/patch/{{                       6617
      p.id }}/edit">编辑</a>                                                    6618
              <form method="post" action="/project/patch/{{                     6619
      p.id }}/delete"                                                           6620
                    onsubmit="return confirm('确定删除图斑「{{                  6621
      p.name }}」吗？\n该图斑的微地质单元、指标数据与评价结果将一并删除，且无法恢复。');">  6622
                {{ form.hidden_tag() }}                                         6623
                <button type="submit" class="link-btn                           6624
      link-btn--danger">删除</button>                                           6625
              </form>                                                           6626
            </span>                                                             6627
          </td>                                                                 6628
        </tr>                                                                   6629
        {% endfor %}                                                            6630
      </tbody>                                                                  6631
    </table>                                                                    6632
    {% endif %}                                                                 6633
</div>                                                                          6634
{% endblock %}                                                                  6635
# ==== app/templates/project/patch_edit.html ====                               6636
{% extends "base.html" %}                                                       6637
{% block title %}{{ "编辑图斑" if patch else "新建图斑" }}{%                    6638
    endblock %}                                                                 6639
{% block content %}                                                             6640
<div class="card" style="max-width:720px;">                                     6641
  <h2>{{ "编辑图斑" if patch else "新建图斑" }} · {{ project.name               6642
    }}</h2>                                                                     6643
  <form method="post">                                                          6644
    {{ form.hidden_tag() }}                                                     6645
    <div class="field">                                                         6646
      <label class="field__label" for="patch-name">图斑名称</lab                6647
    el>                                                                         6648
      <input id="patch-name" type="text" name="name"                            6649
    style="width:100%;" required                                                6650
```

```
            value="{{ patch.name if patch else '' }}">
</div>
<div class="field">
  <label class="field__label" for="admin-region">所在行政区</
label>
  <input id="admin-region" type="text"
name="admin_region" style="width:100%;"
            value="{{ patch.admin_region or '' if patch
else '' }}">
</div>
<div class="field">
  <label class="field__label" for="area-ha">图斑面积（公顷）</la
bel>
  <input id="area-ha" type="number" step="any"
name="area_ha"
            value="{{ patch.area_ha if patch and
patch.area_ha is not none else '' }}">
</div>
<div class="field">
  <label class="field__label" for="mineral-type">开采矿种</l
abel>
  <input id="mineral-type" type="text"
name="mineral_type" style="width:240px;"
            value="{{ patch.mineral_type or '' if patch
else '' }}">
</div>
<div class="field">
  <label class="field__label" for="mining-method">开采方式</
label>
  <input id="mining-method" type="text"
name="mining_method" style="width:240px;"
            value="{{ patch.mining_method or '' if patch
else '' }}">
</div>
<div class="field">
  <label class="field__label" for="stop-years">停产年限</lab
el>
  <input id="stop-years" type="number" step="any"
name="stop_years"
            value="{{ patch.stop_years if patch and
patch.stop_years is not none else '' }}">
</div>
<div class="field">
  <label class="checkbox" for="responsible-missing">
    <input id="responsible-missing" type="checkbox"
name="responsible_entity_missing" value="1"
             {% if not patch or patch.responsible_entity_m
issing %}checked{% endif %}>
    <span>责任主体缺失</span>
  </label>
```

```
      </div>
      <h3>参照生态系统（指标标准化基准）</h3>
      <p class="muted">
        论文 4.2.2 节取值：植被覆盖度 75%、Shannon-Wiener 多样性指数 0.7。
        该基准与图斑绑定管理，可按不同参照生态系统分别设定。
      </p>
      <div class="field">
        <label class="field__label" for="coverage-benchmark">植被覆盖度基准（%）</label>
        <input id="coverage-benchmark" type="number" step="any" name="coverage_benchmark"
               value="{{ reference.coverage_benchmark if reference else 75 }}">
      </div>
      <div class="field">
        <label class="field__label" for="shannon-benchmark">Shannon-Wiener 多样性指数基准</label>
        <input id="shannon-benchmark" type="number" step="any" name="shannon_benchmark"
               value="{{ reference.shannon_benchmark if reference else 0.7 }}">
      </div>
      <div class="field">
        <label class="field__label" for="dominant-species">建群种</label>
        <input id="dominant-species" type="text" name="dominant_species" style="width:100%;"
               value="{{ reference.dominant_species or '' if reference else '' }}">
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/project/{{ project.id }}/patches">返回</a>
      </p>
    </form>
  </div>
  {% if patch %}
  <div class="card" style="max-width:720px;">
    <h2>遥感影像挂接</h2>
    <p class="muted">
      挂接遥感底图或无人机正射影像，用于辅助微地质单元划分与植被覆盖度判读。
      支持 PNG、JPG、GIF、WebP、BMP、TIFF，单个文件不超过 32 MB。
    </p>
    {% if patch.image_path %}
    <figure style="margin:0 0 16px;">
      <img src="{{ url_for('static', filename=patch.image_path) }}"
           alt="图斑「{{ patch.name }}」已挂接的遥感影像"
           style="max-width:100%;border:1px solid
```

```
      var(--line);border-radius:6px;">
      <figcaption class="muted" style="margin-top:6px;">
        {{ patch.image_note or "遥感影像" }}
        {% if patch.image_uploaded_at %}
        · 上传于 {{ patch.image_uploaded_at.strftime("%Y-%m-%d
      %H:%M") }}
        {% endif %}
      </figcaption>
    </figure>
    <form method="post" action="/project/patch/{{ patch.id
      }}/image/delete"
          onsubmit="return confirm('确定移除该遥感影像吗？');">
      {{ form.hidden_tag() }}
      <button type="submit" class="btn btn--sm
      btn--danger">移除影像</button>
    </form>
    <hr style="margin:16px 0;border:none;border-top:1px solid
      var(--line);">
    {% endif %}
    <form method="post" action="/project/patch/{{ patch.id
      }}/image"
          enctype="multipart/form-data">
      {{ form.hidden_tag() }}
      <div class="field">
        <label class="field__label" for="image-file">
          {{ "替换影像文件" if patch.image_path else "选择影像文件" }}
        </label>
        <input id="image-file" type="file" name="image"
               accept=".png,.jpg,.jpeg,.gif,.webp,.bmp,.tif,.t
      iff" required>
      </div>
      <div class="field">
        <label class="field__label" for="image-note">影像说明（如来源、
      时相）</label>
        <input id="image-note" type="text" name="image_note"
      style="width:100%;"
               placeholder="如：2025 年 6 月无人机正射影像"
               value="{{ patch.image_note or '' }}">
      </div>
      <p class="actions">
        <button type="submit" class="btn">{{ "替换影像" if
      patch.image_path else "上传并挂接" }}</button>
      </p>
    </form>
  </div>
  {% endif %}
  {% endblock %}
  # ==== app/templates/unit/list.html ====
  {% extends "base.html" %}
  {% from "_macros.html" import patch_guide %}
```

```
{% block title %}微地质单元{% endblock %}
{% block content %}
<div class="card">
  <h2>微地质单元</h2>
  {{ patch_guide(projects_without_patch) }}
  {% if not rows %}
  <p class="muted">
    暂无矿山图斑。请先到「项目与矿山档案」建立图斑。
  </p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
        <th scope="col">矿山图斑</th>
        <th scope="col">已划分单元</th>
        <th scope="col">占比合计</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rows %}
      <tr>
        <td>{{ r.patch.project.name if r.patch.project else "—" }}</td>
        <td style="text-align:left;">{{ r.patch.name }}</td>
        <td class="num">{{ r.count }} 个</td>
        <td class="num">
          {% if r.count %}
            {{ "%.2f%%"|format(r.ratio_sum * 100) }}
            {% if r.ratio_sum < 0.999 or r.ratio_sum > 1.001 %}
              <span class="warn">（不等于 100%）</span>
            {% endif %}
          {% else %}—{% endif %}
        </td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/unit/{{ r.patch.id }}">划分与维护</a>
            <a class="link-btn" href="/indicator/">录入指标数据</a>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endif %}
</div>
```

```
{% endblock %}
# ==== app/templates/unit/index.html ====
{% extends "base.html" %}
{% block title %}微地质单元 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card">
  <h2>微地质单元 · {{ patch.name }}</h2>
  {% if units %}
  <table>
    <caption class="muted">
      共 {{ units|length }} 个单元；
      面积合计 <span class="num">{{ "%.4f"|format(total_area)
    }}</span>；
      占比合计 <span class="num">{{ "%.2f%%"|format(ratio_sum *
    100) }}</span>
      {% if ratio_sum and (ratio_sum < 0.999 or ratio_sum >
    1.001) %}
      <span class="warn">（占比合计不等于 100%，请检查面积录入）</span>
      {% endif %}
    </caption>
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积</th>
        <th scope="col">面积占比</th>
        <th scope="col">说明</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for u in units %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">{{ u.area if u.area is not none
    else "—" }}</td>
        <td class="num">
          {% if u.ratio is not none %}{{
    "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
        </td>
        <td style="text-align:left;">{{ u.description or
    "—" }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/unit/edit/{{ u.id
    }}">编辑</a>
            <form method="post" action="/unit/delete/{{
    u.id }}"
                  onsubmit="return confirm('确定删除微地质单元「{{
    u.unit_type }}」吗？\n该单元已录入的指标数据将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
```

```
                    <button type="submit" class="link-btn
    link-btn--danger">删除</button>
                  </form>
                </span>
              </td>
            </tr>
            {% endfor %}
          </tbody>
        </table>
        {% else %}
        <p class="muted">尚未划分微地质单元。请在下方添加。</p>
        {% endif %}
        <h3>添加微地质单元</h3>
        <form method="post" action="/unit/{{ patch.id }}/add">
          {{ form.hidden_tag() }}
          <div class="actions">
            <label class="field__label" for="add-unit-type"
          style="margin:0;">类型</label>
            <select id="add-unit-type" name="unit_type">
              {% for t in unit_types %}<option value="{{ t }}">{{
          t }}</option>{% endfor %}
            </select>
            <label class="field__label" for="add-unit-area"
          style="margin:0;">面积</label>
            <input id="add-unit-area" type="number" step="any"
          name="area">
            <label class="field__label" for="add-unit-desc"
          style="margin:0;">说明</label>
            <input id="add-unit-desc" type="text"
          name="description" style="width:220px;">
            <button type="submit" class="btn btn--sm">添加</button>
          </div>
        </form>
        <p class="actions" style="margin-top:20px;">
          <a class="btn btn--sm btn--ghost" href="/project/{{
          patch.project_id }}/patches">返回图斑列表</a>
          <a class="btn btn--sm btn--ghost"
          href="/indicator/">去录入指标数据</a>
        </p>
      </div>
    {% endblock %}
    # ==== app/templates/unit/edit.html ====
    {% extends "base.html" %}
    {% block title %}编辑微地质单元{% endblock %}
    {% block content %}
    <div class="card">
      <h2>编辑微地质单元 · {{ patch.name }}</h2>
      <p class="muted">修改面积后，系统会自动重算本图斑各单元的面积占比。</p>
      <form method="post">
        {{ form.hidden_tag() }}
```

```
      <div class="field">
        <label class="field__label" for="unit-type">微地质单元类型</l
      abel>
        <select id="unit-type" name="unit_type">
          {% for t in unit_types %}
          <option value="{{ t }}" {% if unit.unit_type == t
      %}selected{% endif %}>{{ t }}</option>
          {% endfor %}
        </select>
      </div>
      <div class="field">
        <label class="field__label" for="unit-area">面积</label>
        <input id="unit-area" type="number" step="any"
      name="area"
               value="{{ unit.area if unit.area is not none
      else '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="unit-desc">说明</label>
        <input id="unit-desc" type="text" name="description"
      style="width:420px;"
               value="{{ unit.description or '' }}">
      </div>
      <p class="actions" style="margin-top:16px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/unit/{{
      unit.patch_id }}">返回</a>
      </p>
    </form>
</div>
{% endblock %}
#  ==== app/templates/standard/index.html ====
{% extends "base.html" %}
{% block title %}政策与标准知识库{% endblock %}
{% block content %}
<div class="card">
    <h2>政策与标准知识库</h2>  <p class="actions"
      style="margin-bottom:14px;">
      <a class="btn btn--sm" href="/standard/add">添加标准</a>
      <a class="btn btn--sm btn--ghost"
      href="/standard/compare">省际口径比对</a>
    </p>
    <h3>地方标准</h3>
    {% if local %}
    <table>
      <thead>
        <tr>
          <th scope="col">省份</th>
          <th scope="col">名称</th>
          <th scope="col">编号</th>
```

```
          <th scope="col">规则数</th>                                    7001
          <th scope="col">操作</th>                                      7002
        </tr>                                                            7003
      </thead>                                                           7004
      <tbody>                                                            7005
        {% for s in local %}                                             7006
        <tr>                                                             7007
          <td>{{ s.province }}</td>                                      7008
          <td style="text-align:left;">{{ s.name }}</td>                 7009
          <td class="mono">{{ s.doc_number or "—" }}</td>                7010
          <td class="num">{{ s.rules|length }}</td>                      7011
          <td class="nowrap">                                            7012
            <span class="row-actions">                                   7013
              <a class="link-btn" href="/standard/{{ s.id                7014
    }}/rules">规则</a>                                                   7015
              <a class="link-btn" href="/standard/{{ s.id                7016
    }}/edit">编辑</a>                                                    7017
              <form method="post" action="/standard/{{ s.id              7018
    }}/delete"                                                           7019
                    onsubmit="return confirm('确定删除「{{               7020
    s.province }} {{ s.name }}」吗？\n该标准下的 {{ s.rules|length       7021
    }} 条规则将一并删除，且无法恢复。');">                               7022
                {{ form.hidden_tag() }}                                  7023
                <button type="submit" class="link-btn                    7024
    link-btn--danger">删除</button>                                      7025
              </form>                                                    7026
            </span>                                                      7027
          </td>                                                          7028
        </tr>                                                            7029
        {% endfor %}                                                     7030
      </tbody>                                                           7031
    </table>                                                             7032
    {% else %}                                                           7033
    <p class="muted">暂无地方标准。运行 <span class="mono">python       7034
      seed_demo.py</span> 可载入论文表 1 的 11 项标准与政策文件。</p>   7035
    {% endif %}                                                          7036
    <h3>政策文件</h3>                                                    7037
    {% if policy %}                                                      7038
    <table>                                                              7039
      <thead>                                                            7040
        <tr>                                                             7041
          <th scope="col">省份</th>                                      7042
          <th scope="col">名称</th>                                      7043
          <th scope="col">文号</th>                                      7044
          <th scope="col">规则数</th>                                    7045
          <th scope="col">操作</th>                                      7046
        </tr>                                                            7047
      </thead>                                                           7048
      <tbody>                                                            7049
        {% for s in policy %}                                            7050
```

```
        <tr>                                                                    7051
          <td>{{ s.province }}</td>                                             7052
          <td style="text-align:left;">{{ s.name }}</td>                        7053
          <td class="mono">{{ s.doc_number or "—" }}</td>                       7054
          <td class="num">{{ s.rules|length }}</td>                             7055
          <td class="nowrap">                                                   7056
            <span class="row-actions">                                          7057
              <a class="link-btn" href="/standard/{{ s.id                       7058
    }}/rules">规则</a>                                                        7059
              <a class="link-btn" href="/standard/{{ s.id                       7060
    }}/edit">编辑</a>                                                         7061
              <form method="post" action="/standard/{{ s.id                     7062
    }}/delete"                                                                  7063
                    onsubmit="return confirm('确定删除「{{                      7064
    s.province }} {{ s.name }}」吗？\n该文件下的 {{ s.rules|length              7065
    }} 条规则将一并删除，且无法恢复。');">                                      7066
                {{ form.hidden_tag() }}                                         7067
                <button type="submit" class="link-btn                           7068
    link-btn--danger">删除</button>                                           7069
              </form>                                                           7070
            </span>                                                             7071
          </td>                                                                 7072
        </tr>                                                                   7073
        {% endfor %}                                                            7074
      </tbody>                                                                  7075
    </table>                                                                    7076
    {% else %}                                                                  7077
    <p class="muted">暂无政策文件。</p>                                         7078
    {% endif %}                                                                 7079
</div>                                                                          7080
{% endblock %}                                                                  7081
# ==== app/templates/standard/edit.html ====                                    7082
{% extends "base.html" %}                                                       7083
{% block title %}{{ "编辑标准" if standard else "添加标准" }}{%                 7084
    endblock %}                                                                 7085
{% block content %}                                                             7086
<div class="card" style="max-width:800px;">                                     7087
  <h2>{{ "编辑标准" if standard else "添加标准" }}</h2>                         7088
  <form method="post">                                                          7089
    {{ form.hidden_tag() }}                                                     7090
    <div class="field">                                                         7091
      <label class="field__label" for="province">省份</label>                   7092
      <input id="province" type="text" name="province"                          7093
    style="width:160px;" required                                               7094
            value="{{ standard.province if standard else                        7095
    '' }}">                                                                     7096
    </div>                                                                      7097
    <div class="field">                                                         7098
      <label class="field__label" for="std-name">名称</label>                   7099
      <input id="std-name" type="text" name="name"                              7100
```

```
style="width:100%;" required                                    7101
          value="{{ standard.name if standard else ''           7102
}}">                                                            7103
</div>                                                          7104
<div class="field">                                             7105
  <label class="field__label" for="doc-number">标准编号 /       7106
文号</label>                                                    7107
  <input id="doc-number" type="text" name="doc_number"          7108
style="width:320px;"                                            7109
          value="{{ standard.doc_number or '' if                7110
standard else '' }}">                                           7111
</div>                                                          7112
<div class="field">                                             7113
  <label class="field__label" for="category">类别</label>       7114
  <select id="category" name="category">                        7115
    {% for c in ["地方标准", "政策文件"] %}                     7116
    <option value="{{ c }}" {% if standard and                  7117
standard.category == c %}selected{% endif %}>{{ c               7118
}}</option>                                                     7119
    {% endfor %}                                                7120
  </select>                                                     7121
</div>                                                          7122
<div class="field">                                             7123
  <label class="field__label" for="publish-date">发布时间</l    7124
abel>                                                           7125
  <input id="publish-date" type="text"                          7126
name="publish_date" style="width:200px;"                        7127
          placeholder="如 2023-03-30"                           7128
          value="{{ standard.publish_date or '' if              7129
standard else '' }}">                                           7130
</div>                                                          7131
<div class="field">                                             7132
  <label class="field__label" for="implement-date">实施时间<    7133
/label>                                                         7134
  <input id="implement-date" type="text"                        7135
name="implement_date" style="width:200px;"                      7136
          placeholder="如 2023-04-30（未明确可留空）"           7137
          value="{{ standard.implement_date or '' if            7138
standard else '' }}">                                           7139
</div>                                                          7140
<div class="field">                                             7141
  <label class="field__label" for="summary">特点 /              7142
判定标准</label>                                                7143
  <textarea id="summary" name="summary"                         7144
style="min-height:96px;">{{ standard.summary or '' if           7145
standard else '' }}</textarea>                                  7146
</div>                                                          7147
<p class="actions" style="margin-top:18px;">                    7148
  <button type="submit" class="btn">保存</button>               7149
  {% if standard %}                                             7150
```

```
        <form method="post" action="/standard/{{ standard.id
    }}/delete" style="display:inline;"
              onsubmit="return confirm('确定删除「{{
    standard.province }} {{ standard.name
    }}」吗？\n该标准下的规则将一并删除，且无法恢复。');">
          {{ form.hidden_tag() }}
          <button type="submit" class="btn
    btn--danger">删除该标准</button>
        </form>
        {% endif %}
        <a class="btn btn--ghost" href="/standard/">返回</a>
    </p>
  </form>
</div>
{% endblock %}
# ==== app/templates/standard/rules.html ====
{% extends "base.html" %}
{% block title %}结构化规则 · {{ standard.province }}{% endblock
    %}
{% block content %}
<div class="card">
  <h2>结构化规则 · {{ standard.province }} {{ standard.name
    }}</h2>
  {% if rules %}
  <table>
    <caption class="muted">共 {{ rules|length }}
    条规则</caption>
    <thead>
      <tr>
        <th scope="col">判定项</th>
        <th scope="col">条件</th>
        <th scope="col">条件类型</th>
        <th scope="col">适用分区</th>
        <th scope="col">备注</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rules %}
      <tr>
        <td style="text-align:left;">{{ r.item }}</td>
        <td class="mono nowrap">{{ r.operator }} {{
    r.threshold or "" }}</td>
        <td>{{ r.condition_type }}</td>
        <td>{{ r.region or "—" }}</td>
        <td style="text-align:left;">{{ r.note or "—"
    }}</td>
        <td>
          <form method="post" action="/standard/rule/{{
    r.id }}/delete"
```

```
                onsubmit="return confirm('确定删除规则「{{ r.item }} {{ r.operator }} {{ r.threshold or '' }}」吗？');">
                {{ form.hidden_tag() }}
                <button type="submit" class="btn btn--sm btn--danger">删除</button>
            </form>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">该标准暂无规则。请在下方添加第一条规则。</p>
  {% endif %}

  <h3>添加规则</h3>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="actions">
      <div>
        <label class="field__label" for="r-item">判定项</label>
        <input id="r-item" type="text" name="item" placeholder="如：植被覆盖度" required>
      </div>
      <div>
        <label class="field__label" for="r-op">比较符</label>
        <select id="r-op" name="operator">
          {% for o in [">=", ">", "<=", "<", "=="] %}<option value="{{ o }}">{{ o }}</option>{% endfor %}
        </select>
      </div>
      <div>
        <label class="field__label" for="r-thr">阈值</label>
        <input id="r-thr" type="text" name="threshold" placeholder="如：50" style="width:110px;">
      </div>
      <div>
        <label class="field__label" for="r-cond">条件类型</label>
        <select id="r-cond" name="condition_type">
          <option value="必要条件">必要条件</option>
          <option value="充分条件">充分条件</option>
        </select>
      </div>
      <div>
        <label class="field__label" for="r-region">适用分区</label>
        <input id="r-region" type="text" name="region" placeholder="如宁夏分区" style="width:130px;">
      </div>
      <div>
```

```
            <label class="field__label" for="r-note">备注</label>
            <input id="r-note" type="text" name="note"
      style="width:220px;">
          </div>
          <div>
            <button type="submit" class="btn">添加规则</button>
          </div>
        </div>
      </form>
      <p class="actions" style="margin-top:20px;">
        <a class="btn btn--sm btn--ghost"
        href="/standard/">返回标准列表</a>
        <a class="btn btn--sm btn--ghost"
        href="/standard/compare">省际口径比对</a>
      </p>
    </div>
    {% endblock %}
    # ==== app/templates/standard/compare.html ====
    {% extends "base.html" %}
    {% block title %}省际口径比对{% endblock %}
    {% block content %}
    <div class="card">
      <h2>省际口径比对</h2>
      {% if not groups %}
      <p class="muted">
        暂无规则数据。请先在「标准知识库」中为标准录入规则，
        或运行 <span class="mono">python seed_demo.py</span> 载入论文表
        1 数据。
      </p>
      {% else %}
      {% for item, rs in groups.items() %}
      <h3>{{ item }}</h3>
      <table>
        <thead>
          <tr>
            <th scope="col">省份</th>
            <th scope="col">标准 / 文件</th>
            <th scope="col">条件</th>
            <th scope="col">条件类型</th>
            <th scope="col">备注</th>
          </tr>
        </thead>
        <tbody>
          {% for r in rs %}
          <tr>
            <td>{{ r.standard.province }}</td>
            <td style="text-align:left;">{{ r.standard.name
        }}</td>
            <td class="mono nowrap">{{ r.operator }} {{
        r.threshold or "" }}</td>
```

```
            <td>{{ r.condition_type }}</td>                                  7301
            <td style="text-align:left;">{{ r.note or "—"                    7302
        }}</td>                                                              7303
          </tr>                                                              7304
          {% endfor %}                                                       7305
        </tbody>                                                             7306
      </table>                                                               7307
      {% endfor %}                                                           7308
      {% endif %}                                                            7309
      <p class="actions" style="margin-top:16px;">                           7310
        <a class="btn btn--ghost" href="/standard/">返回标准列表</a>          7311
      </p>                                                                   7312
</div>                                                                       7313
{% endblock %}                                                               7314
# ==== app/templates/constraint/index.html ====                              7315
{% extends "base.html" %}                                                    7316
{% from "_macros.html" import patch_guide %}                                 7317
{% block title %}限制性指标判定{% endblock %}                                7318
{% block content %}                                                          7319
<div class="card">                                                           7320
  <h2>限制性指标判定</h2>                                                    7321
  {{ patch_guide(projects_without_patch) }}                                  7322
  {% if not patches %}                                                       7323
  <p class="muted">暂无矿山图斑。请先在「项目与矿山档案」中建立图斑。</p>   7324
  {% else %}                                                                 7325
  <table>                                                                    7326
    <thead>                                                                  7327
      <tr>                                                                   7328
        <th scope="col">所属项目</th>                                        7329
        <th scope="col">矿山图斑</th>                                        7330
        <th scope="col">判定结果</th>                                        7331
        <th scope="col">操作</th>                                            7332
      </tr>                                                                  7333
    </thead>                                                                 7334
    <tbody>                                                                  7335
      {% for p in patches %}                                                 7336
      {% set r = latest.get(p.id) %}                                         7337
      <tr>                                                                   7338
        <td>{{ p.project.name if p.project else "—" }}</td>                  7339
        <td style="text-align:left;">{{ p.name }}</td>                       7340
        <td style="text-align:left;">                                        7341
          {% if r %}                                                         7342
            {% if r.passed %}                                                7343
              <span class="tag tag--ok">通过</span>                          7344
            {% else %}                                                       7345
              <span class="tag tag--warn">不通过</span>                      7346
            {% endif %}                                                      7347
            <span class="muted">{{ r.reason }}</span>                        7348
          {% else %}                                                         7349
            <span class="tag tag--muted">未判定</span>                       7350
```

```
            {% endif %}                                                    7351
          </td>                                                            7352
          <td class="nowrap">                                              7353
            <span class="row-actions">                                     7354
              <a class="link-btn" href="/constraint/{{ p.id                7355
    }}">执行判定</a>                                                       7356
            </span>                                                        7357
          </td>                                                            7358
        </tr>                                                              7359
        {% endfor %}                                                       7360
      </tbody>                                                             7361
    </table>                                                               7362
    {% endif %}                                                            7363
</div>                                                                     7364
{% endblock %}                                                             7365
# ==== app/templates/constraint/check.html ====                            7366
{% extends "base.html" %}                                                  7367
{% block title %}限制性条件判定 · {{ patch.name }}{% endblock %}           7368
{% block content %}                                                        7369
<div class="card" style="max-width:760px;">                                7370
  <h2>限制性条件判定 · {{ patch.name }}</h2>                               7371
  <form method="post">                                                     7372
    {{ form.hidden_tag() }}                                                7373
    {% for field, name, val in field_states %}                             7374
    <fieldset style="border:1px solid                                      7375
    var(--line);border-radius:6px;padding:10px                             7376
    14px;margin:0 0 10px;">                                                7377
      <legend style="padding:0 6px;font-weight:600;">{{                    7378
    name }}</legend>                                                       7379
      <div class="actions">                                                7380
        <label class="checkbox">                                           7381
          <input type="radio" name="{{ field }}" value="1"                 7382
                 {% if val is none or val %}checked{% endif                7383
    %}>                                                                    7384
          <span>通过</span>                                                7385
        </label>                                                           7386
        <label class="checkbox">                                           7387
          <input type="radio" name="{{ field }}" value="0"                 7388
                 {% if val is not none and not val                         7389
    %}checked{% endif %}>                                                  7390
          <span>不通过</span>                                              7391
        </label>                                                           7392
      </div>                                                               7393
    </fieldset>                                                            7394
    {% endfor %}                                                           7395
    <p class="actions" style="margin-top:16px;">                           7396
      <button type="submit" class="btn">保存判定</button>                  7397
      <a class="btn btn--ghost" href="/constraint/">返回</a>               7398
    </p>                                                                   7399
  </form>                                                                  7400
```

```
  {% if record %}                                                    7401
  <div style="margin-top:18px;padding:14px;border-radius:6px 7402
    ;border-left:4px solid;                                          7403
                {% if record.passed %}background:var(--ok-soft 7404
    );border-left-color:var(--ok);                                   7405
                {% else %}background:var(--danger-soft);border 7406
    -left-color:var(--danger);{% endif %}">                          7407
    <p style="margin:0 0 4px;font-weight:600;">                      7408
      判定结论：                                                     7409
      {% if record.passed %}                                         7410
      <span class="ok">通过，可进入定量评价</span>                   7411
      {% else %}                                                     7412
      <span class="warn">不通过 —— 建议生态重建或人工辅助</span>     7413
      {% endif %}                                                    7414
    </p>                                                             7415
    <p class="muted" style="margin:0;">{{ record.reason              7416
    }}</p>                                                           7417
  </div>                                                             7418
  {% endif %}                                                        7419
</div>                                                               7420
{% endblock %}                                                       7421
# ==== app/templates/ahp/index.html ====                             7422
{% extends "base.html" %}                                            7423
{% block title %}AHP 权重计算{% endblock %}                          7424
{% block content %}                                                  7425
<div class="card" style="max-width:720px;">                          7426
  <h2>AHP 权重计算 · 选择指标层级</h2>                               7427
  <form method="post">                                               7428
    {{ form.hidden_tag() }}                                          7429
    <div class="field">                                              7430
      <label class="field__label" for="{{ form.unit_type.id 7431
    }}">微地质单元类型</label>                                       7432
      {{ form.unit_type(id=form.unit_type.id) }}                     7433
    </div>                                                           7434
    <div class="field">                                              7435
      <label class="field__label" for="{{ form.level.id      7436
    }}">指标层级</label>                                             7437
      {{ form.level(id=form.level.id) }}                             7438
    </div>                                                           7439
    <div class="field">                                              7440
      <label class="field__label" for="{{ form.l1.id }}">     7441
        一级指标（层级选「二级指标」时需选择）                       7442
      </label>                                                       7443
      {{ form.l1(id=form.l1.id) }}                                   7444
    </div>                                                           7445
    <p class="actions" style="margin-top:16px;">                     7446
      {{ form.submit(class_="btn") }}                                7447
      <a class="btn btn--ghost" href="/ahp/indicators">指标定义维 7448
    护</a>                                                           7449
      <a class="btn btn--ghost" href="/ahp/experts">多专家矩阵</a 7450
```

```
      >                                                                    7451
      </p>                                                                 7452
    </form>                                                                7453
</div>                                                                     7454
<div class="card">                                                         7455
    <h2>各微地质单元当前生效权重</h2>                                      7456
    <table>                                                                7457
      <thead>                                                              7458
        <tr>                                                               7459
          <th scope="col">微地质单元</th>                                  7460
          <th scope="col">当前权重来源</th>                                7461
          <th scope="col">操作</th>                                        7462
        </tr>                                                              7463
      </thead>                                                             7464
      <tbody>                                                              7465
        {% for r in weight_rows %}                                         7466
        <tr>                                                               7467
          <td style="text-align:left;">{{ r.unit_type }}</td>              7468
          <td style="text-align:left;">                                    7469
            {% if r.custom %}                                              7470
            <span class="tag tag--warn">{{ r.label }}</span>               7471
            {% else %}                                                     7472
            <span class="tag tag--muted">{{ r.label }}</span>              7473
            {% endif %}                                                    7474
          </td>                                                            7475
          <td class="nowrap">                                              7476
            <span class="row-actions">                                     7477
              <a class="link-btn" href="/ahp/matrix?unit_type=             7478
      {{ r.unit_type }}&level=1">一级指标权重</a>                          7479
              <a class="link-btn" href="/ahp/matrix?unit_type=             7480
      {{ r.unit_type }}&level=1&calc=1">查看一致性</a>                     7481
              {% if r.custom %}                                            7482
              <form method="post" action="/ahp/weights/reset"              7483
                    onsubmit="return confirm('确定把「{{                   7484
      r.unit_type }}」恢复为论文表 3 的默认权重吗？');">                   7485
                <input type="hidden" name="csrf_token"                     7486
      value="{{ csrf_token() }}">                                          7487
                <input type="hidden" name="unit_type"                      7488
      value="{{ r.unit_type }}">                                           7489
                <button type="submit" class="link-btn                      7490
      link-btn--danger">恢复默认</button>                                  7491
              </form>                                                      7492
              {% endif %}                                                  7493
            </span>                                                        7494
          </td>                                                            7495
        </tr>                                                              7496
        {% endfor %}                                                       7497
      </tbody>                                                             7498
    </table>                                                               7499
</div>                                                                     7500
```

```
{% endblock %}
# ==== app/templates/ahp/matrix.html ====
{% extends "base.html" %}
{% block title %}判断矩阵录入{% endblock %}
{% block content %}
<div class="card">
  <h2>
    判断矩阵录入
    · {{ unit_type }}
    · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{%
    endif %}
  </h2>
  <p class="muted">
    采用 1–9 标度：1 同等重要、3 稍微重要、5 明显重要、7 强烈重要、9 极端重要，
    2、4、6、8 为相邻判断的中间值。只需录入上三角，对角线为 1，下三角自动取倒数。
    矩阵已按论文表 3 的权重预填。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    <table style="width:auto;">
      <thead>
        <tr>
          <th></th>
          {% for name in names %}<th scope="col">{{ name
}}</th>{% endfor %}
        </tr>
      </thead>
      <tbody>
        {% for i in range(names|length) %}
        <tr>
          <th scope="row">{{ names[i] }}</th>
          {% for j in range(names|length) %}
            {% if i == j %}
            <td class="muted num">1</td>
            {% elif i < j %}
            <td>
              <label class="muted" style="position:absolute;
left:-9999px;"
                     for="m-{{ i }}-{{ j }}">{{ names[i] }}
相对 {{ names[j] }} 的重要性</label>
              <input id="m-{{ i }}-{{ j }}" type="number"
step="0.01" min="0.01"
                     name="m_{{ i }}_{{ j }}" value="{{
default[i][j] }}">
            </td>
            {% else %}
            <td class="muted num">{{ default[i][j] }}</td>
            {% endif %}
          {% endfor %}
        </tr>
```

```
        {% endfor %}
      </tbody>
    </table>
    <p class="actions">
      <button type="submit" class="btn" name="action"
    value="calc">计算权重</button>
      {% if result and result.consistent %}
      <button type="submit" class="btn" name="action"
    value="save">保存并生效</button>
      {% endif %}
      <a class="btn btn--ghost" href="/ahp/">重新选择</a>
    </p>
  </form>
  <p class="muted" style="margin-top:10px;">
    当前「{{ unit_type }}」生效权重：<strong>{{ source_label
    }}</strong>。
    点击「保存并生效」后，本单元类型的适宜性评价将采用此处算出的权重；
    如需回到论文取值，可在 <a href="/ahp/">权重首页</a> 恢复默认。
  </p>
  {% if result %}
  <hr style="margin:22px 0;border:none;border-top:1px solid
    var(--line);">
  <h3>计算结果</h3>
  <table style="width:auto;">
    <thead>
      <tr>
        <th scope="col">指标</th>
        {% for name in names %}<th scope="col">{{ name
    }}</th>{% endfor %}
      </tr>
    </thead>
    <tbody>
      <tr>
        <th scope="row">权重</th>
        {% for w in result.weights %}<td class="mono">{{ w
    }}</td>{% endfor %}
      </tr>
    </tbody>
  </table>
  <div class="stat-row">
    <div>
      <div class="stat__label">最大特征值 λmax</div>
      <div class="stat__value mono">{{ result.lambda_max
    }}</div>
    </div>
    <div>
      <div class="stat__label">一致性指标 CI</div>
      <div class="stat__value mono">{{
    "%.4f"|format(result.ci) }}</div>
    </div>
```

```
      <div>                                                                7601
        <div class="stat__label">一致性比例 CR</div>                        7602
        <div class="stat__value mono">{{                                   7603
      "%.4f"|format(result.cr) }}</div>                                    7604
      </div>                                                               7605
      <div>                                                                7606
        <div class="stat__label">一致性检验（要求 CR &lt; 0.1）</div>       7607
        <div class="stat__value">                                          7608
          {% if result.consistent %}<span class="ok">通过</span>           7609
          {% else %}<span class="warn">不通过</span>{% endif %}            7610
        </div>                                                             7611
      </div>                                                               7612
    </div>                                                                 7613
    {% if not result.consistent and result.most_inconsistent               7614
      %}                                                                   7615
    <p style="margin-top:14px;padding:12px;border-radius:6px;b             7616
      ackground:var(--danger-soft);                                        7617
                 border-left:4px solid var(--danger);">                    7618
      <strong class="warn">建议优先调整：</strong>                         7619
      第 {{ result.most_inconsistent.i + 1 }} 行、第 {{                    7620
      result.most_inconsistent.j + 1 }} 列的比较                           7621
      （当前 <span class="mono num">{{ result.most_inconsistent.v          7622
      alue|round(3) }}</span>，                                            7623
      理想约 <span class="mono num">{{ result.most_inconsistent.e          7624
      xpected|round(3) }}</span>）                                         7625
    </p>                                                                   7626
    {% endif %}                                                            7627
    {% endif %}                                                            7628
</div>                                                                     7629
{% endblock %}                                                             7630
# ==== app/templates/ahp/indicators.html ====                              7631
{% extends "base.html" %}                                                  7632
{% block title %}指标定义维护{% endblock %}                                7633
{% block content %}                                                        7634
<div class="card">                                                         7635
  <h2>指标体系维护</h2>                                                    7636
  <p class="actions" style="margin-bottom:14px;">                          7637
    <a class="btn btn--sm btn--ghost"                                      7638
    href="/ahp/">返回权重计算</a>                                          7639
    <a class="btn btn--sm btn--ghost"                                      7640
    href="/ahp/experts">多专家矩阵</a>                                     7641
    <form method="post" action="/ahp/indicators/reset"                     7642
    style="display:inline;"                                                7643
          onsubmit="return confirm('将按论文取值重置全部指标定义，自定义修改会丢失。 7644
    确定继续？');">                                                        7645
      {{ form.hidden_tag() }}                                              7646
      <button type="submit" class="btn btn--sm                             7647
    btn--danger">按论文取值重置</button>                                   7648
    </form>                                                                7649
  </p>                                                                     7650
```

```
{% for level, title in [(1, "一级指标"), (2, "二级指标"), (3, "三级指标")] %}
{% set level_items = all_ind|selectattr("level", "equalto", level)|list %}
{% if level_items %}
<h3>{{ title }}（{{ level_items|length }} 项）</h3>
<table>
  <thead>
    <tr>
      <th scope="col">指标</th>
      <th scope="col">上级指标</th>
      <th scope="col">单位</th>
      <th scope="col">标准化类型</th>
      <th scope="col">理想取值</th>
      <th scope="col">启用单元</th>
      <th scope="col">限制性</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for i in level_items %}
    <tr>
      <td style="text-align:left;">
        <span class="mono muted">{{ i.code }}</span><br>{{ i.name }}
      </td>
      <td>{{ (all_ind|selectattr("code", "equalto", i.parent_code)|list|first).name if i.parent_code else "—" }}</td>
      <td>{{ i.unit or "—" }}</td>
      <td>{{ type_label(i.ideal_type) }}</td>
      <td class="mono">{{ i.ideal_value or "—" }}</td>
      <td style="text-align:left;">{{ i.enabled_units or "—" }}</td>
      <td>
        {% if i.is_constraint %}<span class="tag tag--warn">是</span>
        {% else %}<span class="tag tag--muted">否</span>{% endif %}
      </td>
      <td class="nowrap">
        <span class="row-actions">
          <a class="link-btn" href="/ahp/indicators/{{ i.code }}/edit">编辑</a>
        </span>
      </td>
    </tr>
    {% endfor %}
  </tbody>
</table>
```

```
  {% endif %}                                                                  7701
  {% endfor %}                                                                 7702
</div>                                                                         7703
{% endblock %}                                                                 7704
# ==== app/templates/ahp/indicator_edit.html ====                              7705
{% extends "base.html" %}                                                      7706
{% block title %}编辑指标 · {{ ind.name }}{% endblock %}                       7707
{% block content %}                                                            7708
<div class="card" style="max-width:680px;">                                    7709
  <h2>编辑指标 · {{ ind.name }}</h2>                                           7710
  <p class="muted">                                                            7711
    指标编码 <span class="mono">{{ ind.code }}</span>、层级 {{                 7712
    ind.level }} 级，                                                          7713
    由论文指标体系确定，不可修改。                                             7714
  </p>                                                                         7715
  <form method="post">                                                         7716
    {{ form.hidden_tag() }}                                                    7717
    <div class="field">                                                        7718
      <label class="field__label" for="name">指标名称</label>                  7719
      <input id="name" type="text" name="name"                                 7720
    style="width:100%;" required                                               7721
             value="{{ ind.name }}">                                           7722
    </div>                                                                     7723
    <div class="field">                                                        7724
      <label class="field__label" for="unit">计量单位</label>                  7725
      <input id="unit" type="text" name="unit"                                 7726
    style="width:180px;"                                                       7727
             value="{{ ind.unit or '' }}">                                     7728
    </div>                                                                     7729
    <div class="field">                                                        7730
      <label class="field__label" for="ideal-type">标准化类型</la 7731
    bel>                                                                       7732
      <select id="ideal-type" name="ideal_type">                               7733
        {% for code, label in [("benefit","越大越优"),("cost","越 7734
    小越优"),("mid","适中型"),("discrete","离散赋值")] %}                      7735
        <option value="{{ code }}" {% if ind.ideal_type ==  7736
    code %}selected{% endif %}>{{ label }}</option>                            7737
        {% endfor %}                                                           7738
      </select>                                                                7739
    </div>                                                                     7740
    <div class="field">                                                        7741
      <label class="field__label" for="ideal-value">理想取值</la 7742
    bel>                                                                       7743
      <input id="ideal-value" type="text"                                      7744
    name="ideal_value" style="width:100%;"                                     7745
             value="{{ ind.ideal_value or '' }}">                             7746
      <p class="muted" style="margin:4px 0 0;">                                7747
        如土石比 0.8、土层厚度 20、pH 5~9、植被覆盖度 75。                     7748
      </p>                                                                     7749
    </div>                                                                     7750
```

```
      <fieldset style="border:1px solid                                    7751
      var(--line);border-radius:6px;padding:10px                           7752
      14px;margin:0 0 12px;">                                              7753
        <legend style="padding:0 6px;font-weight:600;">启用该指标的微 7754
      地质单元</legend>                                                    7755
        {% for t in unit_types %}                                          7756
        <label class="checkbox" style="margin-right:18px;">                7757
          <input type="checkbox" name="enabled_units"                      7758
      value="{{ t }}"                                                      7759
                 {% if t in ind.units_list() %}checked{%                   7760
      endif %}>                                                            7761
          <span>{{ t }}</span>                                             7762
        </label>                                                           7763
        {% endfor %}                                                       7764
        <p class="muted" style="margin:8px 0 0;">                          7765
          论文设定：岩质边坡不启用土壤指标，平台不启用地形指标。                 7766
        </p>                                                               7767
      </fieldset>                                                          7768
      <div class="field">                                                  7769
        <label class="checkbox" for="is-constraint">                       7770
          <input id="is-constraint" type="checkbox"                        7771
      name="is_constraint" value="1"                                       7772
                 {% if ind.is_constraint %}checked{% endif %}>             7773
          <span>同时作为限制性指标（参与一票否决判定）</span>                    7774
        </label>                                                           7775
      </div>                                                               7776
      <div class="field">                                                  7777
        <label class="field__label" for="description">取值依据说明</ 7778
      label>                                                               7779
        <textarea id="description" name="description"                      7780
      style="min-height:88px;">{{ ind.description or ''                    7781
      }}</textarea>                                                        7782
      </div>                                                               7783
      <p class="actions" style="margin-top:16px;">                         7784
        <button type="submit" class="btn">保存</button>                    7785
        <a class="btn btn--ghost" href="/ahp/indicators">返回指标列 7786
      表</a>                                                               7787
      </p>                                                                 7788
    </form>                                                                7789
</div>                                                                     7790
{% endblock %}                                                             7791
# ==== app/templates/ahp/experts.html ====                                 7792
{% extends "base.html" %}                                                  7793
{% block title %}多专家判断矩阵{% endblock %}                                7794
{% block content %}                                                        7795
<div class="card">                                                         7796
  <h2>多专家判断矩阵 · 录入与汇总</h2>                                        7797
  <form method="get" class="actions" style="margin-bottom:14 7798
    px;">                                                                  7799
    <div>                                                                  7800
```

```
      <label class="field__label" for="unit-type">微地质单元类型</l 7801
    abel>                                                          7802
      <select id="unit-type" name="unit_type"                      7803
    onchange="this.form.submit()">                                 7804
        {% for t in ["岩质边坡", "碎石边坡", "平台"] %}              7805
        <option value="{{ t }}" {% if unit_type == t               7806
    %}selected{% endif %}>{{ t }}</option>                         7807
        {% endfor %}                                               7808
      </select>                                                    7809
    </div>                                                         7810
    <div>                                                          7811
      <label class="field__label" for="level">指标层级</label>     7812
      <select id="level" name="level"                              7813
    onchange="this.form.submit()">                                 7814
        <option value="1" {% if level == 1 %}selected{%            7815
    endif %}>一级指标</option>                                     7816
        <option value="2" {% if level == 2 %}selected{%            7817
    endif %}>二级指标</option>                                     7818
      </select>                                                    7819
    </div>                                                         7820
  </form>                                                          7821
  <p class="muted">                                                7822
    当前范围：<strong>{{ unit_type }}</strong> ·                   7823
    {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif 7824
    %}。                                                           7825
    若选二级指标，请先到                                           7826
    <a href="/ahp/">权重计算</a> 选择一级指标后进入本页。          7827
  </p>                                                             7828
  {% if rows %}                                                    7829
  <table>                                                          7830
    <caption class="muted">已录入 {{ rows|length }}                7831
    位专家的矩阵</caption>                                         7832
    <thead>                                                        7833
      <tr>                                                         7834
        <th scope="col">专家</th>                                  7835
        <th scope="col">矩阵规模</th>                              7836
        <th scope="col">该矩阵 CR</th>                             7837
        <th scope="col">一致性</th>                                7838
        <th scope="col">录入时间</th>                              7839
        <th scope="col">操作</th>                                  7840
      </tr>                                                        7841
    </thead>                                                       7842
    <tbody>                                                        7843
      {% for r in rows %}                                          7844
      <tr>                                                         7845
        <td>{{ r.expert_name }}</td>                               7846
        <td class="num">{{ n }} × {{ n }}</td>                     7847
        <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is         7848
    not none else "—" }}</td>                                      7849
        <td>                                                       7850
```

```
          {% if r.cr is not none and r.cr < 0.1 %}<span
    class="tag tag--ok">通过</span>
          {% else %}<span class="tag tag--warn">不通过</span>{%
    endif %}
        </td>
        <td class="mono nowrap">{{ r.created_at.strftime("%Y
    -%m-%d %H:%M") if r.created_at else "—" }}</td>
        <td>
          <form method="post" action="/ahp/experts/{{ r.id
    }}/delete"
                onsubmit="return confirm('确定删除专家「{{
    r.expert_name }}」的矩阵吗？');">
            {{ form.hidden_tag() }}
            <button type="submit" class="btn btn--sm
    btn--danger">删除</button>
          </form>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  <p class="actions" style="margin-bottom:10px;">
    <a class="btn" href="/ahp/experts/aggregate?unit_type={{
    unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{%
    endif %}">
      汇总计算（几何平均）
    </a>
  </p>
  {% else %}
  <p class="muted">尚未录入任何专家矩阵。请在下方录入第一位专家的判断矩阵。</p>
  {% endif %}
  <h3>录入一位专家的判断矩阵</h3>
  <form method="post">
    {{ form.hidden_tag() }}
    <input type="hidden" name="action" value="add">
    <input type="hidden" name="unit_type" value="{{
    unit_type }}">
    <input type="hidden" name="level" value="{{ level }}">
    {% if l1 %}<input type="hidden" name="l1" value="{{ l1
    }}">{% endif %}
    <div class="field">
      <label class="field__label" for="expert-name">专家姓名</la
    bel>
      <input id="expert-name" type="text" name="expert_name"
             style="width:220px;" placeholder="如：张工">
    </div>
    <table style="width:auto;">
      <thead>
        <tr>
          <th></th>
```

```
            {% for name in names %}<th scope="col">{{ name }}</th>{% endfor %}
          </tr>
        </thead>
        <tbody>
          {% for i in range(n) %}
          <tr>
            <th scope="row">{{ names[i] }}</th>
            {% for j in range(n) %}
              {% if i == j %}
              <td class="muted num">1</td>
              {% elif i < j %}
              <td>
                <label class="muted" style="position:absolute;left:-9999px;"
                       for="e-{{ i }}-{{ j }}">{{ names[i] }}相对 {{ names[j] }} 的重要性</label>
                <input id="e-{{ i }}-{{ j }}" type="number" step="0.01" min="0.01"
                       name="m_{{ i }}_{{ j }}" value="{{ default[i][j] }}">
              </td>
              {% else %}
              <td class="muted num">{{ default[i][j] }}</td>
              {% endif %}
            {% endfor %}
          </tr>
          {% endfor %}
        </tbody>
      </table>
      <p class="actions" style="margin-top:12px;">
        <button type="submit" class="btn">录入该专家矩阵</button>
        <a class="btn btn--ghost" href="/ahp/indicators">指标维护</a>
        <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/ahp/experts_aggregate.html ====
{% extends "base.html" %}
{% block title %}多专家矩阵汇总{% endblock %}
{% block content %}
<div class="card">
  <h2>多专家矩阵汇总 · {{ unit_type }} · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif %}</h2>
  {% if not result %}
  <p class="muted">
    该范围内尚未录入专家矩阵。
    <a href="/ahp/experts?unit_type={{ unit_type
```

```
    }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif
    %}">去录入</a>
  </p>
  {% else %}
  <div class="stat-row">
    <div>
      <div class="stat__label">参与汇总的专家</div>
      <div class="stat__value">{{ result.count }} 位</div>
    </div>
    <div>
      <div class="stat__label">λmax</div>
      <div class="stat__value mono">{{ result.lambda_max
    }}</div>
    </div>
    <div>
      <div class="stat__label">CI</div>
      <div class="stat__value mono">{{
    "%.4f"|format(result.ci) }}</div>
    </div>
    <div>
      <div class="stat__label">CR（要求 &lt; 0.1）</div>
      <div class="stat__value mono">{{
    "%.4f"|format(result.cr) }}</div>
    </div>
    <div>
      <div class="stat__label">一致性检验</div>
      <div class="stat__value">
        {% if result.consistent %}<span class="ok">通过</span>
        {% else %}<span class="warn">不通过</span>{% endif %}
      </div>
    </div>
  </div>
  <h3>汇总后权重</h3>
  <table style="width:auto;">
    <thead>
      <tr>
        <th scope="col">指标</th>
        {% for name in names %}<th scope="col">{{ name
    }}</th>{% endfor %}
      </tr>
    </thead>
    <tbody>
      <tr>
        <th scope="row">权重</th>
        {% for w in result.weights %}<td class="mono">{{ w
    }}</td>{% endfor %}
      </tr>
    </tbody>
  </table>
  <h3>各专家矩阵的一致性</h3>
```

```
<table style="width:auto;">
  <thead>
    <tr><th scope="col">专家</th><th scope="col">该矩阵CR</th><th scope="col">判定</th></tr>
  </thead>
  <tbody>
    {% for r in rows %}
    <tr>
      <td>{{ r.expert_name }}</td>
      <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is not none else "—" }}</td>
      <td>
        {% if r.cr is not none and r.cr < 0.1 %}<span class="ok">通过</span>
        {% else %}<span class="warn">不通过</span>{% endif %}
      </td>
    </tr>
    {% endfor %}
  </tbody>
</table>
{% if not result.consistent and result.most_inconsistent %}
<p style="padding:12px;border-radius:6px;background:var(--danger-soft);
          border-left:4px solid var(--danger);">
  <strong class="warn">汇总矩阵一致性不通过，建议优先调整：</strong>
  第 {{ result.most_inconsistent.i + 1 }} 行、第 {{ result.most_inconsistent.j + 1 }} 列的比较
  （当前 <span class="mono">{{ result.most_inconsistent.value|round(3) }}</span>，
  理想约 <span class="mono">{{ result.most_inconsistent.expected|round(3) }}</span>）。
  该偏差通常来自个别专家打分，可先核对一致性较差的专家矩阵。
</p>
{% endif %}
{% endif %}
<p class="actions" style="margin-top:16px;">
  <form method="post"
        action="/ahp/experts?unit_type={{ unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}"
        style="display:inline;"
        onsubmit="return confirm('保存后，「{{ unit_type }}」的适宜性评价将改用这组汇总权重计算。确定继续吗？');">
    {{ form.hidden_tag() }}
    <input type="hidden" name="action" value="save_agg">
    <button type="submit" class="btn" {% if not result.consistent %}disabled{% endif %}>
      保存为生效权重
    </button>
  </form>
```

```
      <a class="btn btn--ghost" href="/ahp/experts?unit_type={ 8051
      { unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1      8052
      }}{% endif %}">返回录入</a>                                8053
      <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>      8054
   </p>                                                          8055
</div>                                                           8056
{% endblock %}                                                   8057
# ==== app/templates/indicator/index.html ====                   8058
{% extends "base.html" %}                                        8059
{% from "_macros.html" import patch_guide %}                     8060
{% block title %}指标数据采集{% endblock %}                        8061
{% block content %}                                              8062
<div class="card">                                               8063
   <h2>指标数据采集与管理</h2>                                     8064
   <p class="actions" style="margin-bottom:12px;">               8065
      <a class="btn btn--sm" href="/indicator/import">批量导入（CSV 8066
      / Excel）</a>                                              8067
   </p>                                                          8068
   {{ patch_guide(projects_without_patch) }}                     8069
   {% if not patches %}                                          8070
   <p class="muted">                                             8071
      暂无矿山图斑。请先在「项目与矿山档案」中建立图斑，              8072
      或运行 <span class="mono">python seed_demo.py</span>       8073
      载入论文案例。                                             8074
   </p>                                                          8075
   {% else %}                                                    8076
   {% for p in patches %}                                        8077
   <div class="patch-panel">                                     8078
      <div class="patch-panel__head">                            8079
         <h3 class="patch-panel__name">{{ p.name }}</h3>         8080
         <span class="patch-panel__meta">                        8081
            所属项目：{{ p.project.name if p.project else "—" }}  8082
         </span>                                                 8083
         <span class="patch-panel__meta">                        8084
            微地质单元：{{ p.units.count() }} 个                  8085
         </span>                                                 8086
         {% if p.units.count() %}                                8087
         <span class="patch-panel__meta patch-panel__tail">      8088
            <a class="link-btn" href="/indicator/import">批量导入本图斑 8089
      数据</a>                                                   8090
         </span>                                                 8091
         {% endif %}                                             8092
      </div>                                                     8093
      {% if not p.units.count() %}                               8094
      <div class="notice notice--warn">                          8095
         <p class="notice__title">该图斑尚未划分微地质单元，暂不能录入指标数据。</p>  8096
         <p style="margin:0;">                                   8097
            指标数据是按微地质单元（岩质边坡 / 碎石边坡 / 平台）分别组织的，  8098
            请先完成划分，再回到本页录入。                        8099
         </p>                                                    8100
```

```
        <p style="margin:10px 0 0;">                                                    8101
          <a class="btn btn--sm" href="/unit/{{ p.id                                    8102
      }}">去划分微地质单元</a>                                                          8103
        </p>                                                                            8104
      </div>                                                                            8105
      {% else %}                                                                        8106
      <table>                                                                           8107
        <thead>                                                                         8108
          <tr>                                                                          8109
            <th scope="col">微地质单元</th>                                             8110
            <th scope="col">面积占比</th>                                               8111
            <th scope="col">数据状态</th>                                               8112
            <th scope="col">操作</th>                                                   8113
          </tr>                                                                         8114
        </thead>                                                                        8115
        <tbody>                                                                         8116
          {% for u in p.units %}                                                        8117
          <tr>                                                                          8118
            <td>{{ u.unit_type }}</td>                                                  8119
            <td class="num">                                                            8120
              {% if u.ratio is not none %}{{                                            8121
      "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}                           8122
            </td>                                                                       8123
            <td>                                                                        8124
              {% if u.indicator_data %}                                                 8125
              <span class="tag tag--ok">已录入</span>                                   8126
              {% else %}                                                                8127
              <span class="tag tag--muted">未录入</span>                                8128
              {% endif %}                                                               8129
            </td>                                                                       8130
            <td class="nowrap">                                                         8131
              <span class="row-actions">                                                8132
                <a class="link-btn" href="/indicator/{{ u.id                            8133
      }}">录入 / 编辑</a>                                                               8134
              </span>                                                                   8135
            </td>                                                                       8136
          </tr>                                                                         8137
          {% endfor %}                                                                  8138
        </tbody>                                                                        8139
      </table>                                                                          8140
      {% endif %}                                                                       8141
    </div>                                                                              8142
    {% endfor %}                                                                        8143
    {% endif %}                                                                         8144
</div>                                                                                  8145
{% endblock %}                                                                          8146
# ==== app/templates/indicator/edit.html ====                                           8147
{% extends "base.html" %}                                                               8148
{% block title %}指标数据录入 · {{ unit.unit_type }}{% endblock %}                      8149
{% block content %}                                                                     8150
```

```
<div class="card" style="max-width:820px;">
  <h2>指标数据录入 · {{ unit.unit_type }}</h2>
  <p class="muted">
    输入框留空表示该指标「不适用」，与「实测值为 0」含义不同，计算时会被跳过。
    岩质边坡不设土壤指标、平台不设地形指标，系统已按单元类型隐藏不相关字段。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    {% for group_name, fields, has_hm in groups %}
    <h3>{{ group_name }}</h3>
    {% for fname, cname, unit_ in fields %}
    <div class="field">
      <label class="field__label" for="f-{{ fname }}">
        {{ cname }}{% if unit_ %} ({{ unit_ }}) {% endif %}
      </label>
      {% if fname == "aspect" %}
      <select id="f-{{ fname }}" name="aspect">
        <option value="">— 不适用 —</option>
        {% for a in aspect_types %}
        <option value="{{ a }}" {% if current.get("aspect")
    == a %}selected{% endif %}>{{ a }}</option>
        {% endfor %}
      </select>
      {% else %}
      <input id="f-{{ fname }}" type="number" step="any"
    name="{{ fname }}"
             value="{{ current.get(fname) if
    current.get(fname) is not none else '' }}">
      {% endif %}
    </div>
    {% endfor %}
    {% endfor %}
    {% for group_name, fields, has_hm in groups %}
    {% if has_hm %}
    <h3>重金属含量（mg/kg）</h3>
    <p class="muted">未检测的金属可留空，系统按论文注释视为未受污染处理。</p>
    <table style="width:auto;">
      <thead>
        <tr>
          {% for m in heavy_metals %}<th scope="col">{{ m
    }}</th>{% endfor %}
        </tr>
      </thead>
      <tbody>
        <tr>
          {% for m in heavy_metals %}
          <td>
            <label class="muted" style="position:absolute;
    left:-9999px;"
                   for="hm-{{ m }}">{{ m }} 含量</label>
```

```
                    <input id="hm-{{ m }}" type="number"
      step="any" name="hm_{{ m }}"
                           value="{{ current.get('heavy',
      {}).get(m) if current.get('heavy', {}).get(m) is not
      none else '' }}">
                </td>
                {% endfor %}
              </tr>
            </tbody>
          </table>
          {% endif %}
      {% endfor %}
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/indicator/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/indicator/import.html ====
{% extends "base.html" %}
{% block title %}批量导入指标数据{% endblock %}
{% block content %}
<div class="card" style="max-width:860px;">
    <h2>批量导入指标数据</h2>
    <p class="muted">
      支持 CSV 与 Excel（.xlsx/.xls）。文件每行对应一个微地质单元，
      系统按「单元类型」匹配图斑内已划分的单元后写入数据。
      先下载模板按格式填写，可减少出错。
    </p>
    <p class="actions" style="margin-bottom:14px;">
      <a class="btn btn--sm btn--ghost"
      href="/indicator/import/template">下载导入模板（CSV）</a>
      <a class="btn btn--sm btn--ghost"
      href="/indicator/">返回指标数据</a>
    </p>
    <form method="post" enctype="multipart/form-data">
      {{ form.hidden_tag() }}
      <div class="field">
        <label class="field__label" for="patch-id">目标矿山图斑</lab
      el>
        <select id="patch-id" name="patch_id" required>
          <option value="">— 请选择 —</option>
          {% for p in patches %}
          <option value="{{ p.id }}">{{ p.name }} ({{
      p.units.count() }} 个单元）</option>
          {% endfor %}
        </select>
      </div>
      <div class="field">
```

```
        <label class="field__label" for="datafile">数据文件</label
      >
        <input id="datafile" type="file" name="datafile"
               accept=".csv,.xlsx,.xls" required>
      </div>
      <p class="actions" style="margin-top:16px;">
        <button type="submit" class="btn">开始导入</button>
      </p>
    </form>
    <h3>模板列说明</h3>
    <p class="muted">
      共 {{ headers|length }} 列。留空表示该指标「不适用」。
      坡向须填「阳坡 / 阴阳坡 / 阴坡」之一。未检测的重金属留空即可
      （按论文注释视为未受污染）。
    </p>
    <p class="mono muted" style="line-height:1.9;">{{
      headers|join(" , ") }}</p>
    <h3>导入规则</h3>
    <ul class="muted" style="padding-left:20px;line-height:1.9
      ;">
      <li>「单元类型」必须与图斑内已划分的单元一致，否则该行会被跳过并在结果中提示。</li>
      <li>同一单元类型若在文件中出现多行，以最后一行为准。</li>
      <li>已有数据的单元会被覆盖更新；没有数据的单元将新建记录。</li>
      <li>数值列请填纯数字，不要带单位（如填 <span class="mono">10</span> 而非
      <span class="mono">10cm</span>）。</li>
    </ul>
</div>
{% if report %}
<div class="card" style="max-width:860px;">
    <h2>导入结果</h2>
    <div class="stat-row">
      <div>
        <div class="stat__label">目标图斑</div>
        <div style="font-size:15px;">{{ report.patch }}</div>
      </div>
      <div>
        <div class="stat__label">成功导入</div>
        <div class="stat__value ok">{{ report.saved }}
      个单元</div>
      </div>
      <div>
        <div class="stat__label">问题行</div>
        <div class="stat__value {% if report.errors %}warn{%
      endif %}">{{ report.errors|length }}</div>
      </div>
    </div>
    {% if report.errors %}
    <h3>未导入的行</h3>
    <ul class="muted" style="padding-left:20px;line-height:1.9
      ;">
```

```
      {% for e in report.errors %}<li>{{ e }}</li>{% endfor %}
    </ul>
    {% else %}
    <p class="ok" style="margin-top:12px;">全部数据行均已成功导入。</p>
    {% endif %}
    <p class="actions" style="margin-top:16px;">
      <a class="btn btn--ghost" href="/indicator/">查看指标数据</a>
    </p>
  </div>
  {% endif %}
  {% endblock %}
  # ==== app/templates/evaluate/index.html ====
  {% extends "base.html" %}
  {% from "_macros.html" import patch_guide %}
  {% block title %}适宜性定量评价{% endblock %}
  {% block content %}
  <div class="card">
    <h2>适宜性定量评价</h2>
    {{ patch_guide(projects_without_patch) }}
    {% if not patches %}
    <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
    {% else %}
    <table>
      <thead>
        <tr>
          <th scope="col">所属项目</th>
          <th scope="col">矿山图斑</th>
          <th scope="col">前置条件</th>
          <th scope="col">最近评价结果</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for p in patches %}
        {% set pr = progress.get(p.id) %}
        <tr>
          <td>{{ p.project.name if p.project else "—" }}</td>
          <td style="text-align:left;">{{ p.name }}</td>
          <td style="text-align:left;">
            {% if pr and pr.ready %}
            <span class="tag tag--ok">已具备</span>
            {% elif pr %}
            <span class="tag tag--warn">缺：{{
      pr.blocking[0].label }}</span>
            <span class="muted">{{ pr.blocking[0].reason
      }}</span>
            {% endif %}
          </td>
          <td>
            {% if latest.get(p.id) %}
```

```
            F = <span class="mono num">{{                                    8351
    latest[p.id].final_score }}</span>,                                      8352
            <span class="tag tag--ok">{{ latest[p.id].grade                  8353
    }}</span>                                                                8354
            {% else %}                                                       8355
            <span class="tag tag--muted">未评价</span>                        8356
            {% endif %}                                                      8357
          </td>                                                              8358
          <td class="nowrap">                                                8359
            <span class="row-actions">                                       8360
              <a class="link-btn" href="/evaluate/{{ p.id                    8361
    }}">执行评价</a>                                                         8362
              {% if latest.get(p.id) %}                                      8363
              <a class="link-btn" href="/evaluate/result/{{                  8364
    latest[p.id].id }}">查看结果</a>                                         8365
              {% endif %}                                                    8366
            </span>                                                          8367
          </td>                                                              8368
        </tr>                                                                8369
        {% endfor %}                                                         8370
      </tbody>                                                               8371
    </table>                                                                 8372
    {% endif %}                                                              8373
</div>                                                                       8374
{% endblock %}                                                               8375
# ==== app/templates/evaluate/run.html ====                                  8376
{% extends "base.html" %}                                                    8377
{% import "_macros.html" as ui %}                                            8378
{% block title %}执行评价 · {{ patch.name }}{% endblock %}                   8379
{% block content %}                                                          8380
<div class="card" style="max-width:880px;">                                  8381
  <h2>执行适宜性评价 · {{ patch.name }}</h2>                                 8382
  {{ ui.workflow_steps(progress) }}                                          8383
  {{ ui.workflow_blocking(progress) }}                                       8384
  {% if progress.ready %}                                                    8385
  <form method="post">                                                       8386
    {{ form.hidden_tag() }}                                                  8387
    <p class="actions">                                                      8388
      <button type="submit" class="btn">开始评价</button>                    8389
      <a class="btn btn--ghost" href="/evaluate/">返回</a>                   8390
    </p>                                                                     8391
  </form>                                                                    8392
  {% else %}                                                                 8393
  <p class="actions">                                                        8394
    <button type="button" class="btn" disabled>开始评价</button>             8395
    <a class="btn btn--ghost" href="/evaluate/">返回</a>                     8396
  </p>                                                                       8397
  {% endif %}                                                                8398
</div>                                                                       8399
<div class="card" style="max-width:880px;">                                  8400
```

```
  <h2>当前流程状态</h2>                                                    8401
  <table>                                                                  8402
    <thead>                                                                8403
      <tr><th scope="col">步骤</th><th scope="col">状态</th><th            8404
    scope="col">说明</th></tr>                                             8405
    </thead>                                                               8406
    <tbody>                                                                8407
      <tr>                                                                 8408
        <td>限制性条件判定</td>                                            8409
        <td>                                                               8410
          {% if progress.constraint.passed %}<span                         8411
    class="tag tag--ok">已通过</span>                                      8412
          {% elif progress.constraint.done %}<span                         8413
    class="tag tag--warn">未通过</span>                                    8414
          {% else %}<span class="tag tag--muted">待完成</span>{            8415
    % endif %}                                                             8416
        </td>                                                              8417
        <td style="text-align:left;">{{                                    8418
    progress.constraint.detail }}</td>                                     8419
      </tr>                                                                8420
      <tr>                                                                 8421
        <td>划分微地质单元</td>                                            8422
        <td>                                                               8423
          {% if progress.units.done %}<span class="tag                     8424
    tag--ok">已完成</span>                                                 8425
          {% else %}<span class="tag tag--muted">待完成</span>{            8426
    % endif %}                                                             8427
        </td>                                                              8428
        <td style="text-align:left;">{{                                    8429
    progress.units.detail }}</td>                                          8430
      </tr>                                                                8431
      <tr>                                                                 8432
        <td>录入指标数据</td>                                              8433
        <td>                                                               8434
          {% if progress.data.ready %}<span class="tag                     8435
    tag--ok">已完成</span>                                                 8436
          {% else %}<span class="tag tag--muted">待完成</span>{            8437
    % endif %}                                                             8438
        </td>                                                              8439
        <td style="text-align:left;">{{                                    8440
    progress.data.detail }}</td>                                           8441
      </tr>                                                                8442
    </tbody>                                                               8443
  </table>                                                                 8444
</div>                                                                     8445
{% if weight_sources %}                                                    8446
<div class="card" style="max-width:880px;">                                8447
  <h2>本次将采用的指标权重</h2>  <table style="max-width:720px;">          8448
    <thead>                                                                8449
      <tr><th scope="col">微地质单元</th><th                               8450
```

```
      scope="col">权重来源</th></tr>                                  8451
      </thead>                                                        8452
      <tbody>                                                         8453
        {% for ut, label in weight_sources.items() %}                 8454
        <tr>                                                          8455
          <td style="text-align:left;">{{ ut }}</td>                  8456
          <td style="text-align:left;">                               8457
            {% if "自定义" in label %}                                8458
            <span class="tag tag--warn">{{ label }}</span>            8459
            {% else %}                                                8460
            <span class="tag tag--muted">{{ label }}</span>           8461
            {% endif %}                                               8462
          </td>                                                       8463
        </tr>                                                         8464
        {% endfor %}                                                  8465
      </tbody>                                                        8466
    </table>                                                          8467
</div>                                                                8468
{% endif %}                                                           8469
{% endblock %}                                                        8470
# ==== app/templates/evaluate/result.html ====                        8471
{% extends "base.html" %}                                             8472
{% block title %}评价结果{% endblock %}                               8473
{% block content %}                                                   8474
<div class="card">                                                    8475
  <h2>适宜性评价结果 · {{ patch.name if patch else "" }}</h2>         8476
  <div class="stat-row">                                              8477
    <div>                                                             8478
      <div class="stat__label">最终评价因子 F</div>                   8479
      <div class="stat__value mono">{{ r.final_score                  8480
    }}</div>                                                          8481
    </div>                                                            8482
    <div>                                                             8483
      <div class="stat__label">适宜性等级</div>                       8484
      <div class="stat__value ok">{{ r.grade }}</div>                 8485
    </div>                                                            8486
    <div style="flex:1;min-width:240px;">                             8487
      <div class="stat__label">推荐措施</div>                         8488
      <div style="font-size:15px;">{{ r.measures }}</div>             8489
    </div>                                                            8490
  </div>                                                              8491
  {% if unit_scores %}                                                8492
  <h3>各微地质单元得分</h3>                                           8493
  <table style="max-width:860px;">                                    8494
    <thead>                                                           8495
      <tr>                                                            8496
        <th scope="col">微地质单元</th>                               8497
        <th scope="col">面积占比</th>                                 8498
        <th scope="col">单元评价因子 F′</th>                          8499
        <th scope="col">加权贡献</th>                                 8500
```

```
            <th scope="col">权重来源</th>
          </tr>
        </thead>
        <tbody>
          {% for u in unit_scores %}
          <tr>
            <td>{{ u.unit_type }}</td>
            <td class="num">{{ "%.2f%%"|format(u.ratio * 100)
        }}</td>
            <td class="mono">{{ u.unit_score }}</td>
            <td class="mono">{{ "%.4f"|format(u.ratio *
        u.unit_score) }}</td>
            <td class="muted">{{ u.weight_source or "论文表 3
        默认权重" }}</td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      {% endif %}
      <p class="actions" style="margin-top:18px;">
        <a class="btn btn--ghost" href="/report/{{ r.patch_id
        }}">查看得分明细</a>
        <a class="btn btn--ghost" href="/evaluate/">返回评价列表</a>
      </p>
    </div>
    {% endblock %}
    # ==== app/templates/report/index.html ====
    {% extends "base.html" %}
    {% from "_macros.html" import patch_guide %}
    {% block title %}报告输出{% endblock %}
    {% block content %}
    <div class="card">
      <h2>报告与结果输出</h2>
      {{ patch_guide(projects_without_patch) }}
      {% if not patches %}
      <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
      {% else %}
      <table>
        <thead>
          <tr>
            <th scope="col">所属项目</th>
            <th scope="col">矿山图斑</th>
            <th scope="col">可生成报告</th>
            <th scope="col">操作</th>
          </tr>
        </thead>
        <tbody>
          {% for p in patches %}
          {% set pr = progress.get(p.id) %}
          <tr>
```

```
            <td>{{ p.project.name if p.project else "—" }}</td>
            <td style="text-align:left;">{{ p.name }}</td>
            <td style="text-align:left;">
              {% if pr and pr.ready %}
              <span class="tag tag--ok">可以</span>
              {% elif pr %}
              <span class="tag tag--warn">暂不可</span>
              <span class="muted">{{ pr.blocking[0].reason
      }}</span>
              {% endif %}
            </td>
            <td class="nowrap">
              {% if pr and pr.ready %}
              <span class="row-actions">
                <a class="link-btn" href="/report/{{ p.id
      }}">查看得分明细</a>
                <a class="link-btn" href="/report/{{ p.id
      }}/pdf">导出 PDF</a>
                <a class="link-btn" href="/report/{{ p.id
      }}/word">导出 Word</a>
              </span>
              {% else %}
              <span class="muted">—</span>
              {% endif %}
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      {% endif %}
</div>
{% endblock %}
# ==== app/templates/report/detail.html ====
{% extends "base.html" %}
{% block title %}得分明细 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card">
  <h2>得分明细 · {{ patch.name }}</h2>
  <div class="stat-row">
    <div>
      <div class="stat__label">最终评价因子 F</div>
      <div class="stat__value mono">{{
    "%.4f"|format(result.final_score) }}</div>
    </div>
    <div>
      <div class="stat__label">适宜性等级</div>
      <div class="stat__value ok">{{ result.grade }}</div>
    </div>
    <div style="flex:1;min-width:240px;">
      <div class="stat__label">推荐措施</div>
```

```
        <div style="font-size:15px;">{{                                    8601
      result.grade_info.measure }}</div>                                   8602
      </div>                                                               8603
   </div>                                                                  8604
   <p class="muted" style="margin-top:12px;">                              8605
      本页与导出报告均取自                                                 8606
      <strong>{% if record %}{{ record.created_at.strftime("%Y             8607
      -%m-%d %H:%M:%S") }}{% endif %}                                      8608
      记录的那次评价结果</strong>（评价记录编号 {{ record.id if record else 8609
      "—" }}），                                                           8610
      不会因事后修改指标数据而变化。                                       8611
   </p>                                                                    8612
   <p class="actions" style="margin-top:16px;">                            8613
      <a class="btn" href="/report/{{ patch.id }}/pdf">导出 PDF            8614
      报告</a>                                                             8615
      <a class="btn btn--ghost" href="/report/{{ patch.id                  8616
      }}/word">导出 Word 报告</a>                                          8617
      <a class="btn btn--ghost" href="/report/">返回</a>                   8618
   </p>                                                                    8619
</div>                                                                     8620
<div class="card">                                                         8621
   <h2>结果图表</h2>                                                       8622
   <h3>各微地质单元得分对比</h3>                                           8623
   <div>{{ chart_units|safe }}</div>                                       8624
   <h3>各一级指标平均得分</h3>                                             8625
   <p class="muted">反映各一级指标对评价结果的实际影响，便于识别关键控制项。</p>  8626
   <div>{{ chart_weight|safe }}</div>                                      8627
</div>                                                                     8628
{% for u in result.unit_results %}                                         8629
<div class="card">                                                         8630
   <h2>{{ u.unit_type }} · 占比 {{ "%.2f%%"|format(u.ratio *               8631
      100) }} · 单元得分 {{ "%.4f"|format(u.unit_score) }}</h2>            8632
   {% if u.weight_source %}                                                8633
   <p class="muted" style="margin-top:-6px;">权重来源：{{                  8634
      u.weight_source }}</p>                                               8635
   {% endif %}                                                             8636
   <table>                                                                 8637
      <thead>                                                              8638
         <tr>                                                              8639
            <th scope="col">一级指标</th>                                  8640
            <th scope="col">二级指标</th>                                  8641
            <th scope="col">实测值</th>                                    8642
            <th scope="col">标准化值</th>                                  8643
            <th scope="col">权重</th>                                      8644
            <th scope="col">贡献</th>                                      8645
         </tr>                                                             8646
      </thead>                                                             8647
      <tbody>                                                              8648
         {% for d in u.details %}                                          8649
         <tr>                                                              8650
```

```
          <td>{{ d.l1 }}</td>                                                    8651
          <td style="text-align:left;">{{ d.l2 }}</td>                          8652
          <td class="mono" style="text-align:left;">{{ "—" if                   8653
    d.raw is none else d.raw }}</td>                                             8654
          <td class="mono">{{ "—" if d.std_value is none else                    8655
    "%.4f"|format(d.std_value) }}</td>                                           8656
          <td class="mono">{{ "%.4f"|format(d.weight) }}</td>                    8657
          <td class="mono">{{ "—" if d.contribution is none                      8658
    else "%.4f"|format(d.contribution) }}</td>                                   8659
        </tr>                                                                    8660
        {% endfor %}                                                             8661
    </tbody>                                                                     8662
  </table>                                                                       8663
  <p class="muted" style="margin:0;">                                            8664
    一级指标得分：                                                               8665
    {% for k, v in u.l1_scores.items() %}{{ k }} = <span                         8666
    class="num">{{ "%.4f"|format(v) }}</span>{% if not                           8667
    loop.last %};{% endif %}{% endfor %}                                         8668
  </p>                                                                           8669
</div>                                                                           8670
{% endfor %}                                                                     8671
{% endblock %}                                                                   8672
# ==== app/templates/report/print.html ====                                      8673
<!DOCTYPE html>                                                                  8674
<html lang="zh-CN">                                                              8675
<head>                                                                           8676
<meta charset="UTF-8">                                                           8677
<title>自然恢复适宜性评价报告 · {{ patch.name }}</title>                         8678
<style>                                                                          8679
  @page { size: A4; margin: 18mm 16mm; }                                         8680
  * { box-sizing: border-box; }                                                  8681
  body {                                                                         8682
    margin: 0; color: #23262b; background: #fff;                                 8683
    font-family: "Microsoft YaHei", "SimSun", sans-serif;                        8684
    font-size: 10.5pt; line-height: 1.65;                                        8685
  }                                                                              8686
  h1 { font-size: 17pt; text-align: center; margin: 0 0                          8687
    6px; }                                                                       8688
  h2 { font-size: 13pt; color: #1f3a5f; margin: 20px 0 8px;                      8689
       border-bottom: 1px solid #dfe3e8; padding-bottom:                         8690
    4px; }                                                                       8691
  h3 { font-size: 11.5pt; color: #1f3a5f; margin: 16px 0                         8692
    6px; }                                                                       8693
  .sub { text-align: center; color: #6b7280; font-size:                          8694
    9.5pt; margin-bottom: 18px; }                                                8695
  table { width: 100%; border-collapse: collapse; margin:                        8696
    8px 0 14px;                                                                  8697
          font-variant-numeric: tabular-nums; }                                  8698
  th, td { border: 1px solid #c9ccd1; padding: 4px 8px;                          8699
    text-align: center; }                                                        8700
```

```
  th { background: #eef2f7; font-weight: 600; }
  .left { text-align: left; }
  .mono { font-family: Consolas, "Courier New", monospace; }
  .kv { width: auto; }
  .kv th { width: 30%; text-align: left; }
  .kv td { text-align: left; }
  ul { margin: 6px 0 12px; padding-left: 20px; }
  .note { color: #6b7280; font-size: 9.5pt; }
  .chart { margin: 10px 0 16px; }
  .sig { margin-top: 28px; font-size: 10pt; }
  .avoid-break { break-inside: avoid; }
  svg { max-width: 100%; }
</style>
</head>
<body>
  <h1>历史遗留废弃矿山自然恢复适宜性评价报告</h1>
  <p class="sub">
    {{ patch.name }}
    {% if patch.admin_region %} · {{ patch.admin_region
    }}{% endif %}
    <br>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0 ·
    生成时间 {{ generated_at }}
  </p>
  <h2>一、图斑基本信息</h2>
  <table class="kv">
    <tr><th>图斑名称</td><td>{{ patch.name }}</td></tr>
    <tr><th>所在行政区</td><td>{{ patch.admin_region or "—"
    }}</td></tr>
    <tr><th>图斑面积（公顷）</td><td>{{ patch.area_ha if
    patch.area_ha is not none else "—" }}</td></tr>
    <tr><th>开采矿种</td><td>{{ patch.mineral_type or "—"
    }}</td></tr>
    <tr><th>开采方式</td><td>{{ patch.mining_method or "—"
    }}</td></tr>
    <tr><th>责任主体缺失</td><td>{{ "是" if patch.responsible_entit
    y_missing else "否" }}</td></tr>
    {% if reference %}
    <tr><th>参照生态系统</th>
        <td>植被覆盖度基准 {{ reference.coverage_benchmark }}%,
            多样性指数基准 {{ reference.shannon_benchmark
    }}</td></tr>
    {% endif %}
  </table>
  <h2>二、评价结论</h2>
  <table class="kv">
    <tr><th>最终评价因子 F</th><td class="mono"><strong>{{
    "%.4f"|format(result.final_score) }}</strong></td></tr>
    <tr><th>适宜性等级</td><td><strong>{{ result.grade
    }}</strong></td></tr>
    <tr><th>判定依据</td><td>{{ result.grade_info.meaning
```

```
    }}</td></tr>                                                    8751
    <tr><th>推荐措施</th><td>{{ result.grade_info.measure           8752
    }}</td></tr>                                                    8753
</table>                                                            8754
<h2>三、各微地质单元得分</h2>                                        8755
<table>                                                             8756
    <thead>                                                         8757
      <tr><th>微地质单元</th><th>面积占比</th><th>单元评价因子        8758
    F'</th><th>加权贡献</th></tr>                                    8759
    </thead>                                                        8760
    <tbody>                                                         8761
      {% for u in result.unit_results %}                            8762
      <tr>                                                          8763
        <td>{{ u.unit_type }}</td>                                  8764
        <td class="mono">{{ "%.2f%%"|format(u.ratio * 100)          8765
    }}</td>                                                         8766
        <td class="mono">{{ "%.4f"|format(u.unit_score)             8767
    }}</td>                                                         8768
        <td class="mono">{{ "%.4f"|format(u.weighted)               8769
    }}</td>                                                         8770
      </tr>                                                         8771
      {% endfor %}                                                  8772
    </tbody>                                                        8773
</table>                                                            8774
<div class="chart">{{ chart_units|safe }}</div>                     8775
<h2>四、一级指标权重分布</h2>                                        8776
<div class="chart">{{ chart_weight|safe }}</div>                    8777
<h2>五、得分明细</h2>                                                8778
{% for u in result.unit_results %}                                  8779
<h3>{{ u.unit_type }}（占比 {{ "%.2f%%"|format(u.ratio *            8780
    100) }}，单元得分 {{ "%.4f"|format(u.unit_score) }}）</h3>       8781
<table>                                                             8782
    <thead>                                                         8783
      <tr>                                                          8784
        <th>一级指标</th><th>二级指标</th><th>实测值</th>            8785
        <th>标准化值</th><th>权重</th><th>贡献</th>                  8786
      </tr>                                                         8787
    </thead>                                                        8788
    <tbody>                                                         8789
      {% for d in u.details %}                                      8790
      <tr>                                                          8791
        <td>{{ d.l1 }}</td>                                         8792
        <td class="left">{{ d.l2 }}</td>                            8793
        <td class="mono left">{{ "—" if d.raw is none else          8794
    d.raw }}</td>                                                   8795
        <td class="mono">{{ "—" if d.std_value is none else         8796
    "%.4f"|format(d.std_value) }}</td>                              8797
        <td class="mono">{{ "%.4f"|format(d.weight) }}</td>         8798
        <td class="mono">{{ "—" if d.contribution is none           8799
    else "%.4f"|format(d.contribution) }}</td>                      8800
```

```
        </tr>                                                                    8801
        {% endfor %}                                                             8802
      </tbody>                                                                   8803
    </table>                                                                     8804
    <p class="note">                                                             8805
      一级指标得分：                                                             8806
      {% for k, v in u.l1_scores.items() %}{{ k }} = {{                          8807
      "%.4f"|format(v) }}{% if not loop.last %};{% endif %}{%                    8808
      endfor %}                                                                  8809
    </p>                                                                         8810
    {% endfor %}                                                                 8811
    <h2>六、评价依据说明</h2>                                                    8812
    <ul class="note">                                                            8813
      <li>指标标准化按论文式（8） 执行，按指标性质分为越大越优、越小越优、     8814
        适中型与离散赋值四类。</li>                                              8815
      <li>单元评价因子按论文式（9） 由二级指标权重加权求得；                    8816
        整矿评价因子由各微地质单元按面积占比加权合成。</li>                     8817
      <li>适宜性等级按论文表 4 划分：F&lt;0.3 不适宜、0.3≤F≤0.7 基本适宜、F≥0.7 8818
      适宜。</li>                                                                8819
      <li>指标权重取自论文表 3，按岩质边坡、碎石边坡、平台三类微地质单元分别确定。</li>  8820
    </ul>                                                                        8821
    <p class="sig">                                                              8822
      评价人（签字）：_____　　日期：_____                                      8823
    </p>                                                                         8824
</body>                                                                          8825
</html>                                                                          8826
#  ==== app/templates/search.html ====                                           8827
{% extends "base.html" %}                                                        8828
{% from "_macros.html" import icon %}                                            8829
{% block title %}搜索 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}     8830
{% block content %}                                                              8831
<div class="card">                                                               8832
  <h2>搜索结果</h2>                                                             8833
  {% if not q %}                                                                 8834
  <p class="muted">请输入关键词。可检索项目名称与编号、图斑名称与行政区/矿种、  8835
    标准名称与文号。</p>                                                         8836
  {% else %}                                                                     8837
  <p class="muted">                                                              8838
    关键词「<strong>{{ q }}</strong>」共找到                                     8839
    <span class="num">{{ total }}</span> 条结果。                                8840
  </p>                                                                           8841
  {% if not total %}                                                             8842
  <p class="muted">没有匹配的内容。换个词试试，或检查是否有错别字。</p>         8843
  {% endif %}                                                                    8844
  {% endif %}                                                                    8845
</div>                                                                           8846
{% if projects %}                                                                8847
<div class="card">                                                               8848
  <h2>评价项目（{{ projects|length }}）</h2>                                     8849
  <table>                                                                        8850
```

```
      <thead>                                                                  8851
        <tr>                                                                   8852
          <th scope="col">项目名称</th>                                        8853
          <th scope="col">编号</th>                                            8854
          <th scope="col">状态</th>                                            8855
          <th scope="col">图斑数</th>                                          8856
          <th scope="col">操作</th>                                            8857
        </tr>                                                                  8858
      </thead>                                                                 8859
      <tbody>                                                                  8860
        {% for p in projects %}                                                8861
        <tr>                                                                   8862
          <td style="text-align:left;">{{ p.name }}</td>                       8863
          <td class="mono">{{ p.code or "—" }}</td>                            8864
          <td><span class="tag tag--muted">{{ p.status                         8865
      }}</span></td>                                                           8866
          <td class="num">{{ p.patches.count() }}</td>                         8867
          <td class="nowrap">                                                  8868
            <span class="row-actions">                                         8869
              <a class="link-btn" href="/project/{{ p.id                       8870
      }}/patches">矿山图斑</a>                                                 8871
              <a class="link-btn" href="/project/{{ p.id                       8872
      }}/edit">编辑</a>                                                        8873
            </span>                                                            8874
          </td>                                                                8875
        </tr>                                                                  8876
        {% endfor %}                                                           8877
      </tbody>                                                                 8878
    </table>                                                                   8879
</div>                                                                         8880
{% endif %}                                                                    8881
{% if patches %}                                                               8882
<div class="card">                                                             8883
    <h2>矿山图斑（{{ patches|length }}）</h2>                                  8884
    <table>                                                                    8885
      <thead>                                                                  8886
        <tr>                                                                   8887
          <th scope="col">所属项目</th>                                        8888
          <th scope="col">图斑名称</th>                                        8889
          <th scope="col">行政区</th>                                          8890
          <th scope="col">矿种</th>                                            8891
          <th scope="col">操作</th>                                            8892
        </tr>                                                                  8893
      </thead>                                                                 8894
      <tbody>                                                                  8895
        {% for m in patches %}                                                 8896
        <tr>                                                                   8897
          <td>{{ m.project.name if m.project else "—" }}</td>                  8898
          <td style="text-align:left;">{{ m.name }}</td>                       8899
          <td>{{ m.admin_region or "—" }}</td>                                 8900
```

```
            <td>{{ m.mineral_type or "—" }}</td>                    8901
            <td class="nowrap">                                      8902
              <span class="row-actions">                             8903
                <a class="link-btn" href="/constraint/{{ m.id        8904
      }}">限制性判定</a>                                             8905
                <a class="link-btn" href="/unit/{{ m.id              8906
      }}">微地质单元</a>                                             8907
                <a class="link-btn" href="/evaluate/{{ m.id          8908
      }}">执行评价</a>                                               8909
              </span>                                                8910
            </td>                                                    8911
          </tr>                                                      8912
          {% endfor %}                                               8913
        </tbody>                                                     8914
    </table>                                                         8915
</div>                                                               8916
{% endif %}                                                          8917
{% if standards %}                                                   8918
<div class="card">                                                   8919
    <h2>标准与政策文件（{{ standards|length }}）</h2>                  8920
    <table>                                                          8921
      <thead>                                                        8922
        <tr>                                                         8923
          <th scope="col">类别</th>                                  8924
          <th scope="col">省份</th>                                  8925
          <th scope="col">名称</th>                                  8926
          <th scope="col">文号</th>                                  8927
          <th scope="col">操作</th>                                  8928
        </tr>                                                        8929
      </thead>                                                       8930
      <tbody>                                                        8931
        {% for s in standards %}                                     8932
        <tr>                                                         8933
          <td>{{ s.category }}</td>                                  8934
          <td>{{ s.province }}</td>                                  8935
          <td style="text-align:left;">{{ s.name }}</td>             8936
          <td class="mono">{{ s.doc_number or "—" }}</td>            8937
          <td class="nowrap">                                        8938
            <span class="row-actions">                               8939
              <a class="link-btn" href="/standard/{{ s.id            8940
      }}/rules">规则</a>                                             8941
            </span>                                                  8942
          </td>                                                      8943
        </tr>                                                        8944
        {% endfor %}                                                 8945
      </tbody>                                                       8946
    </table>                                                         8947
</div>                                                               8948
{% endif %}                                                          8949
{% endblock %}                                                       8950
```

```
# ==== seed_demo.py ====
# -*- coding: utf-8 -*-
"""内置玉田县案例：把论文第 5 节案例数据写入数据库。
用于功能验证与操作演示。数据逐字取自论文表 5 与第 5 节。
运行：python seed_demo.py                    幂等，只补写缺失的演示数据，
                                                不会删除用户自建的项目与图斑
      python seed_demo.py --reset               连同标准库与指标定义一并重建
"""
import json
import sys
sys.stdout.reconfigure(encoding="utf-8")
from app import create_app
from app.models import (db, User, Project, MinePatch,
    ReferenceEcosystem,
                        MicroUnit, IndicatorData, Standard,
    StandardRule,
                        ConstraintCheck, EvaluationResult,
    ScoreDetail)
app = create_app()
# 内置演示案例的项目编号，用于区分「演示数据」与「用户自建数据」
DEMO_CODE = "DEMO-001"
# 论文表 1：11 项地方标准与政策文件
STANDARDS = [
    dict(category="地方标准", province="辽宁",
    name="矿山废弃地植被自然恢复技术规程",
        doc_number="DB21/T 3702—2023",
    publish_date="2023-03-30",
        implement_date="2023-04-30",
        summary="首个正式发布的矿山自然恢复地方标准；提出必要条件和充分条件",
        rules=[("植被覆盖度", ">=", "50", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None),
               ("与参照系差距", "<", "20", "充分条件", "%")]),
    dict(category="地方标准", province="湖北",
    name="废弃露天矿山生态环境自然恢复标准",
        doc_number="DB42/T 2421—2025",
    publish_date="2025-08-20",
        implement_date="2025-10-20",
        summary="结合矿山生态需求进行自然恢复适宜性分级判断",
        rules=[("自然恢复适宜性分级", "==", "是", "必要条件", None)]),
    dict(category="地方标准", province="内蒙古",
    name="历史遗留废弃矿山自然恢复判定规范",
        doc_number="DB15/T 4280—2026",
    publish_date="2026-01-30",
        implement_date="2026-03-01",
        summary="调查内容包含参照生态系统调查",
        rules=[("地质灾害发育程度", "<=", "弱发育", "必要条件", None),
               ("地形地貌与周边协调", "==", "是", "必要条件", None),
               ("有利于植被恢复的表层土壤", "==", "是", "必要条件", None),
               ("已形成稳定植被覆盖", "==", "是", "必要条件", None)]),
```

```
    dict(category="地方标准", province="江西",                                    9001
    name="废弃矿山自然恢复技术规范",                                              9002
        doc_number="DB36/T 2221—2026",                                          9003
    publish_date="2026-02-02",                                                  9004
        implement_date="2026-08-01",                                            9005
        summary="对参照生态系统进行了要求；提出自然恢复适宜性指数",                  9006
        rules=[("参照生态系统调查", "==", "是", "必要条件", None),                 9007
               ("自然恢复适宜性指数", ">=", "阈值", "充分条件", None)]),          9008
    dict(category="地方标准", province="宁夏", name="矿山自然恢复认定规程",        9009
        doc_number=None, publish_date="2026-06-18",                             9010
    implement_date=None,                                                        9011
        summary="分区差异化：三个一级生态修复分区及 22 个二级分区",                9012
        rules=[("无地质灾害风险", "==", "是", "必要条件", None),                   9013
               ("植被覆盖度趋势", "==", "稳定或正增长", "必要条件", None),        9014
               ("与参照系差距", "<", "20", "充分条件", "%")]),                    9015
    dict(category="政策文件", province="安徽",                                    9016
        name="安徽省自然资源厅关于规范废弃露天矿山采取自然恢复方式进行治理的通知",  9017
        doc_number="皖自然资修函[2020]34号",                                     9018
    publish_date="2020-03-03", implement_date=None,                             9019
        summary="自然恢复条件：责任主体为政府、无地质灾害、不在三线三边、水土条件良好",9020
        rules=[("责任主体", "==", "政府", "必要条件", None),                       9021
               ("无地质灾害", "==", "是", "必要条件", None),                      9022
               ("无水土污染", "==", "是", "必要条件", None)]),                    9023
    dict(category="政策文件", province="山东",                                    9024
        name="山东省自然资源厅关于进一步规范历史遗留废弃露天开采矿山采取自然恢复方式进行治理的通知",9025
        doc_number="鲁自然资字[2022]104号",                                      9026
    publish_date="2022-08-02", implement_date=None,                             9027
        summary="条件与安徽省相同，另增加自然恢复认定标准",                        9028
        rules=[("无地质灾害", "==", "是", "必要条件", None),                       9029
               ("无水土污染", "==", "是", "必要条件", None)]),                    9030
    dict(category="政策文件", province="湖南",                                    9031
        name="关于印发《湖南省历史遗留矿山自然恢复指导意见（试行）》的通知",      9032
        doc_number="湘自资办发[2023]58号",                                       9033
    publish_date="2023-07-20", implement_date=None,                             9034
        summary="自然恢复认定标准：无地质灾害或无威胁对象、无水土污染、设备拆除、符合规划、植被覆盖度9035
    >75%、连续3年正向增长",                                                      9036
        rules=[("植被覆盖度", ">", "75", "必要条件", "%"),                        9037
               ("植被覆盖率趋势", "==", "连续3年正向增长", "必要条件", None),9038
               ("无地质灾害", "==", "是", "必要条件", None),                      9039
               ("符合国土空间规划", "==", "是", "必要条件", None)]),              9040
    dict(category="政策文件", province="新疆",                                    9041
        name="自治区历史遗留废弃矿山自然恢复方式销号图斑工作指南（试行）",          9042
        doc_number="新自然资源办函〔2024〕219号",                                9043
    publish_date="2024-12-20", implement_date=None,                             9044
        summary="荒漠区沙地/裸土地/裸岩石砾地满足无地质灾害、地形协调、无水土污染即可判定",9045
        rules=[("植被覆盖度", ">=", "周边30%", "必要条件", None),                 9046
               ("无地质灾害", "==", "是", "必要条件", None),                      9047
               ("地形平缓地貌稳定", "==", "是", "必要条件", None)]),              9048
    dict(category="政策文件", province="广西",                                    9049
        name="广西壮族自治区历史遗留矿山自然恢复技术指南",                        9050
```

```
            doc_number="桂自然资办〔2024〕250号",
        publish_date="2024-12-30", implement_date=None,
            summary="除高陡边坡和裸露岩体外，水土水热条件良好、植被自然恢复趋势明显",
            rules=[("无地质灾害", "==", "是", "必要条件", None),
                   ("植被自然恢复趋势", "==", "明显", "必要条件", None),
                   ("生产设备拆除完毕", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="西藏",
            name="西藏自治区历史遗留废弃矿山生态自然恢复认定指导意见（试行）",
            doc_number="藏自然资发〔2025〕19号",
        publish_date="2025-09-24", implement_date=None,
            summary="含修复成本高、生态修复效益不显著等适用条件",
            rules=[("责任主体", "==", "政府", "必要条件", None),
                   ("地形地貌完整与周边协调", "==", "是", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None)]),
]
CASE = {
    "name": "玉田县典型历史遗留废弃矿山案例",
    "patch_name": "玉田县孤树镇白云岩采石场",
    "area_ha": 5.6,
    "mineral_type": "白云岩",
    "units": [
        {
            "unit_type": "岩质边坡", "ratio": 0.1386,
            "data": {"coverage": 68.88, "shannon": 0.00,
                     "slope": 63.33, "aspect": "阴阳坡"},
        },
        {
            "unit_type": "碎石边坡", "ratio": 0.6295,
            "data": {"soil_ratio": 0.34, "thickness": 10,
    "ph": 9.24,
                     "organic_matter": 4.89,
                     "heavy_metals": {"Hg": 0.05, "As":
    5.17, "Ni": 12.60,
                                      "Cu": 11.70, "Cd":
    0.04, "Pb": 6.50},
                     "coverage": 87.92, "shannon": 0.60,
                     "slope": 28.75, "aspect": "阳坡"},
        },
        {
            "unit_type": "平台", "ratio": 0.2319,
            "data": {"soil_ratio": 0.56, "thickness": 30,
    "ph": 8.72,
                     "organic_matter": 10.92,
                     "heavy_metals": {"Hg": 0.02, "As":
    3.13, "Ni": 12.33,
                                      "Cu": 11.14, "Cd":
    0.09, "Pb": 9.98},
                     "coverage": 58.42, "shannon": 0.72},
        },
    ],
```

```
}
def clean_case(reset=False):
    """清理数据。
    默认只清理内置演示案例（code = DEMO-001）自身的数据，**不触碰
    用户自己建立的项目与图斑**。加 --reset 时才额外清空标准库与指标定义，
    以便把演示环境恢复到初始状态。
    注意：不要在此处执行 Project.query.delete()，否则用户新建的项目
    会在每次启动时被删除。
    """
    from app.models import EvaluationResult, ScoreDetail,
    ConstraintCheck
    demo_projects = Project.query.filter_by(code=DEMO_CODE).
    all()
    demo_ids = [p.id for p in demo_projects]
    patch_ids = []
    if demo_ids:
        patch_ids = [p.id for p in MinePatch.query.filter(
            MinePatch.project_id.in_(demo_ids)).all()]
    if patch_ids:
        unit_ids = [u.id for u in MicroUnit.query.filter(
            MicroUnit.patch_id.in_(patch_ids)).all()]
        if unit_ids:
            IndicatorData.query.filter(
                IndicatorData.unit_id.in_(unit_ids)).delete(
                synchronize_session=False)
        result_ids = [r.id for r in EvaluationResult.query.f
    ilter(
            EvaluationResult.patch_id.in_(patch_ids)).all()]
        if result_ids:
            ScoreDetail.query.filter(
                ScoreDetail.result_id.in_(result_ids)).delet
    e(
                synchronize_session=False)
        EvaluationResult.query.filter(
            EvaluationResult.patch_id.in_(patch_ids)).delete
    (
            synchronize_session=False)
        ConstraintCheck.query.filter(
            ConstraintCheck.patch_id.in_(patch_ids)).delete(
            synchronize_session=False)
        ReferenceEcosystem.query.filter(
            ReferenceEcosystem.patch_id.in_(patch_ids)).dele
    te(
            synchronize_session=False)
        MicroUnit.query.filter(MicroUnit.patch_id.in_(patch_
    ids)).delete(
            synchronize_session=False)
        MinePatch.query.filter(MinePatch.id.in_(patch_ids)).
    delete(
            synchronize_session=False)
```

```
    if demo_ids:                                                    9151
        Project.query.filter(Project.id.in_(demo_ids)).delet        9152
    e(                                                              9153
            synchronize_session=False)                              9154
    if reset:                                                       9155
        StandardRule.query.delete()                                 9156
        Standard.query.delete()                                     9157
    #  兜底：清理指向已不存在评价记录的孤儿得分明细，                9158
    #  避免长期运行后 score_details 表堆积无效数据。                 9159
    _purge_orphan_details()                                         9160
    db.session.commit()                                             9161
    return len(demo_ids)                                            9162
def demo_patch():                                                   9163
    """取内置演示案例的图斑。                                       9164
    库中可能存在用户自建项目或验证算例（编号 VERIFY-），            9165
    故不能简单用 MinePatch.query.first()，否则取到的对象不确定。    9166
    没有演示案例时才退回第一个图斑。                                9167
    """                                                             9168
    proj = Project.query.filter_by(code=DEMO_CODE).first()          9169
    if proj is not None:                                            9170
        m = MinePatch.query.filter_by(project_id=proj.id).fi        9171
    rst()                                                           9172
        if m is not None:                                           9173
            return m                                                9174
    return MinePatch.query.first()                                  9175
def _purge_orphan_details():                                        9176
    """清理指向已不存在评价记录的孤儿得分明细。                     9177
    长期运行后 score_details 表可能堆积无效明细，启动时顺带清一次。 9178
    """                                                             9179
    live_ids = [r.id for r in EvaluationResult.query.all()]         9180
    if live_ids:                                                    9181
        ScoreDetail.query.filter(~ScoreDetail.result_id.in_(        9182
    live_ids)).delete(                                              9183
            synchronize_session=False)                              9184
    else:                                                           9185
        ScoreDetail.query.delete()                                  9186
    db.session.commit()                                             9187
def seed(reset=False):                                              9188
    with app.app_context():                                         9189
        db.create_all()          #  全新环境下先建表，保证本脚本可独立运行  9190
        _purge_orphan_details()                                     9191
        admin = User.query.filter_by(username="admin").first        9192
    ()                                                              9193
        if not admin:                                               9194
            from werkzeug.security import                           9195
    generate_password_hash                                          9196
            admin = User(username="admin",                          9197
                         password_hash=generate_password_has        9198
    h("admin123"),                                                  9199
                         role="admin")                              9200
```

```
            db.session.add(admin)                                                    9201
            db.session.commit()                                                      9202
        #  幂等：演示案例已存在就不重复写入，更不清空用户自己的项目                    9203
        existing = Project.query.filter_by(code=DEMO_CODE).f                         9204
irst()                                                                               9205
        if existing and not reset:                                                   9206
            print("演示案例已存在，跳过写入（用户自建项目不受影响）")                  9207
            _report()                                                                9208
            _seed_reference_data(reset=False)                                        9209
            return                                                                   9210
        clean_case(reset=reset)                                                      9211
        project = Project(name=CASE["name"], code=DEMO_CODE,                         9212
                          description="论文第 5 节典型案例",                          9213
created_by=admin.id)                                                                 9214
        db.session.add(project)                                                      9215
        db.session.flush()                                                           9216
        patch = MinePatch(project_id=project.id,                                     9217
name=CASE["patch_name"],                                                             9218
                          admin_region="河北省唐山市玉田县",                          9219
                          area_ha=CASE["area_ha"],                                   9220
                          mineral_type=CASE["mineral_type"],                         9221
                          mining_method="露天开采",                                  9222
stop_years=None,                                                                     9223
                          responsible_entity_missing=True)                           9224
        db.session.add(patch)                                                        9225
        db.session.flush()                                                           9226
        ref = ReferenceEcosystem(patch_id=patch.id,                                  9227
                                 coverage_benchmark=75.0,                            9228
shannon_benchmark=0.7)                                                               9229
        db.session.add(ref)                                                          9230
        for u in CASE["units"]:                                                      9231
            unit = MicroUnit(patch_id=patch.id,                                      9232
unit_type=u["unit_type"],                                                            9233
                             ratio=u["ratio"], area=None)                            9234
            db.session.add(unit)                                                     9235
            db.session.flush()                                                       9236
            d = u["data"]                                                            9237
            ind = IndicatorData(                                                     9238
                unit_id=unit.id,                                                     9239
                soil_ratio=d.get("soil_ratio"),                                      9240
                thickness=d.get("thickness"),                                        9241
                ph=d.get("ph"),                                                      9242
                organic_matter=d.get("organic_matter"),                              9243
                heavy_metals=json.dumps(d["heavy_metals"],                           9244
ensure_ascii=False)                                                                  9245
                                if d.get("heavy_metals") else                        9246
None,                                                                                9247
                coverage=d.get("coverage"),                                          9248
                shannon=d.get("shannon"),                                            9249
                slope=d.get("slope"),                                                9250
```

```
                    aspect=d.get("aspect"),                                  9251
                )                                                            9252
                db.session.add(ind)                                          9253
            #  限制性条件判定记录：演示案例按论文第  5  节描述属「无地质灾害隐患点」、  9254
            #  具备自然恢复条件，故五项全部通过。缺此记录会使后续评价被流程门禁拦住。  9255
            db.session.add(ConstraintCheck(                                  9256
                patch_id=patch.id,                                           9257
                policy_pass=True, location_pass=True,                        9258
    pollution_pass=True,                                                     9259
                geo_pass=True, farmland_pass=True,                           9260
                passed=True, reason="全部通过"))                              9261
            db.session.commit()                                              9262
            print("已写入案例：", CASE["name"])                               9263
            print("   图斑：", CASE["patch_name"])                           9264
            print("   微地质单元：岩质边坡  １３．８６%  /  碎石边坡  ６２．９５%  /  平台  9265
    ２３．１９%")                                                             9266
            print("   管理员账号：admin  /  admin１２３")                     9267
            _seed_reference_data(reset=reset)                                9268
            _report()                                                        9269
def _seed_reference_data(reset=False):                                       9270
    """写入标准库与指标定义。                                                9271
    参考资料只在库为空（或显式  --reset）时写入，避免覆盖用户自行维护          9272
    的标准条目与指标定义。                                                   9273
    """                                                                      9274
    if reset or Standard.query.count() == 0:                                 9275
        seed_standards()                                                     9276
    from app.services.indicator_service import                               9277
    seed_indicators                                                          9278
    n = seed_indicators()                                                    9279
    print("指标定义：%d 项" % n)                                             9280
def _report():                                                               9281
    total_p = Project.query.count()                                          9282
    total_patch = MinePatch.query.count()                                    9283
    print()                                                                  9284
    print("当前库内：项目 %d 个、图斑 %d 个" % (total_p, total_patch))        9285
    if total_p > total_patch:                                                9286
        print("   提示：项目是归档单位，评价针对的是图斑。")                  9287
        print("         新项目需先建立图斑，才会出现在后续各模块中。")        9288
def seed_standards():                                                        9289
    """写入论文表 1 的 11 项标准与政策文件及其结构化规则。"""                 9290
    StandardRule.query.delete()                                              9291
    Standard.query.delete()                                                  9292
    db.session.commit()                                                      9293
    for s in STANDARDS:                                                      9294
        std = Standard(category=s["category"],                               9295
    province=s["province"],                                                  9296
                       name=s["name"],                                       9297
    doc_number=s["doc_number"],                                              9298
                       publish_date=s["publish_date"],                       9299
                       implement_date=s["implement_date"],                   9300
```

```
                                summary=s["summary"])                    9301
            db.session.add(std)                                          9302
            db.session.flush()                                           9303
            for item, op, thr, ctype, note in s["rules"]:                9304
                db.session.add(StandardRule(standard_id=std.id,          9305
        item=item,                                                       9306
                                                    operator=op,         9307
        threshold=thr,                                                   9308
                                                                         9309
        condition_type=ctype, note=note))                                9310
        db.session.commit()                                              9311
        print("已写入标准/政策文件：%d 项，规则 %d 条"                    9312
              % (len(STANDARDS), sum(len(s["rules"]) for s in            9313
        STANDARDS)))                                                     9314
if __name__ == "__main__":                                               9315
        #  --reset：把演示环境恢复到初始状态（清空标准库与指标定义后重建）。    9316
        #  无论是否 --reset，都不会删除用户自建的项目与图斑。                9317
        seed(reset="--reset" in sys.argv[1:])                            9318
#  ==== 测试/test_ahp.py ====                                            9319
#  -*- coding: utf-8 -*-                                                 9320
"""地基自检：验证 AHP 引擎与应用可启动。"""                               9321
import sys                                                               9322
sys.stdout.reconfigure(encoding="utf-8")                                 9323
import numpy as np                                                       9324
import os                                                                9325
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，               9326
#  使 `from app import ...` 与 app/app.db 的定位不受影响。                9327
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil             9328
        e__)))                                                           9329
if ROOT not in sys.path:                                                 9330
        sys.path.insert(0, ROOT)                                         9331
from app.services import ahp_service as ahp                              9332
LINE = "=" * 66                                                          9333
#  ------------------------------------------------------------------    9334
        ------------------                                               9335
#  1） 完全一致矩阵 → 权重应精确还原，CR=0                                9336
#  ------------------------------------------------------------------    9337
        ------------------                                               9338
print(LINE)                                                              9339
print("测试 1  完全一致判断矩阵的权重还原与一致性")                       9340
print(LINE)                                                              9341
#  由权重构造完全一致矩阵：a_ij = w_i / w_j                               9342
w_true = np.array([0.44, 0.31, 0.25])                                    9343
A = np.outer(w_true, 1.0 / w_true)                                       9344
r = ahp.weights_and_consistency(A)                                       9345
print("  判断矩阵 A：")                                                 9346
for row in A:                                                            9347
        print("      " + "  ".join(f"{v:8.4f}" for v in row))            9348
print()                                                                  9349
print("  计算权重 :", "  ".join(f"{v:.4f}" for v in                     9350
```

```
        r["weights"]))                                                    9351
print("   真实权重 :", "  ".join(f"{v:.4f}" for v in w_true))            9352
print("   λmax = %.6f    CI = %.3e    CR = %.3e    一致: %s"              9353
      % (r["lambda_max"], r["ci"], r["cr"],                               9354
        r["consistent"]))                                                 9355
assert np.allclose(r["weights"], w_true, atol=1e-6),                      9356
    "权重未精确还原"                                                      9357
assert r["consistent"], "完全一致矩阵应通过检验"                          9358
# ------------------------------------------------------------------------9359
    ------------------                                                    9360
# 2) 上三角输入 → 完整矩阵 → 一致性定位                                   9361
# ------------------------------------------------------------------------9362
    ------------------                                                    9363
print()                                                                   9364
print(LINE)                                                               9365
print("测试 2  上三角录入与最不一致项定位")                               9366
print(LINE)                                                               9367
# 一个不太一致的 3 阶矩阵：上三角 [2, 5, 3]（a12=2, a13=5, a23=3）        9368
M = ahp.matrix_from_upper([2, 5, 3])                                      9369
r2 = ahp.weights_and_consistency(M)                                       9370
bad = ahp.find_most_inconsistent(M)                                       9371
print("   由 [a12=2, a13=5, a23=3] 构造矩阵：")                          9372
for row in M:                                                             9373
    print("     " + "  ".join(f"{v:8.4f}" for v in row))                  9374
print("   权重:", "  ".join(f"{v:.4f}" for v in r2["weights"]),           9375
      "  CR = %.3f  一致: %s" % (r2["cr"], r2["consistent"]))             9376
print("   最不一致比较项: 第(%d,%d) 项，当前 %.3f，理想 %.3f"             9377
      % (bad["i"] + 1, bad["j"] + 1, bad["value"],                        9378
    bad["expected"]))                                                     9379
# ------------------------------------------------------------------------9380
    ------------------                                                    9381
# 3) 多专家聚合（几何平均）                                               9382
# ------------------------------------------------------------------------9383
    ------------------                                                    9384
print()                                                                   9385
print(LINE)                                                               9386
print("测试 3  多专家判断矩阵聚合（几何平均）")                           9387
print(LINE)                                                               9388
m1 = ahp.matrix_from_upper([2, 4, 3])                                     9389
m2 = ahp.matrix_from_upper([3, 5, 2])                                     9390
agg = ahp.aggregate_matrices([m1, m2])                                    9391
print("   专家1 上三角 [2,4,3]，专家2 上三角 [3,5,2]")                  9392
print("   聚合后上三角: a12=%.4f a13=%.4f a23=%.4f"                      9393
      % (agg[0, 1], agg[0, 2], agg[1, 2]))                                9394
assert abs(agg[0, 1] - np.sqrt(2 * 3)) < 1e-9, "几何平均错误"             9395
# ------------------------------------------------------------------------9396
    ------------------                                                    9397
# 4) 论文表 3 权重可还原性                                                9398
# ------------------------------------------------------------------------9399
    ------------------                                                    9400
```

```
print()
print(LINE)
print("测试 4  论文表 3 碎石边坡一级权重 (0.44, 0.31, 0.25)")
print(LINE)
j = ahp.judge_from_weights([0.44, 0.31, 0.25])
print("  λmax = %.6f    CR = %.3e    一致: %s" %
      (j["lambda_max"], j["cr"], j["consistent"]))
assert j["consistent"], "论文权重应能作为完全一致矩阵主特征向量"
# ------------------------------------------------------------------
      ------------------
# 5) 应用可启动 + 建表
# ------------------------------------------------------------------
      ------------------
print()
print(LINE)
print("测试 5  Flask 应用启动与建表")
print(LINE)
from app import create_app
from app.models import db
app = create_app()
with app.app_context():
      db.create_all()
      tables = sorted(db.metadata.tables.keys())
print("  已创建 %d 张表:" % len(tables))
for t in tables:
      print("    -", t)
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_evaluate.py ====
# -*- coding: utf-8 -*-
"""评价引擎自检:用论文玉田县案例跑通完整计算链。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
# 本脚本位于 测试/ 子目录,先把仓库根目录加入模块搜索路径,
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
      e__)))
if ROOT not in sys.path:
      sys.path.insert(0, ROOT)
from app.services import evaluation_service as ev
from app.services import standardization as std
LINE = "=" * 70
# ------------------------------------------------------------------
      ------------------
# 论文第 5 节案例数据(表 5,逐字抄录)
# ------------------------------------------------------------------
      ------------------
```

```
#  三类单元面积占比：岩质边坡  １３．８６%、碎石边坡  ６２．９５%、平台  ２３．１９%
#  注：岩质边坡无土壤指标、平台无地形指标，对应"不适用"。
case = [
    {
        "unit_type": "岩质边坡",
        "ratio": 0.1386,
        "data": {
            "coverage": 68.88, "shannon": 0.00,
            "slope": 63.33, "aspect": "阴阳坡",
        },
    },
    {
        "unit_type": "碎石边坡",
        "ratio": 0.6295,
        "data": {
            "soil_ratio": 0.34, "thickness": 10, "ph":
    9.24, "organic_matter": 4.89,
            "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
    12.60,
                             "Cu": 11.70, "Cd": 0.04, "Pb":
    6.50},
            "coverage": 87.92, "shannon": 0.60,
            "slope": 28.75, "aspect": "阳坡",
        },
    },
    {
        "unit_type": "平台",
        "ratio": 0.2319,
        "data": {
            "soil_ratio": 0.56, "thickness": 30, "ph":
    8.72, "organic_matter": 10.92,
            "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni":
    12.33,
                             "Cu": 11.14, "Cd": 0.09, "Pb":
    9.98},
            "coverage": 58.42, "shannon": 0.72,
        },
    },
]
#  ----------------------------------------------------------------------
    ------------------
#  逐单元输出标准化与得分
#  ----------------------------------------------------------------------
    ------------------
print(LINE)
print("玉田县案例 · 评价引擎完整计算")
print(LINE)
for u in case:
    r = ev.score_unit(u["unit_type"], u["data"])
    print()
```

```
        print("   【%s】   （面积占比 %.2f%%）" % (u["unit_type"],
        u["ratio"] * 100))
        print("   " + "-" * 60)
        for l1, score in r["l1_scores"].items():
            print("      %s 得分 = %.4f" % (l1, score))
        print("      单元评价因子 F' = %.4f" % r["unit_score"])
print()
print(LINE)
print("面积加权合成（式(9)）")
print(LINE)
res = ev.score_mine(case)
total = res["final_score"]
print()
for ur in res["unit_results"]:
        print("   %s  %.2f%% × %.4f = %.4f"
              % (ur["unit_type"], ur["ratio"] * 100,
        ur["unit_score"], ur["weighted"]))
print("   " + "-" * 60)
print("   最终评价因子 F = %.4f" % total)
print()
print("   适宜性等级：%s" % res["grade"])
print("   判定依据   ：%s" % res["grade_info"]["meaning"])
print("   推荐措施   ：%s" % res["grade_info"]["measure"])
print()
print(LINE)
print("结论")
print(LINE)
print("   - 评价引擎按论文式(8)(9) 与表 3 权重完整跑通")
print("   - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")
print("   - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"
      % (total, res["grade"]))
print("   - 论文表 5 所载最终得分为 0.76；差异集中在坡度、坡向的标准化口径，")
print("     该口径在 standardization.std_slope / aspect_score
     中集中定义、可配置")
print(LINE)
#  ==== 测试/test_web_ahp.py ====
#  -*- coding: utf-8 -*-
"""AHP 网页界面自检：走通选择→录入→计算 全流程。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
#  测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）
```

```
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
        db.session.add(User(username="admin",
                            password_hash=generate_password_
    hash("admin123"),
                            role="admin"))
        db.session.commit()
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1　首页与 AHP 入口")
print(LINE)
r = client.get("/")
print("　首页状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
r = client.get("/ahp/")
print("　AHP 入口状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
assert "微地质单元类型" in r.get_data(as_text=True)
print()
print(LINE)
print("测试 2　选择碎石边坡一级指标 → 跳转矩阵页")
print(LINE)
r = client.post("/ahp/", data={
    "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("　跳转后页面含「判断矩阵录入」：", "判断矩阵录入" in body)
print("　含三个一级指标（土壤/植被/地形）：", all(n in body for n in
    ("土壤指标", "植被指标", "地形指标")))
assert "判断矩阵录入" in body
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))
print()
print(LINE)
print("测试 3　提交论文表 3 一致矩阵 → 还原权重")
print(LINE)
# 碎石边坡一级权重 （0.44, 0.31, 0.25） 对应的完全一致矩阵上三角：
# a12 = 0.44/0.31 = 1.4194, a13 = 0.44/0.25 = 1.76, a23 =
```

```
    0.31/0.25 = 1.24
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  结果页含「计算结果」:", "计算结果" in body)
print("  含权重 0.44:", "0.44" in body, " 含 0.31:", "0.31" in
    body, " 含 0.25:", "0.25" in body)
print("  含一致性检验通过:", "通过" in body)
assert "计算结果" in body
assert "0.44" in body and "0.31" in body and "0.25" in body
print()
print(LINE)
print("测试 4  不一致矩阵应提示不通过并定位")
print(LINE)
# 明显不一致：a12=9, a13=1, a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  含「不通过」:", "不通过" in body)
print("  含「建议优先调整」:", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
print()
print(LINE)
print("网页界面全部测试通过 √")
print(LINE)
# ==== 测试/test_web_modules.py ====
# -*- coding: utf-8 -*-
"""模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
```

```
client = app.test_client()
print(LINE)
print("测试 1  未登录应被重定向到登录页")
print(LINE)
r = client.get("/project/")
print("  /project/ 状态码:", r.status_code, " 重定向到:",
    r.headers.get("Location"))
assert r.status_code == 302 and "login" in
    r.headers.get("Location", "")
print()
print(LINE)
print("测试 2  登录")
print(LINE)
r = client.post("/login", data={"username": "admin",
    "password": "admin123"},
                follow_redirects=True)
body = r.get_data(as_text=True)
# 工作台:页首横幅 + 评价流程分区 + 数据概览
print("  登录后含「评价流程」:", "评价流程" in body)
print("  含用户名 admin:", "admin" in body)
print("  含页首横幅:", 'class="hero"' in body)
print("  含数据概览:", 'class="stat-card"' in body)
assert "评价流程" in body and 'class="hero"' in body
print()
print(LINE)
print("测试 3  项目与矿山档案")
print(LINE)
r = client.get("/project/")
body = r.get_data(as_text=True)
print("  项目列表状态码:", r.status_code)
print("  含案例项目:", "玉田县典型历史遗留废弃矿山案例" in body)
assert "玉田县典型历史遗留废弃矿山案例" in body
# 新建项目
client.post("/project/add", data={"name": "测试项目A", "code":
    "T-001"})
r = client.get("/project/")
print("  新建项目后含「测试项目A」:", "测试项目A" in r.get_data(as_text=True
    ))
assert "测试项目A" in r.get_data(as_text=True)
# 清理测试数据,避免污染演示库
with app.app_context():
    from app.models import db, Project
    for p in Project.query.filter_by(name="测试项目A").all():
        db.session.delete(p)
    db.session.commit()
print("  已清理测试项目")
print()
print(LINE)
print("测试 4  微地质单元(面积占比自动计算)")
print(LINE)
```

```
from app.models import MinePatch
with app.app_context():
    patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").first()
    pid = patch.id
r = client.get("/unit/%d" % pid)
body = r.get_data(as_text=True)
print("  单元列表含三类单元：", all(t in body for t in ("岩质边坡", "碎石边坡", "平台")))
print("  含占比合计：", "占比合计" in body)
assert "岩质边坡" in body and "占比合计" in body
print()
print(LINE)
print("测试 5  标准知识库与省际口径比对")
print(LINE)
r = client.get("/standard/")
body = r.get_data(as_text=True)
print("  含辽宁/湖南/新疆标准：", all(p in body for p in ("辽宁", "湖南", "新疆")))
assert "辽宁" in body
r = client.get("/standard/compare")
body = r.get_data(as_text=True)
print("  口径比对页含「植被覆盖度」分组：", "植被覆盖度" in body)
print("  含多省份并列：", "湖南" in body and "新疆" in body)
assert "植被覆盖度" in body
print()
print(LINE)
print("测试 6  限制性指标判定（一票否决）")
print(LINE)
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  含五项条件：", all(x in body for x in ("政策约束", "区位约束", "土壤污染", "地质灾害", "耕地属性")))
assert all(x in body for x in ("政策约束", "区位约束", "土壤污染", "地质灾害", "耕地属性"))
# 全部通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  全通过后显示「通过，可进入定量评价」：", "可进入定量评价" in body)
assert "可进入定量评价" in body
# 一项不通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "0",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
```

```
body = r.get_data(as_text=True)
print("  一项不通过后显示「不通过」:", "不通过" in body)
print("  含失败项「区位约束」:", "区位约束" in body)
assert "不通过" in body
# 恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
print("  已恢复为全部通过:", "可进入定量评价" in r.get_data(as_text=True))
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_web_evaluate.py ====
# -*- coding: utf-8 -*-
"""端到端自检：录数据 → 执行评价 → 查看结果。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  指标数据列表")
print(LINE)
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("  状态码:", r.status_code)
print("  含案例图斑:", "玉田县孤树镇白云岩采石场" in body)
print("  含三个单元:", all(n in body for n in ("岩质边坡", "碎石边坡",
```

```
    "平台"))) 9801
assert r.status_code == 200 9802
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body 9803
print() 9804
print(LINE) 9805
print("测试 2  指标数据录入页（按单元类型显示字段）") 9806
print(LINE) 9807
from app.models import MicroUnit, MinePatch 9808
with app.app_context(): 9809
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡").9810
    first() 9811
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡"9812
    ).first() 9813
r = client.get("/indicator/%d" % unit_rock.id) 9814
b_rock = r.get_data(as_text=True) 9815
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）") 9816
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock) 9817
assert "土石比" not in b_rock and "植被覆盖度" in b_rock 9818
r = client.get("/indicator/%d" % unit_gravel.id) 9819
b_gravel = r.get_data(as_text=True) 9820
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel) 9821
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in 9822
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))) 9823
assert "土石比" in b_gravel and all(m in b_gravel for m in 9824
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")) 9825
print() 9826
print(LINE) 9827
print("测试 3  执行评价") 9828
print(LINE) 9829
with app.app_context(): 9830
    import seed_demo 9831
    patch = seed_demo.demo_patch() 9832
r = client.post("/evaluate/%d" % patch.id, data={}, 9833
    follow_redirects=True) 9834
body = r.get_data(as_text=True) 9835
print("  跳转至结果页：", "适宜性评价结果" in body) 9836
print("  含最终评价因子 F：", "最终评价因子 F" in body) 9837
print("  含等级「适宜」：", "适宜" in body) 9838
print("  含推荐措施「自然恢复」：", "自然恢复" in body) 9839
assert "最终评价因子 F" in body and "自然恢复" in body 9840
print() 9841
print(LINE) 9842
print("测试 4  结果持久化") 9843
print(LINE) 9844
from app.models import EvaluationResult 9845
with app.app_context(): 9846
    res = EvaluationResult.query.order_by(EvaluationResult.i9847
    d.desc()).first() 9848
    print("  最终得分 F =", res.final_score) 9849
    print("  等级 =", res.grade) 9850
```

```
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("端到端全部测试通过 √")
print(LINE)
# ==== 测试/test_web_report.py ====
# -*- coding: utf-8 -*-
"""报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
from app.models import MinePatch, EvaluationResult
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
    pid = patch.id
# 报告取自已保存的评价记录，因此先按网页路径真实执行一次评价
from app.views.evaluate import load_patch_input,
    save_evaluation
with app.app_context():
    units = load_patch_input(pid)
    rec = save_evaluation(pid, units)
    rid = rec.id
    print("前置：已执行评价并保存记录 id=%d，F=%.4f（%s）"
          % (rid, rec.final_score, rec.grade))
    from app.models import ScoreDetail
    n_detail = ScoreDetail.query.filter_by(result_id=rid).co
    unt()
```

```
        print("          逐层得分明细已落库：%d 条" % n_detail)
print()
print(LINE)
print("测试 1  报告列表")
print(LINE)
r = client.get("/report/")
print("  状态码：", r.status_code)
body = r.get_data(as_text=True)
print("  含「所属项目」列：", "所属项目" in body)
assert r.status_code == 200
print()
print(LINE)
print("测试 2  得分明细页")
print(LINE)
r = client.get("/report/%d" % pid)
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含「得分明细」：", "得分明细" in body)
print("  含「最终评价因子 F」：", "最终评价因子 F" in body)
print("  含二级指标明细（植被覆盖度/坡度/坡向）：", all(n in body for n in
      ("植被覆盖度", "坡度", "坡向")))
print("  含权重/标准化值列：", "标准化值" in body and "贡献" in body)
assert r.status_code == 200
assert "植被覆盖度" in body and "标准化值" in body
print()
print(LINE)
print("测试 3  Word 报告导出")
print(LINE)
r = client.get("/report/%d/word" % pid)
print("  状态码：", r.status_code)
print("  Content-Type:", r.headers.get("Content-Type",
      "")[:60])
data = r.data
print("  文件大小：", len(data), "字节")
print("  是否为有效 docx（PK 头）：", data[:2] == b"PK")
assert r.status_code == 200 and data[:2] == b"PK"
# 用 python-docx 打开验证内容
import docx
d = docx.Document(io.BytesIO(data))
text = "\n".join(p.text for p in d.paragraphs)
print("  含报告标题：", "自然恢复适宜性评价报告" in text)
print("  含评价结论章节：", "评价结论" in text)
print("  含得分明细章节：", "得分明细" in text)
print("  段落数：", len(d.paragraphs), " 表格数：", len(d.tables))
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3
print()
print(LINE)
print("报告输出全部测试通过 √")
print(LINE)
# ==== 测试/test_ui_fixes.py ====
```

```
# -*- coding: utf-8 -*-
"""界面修复自检：「记住我」与各模块删除功能。"""
import os
import re
import sys
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
sys.stdout.reconfigure(encoding="utf-8")
LINE = "=" * 70
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    REMEMBER_COOKIE_DURATION = __import__("datetime").timede
    lta(days=30)
app = create_app(TestConfig)
client = app.test_client()
# ------------------------------------------------------------
    ------------------
print(LINE)
print("一、「记住我」")
print(LINE)
html = client.get("/login").get_data(as_text=True)
# 1) 复选框必须有可点击关联的 label（这是原先失效的根因）
#    只判断「复选框与文字是否落在同一个 <label> 内」，不绑定具体类名 ——
#    否则换个样式类就会误报，而真正的回归点其实与类名无关。
labels = re.findall(r"<label[^>]*>(.*?)</label>", html,
    re.S)
wrapped = any(('name="remember"' in b and "记住我" in b) for b
    in labels)
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped
    else "[不符合]"))
assert wrapped, "复选框未与文字关联，点击文字不会勾选"
has_for = 'for="remember"' in html
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else
    "[不符合]"))
# 2) 勾选后应下发带过期时间的 remember_token
c = app.test_client()
r = c.post("/login", data={"username": "admin", "password":
    "admin123",
                           "remember": "y"},
    follow_redirects=False)
```

```
cookies = r.headers.getlist("Set-Cookie")
rt = [h for h in cookies if h.startswith("remember_token=")]
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else
    "[不符合]"))
assert rt, "未下发 remember_token"
exp = re.search(r"Expires=([^;]+)", rt[0])
print("  %s 该 cookie 带过期时间：%s"
      % ("[符合]" if exp else "[不符合]", exp.group(1) if exp
    else "无（会话 cookie）"))
assert exp, "remember_token 无过期时间，关浏览器即失效"
# 3) 不勾选时不应下发 remember_token
c2 = app.test_client()
r2 = c2.post("/login", data={"username": "admin",
    "password": "admin123"},
                follow_redirects=False)
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if
    h.startswith("remember_token=")]
print("  %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2
    else "[不符合]"))
assert not rt2
# 4) 携带 remember_token 的新会话应自动登录
c3 = app.test_client()
c3.set_cookie("remember_token",
              rt[0].split(";")[0].split("=", 1)[1],
    domain="localhost")
r3 = c3.get("/", follow_redirects=False)
print("  %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"
      % ("[符合]" if r3.status_code == 200 else "[不符合]",
    r3.status_code))
assert r3.status_code == 200
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("二、删除功能")
print(LINE)
client.post("/login", data={"username": "admin",
    "password": "admin123"})
# 项目：列表页须有删除按钮
html = client.get("/project/").get_data(as_text=True)
has_del = "/delete" in html and "btn--danger" in html
print("  %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))
assert has_del, "项目列表缺少删除按钮"
print("  %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else
    "[不符合]"))
assert "confirm(" in html
# 图斑列表
with app.app_context():
    from app.models import MinePatch, Project
    import seed_demo
```

```
    patch = seed_demo.demo_patch()
    patch_id = patch.id
html = client.get("/project/%d/patches" %
    patch_id).get_data(as_text=True)
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in
    html and "btn--danger" in html else "[不符合]"))
assert "btn--danger" in html
# 微地质单元
html = client.get("/unit/%d" % patch_id).get_data(as_text=Tr
    ue)
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in
    html else "[不符合]"))
assert "/unit/delete/" in html
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))
# 标准
html = client.get("/standard/").get_data(as_text=True)
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html
    else "[不符合]"))
assert "btn--danger" in html
# 实际执行一次删除：新建项目 → 删除 → 确认消失
client.post("/project/add", data={"name": "待删除测试项目",
    "code": "DEL-1"})
with app.app_context():
    from app.models import db, Project
    tmp = Project.query.filter_by(name="待删除测试项目").first()
    tmp_id = tmp.id
html = client.get("/project/").get_data(as_text=True)
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else
    "[不符合]"))
r = client.post("/project/%d/delete" % tmp_id,
    follow_redirects=True)
html = r.get_data(as_text=True)
gone = "待删除测试项目" not in html
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))
assert gone
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else
    "[不符合]"))
print()
print(LINE)
print("界面修复自检全部通过 √")
print(LINE)
# ==== 测试/test_new_features.py ====
# -*- coding: utf-8 -*-
"""补齐模块自检：操作日志与用户维护、遥感影像、指标与多专家矩阵、
批量导入、图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
```

```
import os
import re
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                        Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    # 固定取内置演示案例的图斑，避免受库中用户自建项目 / 验证算例的影响
    import seed_demo
    patch_id = seed_demo.demo_patch().id
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
def ok(flag, label, extra=""):
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    assert flag, label
# ------------------------------------------------------------
    ----------------
```

```
print(LINE)
print("操作日志与用户维护")
print(LINE)
before = None
with app.app_context():
    before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code":
    "LOG-T"})
with app.app_context():
    after = OperationLog.query.count()
    last = OperationLog.query.order_by(OperationLog.id.desc(
    )).first()
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,
    after))
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
ok("project" in (last.target_type or ""), "日志记录了对象类型",
    last.target_type)
r = c.get("/logs")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "操作日志" in body, "操作日志查看页可访问")
ok("新建项目" in body, "日志页显示操作记录")
# 用户编辑 / 删除
with app.app_context():
    u = User(username="tmp_user", role="viewer",
             password_hash=User.query.first().password_hash)
    db.session.add(u); db.session.commit()
    uid = u.id
r = c.get("/users/%d/edit" % uid)
ok(r.status_code == 200, "用户编辑页可访问")
c.post("/users/%d/edit" % uid, data={"role": "evaluator",
    "active": "1"})
with app.app_context():
    ok(db.session.get(User, uid).role == "evaluator",
    "角色已修改并保存")
# 最后一个管理员保护
with app.app_context():
    admin_id = User.query.filter_by(role="admin").first().id
c.post("/users/%d/edit" % admin_id, data={"role": "viewer",
    "active": "1"})
with app.app_context():
    ok(db.session.get(User, admin_id).role == "admin",
    "拒绝把最后一个管理员降级")
c.post("/users/%d/delete" % uid, follow_redirects=True)
with app.app_context():
    ok(db.session.get(User, uid) is None, "用户删除生效")
# 清理日志核对用的项目
with app.app_context():
    for p in Project.query.filter_by(name="日志核对项目").all():
        db.session.delete(p)
    db.session.commit()
```

```
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("遥感影像挂接")
print(LINE)
doc = fitz.open()
pg = doc.new_page(width=320, height=200)
pg.insert_text((20, 60), "RS image", fontname="china-s",
    fontsize=16)
png = doc[0].get_pixmap(dpi=100).tobytes("png")
doc.close()
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(png), "影像.png"),
                 "image_note": "2025年6月无人机正射影像"},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
    rel = db.session.get(MinePatch, patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
    rel.replace("uploads/", "", 1))
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(b"x"), "bad.txt")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
    follow_redirects=True)
with app.app_context():
    ok(db.session.get(MinePatch, patch_id).image_path is
    None, "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("指标定义维护与多专家矩阵")
print(LINE)
r = c.get("/ahp/indicators")
body = r.get_data(as_text=True)
with app.app_context():
    n_ind = Indicator.query.count()
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d
    项" % n_ind)
```

```
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),
    "指标树按层级展示")
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")
c.post("/ahp/indicators/coverage/edit",
       data={"name": "植被覆盖度", "unit": "%", "ideal_type":
    "benefit",
              "ideal_value": "75", "enabled_units": ["岩质边坡"],
              "description": "自检修改"})
with app.app_context():
    ok(Indicator.query.filter_by(code="coverage").first().de
    scription == "自检修改",
       "指标定义可编辑")
c.post("/ahp/indicators/reset")
with app.app_context():
    ok("自检修改" not in (Indicator.query.filter_by(code="covera
    ge").first().description or ""),
       "可按论文取值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
    d = dict(base); d.update({"action": "add", "unit_type":
    "碎石边坡",
                              "level": "1", "expert_name":
    name})
    c.post("/ahp/experts", data=d)
with app.app_context():
    ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡",
    level=1).count() == 3,
       "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
    "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>",
    body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws
    else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) <
    0.01,
    "汇总权重归一", "、".join(vals))
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
    vals[0] if vals else "")
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
```

```
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
    "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度,坡向\n"
    "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,87.92,0.60,28.75,阳坡\n"
    "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,58.42,0.72,,,\n"
    "不存在单元,1,1,7,1,,,,,,,,,,\n"
)
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(csv_text.encode("utf-8")), "d.csv")},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("导入结果" in body, "导入结果页展示")
ok("不存在单元" in body, "错误行被报告")
with app.app_context():
    u = MicroUnit.query.filter_by(patch_id=patch_id,
    unit_type="碎石边坡").first()
    ind = IndicatorData.query.filter_by(unit_id=u.id).first()
    ok(abs(ind.soil_ratio - 0.34) < 1e-6 and
    abs(ind.coverage - 87.92) < 1e-6,
        "导入数据已写入数据库")
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(b"x"), "d.pdf")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("图表与 PDF 导出")
print(LINE)
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,
    "0.8376")])
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,
    "0.4000")])
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")
#  报告是评价结论的呈现，需先真实执行一次评价才能打开
with app.app_context():
    from app.views.evaluate import load_patch_input,
    save_evaluation
    save_evaluation(patch_id, load_patch_input(patch_id))
```

```
r = c.get("/report/%d" % patch_id)
body = r.get_data(as_text=True)
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %
    body.count("<svg"))
exe = browser.browser_path()
if exe is None:
    print("  [跳过] 未找到浏览器，PDF 导出无法验证")
else:
    r = c.get("/report/%d/pdf" % patch_id)
    data = r.data
    ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %
    len(data))
    d = fitz.open(stream=data, filetype="pdf")
    txt = "".join(d[i].get_text() for i in
    range(d.page_count))
    ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")
    ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")
    ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)
    d.close()
print()
print(LINE)
print("补齐模块自检全部通过 √")
print(LINE)
#  ==== 测试/test_workflow.py ====
#  -*- coding: utf-8 -*-
"""流程门禁自检：前置步骤未完成时，后续模块必须拦得住。
论文 4.1 节确定的评价流程存在先后关系，缺任何一环都不应产出评价结论。
本测试逐个构造「缺前置」的场景，验证系统不会放行。
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult)
from app.services import workflow as wf
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
```

```
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
def new_patch(tag):
    with app.app_context():
        p = Project(name="门禁-" + tag, code="WF-" + tag)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="门禁图斑-" +
    tag)
        db.session.add(patch); db.session.flush()
        pid = patch.id
        db.session.commit()
    return pid
def add_units(pid):
    with app.app_context():
        db.session.add(MicroUnit(patch_id=pid,
    unit_type="碎石边坡",
                                 area=1.0, ratio=1.0))
        db.session.commit()
def add_constraint(pid, passed=True):
    with app.app_context():
        db.session.add(ConstraintCheck(
            patch_id=pid, policy_pass=True,
    location_pass=passed,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=passed, reason="全部通过" if passed else
    "未通过：区位约束"))
        db.session.commit()
def add_data(pid):
    with app.app_context():
        u = MicroUnit.query.filter_by(patch_id=pid).first()
        db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.34, thickness=10,
                                     ph=9.24,
    organic_matter=4.89, coverage=87.92,
                                     shannon=0.60,
    slope=28.75, aspect="阳坡"))
        db.session.commit()
def cleanup(pid, tag):
    with app.app_context():
```

```
        from app.models import ScoreDetail
        # 先删得分明细，再删评价记录，避免留下孤儿明细
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delet
    e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(pat
    h_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id
    .in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        pr = Project.query.filter_by(name="门禁-" +
    tag).first()
        if pr:
            db.session.delete(pr)
        db.session.commit()
def results_of(pid):
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ).count()
def post_eval(pid):
    return c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
def get_report(pid):
    return c.get("/report/%d" % pid, follow_redirects=False)
# ----------------------------------------------------------------
    ----------------
print(LINE)
print("一、前置条件缺失时，评价必须被拦住")
print(LINE)
print()
print("场景 1：连微地质单元都没划分")
pid = new_patch("1")
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未划分单元 → 不产生评价结果",
   "HTTP %d，结果数 %d" % (r.status_code, n))
with app.app_context():
    pr = wf.patch_progress(pid)
```

```
steps = [b["step"] for b in pr["blocking"]]
ok(not pr["ready"] and "units" in steps,
   "拦原因包含「划分微地质单元」", "、".join(steps))
ok(steps == ["constraint", "units"],
   "拦截顺序符合论文 4.1 节：限制性判定在前，划分单元在后",
   "、".join(steps))
cleanup(pid, "1")
print()
print("场景 2：有单元，但未做限制性条件判定")
pid = new_patch("2"); add_units(pid); add_data(pid)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未做限制性判定 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "constraint" for b in pr["blocking"]),
   "拦原因指向「执行限制性条件判定」")
cleanup(pid, "2")
print()
print("场景 3：限制性条件判定【未通过】")
pid = new_patch("3"); add_units(pid); add_data(pid);
    add_constraint(pid, passed=False)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "判定未通过 → 不进入定量评价",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(not pr["constraint"]["passed"], "流程状态显示判定未通过")
blocked = [b for b in pr["blocking"] if b["step"] ==
    "constraint"]
# 提示只写「缺什么」：label 说明未通过，reason 直接引用判定记录里没过的项
ok(bool(blocked) and "未通过" in blocked[0]["label"],
   "拦提示指明限制性条件判定未通过", blocked[0]["label"] if blocked else
    "")
ok(bool(blocked) and "区位约束" in (blocked[0]["reason"] or ""),
   "并把具体没通过的条件带出来", blocked[0]["reason"] if blocked else "")
cleanup(pid, "3")
print()
print("场景 4：判定通过，但指标数据全空")
pid = new_patch("4"); add_units(pid); add_constraint(pid,
    passed=True)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "数据未录入 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "data" for b in pr["blocking"]),
```

```
    "拦原因指向「录入指标数据」")
ok(pr["data"]["missing"] == ["碎石边坡"],
    "指出具体缺哪个单元的数据", "、".join(pr["data"]["missing"]))
cleanup(pid, "4")
print()
print("场景 5：单元面积占比合计不为 100%")
pid = new_patch("5"); add_constraint(pid, passed=True)
with app.app_context():
    u = MicroUnit(patch_id=pid, unit_type="碎石边坡", area=1.0,
    ratio=0.5)
    db.session.add(u); db.session.commit()
    db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.3, thickness=10,
                                 ph=8.0,
    organic_matter=5.0, coverage=80.0,
                                 shannon=0.6, slope=20.0,
    aspect="阳坡"))
    db.session.commit()
r = post_eval(pid)
ok(results_of(pid) == 0, "占比不足 100% → 不产生评价结果")
cleanup(pid, "5")
print()
print(LINE)
print("二、前置条件齐备时才放行")
print(LINE)
pid = new_patch("OK"); add_units(pid); add_constraint(pid,
    passed=True); add_data(pid)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(pr["ready"], "单元 + 判定通过 + 数据齐备 → 判定为可执行")
r = post_eval(pid)
n = results_of(pid)
ok(n == 1, "此时执行评价成功产生结果", "结果数 %d" % n)
print("  重定向到:", r.headers.get("Location"))
print()
print(LINE)
print("三、报告不得先于评价产生")
print(LINE)
pid2 = new_patch("R")    # 什么都不做
r = get_report(pid2)
ok(r.status_code == 302 and "/report/" ==
    (r.headers.get("Location") or ""),
    "前置缺失时 /report/<id> 被退回列表", "HTTP %d" % r.status_code)
for suffix in ("/pdf", "/word"):
    rr = c.get("/report/%d%s" % (pid2, suffix),
    follow_redirects=False)
    ok(rr.status_code == 302, "导出 %s 同样被拦住" % suffix, "HTTP
    %d" % rr.status_code)
cleanup(pid2, "R")
cleanup(pid, "OK")
```

```
print()
print(LINE)
print("四、前置齐备但从未执行评价时，报告仍不得产生")
print(LINE)
#  报告是评价结论的呈现：即使前置步骤都做完了，只要没真正执行过定量评价，
#  也不应凭空推算出结论。
pid3 = new_patch("N")
add_units(pid3); add_constraint(pid3, passed=True);
    add_data(pid3)
with app.app_context():
    pr = wf.report_progress(pid3)
ok(not pr["ready"], "无评价记录 → 报告流程判定为未就绪")
ok(bool(pr["blocking"]) and pr["blocking"][0]["step"] ==
    "evaluate",
   "拦原因指向「执行适宜性评价」")
r = get_report(pid3)
ok(r.status_code == 302 and results_of(pid3) == 0,
   "直接访问 /report/<id> 被退回，且不产生任何评价结果",
   "HTTP %d" % r.status_code)
#  真正执行评价之后，报告才放行，且取值与评价记录一致
r = post_eval(pid3)
ok(results_of(pid3) == 1, "先执行评价 → 产生评价记录")
with app.app_context():
    pr = wf.report_progress(pid3)
    rec = pr["result"]
ok(pr["ready"] and rec is not None, "有评价记录后，报告流程就绪")
rr = get_report(pid3)
body = rr.get_data(as_text=True)
ok(rr.status_code == 200, "此时可打开得分明细页", "HTTP %d" %
    rr.status_code)
ok(("%.4f" % rec.final_score) in body,
   "报告数值取自评价记录，而非重新计算",
   "F=%.4f" % rec.final_score)
cleanup(pid3, "N")
print()
print(LINE)
print("五、内置演示数据的清理不得波及用户自建数据")
print(LINE)
#  回归保护：seed_demo.clean_case() 曾执行 Project.query.delete(),
#  导致用户新建的项目在每次启动时被连带删除。
import seed_demo as sd
with app.app_context():
    keep_p = Project(name="用户自建项目-守护", code="KEEP-001")
    db.session.add(keep_p); db.session.flush()
    keep_patch = MinePatch(project_id=keep_p.id,
    name="用户自建图斑-守护")
    db.session.add(keep_patch); db.session.flush()
    keep_pid = keep_p.id
    keep_patch_id = keep_patch.id
    db.session.commit()
```

```
with app.app_context():
    sd.clean_case(reset=False)
with app.app_context():
    still_p = db.session.get(Project, keep_pid)
    still_patch = db.session.get(MinePatch, keep_patch_id)
ok(still_p is not None, "清理演示数据后，用户自建项目仍存在")
ok(still_patch is not None, "清理演示数据后，用户自建图斑仍存在")
#  演示数据清掉后应能重新写入，保证演示环境可恢复
sd.seed()
with app.app_context():
    demo_back = Project.query.filter_by(code=sd.DEMO_CODE).first()
ok(demo_back is not None, "演示案例可重新写入（幂等恢复）")
with app.app_context():
    m = db.session.get(MinePatch, keep_patch_id)
    if m:
        db.session.delete(m)
    p = db.session.get(Project, keep_pid)
    if p:
        db.session.delete(p)
    db.session.commit()
print()
print(LINE)
print("六、同一图斑不得重复划分同类型微地质单元")
print(LINE)
#  论文 4.1 节把图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
#  允许重复会让面积占比、按类型组织的指标数据与批量导入对不上号。
pid4 = new_patch("D")
c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "1.0"},
       follow_redirects=True)
with app.app_context():
    n1 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n1 == 1, "首次添加「平台」成功", "单元数 %d" % n1)
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "2.0"},
           follow_redirects=True)
with app.app_context():
    n2 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n2 == 1, "重复添加同类型被拒绝", "单元数 %d" % n2)
ok("同一类型只能划分一个" in r.get_data(as_text=True), "给出重复提示")
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "火山口",
    "area": "1.0"},
           follow_redirects=True)
with app.app_context():
    n3 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n3 == 1, "非法单元类型被拒绝", "单元数 %d" % n3)
c.post("/unit/%d/add" % pid4, data={"unit_type": "碎石边坡",
    "area": "3.0"},
       follow_redirects=True)
```

```
with app.app_context():
    uid = MicroUnit.query.filter_by(patch_id=pid4,
                                    unit_type="碎石边坡").first(
    ).id
r = c.post("/unit/edit/%d" % uid,
           data={"unit_type": "平台", "area": "3.0"},
    follow_redirects=True)
with app.app_context():
    types = sorted(u.unit_type for u in
                   MicroUnit.query.filter_by(patch_id=pid4)
    all())
ok(types == ["平台", "碎石边坡"], "改类型时同样拦截重复", "、".join(types))
cleanup(pid4, "D")
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("流程门禁自检全部通过 √")
print(LINE)
#  ==== 测试/test_weights.py ====
#  -*- coding: utf-8 -*-
"""权重生效链路自检：AHP 算出的权重必须真正参与评价计算。
此前 weight_schemes / weight_items 两张表从未被写入，评价引擎也只用
代码里的论文表 3 默认权重，相当于权重模块算完即弃。本测试守住这条链路：
    默认 → 论文表 3
    保存 AHP 权重 → 评价改用新权重，结果随之变化
    恢复默认 → 结果回到论文取值
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    WeightScheme,
                        WeightItem)
from app.services import workflow as wf
from app.services import weight_service as ws
LINE = "=" * 72
BASE = ROOT
```

```
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
TAG = "W"
UNIT_TYPE = "碎石边坡"
def build_case():
    """建一个只含单一微地质单元的图斑，便于观察权重变化对得分的影响。"""
    with app.app_context():
        p = Project(name="权重测试-%s" % TAG, code="WEIGHT-%s"
    % TAG)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="权重测试图斑-%s"
    % TAG)
        db.session.add(patch); db.session.flush()
        db.session.add(MicroUnit(patch_id=patch.id,
    unit_type=UNIT_TYPE,
                                 area=1.0, ratio=1.0))
        db.session.flush()
        u = MicroUnit.query.filter_by(patch_id=patch.id).fir
    st()
        db.session.add(IndicatorData(
            unit_id=u.id, soil_ratio=0.34, thickness=10,
    ph=9.24,
            organic_matter=4.89, coverage=87.92,
    shannon=0.60,
            slope=28.75, aspect="阳坡"))
        db.session.add(ConstraintCheck(
            patch_id=patch.id, policy_pass=True,
    location_pass=True,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=True, reason="全部通过"))
        db.session.commit()
        return patch.id
def cleanup(pid):
    with app.app_context():
```

```
        from app.models import ScoreDetail
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delet
    e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(pat
    h_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id
    .in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        p = Project.query.filter_by(code="WEIGHT-%s" %
    TAG).first()
        if p:
            db.session.delete(p)
        db.session.commit()
def run_eval(pid):
    c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ) \
            .order_by(EvaluationResult.id.desc()).first().fi
    nal_score
print(LINE)
print("一、默认应使用论文表 3 的权重")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
    w, label, scheme = ws.effective_weights(UNIT_TYPE)
ok(scheme is None and "论文表" in label, "未配置时权重来源为论文表 3",
    label)
ok(abs(w["l1"]["土壤指标"] - 0.44) < 1e-9, "一级权重取自论文表 3",
    "土壤指标 %.2f" % w["l1"]["土壤指标"])
pid = build_case()
base_score = run_eval(pid)
print("  默认权重下的评价值 F = %.4f" % base_score)
print()
print(LINE)
print("二、保存 AHP 算出的权重后，评价应改用新权重")
```

```
print(LINE)
with app.app_context():
    # 模拟 AHP 计算结果：把土壤指标权重提高、植被与地形降低（仍和为 1）
    ws.save_weights(UNIT_TYPE, [("soil", 0.70), ("veg",
    0.20),
                                ("terrain", 0.10)])
    n_items = WeightItem.query.count()
ok(n_items >= 3, "weight_items 表已写入权重条目", "%d 条" % n_items)
with app.app_context():
    w2, label2, scheme2 = ws.effective_weights(UNIT_TYPE)
ok(scheme2 is not None and "自定义" in label2, "权重来源变为自定义方案",
    label2)
ok(abs(w2["l1"]["土壤指标"] - 0.70) < 1e-9, "新权重已生效",
   "土壤指标 %.2f" % w2["l1"]["土壤指标"])
ok(abs(w2["l1"]["植被指标"] - 0.20) < 1e-9, "同批保存的其他项也已生效")
ok(abs(w2["l2"]["土壤指标"]["土石比"] - 0.25) < 1e-9,
   "未覆盖的二级权重仍保留论文取值",
   "土石比 %.2f" % w2["l2"]["土壤指标"]["土石比"])
new_score = run_eval(pid)
print("  新权重下的评价值 F = %.4f" % new_score)
ok(abs(new_score - base_score) > 1e-4,
   "权重变化确实改变了评价结果",
   "%.4f → %.4f" % (base_score, new_score))
print()
print(LINE)
print("三、结果应记录所用权重来源")
print(LINE)
import json
with app.app_context():
    rec = EvaluationResult.query.filter_by(patch_id=pid) \
        .order_by(EvaluationResult.id.desc()).first()
    stored = json.loads(rec.unit_scores)
ok(any(u.get("weight_source") for u in stored),
   "评价记录中标注了权重来源",
   stored[0].get("weight_source") or "")
print()
print(LINE)
print("四、恢复默认后应回到论文取值")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
    w3, label3, scheme3 = ws.effective_weights(UNIT_TYPE)
ok(scheme3 is None and "论文表" in label3, "已恢复为论文表 3 权重",
    label3)
back_score = run_eval(pid)
print("  恢复后评价值 F = %.4f（默认时 %.4f）" % (back_score,
    base_score))
ok(abs(back_score - base_score) < 1e-9, "评价结果与默认时一致")
print()
print(LINE)
```

```
print("五、权重首页概览")
print(LINE)
with app.app_context():
    rows = ws.summary()
ok(len(rows) == 3, "三种微地质单元均有权重来源说明", "%d 行" % len(rows))
r = c.get("/ahp/")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "当前生效权重" in body, "权重首页正常展示")
print()
print(LINE)
print("六、一致性不通过的矩阵不得保存生效")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
n = 3
# 明显矛盾的矩阵：a12=9, a13=1/9, a23=9 —— CR 必然超限
mats = [(0, 1, "9"), (0, 2, "0.1111"), (1, 2, "9")]
data = {"csrf_token": "", "action": "save"}
for i, j, v in mats:
    data["m_%d_%d" % (i, j)] = v
r = c.post("/ahp/matrix?unit_type=%s&level=1" % UNIT_TYPE,
    data=data,
           follow_redirects=True)
with app.app_context():
    _, label4, scheme4 = ws.effective_weights(UNIT_TYPE)
ok(scheme4 is None, "CR 超限的矩阵未被保存生效", label4)
ok("一致性" in r.get_data(as_text=True), "页面给出了未通过一致性的提示")
cleanup(pid)
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("权重生效链路自检全部通过 √")
print(LINE)
# ==== app/services/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/utils/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/views/__init__.py ====
# -*- coding: utf-8 -*-
```